Dynamical Systems, Ergodic Theory, and Reservoir Computing

An Introduction

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Part I

Dynamical Systems

Chapter 1

Introduction to Dynamical Systems

1.1 What Is a Dynamical System?

A dynamical system is, at its core, a mathematical framework for describing how things change over time. The "things" in question could be populations of organisms, the positions and velocities of planets, the concentrations of chemicals in a reaction, or the state of the atmosphere. What unifies all of these is the idea that the present state of the system determines its future states according to some fixed rule.

Before giving the formal definition, let us fix some intuition. Imagine tracking a point moving through some space. At each moment, the point occupies a definite position. The rule governing its motion tells us, given the current position, where the point will be at any future time. The space of all possible positions is the **state space** (or **phase space**), and the rule of evolution is the **dynamics**.

Definition 1.1 (Dynamical System)**.** A *dynamical system* is a triple (T, X, Φ) where:

1. T is a **time set**, a subset of $\mathbb{R}$ (or $\mathbb{Z}$) that is closed under addition and contains 0; 2. X is a **state space** (or **phase space**), a set equipped with some structure (typically a metric space, a topological space, or a smooth manifold); 3. $\Phi : T \times X \to X$ is an **evolution operator** satisfying: - $\Phi(0, x) = x$ for all $x \in X$ (the identity condition), - $\Phi(t + s, x) = \Phi(t, \Phi(s, x))$ for all $t, s \in T$ and $x \in X$ (the semigroup condition).

The semigroup condition encodes a fundamental property: evolving for time s and then for time t is the same as evolving for time $t + s$. This is sometimes called the **deterministic** property — the evolution depends only on the current state, not on the history of how the system arrived there.

When T is the full real line $\mathbb{R}$ or the full integers $\mathbb{Z}$, the system is **invertible**: we can run it backward as well as forward. In this case, for each fixed t, the map $\Phi(t, \cdot) : X \to X$ is a bijection with inverse $\Phi(-t, \cdot)$, and $\{\Phi(t, \cdot)\}_{t \in T}$ forms a **group** of transformations on X. When $T = [0, \infty)$ or $T = \mathbb{N}_0 = \{0, 1, 2, \ldots\}$, the system may not be invertible, and we have only a **semigroup**.

Remark 1.1. Some authors define a dynamical system more restrictively, requiring X to be a smooth manifold and Φ to be smooth (or at least continuous). Others work in full generality

with measurable spaces. The appropriate level of structure depends on the questions one wishes to ask. We will see all of these flavors in the chapters to come.

1.2 Discrete Time vs. Continuous Time

The time set T in Definition 1.1 comes in two fundamental varieties.

Discrete-time dynamical systems. Here $T = \mathbb{Z}$ (invertible) or $T = \mathbb{N}_0$ (non-invertible). The evolution operator is determined entirely by the single map $f : X \to X$ defined by $f(x) = \Phi(1, x)$. Then

$$\Phi(n, x) = f^n(x) = \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}(x).$$

A discrete-time dynamical system is thus nothing more than a set X and a map $f : X \to X$, together with the rule of iteration. The study of such systems is sometimes called the **theory of iterated maps**.

Discrete-time systems arise naturally when:

- observations or measurements are made at fixed intervals (e.g., census data taken yearly),
- the dynamics are inherently stepwise (e.g., a generation-based population model),
- one reduces a continuous-time system to a discrete one via a Poincaré section (Chapter 4).

Continuous-time dynamical systems. Here $T = \mathbb{R}$ or $T = [0, \infty)$. The evolution is typically specified by an ordinary differential equation (ODE). Given a vector field $F : X \to TX$ on a manifold X (or more concretely, $F : \mathbb{R}^n \to \mathbb{R}^n$), the ODE

$$\frac{dx}{dt} = F(x)$$

determines the evolution by the rule: the trajectory through x_0 is the solution $x(t)$ satisfying $x(0) = x_0$. If solutions exist and are unique for all time, the map $\Phi(t, x_0) = x(t)$ defines a **flow** on X, meaning a one-parameter group (or semigroup) of transformations $\{\varphi_t\}_{t \in T}$ where $\varphi_t(x) = \Phi(t, x)$.

Continuous-time systems arise naturally when modeling physical processes governed by differential laws — Newton's second law, the Navier-Stokes equations, chemical kinetics, and so on.

The relationship between the two. Discrete and continuous dynamics are not separate worlds. They interact in deep ways:

- Given a flow $\{\varphi_t\}$, fixing any $\tau > 0$ and setting $f = \varphi_\tau$ yields a discrete-time system. This is called **time-τ sampling** of the flow.
- Conversely, one might ask whether a given map f can be realized as the time-one map of some flow. This is the **embedding problem**, and the answer is not always yes.
- The Poincaré section technique (Chapter 4) systematically reduces the study of a continuous system to a discrete one, often making the analysis more tractable.

Both settings share much of the same conceptual vocabulary — fixed points, periodic orbits, stability, chaos — though the technical details differ. We develop discrete-time systems in

Chapter 2 and continuous-time systems in Chapter 3.

1.3 State Space, Orbits, and Trajectories

Definition 1.2 (Orbit)**.** Let (T, X, Φ) be a dynamical system. The **orbit** (or **trajectory**) of a point $x \in X$ is the set

$$\mathcal{O}(x) = \{\Phi(t, x) : t \in T\}.$$

When $T = \mathbb{N}_0$ and the dynamics are given by iteration of $f : X \to X$, the **forward orbit** of x is the sequence

$$\mathcal{O}^+(x) = \{x, f(x), f^2(x), f^3(x), \ldots\}.$$

If f is invertible, the **backward orbit** is

$$\mathcal{O}^-(x) = \{x, f^{-1}(x), f^{-2}(x), \ldots\},$$

and the **full orbit** is $\mathcal{O}(x) = \mathcal{O}^+(x) \cup \mathcal{O}^-(x)$.

For continuous-time systems, the orbit of x_0 is the curve $\{x(t) : t \in T\}$ traced out in state space by the solution of the ODE passing through x_0. This curve is called a **phase curve** or **trajectory**.

The state space X is decomposed by the dynamics into a collection of disjoint orbits. Understanding the geometry and topology of this decomposition — how orbits are arranged in phase space — is one of the central goals of dynamical systems theory.

Definition 1.3 (Fixed Point)**.** A point $x^* \in X$ is a **fixed point** (or **equilibrium point**) if $\Phi(t, x^*) = x^*$ for all $t \in T$. For a map f, this means $f(x^*) = x^*$. For a flow generated by $\dot{x} = F(x)$, this means $F(x^*) = 0$: the vector field vanishes at x^*.

Definition 1.4 (Periodic Orbit)**.** A point x has a **periodic orbit** of period $p > 0$ if $\Phi(p, x) = x$ and $\Phi(t, x) \neq x$ for $0 < t < p$. For a map, x is a **periodic point** of (minimal) period n if $f^n(x) = x$ and $f^k(x) \neq x$ for $0 < k < n$. The orbit $\{x, f(x), \ldots, f^{n-1}(x)\}$ is called an **n-cycle**.

Fixed points and periodic orbits are the simplest types of long-term behavior. But as we shall see, dynamical systems can exhibit far more complicated behavior: quasiperiodic orbits, homoclinic tangles, strange attractors, and chaos.

The phase portrait. For low-dimensional systems (especially in $\mathbb{R}^2$), one often studies the dynamics by drawing the **phase portrait**: a picture of the state space showing representative orbits, fixed points, periodic orbits, and the arrows indicating the direction of time. Phase portraits are one of the most powerful tools for building intuition about a system's behavior. We will make extensive use of them in Chapters 2 and 3.

1.4 Motivating Examples

The best way to appreciate the scope and power of dynamical systems theory is through examples. We present five here, ranging from discrete population models to continuous models of celestial mechanics, ecology, and atmospheric dynamics.

1.4.1 The Logistic Map (Discrete Time)

One of the simplest dynamical systems that exhibits astonishingly rich behavior is the **logistic map**:

$$f_r(x) = rx(1-x), \quad x \in [0,1], \quad r \in [0,4].$$

This map was popularized by the biologist Robert May [May, 1976] as a simple model for population dynamics. Here $x_n \in [0,1]$ represents the population at generation n as a fraction of the maximum carrying capacity, and r is a growth rate parameter. The factor $(1-x)$ provides a density-dependent brake on growth: when the population is near capacity ($x \approx 1$), the growth rate drops to zero.

The dynamics are given by iteration:

$$x_{n+1} = rx_n(1-x_n).$$

Fixed points. Setting $f_r(x) = x$, we get $rx(1-x) = x$, which gives $x = 0$ or $x = 1 - 1/r$ (the latter existing in $[0,1]$ only for $r \geq 1$). Let us denote $x^* = 1 - 1/r$.

Stability of fixed points. For a one-dimensional map, a fixed point x^* is (locally) stable if $|f'(x^*)| < 1$ and unstable if $|f'(x^*)| > 1$. We have $f_r'(x) = r(1-2x)$.

At $x = 0$: $f_r'(0) = r$. So $x = 0$ is stable for $r < 1$ and unstable for $r > 1$.

At $x^* = 1 - 1/r$: $f_r'(x^*) = r(1 - 2(1 - 1/r)) = r(2/r - 1) = 2 - r$. So x^* is stable when $|2 - r| < 1$, i.e., when $1 < r < 3$.

What happens for $r > 3$? As r increases past 3, the fixed point x^* loses stability, and a stable 2-cycle is born. As r increases further, this 2-cycle loses stability and a 4-cycle appears, then an 8-cycle, and so on in a cascade of **period-doubling bifurcations**. The parameter values at which these bifurcations occur converge geometrically to a limit $r_\infty \approx 3.56995\ldots$, with the ratio of successive intervals converging to the **Feigenbaum constant** $\delta \approx 4.6692\ldots$, a universal constant that appears in a wide class of one-dimensional maps [Feigenbaum, 1978].

For $r > r_\infty$, the dynamics become **chaotic** for many parameter values: orbits are aperiodic, sensitive to initial conditions, and yet confined to a bounded region. At $r = 4$, the logistic map is conjugate to the **tent map** and to the **doubling map** on the circle, and its dynamics can be analyzed completely using symbolic dynamics.

We will study the logistic map in much greater depth in Chapters 2, 5, and 6.

1.4.2 The Simple Pendulum (Continuous Time)

Consider a rigid pendulum of length ℓ and mass m swinging under gravity g in a plane, with no friction. Let θ denote the angle from the downward vertical. Newton's second law gives

$$m\ell\ddot{\theta} = -mg\sin\theta,$$

or equivalently,

$$\ddot{\theta} = -\frac{g}{\ell}\sin\theta.$$

This is a second-order ODE. To cast it as a dynamical system in the sense of Definition 1.1, we introduce the angular velocity $\omega = \dot{\theta}$ and write the system as two first-order equations:

$$\dot{\theta} = \omega, \qquad \dot{\omega} = -\frac{g}{\ell}\sin\theta.$$

The state space is the **cylinder** $X = S^1 \times \mathbb{R}$ (since θ is an angle, periodic modulo 2π, and ω is unrestricted). The state at any time is the pair (θ, ω), which determines the future evolution.

Fixed points. Setting $\dot{\theta} = 0$ and $\dot{\omega} = 0$, we get $\omega = 0$ and $\sin\theta = 0$, giving two equilibria:

- $(\theta, \omega) = (0, 0)$: the pendulum hanging straight down (stable equilibrium).
- $(\theta, \omega) = (\pi, 0)$: the pendulum balanced upright (unstable equilibrium).

Energy and orbits. The total energy

$$E(\theta, \omega) = \frac{1}{2}\omega^2 - \frac{g}{\ell}\cos\theta$$

is conserved (i.e., $\dot{E} = 0$ along solutions), so orbits lie on level curves of E. These level curves give the phase portrait:

- For $E < g/\ell$, the orbits are **closed curves** surrounding $(0, 0)$: the pendulum oscillates back and forth (librations).
- For $E = g/\ell$, there are special orbits called **separatrices** (or **homoclinic orbits**) that connect the unstable equilibrium $(\pi, 0)$ to itself.
- For $E > g/\ell$, the orbits are curves that wind all the way around the cylinder: the pendulum spins continuously (rotations).

The pendulum is one of the oldest and most studied dynamical systems. Despite its simplicity, it already illustrates several important themes: the role of conserved quantities (energy), the coexistence of qualitatively different orbit types, and the presence of both stable and unstable equilibria.

1.4.3 The Two-Body Problem (Continuous Time)

The motion of two point masses interacting through Newtonian gravitation is one of the great successes of classical mechanics. After reducing to center-of-mass coordinates, the problem reduces to a single body of reduced mass μ moving in a central force field:

$$\ddot{\mathbf{r}} = -\frac{GM}{|\mathbf{r}|^3}\mathbf{r},$$

where $\mathbf{r} \in \mathbb{R}^3$ is the relative position vector, M is the total mass, and G is the gravitational constant.

The state space is (a subset of) $\mathbb{R}^3 \times \mathbb{R}^3$ with coordinates $(\mathbf{r}, \dot{\mathbf{r}})$. Conservation of angular

momentum $\mathbf{L} = \mu \mathbf{r} \times \dot{\mathbf{r}}$ restricts the motion to a plane when $\mathbf{L} \neq 0$, reducing the problem to four dimensions. Conservation of energy further constrains orbits to three-dimensional surfaces in phase space. After exploiting all symmetries, the problem is completely integrable, and the solutions are the **conic sections**: ellipses (bound orbits), parabolas, and hyperbolas (unbound orbits).

Kepler's laws — (1) orbits are ellipses with the sun at one focus, (2) equal areas are swept in equal times, (3) the square of the period is proportional to the cube of the semi-major axis — are consequences of this analysis.

The two-body problem illustrates the power of conserved quantities (energy, angular momentum) in constraining and solving a dynamical system. It is also a cautionary tale: the three-body problem, obtained by adding just one more mass, is **not** integrable in general and exhibits chaotic dynamics. This was Poincaré's great discovery, which launched much of modern dynamical systems theory (Section 1.5).

1.4.4 The Lotka-Volterra Predator-Prey Model (Continuous Time)

In 1925-1926, Alfred Lotka and Vito Volterra independently proposed a simple model for the interaction of a predator species and a prey species:

$$\dot{x} = \alpha x - \beta x y, \qquad \dot{y} = \delta x y - \gamma y,$$

where $x(t) \geq 0$ is the prey population, $y(t) \geq 0$ is the predator population, and $\alpha, \beta, \gamma, \delta > 0$ are parameters representing birth rates, predation rates, and death rates.

The state space is the first quadrant $X = \{(x, y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0\}$.

Fixed points. Setting both equations to zero:

- $(0, 0)$: both species extinct.
- $(x^*, y^*) = (\gamma/\delta,\ \alpha/\beta)$: a coexistence equilibrium.

Analysis at the coexistence equilibrium. The Jacobian of the vector field at (x^*, y^*) is

$$J = \begin{pmatrix} 0 & -\beta x^* \\ \delta y^* & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\beta\gamma/\delta \\ \alpha\delta/\beta & 0 \end{pmatrix}.$$

The eigenvalues are $\lambda = \pm i\sqrt{\alpha\gamma}$, which are purely imaginary. The equilibrium is a **center** (in the linear approximation), and a conserved quantity argument shows it is also a center for the nonlinear system: the function

$$H(x, y) = \delta x - \gamma \ln x + \beta y - \alpha \ln y$$

is constant along solutions (verify: $\dot{H} = 0$). Thus, orbits in the interior of the first quadrant are **closed curves** encircling (x^*, y^*), representing periodic oscillations of the predator and prey populations.

This model, while idealized, captures the essential qualitative feature observed in many predator-prey systems: cyclic oscillations in population sizes, with the predator cycle lagging behind the prey cycle.

1.4.5 The Lorenz System (Continuous Time)

In 1963, the meteorologist Edward Lorenz published a paper that would reshape our understanding of determinism, predictability, and the nature of chaos [Lorenz, 1963]. Lorenz derived a simplified model for atmospheric convection — a system of three coupled ODEs:

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = x(\rho - z) - y, \qquad \dot{z} = xy - \beta z,$$

where σ, ρ, $\beta > 0$ are parameters. The variables x, y, z are related to the intensity of convective motion and the temperature distribution. Lorenz used the values $\sigma = 10$, $\rho = 28$, $\beta = 8/3$, which correspond to a physically relevant regime.

The state space is $X = \mathbb{R}^3$.

Fixed points. Setting $\dot{x} = \dot{y} = \dot{z} = 0$:

From $\dot{x} = 0$: $y = x$.

From $\dot{y} = 0$: $x(\rho - z) - x = 0$, so $x(\rho - 1 - z) = 0$.

From $\dot{z} = 0$: $x^2 = \beta z$ (using $y = x$).

This gives three equilibria:

- The **origin** $(0, 0, 0)$, which represents no convection.
- The points $C^{\pm} = (\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1)$, which exist for $\rho > 1$ and represent steady convective rolls.

For the classical parameter values, all three equilibria are **unstable**. Where, then, do the orbits go? They are attracted to a complicated, fractal-like set now called the **Lorenz attractor** — a **strange attractor** with the following remarkable properties:

1. **Bounded:** All orbits eventually enter and remain within a bounded region of $\mathbb{R}^3$.
2. **Sensitive dependence on initial conditions:** Two orbits starting from nearby points diverge exponentially fast (on average), even though both remain on the attractor.
3. **Aperiodic:** Typical orbits on the attractor never exactly repeat; they are not periodic.
4. **Geometric complexity:** The attractor has a fractal structure with Hausdorff dimension approximately 2.06, meaning it is "almost" a surface but has fine-scale structure at all magnifications.

The Lorenz system was the first concrete, physically motivated example of deterministic chaos. Lorenz's discovery showed that even simple, deterministic systems can be effectively unpredictable in the long run — not because of any randomness in the equations, but because of the exponential amplification of small uncertainties. This has profound implications for weather prediction and for the foundations of classical mechanics more broadly. We will return to the Lorenz system in Chapter 6.

1.5 Historical Context

The theory of dynamical systems has deep roots in classical mechanics, but its development as a distinct mathematical discipline is largely a story of the twentieth century.

Henri Poincaré (1854–1912) is widely regarded as the founder of the qualitative theory of dynamical systems. In his work on the three-body problem in celestial mechanics [Poincaré,

1890], Poincaré introduced a battery of concepts and techniques that remain central to the field: the use of topology and geometry (rather than explicit formulas) to understand the structure of solutions; the Poincaré section and first-return map; the notion of homoclinic orbits and their role in generating complicated dynamics; and the concept of recurrence, which he proved in the context of volume-preserving flows (the Poincaré recurrence theorem, Chapter 4). Poincaré recognized that many dynamical systems cannot be solved in closed form, and that the right questions are qualitative: What is the long-term behavior of typical orbits? What structures (fixed points, periodic orbits, invariant surfaces) organize the phase portrait?

George David Birkhoff (1884–1944) extended Poincaré's work in several directions. His pointwise ergodic theorem [Birkhoff, 1931] is one of the foundational results of ergodic theory, establishing that time averages of observables along orbits converge for almost every initial condition. This theorem provides the rigorous justification for replacing time averages with space averages — a procedure used routinely in statistical mechanics and throughout the physical sciences. We will prove and study this theorem in Chapter 9.

The Soviet school. In the mid-twentieth century, a number of mathematicians in the Soviet Union made fundamental contributions to both dynamical systems and ergodic theory. Andrei Kolmogorov developed the measure-theoretic foundations of probability and introduced the concept of metric entropy (Kolmogorov-Sinai entropy) for dynamical systems. Kolmogorov also proved the first version of the KAM theorem (Chapter 3 of advanced treatments), showing that many quasi-periodic motions in Hamiltonian systems persist under small perturbations. Yakov Sinai carried forward Kolmogorov's work, developing the theory of entropy, establishing deep connections between hyperbolic dynamics and statistical mechanics, and introducing the concept of Sinai-Ruelle-Bowen (SRB) measures.

Stephen Smale (b. 1930) brought the tools of differential topology to dynamical systems in the 1960s. His introduction of the horseshoe map [Smale, 1967] provided a clean geometric mechanism for chaos and showed that chaotic dynamics are structurally stable — they persist under small perturbations of the system. Smale's program of classifying dynamical systems up to topological equivalence (the program of **hyperbolic dynamics**) shaped much of the field for decades.

Edward Lorenz (1917–2008) was not a mathematician but a meteorologist. His 1963 paper, arising from a numerical experiment on a simplified weather model, provided the first concrete example of sensitive dependence on initial conditions in a physically motivated system. Lorenz's work brought deterministic chaos to the attention of scientists across disciplines and fundamentally changed our understanding of predictability. His description of what is now called the **butterfly effect** — the idea that small perturbations can lead to vastly different outcomes — has become one of the most widely known ideas in all of science.

These are only a few of the many contributors. The field has been enriched by the work of Lyapunov (stability theory), Hadamard (geodesic flows on negatively curved surfaces), Kolmogorov, Arnold, and Moser (the KAM theorem), Ruelle and Takens (turbulence and strange attractors), Milnor (one-dimensional dynamics), and many others. Throughout this textbook, we will encounter their ideas repeatedly.

1.6 The Central Questions

Dynamical systems theory organizes itself around a few recurring questions. These questions can be stated simply, but answering them — even for specific, concrete systems — can require the full machinery of analysis, algebra, topology, and measure theory.

Question 1: What happens in the long run?

Given an initial state $x_0 \in X$, what can we say about $\Phi(t, x_0)$ as $t \to \infty$? Does the orbit converge to a fixed point? Does it settle onto a periodic orbit or a more complicated attractor? Does it escape to infinity? The **ω-limit set** of x_0,

$$\omega(x_0) = \bigcap_{N \geq 0} \overline{\{\Phi(t, x_0) : t \geq N\}},$$

captures the set of accumulation points of the orbit as $t \to \infty$. Understanding ω-limit sets — their structure, their dependence on x_0, and how they are organized in phase space — is a central theme.

For the logistic map with $r = 2.5$, every orbit in $(0, 1)$ converges to the fixed point $x^* = 0.6$: the ω-limit set is a single point. For the Lotka-Volterra system, the ω-limit set of a typical orbit is a closed curve. For the Lorenz system, the ω-limit set of a typical orbit is a strange attractor.

Question 2: Are there stable states, and which orbits approach them?

A fixed point or periodic orbit is **stable** (informally) if nearby orbits remain nearby for all future time, and **asymptotically stable** if nearby orbits actually converge to it. More precisely:

Definition 1.5 (Lyapunov Stability)**.** A fixed point x^* is **Lyapunov stable** if for every $\epsilon > 0$ there exists $\delta > 0$ such that $d(x_0, x^*) < \delta$ implies $d(\Phi(t, x_0), x^*) < \epsilon$ for all $t \geq 0$.

Definition 1.6 (Asymptotic Stability)**.** A fixed point x^* is **asymptotically stable** if it is Lyapunov stable and there exists $\delta > 0$ such that $d(x_0, x^*) < \delta$ implies $\Phi(t, x_0) \to x^*$ as $t \to \infty$.

The **basin of attraction** of an asymptotically stable fixed point x^* is the set of all initial conditions whose orbits converge to x^*:

$$\mathcal{B}(x^*) = \{x_0 \in X : \Phi(t, x_0) \to x^* \text{ as } t \to \infty\}.$$

Understanding the basins of attraction — their geometry, their boundaries, and how they change as parameters vary — is crucial in applications, where one often wants to know which initial conditions lead to desirable long-term outcomes.

Question 3: How sensitive is the system to initial conditions?

This question, sharpened by Lorenz's work, is now understood to be one of the most fundamental. Two initial conditions that are very close in state space may produce orbits that diverge exponentially fast. This is quantified by **Lyapunov exponents**:

Definition 1.7 (Lyapunov Exponent, Informal)**.** For a differentiable map $f : \mathbb{R}^n \to \mathbb{R}^n$ and an initial condition x_0, the **maximal Lyapunov exponent** is

$$\lambda(x_0) = \lim_{n \to \infty} \frac{1}{n} \ln \|Df^n(x_0)\|,$$

when this limit exists. Here $Df^n(x_0)$ is the derivative (Jacobian) of the n-th iterate f^n evaluated at x_0.

A positive maximal Lyapunov exponent indicates exponential divergence of nearby orbits — the hallmark of chaos. A negative exponent indicates convergence. The precise definition and properties of Lyapunov exponents will be developed in Chapter 6.

For the logistic map at $r = 4$, a computation (or appeal to the theory of Chapter 6) shows that the Lyapunov exponent is $\lambda = \ln 2 > 0$: the system is chaotic. For $r = 2.5$, orbits converge to a stable fixed point and $\lambda < 0$.

Question 4: How does the behavior change as parameters vary?

Most dynamical systems of interest depend on parameters (the growth rate r in the logistic map, the Rayleigh number ρ in the Lorenz system, etc.). As these parameters change, the qualitative behavior of the system can change dramatically: fixed points can appear, disappear, or exchange stability; periodic orbits can emerge; and chaos can set in. These qualitative changes are called **bifurcations**, and their systematic study is the subject of Chapter 5.

Question 5: What can we say about "typical" or "average" behavior?

Individual orbits of a chaotic system may be unpredictable, but the statistical behavior of an ensemble of orbits can be highly regular. Ergodic theory provides the framework for studying this statistical behavior. The key idea is to equip the state space with a measure (often an invariant measure preserved by the dynamics) and ask about the statistical distribution of orbits with respect to this measure. The ergodic theorems (Chapters 9 and 10) assert that, under appropriate conditions, time averages along individual orbits equal space averages with respect to the invariant measure — giving rigorous meaning to the informal physicist's idea of "typical" or "average" behavior.

1.7 Where This Textbook Is Headed

This chapter has set up the basic language and given a glimpse of the diversity of phenomena that dynamical systems can exhibit. The rest of the textbook develops this material in three parts.

Part I: Dynamical Systems (Chapters 2–6) develops the theory systematically. We begin with discrete-time systems (Chapter 2): iterated maps, fixed points, periodic orbits, stability, and the logistic map as a running example. Chapter 3 treats continuous-time systems: flows, vector fields, phase portraits, and the linearization of nonlinear systems near equilibria. Chapter 4 introduces Poincaré sections, which connect the discrete and continuous theories, and proves the Poincaré recurrence theorem. Chapter 5 studies bifurcations: the mechanisms by which qualitative behavior changes as parameters vary. Chapter 6 tackles chaos head-on: sensitive dependence, Lyapunov exponents, strange attractors, and rigorous definitions of what "chaos" means.

Part II: Ergodic Theory (Chapters 7–12) introduces the measure-theoretic framework for studying the statistical behavior of dynamical systems. After a concise review of measure theory (Chapter 7), we define measure-preserving transformations and invariant measures (Chapter 8). The ergodic theorems of von Neumann and Birkhoff (Chapter 9) are the cor-

nerstones of the theory. Chapters 10 and 11 develop mixing, spectral theory, and entropy, culminating in Chapter 12 with a synthesis connecting ergodic theory back to the dynamical systems of Part I, particularly through SRB measures and the statistical properties of chaotic systems.

Part III: Reservoir Computing (Chapters 13–17) applies the ideas of Parts I and II to a modern topic at the interface of dynamical systems theory and machine learning. Reservoir computing exploits the dynamics of a high-dimensional nonlinear system (the "reservoir") to perform computation, with only a simple readout layer trained by linear regression. The echo state property, which guarantees that the reservoir forgets its initial condition and responds only to its input, is intimately connected to the stability and ergodic-theoretic properties developed in the earlier parts. We will see how Lyapunov exponents, invariant measures, and entropy all play roles in understanding when and why reservoir computers work.

The unifying thread is that dynamical systems are not just objects of mathematical study — they are computational substrates. The same properties that make a system rich enough to exhibit complex, chaotic behavior also make it rich enough to process and represent information. Ergodic theory provides the tools for understanding this connection at a rigorous level.

Exercises

Exercise 1.1. Consider the map $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$.

(a) Find all fixed points of f.

(b) Determine the orbit $\mathcal{O}^+(x_0)$ for $x_0 = 1/2$, $x_0 = 1$, $x_0 = 2$, and $x_0 = -1$. (You may describe the qualitative behavior rather than listing every element.)

(c) For which initial conditions $x_0 \in \mathbb{R}$ does the orbit converge to 0? To ∞? Describe the basins of attraction.

—

Exercise 1.2. Let $R_\alpha : S^1 \to S^1$ be the rotation of the circle by angle α, defined by $R_\alpha(\theta) = \theta + \alpha \pmod{2\pi}$.

(a) Show that R_α satisfies the semigroup property: $R_\alpha^n(\theta) = \theta + n\alpha \pmod{2\pi}$.

(b) Show that every point is a periodic point if and only if $\alpha/(2\pi)$ is rational.

(c) Show that if $\alpha/(2\pi)$ is irrational, then the orbit of every point is dense in S^1. (Hint: show that the orbit is infinite, and use the pigeonhole principle to find orbit points arbitrarily close together, then use the group structure of the circle.)

—

Exercise 1.3. For the logistic map $f_r(x) = rx(1 - x)$:

(a) Verify that f_r maps the interval $[0, 1]$ into itself for $r \in [0, 4]$.

(b) Find the 2-cycles of f_r by solving $f_r(f_r(x)) = x$ and removing the fixed points. Show that 2-cycles exist for $r > 3$.

(c) Show that the 2-cycle found in (b) is stable for $3 < r < 1 + \sqrt{6}$.

—

Exercise 1.4. For the pendulum system $\dot{\theta} = \omega$, $\dot{\omega} = -(g/\ell)\sin\theta$:

(a) Verify that $E(\theta, \omega) = \frac{1}{2}\omega^2 - (g/\ell)\cos\theta$ is a conserved quantity by showing $\dot{E} = 0$ along solutions.

(b) Linearize the system at the equilibrium $(\theta, \omega) = (0, 0)$ and classify the equilibrium of the linearized system. (What are the eigenvalues of the Jacobian?)

(c) Repeat for the equilibrium $(\pi, 0)$.

—

Exercise 1.5. Consider the Lotka-Volterra system

$$\dot{x} = \alpha x - \beta xy, \qquad \dot{y} = \delta xy - \gamma y$$

with $\alpha, \beta, \gamma, \delta > 0$.

(a) Verify that $H(x, y) = \delta x - \gamma \ln x + \beta y - \alpha \ln y$ satisfies $\dot{H} = 0$ along solutions, confirming it is a conserved quantity.

(b) Show that the axes $x = 0$ and $y = 0$ are invariant: if an orbit starts on an axis, it remains on that axis. Describe the dynamics on each axis.

(c) Explain why the existence of a conserved quantity implies that orbits in the interior of the first quadrant cannot converge to the equilibrium $(\gamma/\delta, \alpha/\beta)$.

—

Exercise 1.6. Consider the Lorenz system with parameters $\sigma = 10$, $\rho = 28$, $\beta = 8/3$.

(a) Find all three equilibrium points.

(b) Compute the Jacobian matrix of the vector field at the origin and find its eigenvalues. Is the origin stable or unstable?

(c) Show that the function $V(x, y, z) = x^2 + y^2 + (z - \rho - \sigma)^2$ satisfies $\dot{V} < 0$ outside a sufficiently large ellipsoid in $\mathbb{R}^3$. (This shows that all orbits eventually enter a bounded region. You do not need to find the optimal ellipsoid.)

—

Exercise 1.7. contraction**: there exists $c \in [0, 1)$ such that $d(f(x), f(y)) \leq c \cdot d(x, y)$ for all $x, y \in X$.

(a) Show that f has a unique fixed point x^*.

(b) Show that for every $x_0 \in X$, the orbit $\{f^n(x_0)\}$ converges to x^*.

(c) Show that $d(f^n(x_0), x^*) \leq \frac{c^n}{1-c} d(x_0, f(x_0))$, giving an explicit rate of convergence.

(d) Relate this to the notion of the basin of attraction: what is $\mathcal{B}(x^*)$ for a contraction?

References

- G. D. Birkhoff, "Proof of the ergodic theorem," *Proceedings of the National Academy of Sciences*, 17(12):656–660, 1931.
- R. L. Devaney, *An Introduction to Chaotic Dynamical Systems*, 3rd edition, CRC Press, 2021.
- M. J. Feigenbaum, "Quantitative universality for a class of nonlinear transformations," *Journal of Statistical Physics*, 19(1):25–52, 1978.

- M. W. Hirsch, S. Smale, and R. L. Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*, 3rd edition, Academic Press, 2013.
- A. Katok and B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*, Cambridge University Press, 1995.
- E. N. Lorenz, "Deterministic nonperiodic flow," *Journal of the Atmospheric Sciences*, 20(2):130–141, 1963.
- R. M. May, "Simple mathematical models with very complicated dynamics," *Nature*, 261:459–467, 1976.
- H. Poincaré, "Sur le problème des trois corps et les équations de la dynamique," *Acta Mathematica*, 13:1–270, 1890.
- S. Smale, "Differentiable dynamical systems," *Bulletin of the American Mathematical Society*, 73(6):747–817, 1967.
- S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd edition, Westview Press, 2015.

Recommended Reading

For a first course in dynamical systems, **Strogatz** [2015] is highly recommended: it is clearly written, rich with examples and applications, and assumes minimal prerequisites. **Hirsch, Smale, and Devaney** [2013] provides a more mathematically rigorous treatment of ODEs and dynamical systems, suitable for students with a stronger analysis background. **Devaney** [2021] focuses on discrete dynamical systems and gives a particularly clean treatment of chaos, symbolic dynamics, and the Sarkovskii theorem.

For the more advanced material that we will develop in later chapters, **Katok and Hasselblatt** [1995] is the standard reference for the modern theory of dynamical systems, including ergodic theory, hyperbolic dynamics, and entropy. It is encyclopedic and demanding, but indispensable.

Lorenz's original paper [Lorenz, 1963] is short, readable, and still worth engaging with directly. Smale's survey [Smale, 1967] gives a sweeping overview of the state of dynamical systems theory in the 1960s and remains a landmark of mathematical exposition.

Chapter 2

Discrete Dynamical Systems

In this chapter we study the simplest class of dynamical systems: iteration of a map. Despite the apparent simplicity of the setup — apply a function, then apply it again, then again — the resulting behavior can be extraordinarily rich. We will develop the basic vocabulary (orbits, fixed points, periodic orbits, stability), carry out detailed analysis of the logistic map, introduce the visual tool of cobweb diagrams, and encounter two of the most remarkable theorems in the subject: Sharkovskii's theorem and the Li–Yorke theorem.

2.1 Maps and Iteration

Definition 2.1. Let X be a set. A *discrete dynamical system* on X is a map $f : X \to X$. The *orbit* of a point $x_0 \in X$ under f is the sequence

$$\mathcal{O}(x_0) = \{x_0,\, f(x_0),\, f^2(x_0),\, f^3(x_0),\, \ldots\},$$

where f^n denotes the n-th iterate of f:

$$f^0(x) = x, \qquad f^{n+1}(x) = f(f^n(x)).$$

We write $x_n = f^n(x_0)$ and call x_n the *state at time n*. The set X is the *state space* or *phase space*.

Remark 2.1. The notation f^n always means n-fold composition, never the n-th power. When we need the n-th power of $f(x)$, we write $(f(x))^n$ or $[f(x)]^n$.

Throughout this chapter, unless stated otherwise, we take $X \subseteq \mathbb{R}$ and assume f is continuous. Many definitions extend to higher dimensions and more general spaces, but one-dimensional maps already contain most of the essential phenomena.

Example 2.2 (Linear map). Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = ax$ for some constant $a \in \mathbb{R}$. Then $f^n(x_0) = a^n x_0$. The orbit of any $x_0 \neq 0$ satisfies:

- If $|a| < 1$: $x_n \to 0$ as $n \to \infty$.
- If $|a| > 1$: $|x_n| \to \infty$ as $n \to \infty$.
- If $a = 1$: every point is fixed. If $a = -1$: every nonzero orbit has period 2.

Simple as it is, this example already illustrates the central question: *given f and x_0, what is the long-term behavior of the orbit?*

Example 2.3 (Rotation on the circle). Let $X = \mathbb{R}/\mathbb{Z}$ (the unit circle, identified with $[0, 1)$) and $f(x) = x + \alpha \pmod 1$ for some $\alpha \in \mathbb{R}$. If $\alpha = p/q$ is rational (in lowest terms), then every orbit is periodic with period q. If α is irrational, then every orbit is dense in $[0, 1)$ — a fact we will prove in Chapter 9 using ergodic theory. This dichotomy between rational and irrational rotation is a prototype for many phenomena in dynamics.

2.2 Fixed Points and Periodic Orbits

Definition 2.2. A point $x^* \in X$ is a *fixed point* of f if $f(x^*) = x^*$.

Definition 2.3. A point x^* is a *periodic point* of f with *period* n (or is n-periodic) if $f^n(x^*) = x^*$ and n is the smallest positive integer with this property. The set $\{x^*, f(x^*), \ldots, f^{n-1}(x^*)\}$ is the *periodic orbit* (or *cycle*) of x^*.

A fixed point is a periodic point of period 1. If $f^n(x^*) = x^*$ but n is not necessarily minimal, we say x^* is an *eventually periodic point of period dividing n*, or simply that x^* is a *point of period n* (with the understanding that the minimal period may be a proper divisor of n). In this text we will be careful to distinguish between the two conventions when it matters.

Observation. The periodic points of period n are exactly the fixed points of f^n. This is important: to find period-n orbits, solve $f^n(x) = x$.

Example 2.1. Consider $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 - 1$. The fixed points satisfy $x^2 - 1 = x$, i.e., $x^2 - x - 1 = 0$, giving

$$x^* = \frac{1 \pm \sqrt{5}}{2}.$$

For period-2 orbits, we solve $f^2(x) = x$. We have $f^2(x) = f(x^2-1) = (x^2-1)^2-1 = x^4-2x^2$. Setting $f^2(x) = x$:

$$x^4 - 2x^2 - x = 0 \implies x(x^3 - 2x - 1) = 0.$$

Since fixed points of f are also fixed points of f^2, the polynomial x^2-x-1 divides x^3-2x-1. Performing the division:

$$x^3 - 2x - 1 = (x^2 - x - 1)(x + 1).$$

So the solutions of $f^2(x) = x$ that are *not* fixed points of f are $x = 0$ and $x = -1$. Check: $f(0) = -1$ and $f(-1) = 0$. Thus $\{0, -1\}$ is a period-2 orbit.

2.3 Stability of Fixed Points

Not all fixed points are created equal. Some attract nearby orbits; others repel them. Making this precise requires the derivative.

Definition 2.4. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable at a fixed point x^*. The quantity $\lambda = f'(x^*)$ is called the *multiplier* (or *eigenvalue*) of the fixed point. We say x^* is:

- *attracting* (or *stable*) if $|\lambda| < 1$, - *repelling* (or *unstable*) if $|\lambda| > 1$, - *neutral* (or *indifferent*) if $|\lambda| = 1$.

The neutral case requires further analysis and will not be our main focus.

2.3.1 Why the Multiplier Determines Stability

The intuition is straightforward. Near x^*, the map f is approximately linear:

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) = x^* + \lambda(x - x^*).$$

Writing $\varepsilon_n = x_n - x^*$ for the displacement from the fixed point, we have $\varepsilon_{n+1} \approx \lambda\,\varepsilon_n$, so $\varepsilon_n \approx \lambda^n \varepsilon_0$. If $|\lambda| < 1$ then $\varepsilon_n \to 0$; if $|\lambda| > 1$ then $|\varepsilon_n| \to \infty$. We now make this rigorous.

Theorem 2.8 (Stability of fixed points). Let $f : \mathbb{R} \to \mathbb{R}$ be C^1 in a neighborhood of a fixed point x^*.

(a) If $|f'(x^*)| < 1$, then x^* is asymptotically stable: there exists $\delta > 0$ such that $|x_0 - x^*| < \delta$ implies $f^n(x_0) \to x^*$ as $n \to \infty$.

(b) If $|f'(x^*)| > 1$, then x^* is unstable: there exists $\delta > 0$ such that for any x_0 with $0 < |x_0 - x^*| < \delta$, the orbit eventually leaves the interval $(x^* - \delta, x^* + \delta)$.

Proof of (a). of (a).* Since f' is continuous and $|f'(x^*)| < 1$, there exists μ with $|f'(x^*)| < \mu < 1$ and an interval $I = (x^* - \delta, x^* + \delta)$ such that $|f'(x)| \le \mu$ for all $x \in I$.

Let $x_0 \in I$. By the mean value theorem, there exists c between x_0 and x^* such that

$$|f(x_0) - x^*| = |f(x_0) - f(x^*)| = |f'(c)| \cdot |x_0 - x^*| \le \mu\,|x_0 - x^*| < |x_0 - x^*| < \delta.$$

So $f(x_0) \in I$, and by induction $f^n(x_0) \in I$ for all $n \ge 0$. Applying the estimate inductively:

$$|f^n(x_0) - x^*| \le \mu^n |x_0 - x^*|.$$

Since $0 < \mu < 1$, we have $\mu^n \to 0$, so $f^n(x_0) \to x^*$. Moreover, convergence is geometric (exponential) with rate μ. □

Proof of (b). of (b).* Since $|f'(x^*)| > 1$, choose μ with $1 < \mu < |f'(x^*)|$ and $\delta > 0$ such that $|f'(x)| \ge \mu$ for all $x \in I = (x^* - \delta, x^* + \delta)$. By the mean value theorem, for any $x \in I$ with $x \ne x^*$:

$$|f(x) - x^*| = |f'(c)| \cdot |x - x^*| \ge \mu\,|x - x^*| > |x - x^*|.$$

Thus the distance from x^* increases by at least a factor of $\mu > 1$ at each step, as long as the orbit remains in I. Since the interval is bounded, the orbit must eventually leave I. □

Remark 2.2. The proof of part (a) is essentially a contraction mapping argument: f restricted to I is a contraction with Lipschitz constant $\mu < 1$, and x^* is the unique fixed point of this

contraction.

2.3.2 Stability of Periodic Orbits

For a periodic orbit of period n, the relevant quantity is the multiplier of the corresponding fixed point of f^n. By the chain rule, if $\{x_0, x_1, \ldots, x_{n-1}\}$ is a period-n orbit, then

$$(f^n)'(x_0) = f'(x_{n-1}) \cdot f'(x_{n-2}) \cdots f'(x_1) \cdot f'(x_0) = \prod_{k=0}^{n-1} f'(x_k).$$

Note that this product is the same regardless of which point x_k in the cycle we start from (the factors are cyclically permuted). So *stability is a property of the entire orbit, not of individual points on it.*

Definition 2.5. A period-n orbit is attracting if $|(f^n)'(x_0)| < 1$, repelling if $|(f^n)'(x_0)| > 1$, and neutral if $|(f^n)'(x_0)| = 1$, where x_0 is any point on the orbit.

2.4 The Logistic Map

The logistic map

$$f_r(x) = rx(1-x), \qquad r \in [0,4], \quad x \in [0,1],$$

is a one-parameter family of maps that serves as the principal example in one-dimensional dynamics. It arises as a simple model of population growth with density-dependent limitation (the factor $(1-x)$ represents the effect of limited resources when the population fraction x is near its carrying capacity). Despite its elementary form — a single quadratic polynomial — the logistic map exhibits the full range of dynamical behaviors, from simple convergence to a fixed point, through period-doubling bifurcations, to chaos.

We restrict to $r \in [0,4]$ because this is the range for which f_r maps $[0,1]$ into itself. (The maximum of f_r on $[0,1]$ occurs at $x = 1/2$ with value $r/4$, so $f_r([0,1]) \subseteq [0,1]$ if and only if $r \leq 4$.)

2.4.1 Fixed Points

Setting $f_r(x) = x$:

$$rx(1-x) = x \implies x[r(1-x)-1] = 0 \implies x[r-1-rx] = 0.$$

This gives two fixed points:

$$x_0^* = 0, \qquad x_1^* = 1 - \frac{1}{r} = \frac{r-1}{r}.$$

The second fixed point x_1^* lies in $(0,1)$ only when $r > 1$. At $r = 1$, the two fixed points collide (both equal 0), and for $r < 1$, the fixed point x_1^* is negative and therefore outside $[0,1]$.

2.4.2 Stability of the Fixed Points

The derivative is $f_r'(x) = r(1-2x)$.

At $x_0^* = 0$: The multiplier is $f_r'(0) = r$.

- For $0 < r < 1$: $|f_r'(0)| = r < 1$, so $x_0^* = 0$ is attracting. (The population dies out.)
- For $r > 1$: $|f_r'(0)| = r > 1$, so $x_0^* = 0$ is repelling.
- At $r = 1$: neutral.

At $x_1^* = (r-1)/r$: The multiplier is

$$f_r'\left(\frac{r-1}{r}\right) = r\left(1 - \frac{2(r-1)}{r}\right) = r \cdot \frac{r-2r+2}{r} = 2 - r.$$

- For $1 < r < 3$: $|2-r| < 1$, so x_1^* is attracting. (The population converges to a nonzero equilibrium.)
- For $r > 3$: $|2-r| = r-2 > 1$, so x_1^* is repelling.
- At $r = 3$: $f_r'(x_1^*) = -1$ (neutral — a *period-doubling bifurcation* occurs here).

Summary so far:

| Parameter range | Attracting fixed point | |—|—| | $0 < r < 1$ | $x_0^* = 0$ | | $1 < r < 3$ | $x_1^* = (r-1)/r$ | | $r > 3$ | Neither fixed point is attracting |

2.4.3 The Period-2 Orbit

For $r > 3$, the fixed point x_1^* loses stability and a period-2 orbit is born. To find it, we solve $f_r^2(x) = x$ and factor out the fixed points.

Compute $f_r^2(x) = f_r(f_r(x)) = r \cdot [rx(1-x)] \cdot [1 - rx(1-x)]$. Setting $f_r^2(x) = x$:

$$r^2 x(1-x)[1 - rx(1-x)] = x.$$

The solutions include the fixed points $x = 0$ and $x = (r-1)/r$. Dividing $f_r^2(x) - x$ by $x(x - (r-1)/r)$, we can find the remaining factor. After algebra (which the reader should verify), the period-2 points satisfy the quadratic equation

$$r^2x^2 - r(r+1)x + (r+1) = 0.$$

By the quadratic formula:

$$x = \frac{r(r+1) \pm r\sqrt{(r+1)(r-3)}}{2r^2} = \frac{(r+1) \pm \sqrt{(r+1)(r-3)}}{2r}.$$

These are real and distinct when $r > 3$ (since $(r+1)(r-3) > 0$). Denote the two period-2 points by

$$p = \frac{(r+1) + \sqrt{(r+1)(r-3)}}{2r}, \qquad q = \frac{(r+1) - \sqrt{(r+1)(r-3)}}{2r}.$$

One verifies that $f_r(p) = q$ and $f_r(q) = p$, so $\{p, q\}$ is indeed a 2-cycle.

Stability of the 2-cycle. The multiplier is

$$(f_r^2)'(p) = f_r'(p) \cdot f_r'(q).$$

Using $f_r'(x) = r(1-2x)$:

$$f_r'(p) \cdot f_r'(q) = r^2(1-2p)(1-2q).$$

From the quadratic $r^2x^2 - r(r+1)x + (r+1) = 0$, the sum and product of roots are:

$$p + q = \frac{r+1}{r}, \qquad pq = \frac{r+1}{r^2}.$$

Therefore:

$$(1-2p)(1-2q) = 1 - 2(p+q) + 4pq = 1 - \frac{2(r+1)}{r} + \frac{4(r+1)}{r^2} = 1 - \frac{2(r+1)}{r} + \frac{4(r+1)}{r^2}.$$

Simplifying with a common denominator of r^2:

$$= \frac{r^2 - 2r(r+1) + 4(r+1)}{r^2} = \frac{r^2 - 2r^2 - 2r + 4r + 4}{r^2} = \frac{-r^2 + 2r + 4}{r^2}.$$

So the multiplier of the 2-cycle is

$$(f_r^2)'(p) = r^2 \cdot \frac{-r^2 + 2r + 4}{r^2} = -r^2 + 2r + 4 = -(r^2 - 2r - 4).$$

The 2-cycle is attracting when $|-r^2 + 2r + 4| < 1$.
At $r = 3$: the multiplier equals $-9 + 6 + 4 = 1$. (The 2-cycle is born with multiplier $+1$.)
The 2-cycle loses stability when the multiplier reaches -1:

$$-r^2 + 2r + 4 = -1 \implies r^2 - 2r - 5 = 0 \implies r = 1 + \sqrt{6} \approx 3.449.$$

So the 2-cycle is attracting for $3 < r < 1 + \sqrt{6}$.
Numerical check at $r = 3.2$:

$$p = \frac{4.2 + \sqrt{4.2 \cdot 0.2}}{6.4} = \frac{4.2 + \sqrt{0.84}}{6.4} \approx \frac{4.2 + 0.9165}{6.4} \approx 0.7995.$$

$$q \approx \frac{4.2 - 0.9165}{6.4} \approx 0.5130.$$

Check: $f_{3.2}(0.7995) = 3.2 \cdot 0.7995 \cdot 0.2005 \approx 0.5130$. Correct.

The multiplier: $-(3.2)^2 + 2(3.2) + 4 = -10.24 + 6.4 + 4 = 0.16$. Since $|0.16| < 1$, the 2-cycle is attracting, as expected.

2.4.4 Preview of the Period-Doubling Cascade

At $r = 1 + \sqrt{6} \approx 3.449$, the period-2 orbit loses stability and gives birth to a period-4 orbit, which is attracting in a further interval of r-values. This period-4 orbit in turn loses stability and creates a period-8 orbit, and so on. This sequence of *period-doubling bifurcations* produces attracting orbits of periods $1, 2, 4, 8, 16, \ldots$ at parameter values $r_1 = 3$, $r_2 = 1 + \sqrt{6}$, r_3, $r_4, \ldots$ The sequence $\{r_n\}$ converges to a value $r_\infty \approx 3.5699$, and the ratios of successive intervals converge to a universal constant:

$$\lim_{n\to\infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n} = \delta \approx 4.6692\ldots,$$

the *Feigenbaum constant*. Remarkably, this constant is the same for *every* smooth unimodal map undergoing period-doubling — it is a universal quantity, independent of the specific form of f. We will study bifurcations systematically in Chapter 5.

For $r > r_\infty$, the behavior is much more complex: there are infinitely many parameter values with periodic orbits of every period, interspersed with parameter values where the dynamics is chaotic (in a sense we will make precise in Chapter 6). The *bifurcation diagram* — plotting the attractor as a function of r — reveals an intricate structure that repays careful study.

2.5 Cobweb Diagrams

A *cobweb diagram* (also called a *staircase diagram*) is a graphical method for visualizing the orbit of a one-dimensional map. It translates the algebraic iteration $x_{n+1} = f(x_n)$ into a geometric construction.

2.5.1 Construction

To construct a cobweb diagram for $f : \mathbb{R} \to \mathbb{R}$ starting from x_0:

1. Plot the graph $y = f(x)$ and the diagonal $y = x$ in the same coordinate system.
2. Start at the point $(x_0, 0)$ on the x-axis. Draw a vertical line up (or down) to the curve $y = f(x)$, reaching the point $(x_0, f(x_0)) = (x_0, x_1)$.
3. Draw a horizontal line from (x_0, x_1) to the diagonal $y = x$, reaching the point (x_1, x_1). This "reflects" the output x_1 to the input axis.
4. Repeat: draw a vertical line to the curve, reaching $(x_1, f(x_1)) = (x_1, x_2)$, then horizontal to the diagonal, reaching (x_2, x_2). Continue.

The resulting zigzag pattern between the curve and the diagonal traces out the orbit.

2.5.2 Interpretation

The geometry of the cobweb diagram directly reveals the dynamics:

- **Attracting fixed point with $0 < f'(x^*) < 1$:** The cobweb spirals inward in a "staircase" pattern, stepping monotonically toward the fixed point (where the curve and diagonal intersect).
- **Attracting fixed point with $-1 < f'(x^*) < 0$:** The cobweb spirals inward, but now it oscillates — the iterates alternate between sides of x^*, converging in a spiral.
- **Repelling fixed point:** The cobweb moves outward, away from the intersection.
- **Period-2 orbit:** The cobweb forms a rectangle (approximately) cycling between the two periodic points.

2.5.3 Worked Example: Logistic Map with $r = 2.8$

Consider $f(x) = 2.8\,x(1 - x)$ with $x_0 = 0.1$.

The fixed point $x^* = 1 - 1/2.8 = 9/14 \approx 0.6429$, with multiplier $2 - 2.8 = -0.8$. Since $|-0.8| < 1$, the fixed point is attracting and the cobweb should spiral inward.

Computing the first several iterates:

$$
\begin{aligned}
x_0 &= 0.1 \\
x_1 &= 2.8 \cdot 0.1 \cdot 0.9 = 0.2520 \\
x_2 &= 2.8 \cdot 0.2520 \cdot 0.7480 \approx 0.5278 \\
x_3 &= 2.8 \cdot 0.5278 \cdot 0.4722 \approx 0.6980 \\
x_4 &= 2.8 \cdot 0.6980 \cdot 0.3020 \approx 0.5902 \\
x_5 &= 2.8 \cdot 0.5902 \cdot 0.4098 \approx 0.6773 \\
x_6 &\approx 0.6116 \\
x_7 &\approx 0.6651
\end{aligned}
$$

The iterates oscillate around $x^* \approx 0.6429$ with decreasing amplitude, consistent with the multiplier -0.8 (negative sign causes oscillation, magnitude less than 1 causes convergence). In the cobweb diagram, the vertical and horizontal line segments form a spiral that converges to the intersection of $y = 2.8\,x(1-x)$ and $y = x$.

2.5.4 Worked Example: Logistic Map with $r = 3.2$

Now $f(x) = 3.2\,x(1-x)$. The fixed point $x^* = (r-1)/r = 2.2/3.2 = 0.6875$ has multiplier $2 - 3.2 = -1.2$. Since $|-1.2| > 1$, the fixed point is repelling, and orbits near it are pushed toward the attracting 2-cycle.

Starting from $x_0 = 0.5$:

$$
\begin{aligned}
x_0 &= 0.5000 \\
x_1 &= 0.8000 \\
x_2 &= 0.5120 \\
x_3 &= 0.7995 \\
x_4 &= 0.5131 \\
x_5 &= 0.7994
\end{aligned}
$$

The orbit quickly settles into alternation between approximately 0.5130 and 0.7995 — the period-2 cycle we computed in Section 2.4.3. In the cobweb diagram, the pattern is a rectangle whose corners lie on the curve and the diagonal, cycling between the two periodic points.

2.6 Topological Conjugacy

When are two dynamical systems "the same"? The right notion of equivalence in dynamics is *topological conjugacy*: two systems are conjugate if there is a continuous invertible change of coordinates that transforms one into the other.

Definition 2.6. Let $f : X \to X$ and $g : Y \to Y$ be continuous maps on topological spaces X and Y. We say f and g are *topologically conjugate* if there exists a homeomorphism $h : X \to Y$ such that

$$h \circ f = g \circ h,$$

or equivalently, $g = h \circ f \circ h^{-1}$. The map h is called a *conjugacy.*

The condition $h \circ f = g \circ h$ means that the following diagram commutes:

$$\begin{array}{c} X\& \xrightarrow{f} \&X \\ \downarrow h\&\& \downarrow h \\ Y\& \xrightarrow{g} \&Y \end{array}$$

Proposition 2.1. *If h is a conjugacy between f and g, then for all $n \geq 1$:*

$$h \circ f^n = g^n \circ h.$$

Proof. * By induction. The base case $n = 1$ is the definition. Assuming $h \circ f^n = g^n \circ h$, then $h \circ f^{n+1} = h \circ f \circ f^n = g \circ h \circ f^n = g \circ g^n \circ h = g^{n+1} \circ h$. □

Corollary 2.12. Topological conjugacy preserves:

- Fixed points: x^* is a fixed point of f if and only if $h(x^*)$ is a fixed point of g.
- Periodic orbits: x^* has period n under f if and only if $h(x^*)$ has period n under g.
- The topology of orbits: dense orbits map to dense orbits, etc.

Proof. * If $f(x^*) = x^*$, then $g(h(x^*)) = h(f(x^*)) = h(x^*)$, so $h(x^*)$ is fixed under g. The period-n statement follows from Proposition 2.11. Density is preserved because h is a homeomorphism. □

Conjugacy is an equivalence relation on dynamical systems. Two conjugate systems have identical orbit structures — they differ only by a continuous change of coordinates. From the viewpoint of dynamics, they are the same system.

□

□

Definition 2.7. If the map h in the above definition is only required to be a continuous surjection (rather than a homeomorphism), we say f is *semiconjugate* to g, and h is a *semiconjugacy.* Semiconjugacy is a weaker relationship: it means g is a "factor" of f (information about orbits of f projects down to orbits of g, but not necessarily vice versa).

2.6.1 Example: The Doubling Map and the Tent Map

The *doubling map* $D : [0, 1) \to [0, 1)$ is defined by

$$D(x) = 2x \pmod 1 = \begin{cases} 2x & \text{if } 0 \leq x < 1/2, \\ 2x - 1 & \text{if } 1/2 \leq x < 1. \end{cases}$$

The *tent map* $T : [0, 1] \to [0, 1]$ is defined by

$$T(x) = \begin{cases} 2x & \text{if } 0 \le x \le 1/2, \\ 2 - 2x & \text{if } 1/2 < x \le 1. \end{cases}$$

These two maps are *not* topologically conjugate (the tent map is continuous on $[0, 1]$ while the doubling map has a discontinuity, and moreover their orbit structures differ in detail). However, there is an important semiconjugacy from the doubling map to a system on symbolic sequences, and the tent map is conjugate to the logistic map at $r = 4$ via an explicit change of coordinates.

2.6.2 Example: Conjugacy Between the Tent Map and the Logistic Map at $r = 4$

Claim. The logistic map $f_4(x) = 4x(1 - x)$ on $[0, 1]$ is topologically conjugate to the tent map T on $[0, 1]$ via the conjugacy

$$h(x) = \frac{2}{\pi} \arcsin(\sqrt{x}).$$

To verify this, we need $h \circ f_4 = T \circ h$, i.e., for all $x \in [0, 1]$:

$$\frac{2}{\pi} \arcsin\left(\sqrt{4x(1 - x)}\right) = T\left(\frac{2}{\pi} \arcsin(\sqrt{x})\right).$$

Let $\theta = \arcsin(\sqrt{x})$, so $x = \sin^2 \theta$ with $\theta \in [0, \pi/2]$. Then:

Left side:

$$\frac{2}{\pi} \arcsin\left(\sqrt{4 \sin^2 \theta \cos^2 \theta}\right) = \frac{2}{\pi} \arcsin(|\sin(2\theta)|) = \frac{2}{\pi} \arcsin(\sin(2\theta)),$$

where the last equality uses $\sin(2\theta) \ge 0$ for $\theta \in [0, \pi/2]$.

Right side: We have $h(x) = (2/\pi)\theta$, so:

- If $\theta \le \pi/4$ (i.e., $h(x) \le 1/2$): $T(h(x)) = 2 \cdot (2\theta/\pi) = 4\theta/\pi$.
- If $\theta > \pi/4$ (i.e., $h(x) > 1/2$): $T(h(x)) = 2 - 4\theta/\pi$.

Checking the left side:

- If $\theta \le \pi/4$: $\arcsin(\sin(2\theta)) = 2\theta$ (since $2\theta \le \pi/2$), so the left side is $4\theta/\pi$. Matches.
- If $\theta > \pi/4$: $\arcsin(\sin(2\theta)) = \pi - 2\theta$ (since $2\theta > \pi/2$), so the left side is $(2/\pi)(\pi - 2\theta) = 2 - 4\theta/\pi$. Matches.

Thus the conjugacy relation holds. The map h is a homeomorphism of $[0, 1]$ (it is continuous, strictly increasing, with $h(0) = 0$ and $h(1) = 1$), so this is a genuine topological conjugacy.

This conjugacy has a profound consequence: every dynamical property of the tent map at full slope transfers to the logistic map at $r = 4$. In particular, the logistic map at $r = 4$ has a dense set of periodic orbits and is topologically transitive — properties we will revisit when we discuss chaos in Chapter 6.

2.7 Sharkovskii's Theorem

One of the most striking results in one-dimensional dynamics concerns which periods *must* coexist. Suppose a continuous map $f : \mathbb{R} \to \mathbb{R}$ has a periodic orbit of some period n. Does it necessarily have periodic orbits of other periods? The answer is given by Sharkovskii's theorem, which provides a complete ordering.

Definition 2.14 (Sharkovskii ordering). Define the following total order on the positive integers:

$$3 \rhd 5 \rhd 7 \rhd 9 \rhd \cdots$$

$$\rhd 2 \cdot 3 \rhd 2 \cdot 5 \rhd 2 \cdot 7 \rhd \cdots$$

$$\rhd 4 \cdot 3 \rhd 4 \cdot 5 \rhd 4 \cdot 7 \rhd \cdots$$

$$\rhd 8 \cdot 3 \rhd 8 \cdot 5 \rhd 8 \cdot 7 \rhd \cdots$$

$$\rhd \cdots \rhd 2^n \cdot 3 \rhd 2^n \cdot 5 \rhd 2^n \cdot 7 \rhd \cdots$$

$$\rhd \cdots \rhd 2^3 \rhd 2^2 \rhd 2 \rhd 1.$$

In words: first all odd numbers ≥ 3, then 2 times all odd numbers ≥ 3, then 4 times all odd numbers ≥ 3, and so on, followed by all powers of 2 in decreasing order $(\ldots, 16, 8, 4, 2, 1)$.

Theorem 2.15 (Sharkovskii, 1964). Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. If f has a periodic orbit of period m, then f has a periodic orbit of period n for every n with $m \rhd n$ in the Sharkovskii ordering.

Moreover, for each m, there exists a continuous map having a periodic orbit of period m but no periodic orbit of any period n with $n \rhd m$.

The theorem says that the set of periods that a continuous map can have is always a "tail" (a downward-closed set) in the Sharkovskii ordering. The number 3 sits at the top of the ordering, so having a period-3 orbit forces orbits of every period. The powers of 2 sit at the bottom: a map can have orbits of periods $1, 2, 4, 8, 16, \ldots$ without having an orbit of period 3 (or 5, or 6, etc.). The logistic map at r values in the period-doubling cascade provides exactly such examples.

Significance. Sharkovskii's theorem is remarkable for several reasons:

1. It applies to *every* continuous map $\mathbb{R} \to \mathbb{R}$ (or $I \to I$ for a closed interval I), with no smoothness assumptions.
2. The ordering is far from obvious. Why should period 3 force period 5? Why should period 6 force period 4 but not conversely? The proof, while elementary (it uses only the intermediate value theorem), requires a delicate combinatorial argument tracking how intervals map over each other.
3. It connects to the onset of chaos: the appearance of a period-3 orbit signals the presence of all other periods.

The original proof appears in Sharkovskii (1964). Accessible English-language proofs can be found in Devaney (1989, Chapter 1.11) and in Block and Coppel (1992). The theorem holds for continuous self-maps of an interval; it does *not* hold for maps of the circle, for discontinuous

maps, or in dimensions higher than one.

2.8 The Li–Yorke Theorem: "Period Three Implies Chaos"

The special case of Sharkovskii's theorem that received the most attention is due to Li and Yorke (1975), published a decade after Sharkovskii's original work (which was not widely known outside the Soviet Union at the time).

Theorem 2.16 (Li–Yorke, 1975). Let $f : [a, b] \to [a, b]$ be continuous. If f has a periodic point of period 3, then:

1. For every positive integer n, f has a periodic point of period n.
2. There exists an uncountable set $S \subseteq [a, b]$ (called a *scrambled set*) such that for every pair $x, y \in S$ with $x \neq y$:

$$\limsup_{n\to\infty} |f^n(x) - f^n(y)| > 0 \qquad \text{and} \qquad \liminf_{n\to\infty} |f^n(x) - f^n(y)| = 0,$$

and for every periodic point p of f and every $x \in S$:

$$\limsup_{n\to\infty} |f^n(x) - f^n(p)| > 0.$$

Part (1) follows immediately from Sharkovskii's theorem (since 3 is first in the Sharkovskii ordering). The real novelty of Li and Yorke's paper is part (2): the existence of an uncountable scrambled set. This means there is a large set of initial conditions whose orbits are neither eventually periodic nor convergent to each other — they are perpetually drawn together and pushed apart, never settling into a regular pattern. This is a rigorous (though relatively weak) form of chaos.

The meaning of "chaos" here. The Li–Yorke definition of chaos (existence of an uncountable scrambled set) is one of several competing definitions. It is weaker than the definition due to Devaney (which additionally requires topological transitivity and density of periodic orbits). A system can be Li–Yorke chaotic without exhibiting the strong mixing behavior one might expect from "chaos" in the colloquial sense. Nevertheless, the title of their paper — "Period Three Implies Chaos" — became one of the most famous slogans in mathematics, and did much to bring the study of chaos to a broad audience.

A concrete illustration. The logistic map $f_r(x) = rx(1 - x)$ has a period-3 orbit for $r = 1 + 2\sqrt{2} \approx 3.8284$ (this is the value at which a period-3 window opens in the bifurcation diagram). For this and all larger values of r at which a 3-cycle exists, the Li–Yorke theorem guarantees periodic orbits of every period and an uncountable scrambled set.

To see a period-3 orbit concretely, consider $r \approx 3.8319$. Numerical iteration from $x_0 = 0.5$ yields an orbit that settles into the approximate cycle $0.1561 \to 0.5047 \to 0.9563 \to 0.1561 \to \cdots$. This orbit visits three distinct values cyclically, confirming the presence of a 3-cycle.

2.9 Bibliographical Notes

The study of iteration of maps goes back at least to Cayley and Schröder in the 19th century, but the modern theory was shaped by the work of Fatou and Julia (in the complex setting), Sharkovskii (1964), and Li and Yorke (1975). The logistic map was popularized as a model of population dynamics by Robert May (1976), whose influential paper in *Nature* brought the richness of nonlinear dynamics to the attention of scientists across disciplines.

The universality of the Feigenbaum constant was discovered independently by Feigenbaum (1978) and Coullet and Tresser (1978). Rigorous proofs of universality came later, notably by Lanford (1982) using computer-assisted methods.

The conjugacy between the logistic map at $r = 4$ and the tent map is a classical result; see Devaney (1989, Section 1.7) for a textbook treatment. The connection to symbolic dynamics (which we will develop in later chapters) provides a powerful framework for understanding such conjugacies.

Recommended Reading

- **R. L. Devaney**, *An Introduction to Chaotic Dynamical Systems*, 3rd edition, CRC Press, 2021. The standard undergraduate text on discrete dynamical systems. Chapters 1–4 and 11–12 cover the material of this chapter in greater depth. Contains a proof of Sharkovskii's theorem.
- **S. H. Strogatz**, *Nonlinear Dynamics and Chaos*, 3rd edition, CRC Press, 2024. An outstanding textbook emphasizing continuous systems but with excellent chapters on one-dimensional maps (Chapter 10) and the logistic map in particular. More applied in flavor.
- **K. T. Alligood, T. D. Sauer, and J. A. Yorke**, *Chaos: An Introduction to Dynamical Systems*, Springer, 1996. A well-balanced text covering both discrete and continuous systems with many computational exercises. The treatment of the logistic map and period-doubling (Chapters 1–3 and 12) is particularly thorough.
- **A. N. Sharkovskii**, "Co-existence of cycles of a continuous mapping of the line into itself," *Ukrainian Mathematical Journal* **16** (1964), 61–71. The original paper proving Theorem 2.15 (in Ukrainian; English translations are available).
- **T.-Y. Li and J. A. Yorke**, "Period three implies chaos," *American Mathematical Monthly* **82** (1975), 985–992. The paper that introduced the term "chaos" to mathematics and proved Theorem 2.16. Remarkably short and readable.
- **R. May**, "Simple mathematical models with very complicated dynamics," *Nature* **261** (1976), 459–467. A landmark expository paper on the logistic map as a model in population biology.

Exercises

Exercise 2.1. Let $f(x) = \cos(x)$. Show that f has a unique fixed point $x^* \in [0, 1]$, and determine whether it is attracting or repelling. Starting from $x_0 = 0$, compute the first 10

iterates numerically and verify your stability conclusion.

Hint: The fixed point satisfies $\cos(x^*) = x^*$. This equation has a unique solution $x^* \approx 0.7391$.

Exercise 2.2. Consider the logistic map $f_r(x) = rx(1-x)$.

(a) Verify that for $r = 2$, the fixed point $x^* = 1/2$ is attracting by computing the multiplier.

(b) Sketch a cobweb diagram (or describe the cobweb geometry) for $r = 2$, $x_0 = 0.1$. Is the convergence to x^* monotone or oscillatory?

(c) For $r = 3.5$, compute the first 20 iterates starting from $x_0 = 0.4$. What is the eventual behavior? What is the period of the attracting cycle?

Exercise 2.3. Let $f(x) = x^2$.

(a) Find all fixed points and determine their stability.

(b) Find all period-2 orbits. (*Hint:* Solve $f^2(x) = x$ and remove the fixed points.)

(c) Describe the orbit of x_0 for each of the following cases: $x_0 \in (0,1)$; $x_0 = 1$; $x_0 > 1$; $x_0 \in (-1,0)$; $x_0 = -1$; $x_0 < -1$.

Exercise 2.4. Let $f(x) = 4x(1-x)$ and $g(x) = 1-2x^2$, both considered as maps on appropriate intervals.

(a) Find a linear change of variables $h(x) = ax + b$ such that $h \circ f = g \circ h$, where f maps $[0,1]$ to itself and g maps $[-1,1]$ to itself. (*Hint:* The conjugacy should map $[0,1]$ onto $[-1,1]$, so try $h(x) = 1-2x$ or $h(x) = 2x-1$.)

(b) Verify the conjugacy relation explicitly by computing both sides for general $x \in [0,1]$.

(c) Use this conjugacy to transfer the period-3 orbit of f at $r = 4$ (if one exists) to g. What are the coordinates of the corresponding period-3 orbit of g?

Exercise 2.5. Consider the Sharkovskii ordering $\triangleright$.

(a) List the first 20 elements of the Sharkovskii ordering (i.e., the 20 "largest" positive integers in this order).

(b) Where does the number 12 fall in the ordering? List five periods that must exist if a continuous map has a period-12 orbit, and five periods that are *not* guaranteed by having a period-12 orbit.

(c) A continuous map $f : [0,1] \to [0,1]$ has periodic orbits of periods $1, 2, 4$, and 8, but no others. Is this consistent with Sharkovskii's theorem? Why or why not?

Exercise 2.6. Let $f(x) = rx(1-x)$ with $r = 3.84$. At this parameter value, the logistic map has a period-3 orbit. Find the three points of this orbit numerically (to at least four decimal places) and compute the multiplier $(f^3)'$ at one of them using the chain rule formula from Section 2.3.2. Is the 3-cycle attracting or repelling?

Exercise 2.7. Define $f : [0,1] \to [0,1]$ by

$$f(x) = \begin{cases} 2x + 1/2 \& \text{if } 0 \le x < 1/4, \\ 3/2 - 2x \& \text{if } 1/4 \le x \le 1/2, \\ f(1-x) \& \text{if } 1/2 < x \le 1. \end{cases}$$

(a) Sketch the graph of f and verify that it is a continuous map from $[0, 1]$ to $[0, 1]$.

(b) Find all fixed points of f.

(c) Show that f has a periodic orbit of period 2. Does Sharkovskii's theorem guarantee a periodic orbit of period 3?

(d) Does f have a periodic orbit of period 4? Justify your answer.

Exercise 2.8. Let $f : [a, b] \to [a, b]$ be a continuous map with a fixed point x^*, and suppose f is differentiable on (a, b) with $|f'(x)| \leq L$ for all $x \in (a, b)$, where $0 < L < 1$. Prove that x^* is the *unique* fixed point of f in $[a, b]$, and that $f^n(x_0) \to x^*$ for every $x_0 \in [a, b]$.

Hint: This is the Banach fixed-point theorem (contraction mapping theorem) applied to the complete metric space $[a, b]$. Prove it directly using the mean value theorem.

Chapter 3

Continuous Dynamical Systems

In Chapter 2, we studied discrete dynamical systems generated by iterating a map $T : X \to X$. We now turn to the continuous-time setting, where the state of a system evolves according to an ordinary differential equation (ODE). This is the natural framework for modeling physical systems governed by Newton's laws, population dynamics in ecology, chemical reaction kinetics, and electrical circuits, among many others.

The shift from discrete to continuous time is more than a change of notation. New phenomena arise—limit cycles, for instance, have no direct analogue in one-dimensional discrete systems—and certain phenomena disappear: we shall see that chaos is impossible for flows on the plane, a striking contrast with the rich dynamics of one-dimensional maps.

3.1 ODEs as Dynamical Systems

3.1.1 Setup and Terminology

An **autonomous ordinary differential equation** on $\mathbb{R}^n$ is a system of the form

$$\dot{x} = f(x), \tag{3.1}$$

where $x \in \mathbb{R}^n$, $\dot{x} = dx/dt$, and $f : U \to \mathbb{R}^n$ is a vector field defined on some open set $U \subseteq \mathbb{R}^n$. The word *autonomous* means that f does not depend explicitly on time; the law governing the evolution depends only on the current state.

A **solution** (or **trajectory**, or **integral curve**) of (3.1) with initial condition $x(0) = x_0$ is a differentiable function $x : I \to \mathbb{R}^n$, defined on some interval I containing 0, such that $x'(t) = f(x(t))$ for all $t \in I$ and $x(0) = x_0$.

A point $x^* \in U$ is called a **fixed point** (or **equilibrium**, or **critical point**) of the system if $f(x^*) = 0$. At such a point, the constant function $x(t) = x^*$ for all t is a solution.

3.1.2 The Flow

When (3.1) has a unique solution for each initial condition x_0, we can define the **flow** of the system. Write $\phi_t(x_0)$ for the value at time t of the solution starting at x_0. This gives a family of maps

$$\phi_t : U \to \mathbb{R}^n, \quad t \in \mathbb{R},$$

satisfying the following properties:

1. **Identity:** $\phi_0(x) = x$ for all $x \in U$.
2. **Group property:** $\phi_{t+s}(x) = \phi_t(\phi_s(x))$ for all s, t for which both sides are defined.
3. **Smoothness:** If f is C^k, then $\phi_t(x)$ is C^k in both t and x.

The group property deserves emphasis. It says that flowing for time s and then for time t is the same as flowing for time $t + s$. In the language of algebra, $\{\phi_t\}_{t\in\mathbb{R}}$ is a one-parameter group of transformations of U. This is precisely the continuous analogue of iterating a map: in a discrete system, $T^{n+m} = T^n \circ T^m$.

Proof of the group property. of the group property.** Fix $x_0 \in U$ and $s \in \mathbb{R}$. Define two functions of t:

- $\alpha(t) = \phi_{t+s}(x_0)$,
- $\beta(t) = \phi_t(\phi_s(x_0))$.

Both satisfy the ODE $\dot{y} = f(y)$ (for α, differentiate with respect to t and use the definition of flow; for β, this holds by definition). At $t = 0$, both equal $\phi_s(x_0)$. By uniqueness of solutions, $\alpha(t) = \beta(t)$ for all t. □

Note the essential role of uniqueness here. Without it, the flow is not well-defined, and the entire framework breaks down.

3.1.3 Continuous vs. Discrete

It is worth pausing to compare the two settings.

| | Discrete: $x_{n+1} = T(x_n)$ | Continuous: $\dot{x} = f(x)$ | |—|—|—| | Time | $\mathbb{Z}$ or $\mathbb{N}$ | $\mathbb{R}$ | | Evolution operator | T^n (iterate) | ϕ_t (flow) | | Group property | $T^{n+m} = T^n \circ T^m$ | $\phi_{t+s} = \phi_t \circ \phi_s$ | | Orbits | Sequences $\{x_n\}$ | Curves $\{x(t)\}$ |

One can always pass from continuous to discrete by **sampling** the flow: set $T = \phi_\tau$ for some fixed time step $\tau > 0$. The map T is then called the **time-τ map** of the flow. This construction will reappear when we discuss Poincaré sections and, later, in the context of reservoir computing.

3.2 Existence and Uniqueness

3.2.1 The Picard–Lindelöf Theorem

The fundamental theorem guaranteeing that solutions exist and are unique is the following.

Theorem 3.1 (Picard–Lindelöf). *Let $U \subseteq \mathbb{R}^n$ be open, and let $f : U \to \mathbb{R}^n$ be continuous. Suppose f is **locally Lipschitz** on U: for every compact set $K \subset U$, there exists a constant $L > 0$ such that*

$$\|f(x) - f(y)\| \leq L\|x - y\| \quad \text{for all } x, y \in K.$$

Then for every $x_0 \in U$, there exists an interval $(-\varepsilon, \varepsilon)$ with $\varepsilon > 0$ and a unique solution $x : (-\varepsilon, \varepsilon) \to U$ of $\dot{x} = f(x)$ with $x(0) = x_0$.

Moreover, this solution can be extended to a **maximal interval of existence** $(t_{\min}, t_{\max})$ with $-\infty \leq t_{\min} < 0 < t_{\max} \leq +\infty$, and if $t_{\max} < \infty$, then the solution eventually leaves every compact subset of U as $t \to t_{\max}^-$ (and similarly for $t_{\min}$).

Sufficient condition. If f is C^1 (i.e., continuously differentiable), then f is locally Lipschitz. This follows immediately from the mean value theorem. In practice, the vector fields one encounters in applications are almost always C^1 or better, so the Picard–Lindelöf theorem applies.

Proof sketch. sketch.** The standard proof proceeds by converting the ODE to an integral equation

$$x(t) = x_0 + \int_0^t f(x(s))\, ds$$

and showing that the map $\mathcal{T}$ defined by $(\mathcal{T}x)(t) = x_0 + \int_0^t f(x(s))\, ds$ is a contraction on an appropriate function space (continuous functions on $[-\varepsilon, \varepsilon]$ with values in a ball around x_0, equipped with the supremum norm). The Banach fixed-point theorem then gives existence and uniqueness of a fixed point, which is the desired solution.

In detail: choose $r > 0$ so that the closed ball $\overline{B}_r(x_0) \subset U$, and let $M = \sup_{\overline{B}_r(x_0)} \|f\|$, L the Lipschitz constant on $\overline{B}_r(x_0)$. Set $\varepsilon = \min(r/M, 1/(2L))$. On the space $C([-\varepsilon, \varepsilon], \overline{B}_r(x_0))$ with the supremum norm, one checks:

1. $\mathcal{T}$ maps this space to itself (because $\|\mathcal{T}x(t) - x_0\| \leq M\varepsilon \leq r$).
2. $\|\mathcal{T}x - \mathcal{T}y\|_\infty \leq L\varepsilon \|x - y\|_\infty \leq \frac{1}{2}\|x - y\|_\infty$.

By the Banach fixed-point theorem, $\mathcal{T}$ has a unique fixed point. □

3.2.2 Failure of Uniqueness

What happens when f is continuous but not Lipschitz? Consider the scalar ODE

$$\dot{x} = x^{2/3}, \quad x(0) = 0. \tag{3.2}$$

The vector field $f(x) = x^{2/3}$ is continuous on $\mathbb{R}$ but is *not* Lipschitz at $x = 0$: for $x > 0$, $f'(x) = \frac{2}{3}x^{-1/3} \to \infty$ as $x \to 0^+$.

One can verify by direct substitution that for any $c \geq 0$, the function

$$x_c(t) = \begin{cases} 0 & \text{if } t \leq c, \\ \left(\frac{t-c}{3}\right)^3 & \text{if } t > c, \end{cases}$$

is a solution of (3.2) with $x_c(0) = 0$. (Check: for $t > c$, $\dot{x}_c = 3 \cdot \frac{1}{27}(t-c)^2 = \frac{(t-c)^2}{9}$, while $x_c^{2/3} = \left(\frac{(t-c)^3}{27}\right)^{2/3} = \frac{(t-c)^2}{9}$.) The trivial solution $x(t) \equiv 0$ is also valid. So uniqueness fails spectacularly: there are uncountably many solutions through the origin.

3.2.3 The Peano Existence Theorem

Even without the Lipschitz condition, continuity alone guarantees existence (though not uniqueness).

Theorem 3.2 (Peano). *Let $f : U \to \mathbb{R}^n$ be continuous on an open set $U \subseteq \mathbb{R}^n$. Then for every $x_0 \in U$, there exists $\varepsilon > 0$ and a solution $x : [0, \varepsilon) \to U$ of $\dot{x} = f(x)$ with $x(0) = x_0$.*

The proof uses the Arzelà–Ascoli theorem and is more delicate than the Picard–Lindelöf proof. Since we will always assume at least Lipschitz regularity in what follows, we state Peano's theorem without proof and refer the reader to Coddington and Levinson (1955) for details.

3.3 Linear Systems

3.3.1 The Matrix Exponential

The simplest class of autonomous ODEs is the **linear** case:

$$\dot{x} = Ax, \tag{3.3}$$

where A is a real $n \times n$ matrix and $x \in \mathbb{R}^n$. These systems can be solved exactly.

Definition 3.1. The **matrix exponential** of $A \in \mathbb{R}^{n\times n}$ is

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

This series converges absolutely for every A (with respect to any matrix norm), since $\|A^k/k!\| \leq \|A\|^k/k!$ and $\sum \|A\|^k/k! = e^{\|A\|} < \infty$.

Proposition 3.1. **The solution of $\dot{x} = Ax$, $x(0) = x_0$ is**

$$x(t) = e^{At}x_0.$$

Proof. * Differentiate term by term:

$$\frac{d}{dt}e^{At} = \frac{d}{dt}\sum_{k=0}^{\infty} \frac{(At)^k}{k!} = \sum_{k=1}^{\infty} \frac{kA^k t^{k-1}}{k!} = A\sum_{k=1}^{\infty} \frac{(At)^{k-1}}{(k-1)!} = Ae^{At}.$$

Term-by-term differentiation is justified by the uniform convergence of the series on compact intervals. So $x(t) = e^{At}x_0$ satisfies $\dot{x} = Ae^{At}x_0 = Ax(t)$, and $x(0) = e^0x_0 = Ix_0 = x_0$. □

The flow of the linear system is therefore $\phi_t(x_0) = e^{At}x_0$. Key properties of the matrix exponential include:

- $e^0 = I$.
- $e^{(t+s)A} = e^{tA}e^{sA}$ (confirming the group property).
- If $AB = BA$, then $e^{A+B} = e^Ae^B$. **Caution:** this fails in general when A and B do not commute.

- $\det(e^A) = e^{\mathrm{tr}(A)}$ (Jacobi's formula). In particular, e^A is always invertible.
- If P is invertible, then $e^{PAP^{-1}} = Pe^AP^{-1}$.

The last property is crucial: it reduces computation of e^{At} to the case where A is in a convenient canonical form.

3.3.2 Computing the Matrix Exponential via Jordan Form

Every real matrix A can be brought (over $\mathbb{C}$) to **Jordan normal form**: there exists an invertible matrix P such that $A = PJP^{-1}$, where J is block-diagonal with Jordan blocks

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix} \in \mathbb{C}^{k\times k}.$$

Since $e^{PJP^{-1}t} = Pe^{Jt}P^{-1}$ and e^{Jt} is block-diagonal with blocks $e^{J_k(\lambda)t}$, we need only compute

$$e^{J_k(\lambda)t} = e^{\lambda t} \begin{pmatrix} 1 & t & t^2/2! & \cdots & t^{k-1}/(k-1)! \\ 0 & 1 & t & \cdots & t^{k-2}/(k-2)! \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & t \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix}. \tag{3.4}$$

This follows by writing $J_k(\lambda) = \lambda I + N$ where N is the nilpotent part (the superdiagonal of ones), noting that λI and N commute, so $e^{J_k(\lambda)t} = e^{\lambda t}e^{Nt}$, and computing e^{Nt} using the fact that $N^k = 0$.

For a diagonalizable matrix (when all Jordan blocks are 1×1), this simplifies: if $A = P\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)P^{-1}$, then $e^{At} = P\,\mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t})P^{-1}$.

3.3.3 Classification of Linear Flows in 2D

Consider $\dot{x} = Ax$ with $A \in \mathbb{R}^{2\times 2}$. The qualitative behavior of the flow is determined entirely by the eigenvalues of A. Let $\tau = \mathrm{tr}(A)$ and $\delta = \det(A)$. The eigenvalues satisfy $\lambda^2 - \tau\lambda + \delta = 0$, so $\lambda = \frac{1}{2}(\tau \pm \sqrt{\tau^2 - 4\delta})$. We classify:

Case 1: Two distinct real eigenvalues λ_1, λ_2 ($\tau^2 - 4\delta > 0$).

After a change of basis, the system becomes $\dot{y}_1 = \lambda_1 y_1$, $\dot{y}_2 = \lambda_2 y_2$, with solutions $y_1(t) = c_1 e^{\lambda_1 t}$, $y_2(t) = c_2 e^{\lambda_2 t}$. Trajectories satisfy $y_2 = C|y_1|^{\lambda_2/\lambda_1}$ for some constant C.

- **Stable node** ($\lambda_1 < \lambda_2 < 0$): All trajectories approach the origin as $t \to \infty$. They are tangent to the "slow" eigendirection (corresponding to the eigenvalue closer to zero, i.e., λ_2) as they approach. The phase portrait looks like a sink, with all arrows pointing inward.
- **Unstable node** ($0 < \lambda_1 < \lambda_2$): Time-reversed stable node. All trajectories escape to infinity. Tangent to the slow eigendirection near the origin.

- **Saddle point** ($\lambda_1 < 0 < \lambda_2$): The eigendirection for λ_1 is the **stable manifold** W^s—trajectories starting on it approach the origin. The eigendirection for λ_2 is the **unstable manifold** W^u—trajectories starting on it escape the origin. All other trajectories approach the unstable manifold as $t \to +\infty$ and the stable manifold as $t \to -\infty$. The phase portrait has a characteristic "X" shape.

Case 2: Complex eigenvalues $\lambda = \alpha \pm i\beta$ with $\beta \neq 0$ ($\tau^2 - 4\delta < 0$).

Here $\alpha = \tau/2$ and $\beta = \sqrt{4\delta - \tau^2}/2$. After a real change of basis, the system becomes

$$\dot{y}_1 = \alpha y_1 - \beta y_2, \qquad \dot{y}_2 = \beta y_1 + \alpha y_2.$$

Switching to polar coordinates $r = \sqrt{y_1^2 + y_2^2}$, $\theta = \arctan(y_2/y_1)$:

$$\dot{r} = \alpha r, \qquad \dot{\theta} = \beta.$$

So $r(t) = r_0 e^{\alpha t}$ and $\theta(t) = \theta_0 + \beta t$. Trajectories are logarithmic spirals (when $\alpha \neq 0$) or circles (when $\alpha = 0$).

- **Stable spiral (focus)** ($\alpha < 0$): Trajectories spiral inward toward the origin.
- **Unstable spiral (focus)** ($\alpha > 0$): Trajectories spiral outward.
- **Center** ($\alpha = 0$, i.e., $\tau = 0$): Trajectories are closed ellipses (circles in the eigenbasis). The origin is Lyapunov stable but not asymptotically stable.

Case 3: Repeated eigenvalue $\lambda = \tau/2$ ($\tau^2 - 4\delta = 0$).

- If A is diagonalizable (i.e., $A = \lambda I$), then $e^{At} = e^{\lambda t} I$ and trajectories are rays from the origin. This is called a **star node**.
- If A is not diagonalizable, the Jordan form is $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$, giving $e^{Jt} = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. Trajectories approach the origin tangent to the eigendirection. This is called a **degenerate (improper) node**.

Case 4: Degenerate cases ($\delta = 0$).

When $\det(A) = 0$, at least one eigenvalue is zero, and the origin is not an isolated fixed point. Instead, there is an entire line (or plane) of fixed points. We exclude this case from the classification since the origin is not an isolated equilibrium.

Summary. The classification is governed by the (τ, δ)-plane:

- The parabola $\tau^2 = 4\delta$ separates nodes from spirals/centers.
- The δ-axis ($\tau = 0$, $\delta > 0$) corresponds to centers.
- The region $\delta < 0$ corresponds to saddle points.
- Stability is determined by the sign of τ: stable if $\tau < 0$, unstable if $\tau > 0$.

Worked Example 3.5. Consider $A = \begin{pmatrix} -1 & 2 \\ 0 & -3 \end{pmatrix}$. The eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = -3$ (read off the diagonal since A is upper-triangular). Both are negative, so the origin is a **stable node**.

The eigenvectors: for $\lambda_1 = -1$, $(A + I)v = 0$ gives $\begin{pmatrix} 0 & 2 \\ 0 & -2 \end{pmatrix} v = 0$, so $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. For $\lambda_2 = -3$, $(A + 3I)v = 0$ gives $\begin{pmatrix} 2 & 2 \\ 0 & 0 \end{pmatrix} v = 0$, so $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

The general solution is $x(t) = c_1 e^{-t}\begin{pmatrix}1\\0\end{pmatrix} + c_2 e^{-3t}\begin{pmatrix}1\\-1\end{pmatrix}$. As $t \to \infty$, the e^{-3t} term decays faster, so trajectories become tangent to the v_1 direction (the slow eigendirection).

Worked Example 3.6. Consider $A = \begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}$. The eigenvalues are $\lambda = \pm i$ (here $\tau = 0$, $\delta = 1$). The origin is a **center**. The solutions are $x_1(t) = c_1 \cos t - c_2 \sin t$, $x_2(t) = c_1 \sin t + c_2 \cos t$—uniform rotation. Trajectories are circles centered at the origin.

3.4 Phase Portraits in 2D

3.4.1 Nullclines

For a two-dimensional system

$$\dot{x}_1 = f_1(x_1, x_2), \qquad \dot{x}_2 = f_2(x_1, x_2),$$

the **x_1-nullcline** is the set $\{(x_1, x_2) : f_1(x_1, x_2) = 0\}$ and the **x_2-nullcline** is $\{(x_1, x_2) : f_2(x_1, x_2) = 0\}$.

On the x_1-nullcline, $\dot{x}_1 = 0$, so the flow is purely vertical. On the x_2-nullcline, $\dot{x}_2 = 0$, so the flow is purely horizontal. Fixed points lie at intersections of the two nullclines. Between the nullclines, the signs of $\dot{x}_1$ and $\dot{x}_2$ are constant, which determines the general direction of the flow in each region.

This gives a systematic method for sketching phase portraits:

1. Find the nullclines.
2. Mark their intersections (fixed points).
3. Determine the signs of $\dot{x}_1$ and $\dot{x}_2$ in each region.
4. Indicate the flow direction on the nullclines (vertical on x_1-nullclines, horizontal on x_2-nullclines).
5. Sketch representative trajectories consistent with this information.

3.4.2 Worked Example: The Simple Pendulum

The equation of motion for a frictionless pendulum of unit length and unit mass under gravity $g = 1$ is

$$\ddot{\theta} + \sin\theta = 0. \tag{3.5}$$

We convert this to a first-order system by setting $x_1 = \theta$ (angle) and $x_2 = \dot{\theta}$ (angular velocity):

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = -\sin x_1. \tag{3.6}$$

Fixed points. Setting both right-hand sides to zero: $x_2 = 0$ and $\sin x_1 = 0$, so $x_1 = k\pi$ for $k \in \mathbb{Z}$. The physically distinct fixed points are $(0, 0)$ (pendulum hanging down) and $(\pi, 0)$ (pendulum balanced upright).

Nullclines. The x_1-nullcline is $x_2 = 0$ (the horizontal axis). The x_2-nullcline is $\sin x_1 = 0$, i.e., $x_1 = k\pi$ (vertical lines).

Linearization at $(0,0)$. The Jacobian (see Section 3.5) is

$$Df(0,0) = \begin{pmatrix} 0 & 1 \\ -\cos 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Eigenvalues: $\lambda = \pm i$. The origin is a **center** of the linearized system.

Linearization at $(\pi,0)$. The Jacobian is

$$Df(\pi,0) = \begin{pmatrix} 0 & 1 \\ -\cos \pi & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Eigenvalues: $\lambda = \pm 1$. This is a **saddle point**.

Energy and the global phase portrait. The system (3.6) is Hamiltonian with energy

$$E(\theta, \dot\theta) = \frac{1}{2}\dot\theta^2 - \cos\theta = \frac{1}{2}x_2^2 - \cos x_1.$$

One verifies that $\frac{d}{dt}E = x_2\dot x_2 + \sin(x_1)\dot x_1 = x_2(-\sin x_1) + \sin(x_1)\cdot x_2 = 0$, so E is conserved along trajectories. The level curves of E are therefore exactly the trajectories.

At the saddle $(\pi,0)$: $E(\pi,0) = 0 + 1 = 1$. The level set $E = 1$ consists of the **separatrices** that connect the saddle points—these are called **heteroclinic orbits** (connecting $(-\pi,0)$ to $(\pi,0)$ and vice versa).

- For $E < 1$ near $(0,0)$: closed orbits (oscillations, the pendulum swings back and forth).
- For $E > 1$: open orbits (rotations, the pendulum goes over the top).
- $E = 1$: the separatrices.

This phase portrait is one of the canonical examples in dynamics. The coexistence of a center, saddle points, and the separatrix structure connecting them illustrates the richness that even simple mechanical systems exhibit.

3.4.3 Worked Example: Lotka–Volterra Predator-Prey

The classical predator-prey model is

$$\dot x = \alpha x - \beta xy, \qquad \dot y = \delta xy - \gamma y, \tag{3.7}$$

where $x \geq 0$ is the prey population, $y \geq 0$ is the predator population, and $\alpha, \beta, \gamma, \delta > 0$ are parameters. For concreteness, take $\alpha = \beta = \gamma = \delta = 1$:

$$\dot x = x(1-y), \qquad \dot y = y(x-1). \tag{3.8}$$

Fixed points. From $\dot x = 0$: $x = 0$ or $y = 1$. From $\dot y = 0$: $y = 0$ or $x = 1$. The fixed points (in the first quadrant and its boundary) are $(0,0)$ and $(1,1)$.

Nullclines. The x-nullcline consists of $x = 0$ (the y-axis) and $y = 1$ (a horizontal line). The y-nullcline consists of $y = 0$ (the x-axis) and $x = 1$ (a vertical line). These divide the first quadrant into four regions:

| Region | $\dot{x}$ | $\dot{y}$ | Flow direction | |——|————|————|————-| | $x < 1, y < 1$ | + | − | right and down | | $x > 1, y < 1$ | + | + | right and up | | $x > 1, y > 1$ | − | + | left and up | | $x < 1, y > 1$ | − | − | left and down |

The flow circulates counterclockwise around $(1, 1)$.

Linearization at $(1, 1)$. The Jacobian of the right-hand side of (3.8) is

$$J = \begin{pmatrix} 1-y & -x \\ y & x-1 \end{pmatrix}.$$

At $(1, 1)$:

$$J(1, 1) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The eigenvalues are $\lambda = \pm i$, so the linearization predicts a **center**.

Conserved quantity. Like the pendulum, this system has a conserved quantity. Define

$$H(x, y) = \delta x - \gamma \ln x + \beta y - \alpha \ln y.$$

For our parameters: $H(x, y) = x - \ln x + y - \ln y$. One can verify $\dot{H} = 0$ along trajectories (exercise). Since H is a strictly convex function with a unique minimum at $(1, 1)$ (check: $H_x = 1 - 1/x = 0$ at $x = 1$, $H_{xx} = 1/x^2 > 0$, similarly for y), its level sets are closed curves surrounding $(1, 1)$. Therefore the trajectories in the first quadrant are closed orbits: both populations oscillate periodically, with predator oscillations lagging behind prey oscillations.

Linearization at $(0, 0)$. The Jacobian at the origin is $J(0, 0) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ with eigenvalues 1 and -1: a **saddle**. The x-axis (prey only, no predators) and y-axis (predators only, no prey) are invariant; along the x-axis prey grow exponentially, along the y-axis predators die off exponentially.

3.5 Nonlinear Systems and Linearization

3.5.1 The Jacobian at a Fixed Point

Consider a general system $\dot{x} = f(x)$ with $f : \mathbb{R}^n \to \mathbb{R}^n$ of class C^1, and suppose x^* is a fixed point: $f(x^*) = 0$. Writing $u = x - x^*$ for the displacement from the fixed point, Taylor expansion gives

$$\dot{u} = f(x^* + u) = \underbrace{f(x^*)}_{=0} + Df(x^*)\, u + O(\|u\|^2),$$

where $Df(x^*)$ is the **Jacobian matrix** of f at x^*:

$$Df(x^*)_{ij} = \left.\frac{\partial f_i}{\partial x_j}\right|_{x=x^*}.$$

The **linearization** of the system at x^* is the linear system $\dot{u} = Df(x^*)\,u$. The natural question is: to what extent does the linearization faithfully represent the local dynamics of the nonlinear system?

3.5.2 Hyperbolic Fixed Points and the Hartman–Grobman Theorem

Definition 3.2. A fixed point x^* is called **hyperbolic** if all eigenvalues of $Df(x^*)$ have nonzero real part.

Equivalently, no eigenvalue lies on the imaginary axis. This excludes centers ($\mathrm{Re}(\lambda) = 0$) and cases with a zero eigenvalue.

Theorem 3.8 (Hartman–Grobman). *Let x^* be a hyperbolic fixed point of $\dot{x} = f(x)$ where f is C^1. Then there exists a neighborhood U of x^* and a homeomorphism $h : U \to V$ (where V is a neighborhood of the origin) that maps trajectories of the nonlinear system to trajectories of the linearized system $\dot{u} = Df(x^*)\,u$, preserving the direction of time.*

In precise terms, h conjugates the flow: $h(\phi_t(x)) = e^{Df(x^*)t}h(x)$ for x near x^* and small $|t|$.

Significance. The theorem says that near a hyperbolic fixed point, the nonlinear flow is *topologically equivalent* to its linearization. The phase portrait of the nonlinear system looks qualitatively the same as that of the linear system. In particular:

- A stable node of the linearization is a stable node of the nonlinear system.
- A saddle of the linearization is a saddle of the nonlinear system.
- A stable spiral of the linearization is a stable spiral of the nonlinear system.
- And so on.

Limitations. The theorem has two important limitations:

1. **The conjugacy h is only a homeomorphism, not a diffeomorphism.** It preserves the topology of the phase portrait but may distort angles, smoothness of trajectories, etc. (In fact, a smooth version—the Sternberg linearization theorem—holds under additional non-resonance conditions on the eigenvalues, but the general statement requires only continuity.)
2. **The theorem says nothing about non-hyperbolic fixed points.** When the Jacobian has eigenvalues on the imaginary axis, the nonlinear terms can qualitatively change the dynamics. The most important example is the **center**: the linearization of the simple pendulum at $(0, 0)$ is a center, and it happens that the nonlinear system also has a center there (due to the conserved energy), but this is *not* guaranteed in general. A small perturbation can turn a linear center into a spiral. The determination of stability for non-hyperbolic fixed points requires more refined tools, such as center manifold theory or Lyapunov functions.

3.5.3 Stable and Unstable Manifolds

Theorem 3.9 (Stable Manifold Theorem). *Let x^* be a hyperbolic fixed point of $\dot{x} = f(x)$ with $f \in C^1$. Let E^s and E^u denote the stable and unstable subspaces of the linearization (the eigenspaces corresponding to eigenvalues with negative and positive real parts, respectively). Then there exist C^1 embedded submanifolds $W^s(x^*)$ and $W^u(x^*)$ of $\mathbb{R}^n$ such that:*

1. *$W^s(x^*)$ is tangent to E^s at x^*, and $\dim W^s = \dim E^s$.*
2. *$W^u(x^*)$ is tangent to E^u at x^*, and $\dim W^u = \dim E^u$.*
3. *$W^s(x^*) = \{x : \phi_t(x) \to x^* \text{ as } t \to +\infty\}$ (locally, near x^*).*
4. *$W^u(x^*) = \{x : \phi_t(x) \to x^* \text{ as } t \to -\infty\}$ (locally, near x^*).*
5. *W^s and W^u are invariant under the flow.*

In the linear case ($f(x) = Ax$), the stable and unstable manifolds are exactly E^s and E^u (linear subspaces). In the nonlinear case, they are curved, but they are tangent to the corresponding linear subspaces at the fixed point.

Worked Example 3.10. Consider the system

$$\dot{x}_1 = -x_1, \qquad \dot{x}_2 = x_2 + x_1^2.$$

The origin is a fixed point. The Jacobian is $Df(0) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, a saddle with $E^s = \{x_2 = 0\}$ and $E^u = \{x_1 = 0\}$.

The stable manifold W^s: on it, $\phi_t(x) \to 0$ as $t \to \infty$. From the first equation, $x_1(t) = c_1 e^{-t}$. Substituting into the second: $\dot{x}_2 = x_2 + c_1^2 e^{-2t}$. This linear ODE has solution $x_2(t) = c_2 e^t - \frac{c_1^2}{3} e^{-2t}$. For $x_2(t) \to 0$ as $t \to \infty$, we need $c_2 = 0$, giving $x_2(t) = -\frac{c_1^2}{3} e^{-2t}$. Since $x_1(t) = c_1 e^{-t}$, we have $x_1^2 = c_1^2 e^{-2t}$ and therefore $x_2 = -x_1^2/3$.

The local stable manifold is the parabola $W^s_{\text{loc}} = \{(x_1, x_2) : x_2 = -x_1^2/3\}$, which is tangent to $E^s = \{x_2 = 0\}$ at the origin, as the theorem predicts.

Similarly, one can compute that the unstable manifold is exactly the x_2-axis: $W^u = \{x_1 = 0\}$ (since $x_1 = 0$ is invariant and $\dot{x}_2 = x_2$ there).

3.6 Limit Cycles and the Poincaré–Bendixson Theorem

3.6.1 Limit Cycles

A **limit cycle** is an isolated periodic orbit. "Isolated" means that nearby trajectories are not periodic—they spiral toward or away from the limit cycle.

- A **stable limit cycle** (attracting) is one where nearby trajectories spiral toward it as $t \to +\infty$.
- An **unstable limit cycle** (repelling) is one where nearby trajectories spiral away.
- A **semi-stable limit cycle** attracts from one side and repels from the other.

Limit cycles are inherently nonlinear phenomena: linear systems in 2D can have periodic orbits (centers), but these are never isolated—they come in continuous families filling an annular region.

3.6.2 The Poincaré–Bendixson Theorem

The following theorem is one of the most important results in planar dynamics. It places severe constraints on what the long-term behavior of a two-dimensional flow can look like.

Theorem 3.11 (Poincaré–Bendixson). *Let $\dot{x} = f(x)$ be a C^1 system on an open subset of $\mathbb{R}^2$. Suppose γ^+ is a forward trajectory that remains in a compact set K containing only finitely many fixed points. Then the ω-limit set $\omega(\gamma^+)$ is one of the following:*

1. *A single fixed point.*
2. *A periodic orbit.*
3. *A finite union of fixed points together with trajectories connecting them (a heteroclinic or homoclinic cycle).*

Proof sketch. sketch.** The key geometric insight is that in two dimensions, a trajectory cannot cross itself (by uniqueness of solutions) and hence must "use up" the available space in a monotone way. More precisely, consider a local cross-section (a short line segment transverse to the flow). The return map to this section (if the trajectory returns) must be monotone. Monotone maps on intervals have simple dynamics: every orbit converges. This forces the ω-limit set to be one of the three types listed. The rigorous proof uses the Jordan curve theorem and requires some careful topological arguments; see Hirsch, Smale, and Devaney (2013), Chapter 10, or Perko (2001), Chapter 3.

Corollary 3.12. *If K is a compact, positively invariant region containing no fixed points, then K contains a periodic orbit.*

Proof. * Take any trajectory in K. It remains in K for all forward time (positive invariance), and K is compact, so the trajectory remains in a compact set. By Poincaré–Bendixson, the ω-limit set is a fixed point, a periodic orbit, or a heteroclinic cycle. Since K has no fixed points, options (1) and (3) are excluded. ■

This corollary provides a powerful technique for *proving the existence* of limit cycles: construct a trapping region with no fixed points.

□

□

3.6.3 No Chaos in Two Dimensions

The Poincaré–Bendixson theorem has a remarkable consequence:

Corollary 3.13. *A continuous-time autonomous system on $\mathbb{R}^2$ (or any subset of $\mathbb{R}^2$) cannot exhibit chaotic behavior.*

The argument is straightforward: chaotic dynamics requires trajectories to have complicated ω-limit sets (e.g., strange attractors), but Poincaré–Bendixson forces ω-limit sets in 2D to be one of three simple types. In particular, sensitive dependence on initial conditions in the sense of chaos is impossible.

This is why one needs at least three dimensions for continuous-time chaos. The Lorenz system, Rössler attractor, and other classic chaotic flows all live in $\mathbb{R}^3$ or higher. This dimensional threshold is a fundamental difference between continuous and discrete dynamics, where even one-dimensional maps (like the logistic map from Chapter 2) can exhibit chaos.

3.6.4 The Van der Pol Oscillator

The **Van der Pol oscillator** is a second-order ODE arising from electrical circuit theory:

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0, \quad \mu > 0. \tag{3.9}$$

Note the damping term $\mu(x^2 - 1)\dot{x}$: when $|x| < 1$, the coefficient $(x^2 - 1)$ is negative, so the "damping" is actually *negative* (energy is pumped in); when $|x| > 1$, the damping is positive (energy is dissipated). This balance gives rise to a stable limit cycle.

Converting to a first-order system with $x_1 = x$, $x_2 = \dot{x}$:

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = -x_1 + \mu(1 - x_1^2)x_2.$$

Fixed point. The only fixed point is the origin. The Jacobian there is

$$Df(0) = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}.$$

The eigenvalues satisfy $\lambda^2 - \mu\lambda + 1 = 0$, so $\lambda = \frac{\mu \pm \sqrt{\mu^2 - 4}}{2}$. For $0 < \mu < 2$, the eigenvalues are complex with positive real part $\mu/2 > 0$: the origin is an **unstable spiral**. For $\mu \geq 2$, both eigenvalues are real and positive: an **unstable node**.

In either case, trajectories are repelled from the origin. One can show (and we take this on faith, though the argument uses a trapping region and the Poincaré–Bendixson theorem) that trajectories starting far from the origin are pushed inward: the dissipative damping for $|x| > 1$ eventually dominates. By constructing a compact annular region that trajectories cannot leave, and noting that it contains no fixed points, Corollary 3.12 guarantees the existence of a periodic orbit.

For small μ, the limit cycle is approximately circular with radius 2 (this can be shown by the method of averaging). For large μ, the limit cycle develops sharp transitions and slow-fast structure, approaching a *relaxation oscillation*.

3.7 Lyapunov Functions

3.7.1 Lyapunov Stability

We have seen several methods for determining the stability of a fixed point: eigenvalue analysis for linear systems, Hartman–Grobman for hyperbolic fixed points of nonlinear systems. But what about non-hyperbolic fixed points? And what about *global* (as opposed to local) stability? Lyapunov functions provide a powerful and flexible approach.

Definition 3.3. Let x^* be a fixed point of $\dot{x} = f(x)$. A C^1 function $V : U \to \mathbb{R}$ defined on a neighborhood U of x^* is called a **Lyapunov function** if:

1. $V(x^*) = 0$ and $V(x) > 0$ for all $x \in U \setminus \{x^*\}$. 2. $\dot{V}(x) \leq 0$ for all $x \in U$, where $\dot{V}$ denotes the time derivative along trajectories:

$$\dot{V}(x) = \nabla V(x) \cdot f(x) = \sum_{i=1}^{n} \frac{\partial V}{\partial x_i} f_i(x).$$

Note that $\dot{V}$ can be computed without solving the ODE; it depends only on V and f.

Theorem 3.15 (Lyapunov Stability Theorem). *Let x^* be a fixed point of $\dot{x} = f(x)$ and let V be a Lyapunov function on a neighborhood of x^*.*

1. *If $\dot{V}(x) \leq 0$ in U, then x^* is **Lyapunov stable** (trajectories starting near x^* remain near x^*).*
2. *If $\dot{V}(x) < 0$ for all $x \in U \setminus \{x^*\}$, then x^* is **asymptotically stable** (trajectories starting near x^* converge to x^*).*

Proof of (1). of (1).* Let $\varepsilon > 0$ be small enough that $\overline{B}_\varepsilon(x^*) \subset U$. Since V is continuous, positive on the compact set $S_\varepsilon = \{x : \|x - x^*\| = \varepsilon\}$, and S_ε does not contain x^*, we have $m = \min_{S_\varepsilon} V > 0$. By continuity of V and $V(x^*) = 0$, there exists $\delta > 0$ such that $V(x) < m$ for all $x \in B_\delta(x^*)$.

Now take any trajectory starting in $B_\delta(x^*)$. Since $\dot{V} \leq 0$, V is non-increasing along the trajectory, so $V(x(t)) \leq V(x(0)) < m$ for all $t \geq 0$. But $V \geq m$ on S_ε, so the trajectory can never reach S_ε. Therefore $x(t) \in B_\varepsilon(x^*)$ for all $t \geq 0$. □

Proof of (2). of (2).* By (1), x^* is Lyapunov stable. It remains to show convergence. Given the stronger hypothesis $\dot{V} < 0$, the function $V(x(t))$ is strictly decreasing, hence converges to some limit $\ell \geq 0$. If $\ell > 0$, then the trajectory stays in the compact annular region $\{x : \ell \leq V(x) \leq V(x(0))\}$, which does not contain x^*. On this compact set, $\dot{V}$ achieves a maximum $-c < 0$ (since $\dot{V}$ is continuous and strictly negative away from x^*). Then $V(x(t)) \leq V(x(0)) - ct \to -\infty$, contradicting $V \geq 0$. So $\ell = 0$ and $V(x(t)) \to 0$, which forces $x(t) \to x^*$. ■

□

3.7.2 Example: Energy as a Lyapunov Function

Consider the damped pendulum

$$\ddot{\theta} + b\dot{\theta} + \sin\theta = 0, \quad b > 0,$$

or equivalently $\dot{x}_1 = x_2$, $\dot{x}_2 = -\sin x_1 - bx_2$, with the equilibrium at $(0, 0)$.

Define the energy $V(x_1, x_2) = \frac{1}{2}x_2^2 + (1 - \cos x_1)$.

- $V(0, 0) = 0$, and $V(x_1, x_2) > 0$ for (x_1, x_2) in a neighborhood of the origin (since $1 - \cos x_1 \geq 0$ with equality only at $x_1 = 2k\pi$, and $\frac{1}{2}x_2^2 \geq 0$; both vanish simultaneously only at the origin, within $|x_1| < 2\pi$).
- $\dot{V} = x_2\dot{x}_2 + \sin(x_1)\dot{x}_1 = x_2(-\sin x_1 - bx_2) + \sin(x_1) \cdot x_2 = -bx_2^2 \leq 0$.

So V is a Lyapunov function, and the origin is Lyapunov stable.

Can we conclude asymptotic stability? We have $\dot{V} = -bx_2^2$, which is zero not only at the origin but on the entire x_1-axis (where $x_2 = 0$). So condition (2) of the theorem is not met. However, a trajectory cannot remain on $\{x_2 = 0\}$ unless it is at a fixed point, because if $x_2 = 0$

and $x_1 \neq 0$ (and $x_1 \neq k\pi$), then $\dot{x}_2 = -\sin x_1 \neq 0$ forces the trajectory off the x_1-axis. This is the idea behind **LaSalle's invariance principle**, which strengthens the Lyapunov theorem:

Theorem 3.16 (LaSalle's Invariance Principle). *Under the hypotheses of Theorem 3.15(1), let $S = \{x \in U : \dot{V}(x) = 0\}$, and let M be the largest invariant subset of S. Then every trajectory starting in a compact level set $\{V \leq c\} \subset U$ converges to M as $t \to \infty$.*

For the damped pendulum, $S = \{x_2 = 0\}$ and the only invariant subset of S near the origin is $M = \{(0, 0)\}$. By LaSalle's principle, $(0, 0)$ is asymptotically stable.

3.7.3 Remarks on Finding Lyapunov Functions

There is no general algorithm for constructing Lyapunov functions, and this is the main practical difficulty. However, physical intuition often helps:

- **Mechanical systems:** Total energy (kinetic + potential) is a natural candidate.
- **Gradient systems** $\dot{x} = -\nabla V(x)$: the function V itself is a Lyapunov function, since $\dot{V} = \nabla V \cdot (-\nabla V) = -\|\nabla V\|^2 \leq 0$.
- **Quadratic forms:** For linear systems $\dot{x} = Ax$ with A stable (all eigenvalues in the left half-plane), the Lyapunov equation $A^T P + PA = -Q$ has a unique positive-definite solution P for any positive-definite Q, and $V(x) = x^T P x$ is a Lyapunov function.

3.8 Looking Ahead

In this chapter we have developed the theory of continuous dynamical systems on $\mathbb{R}^n$, with particular attention to the plane. We saw that the Poincaré–Bendixson theorem constrains planar flows to have simple ω-limit sets, ruling out chaos. In Chapter 4, we will venture into three and higher dimensions, where the Poincaré–Bendixson theorem no longer applies and the door opens to chaotic attractors, sensitive dependence on initial conditions, and the rich phenomena that motivate ergodic theory.

Exercises

Exercise 3.1. Consider the linear system $\dot{x} = Ax$ where $A = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix}$.

(a) Find the eigenvalues and classify the fixed point at the origin.

(b) Find the general solution using the matrix exponential.

(c) Sketch the phase portrait.

Exercise 3.2. (Failure of uniqueness.) Show that the initial value problem $\dot{x} = \sqrt{|x|}$, $x(0) = 0$ has infinitely many solutions. Verify that each of the functions

$$x_a(t) = \begin{cases} 0 & t \leq a, \\ \frac{1}{4}(t-a)^2 & t > a \end{cases}$$

is a solution for every $a \geq 0$. Why does the Picard–Lindelöf theorem not apply?

Exercise 3.3. (Nullcline analysis.) For the system

$$\dot{x} = x(3 - x - 2y), \qquad \dot{y} = y(2 - x - y),$$

(a) Find all fixed points in the first quadrant (including the boundary).

(b) Sketch the nullclines and determine the direction of the flow in each region.

(c) Determine the stability of each fixed point using linearization.

(d) Sketch the phase portrait in the first quadrant.

Exercise 3.4. (Lyapunov function.) Consider the system $\dot{x}_1 = -x_1 + x_1 x_2^2$, $\dot{x}_2 = -x_2 + x_1^2 x_2$. Show that $V(x_1, x_2) = x_1^2 + x_2^2$ is a Lyapunov function in the region $\{x_1^2 + x_2^2 < 1\}$ and conclude that the origin is asymptotically stable.

Exercise 3.5. (Stable manifold computation.) For the system

$$\dot{x}_1 = x_1 + x_2^2, \qquad \dot{x}_2 = -x_2,$$

(a) Find the linearization at the origin and identify E^s and E^u.

(b) Compute the stable manifold W^s exactly (proceed as in Worked Example 3.10).

(c) Compute the unstable manifold W^u by a power-series expansion $x_2 = h(x_1) = ax_1 + bx_1^2 + \cdots$ and matching coefficients.

Exercise 3.6. (Poincaré–Bendixson in action.) Consider the system in polar coordinates

$$\dot{r} = r(1 - r^2) + \mu r \cos\theta, \qquad \dot{\theta} = 1,$$

for small $\mu > 0$.

(a) Show that for $\mu = 0$, the unit circle $r = 1$ is a stable limit cycle.

(b) For small $\mu > 0$, construct a trapping region (an annulus $r_1 \le r \le r_2$ with appropriate r_1, r_2) and use the Poincaré–Bendixson theorem to prove that a periodic orbit still exists.

Exercise 3.7. (Jordan blocks and polynomial growth.) Consider the 3×3 system $\dot{x} = Jx$ where

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

(a) Compute e^{Jt} directly from the series definition (note J is nilpotent: find the smallest k with $J^k = 0$).

(b) Write down the general solution and describe the trajectories. Show that $\|x(t)\|$ grows polynomially (not exponentially) in time.

(c) Is the origin stable? Lyapunov stable? Asymptotically stable?

Exercise 3.8. (Lotka–Volterra conserved quantity.) For the system (3.8), verify that $H(x, y) = x - \ln x + y - \ln y$ satisfies $\dot{H} = 0$ along trajectories. Conclude that all orbits in the open first quadrant are periodic.

—

Recommended Reading

- **Hirsch, M. W., Smale, S., and Devaney, R. L.** *Differential Equations, Dynamical Systems, and an Introduction to Chaos*, 3rd edition. Academic Press, 2013. An excellent undergraduate text; Chapters 1–10 cover all the material in this chapter and more, with many examples and clear exposition.
- **Strogatz, S. H.** *Nonlinear Dynamics and Chaos*, 2nd edition. Westview Press, 2015. The standard "first course" in nonlinear dynamics. Highly readable, with wonderful physical intuition and examples. Chapters 5–8 are particularly relevant.
- **Perko, L.** *Differential Equations and Dynamical Systems*, 3rd edition. Springer, 2001. More rigorous and comprehensive than Strogatz; includes proofs of the Hartman–Grobman theorem, the stable manifold theorem, and the Poincaré–Bendixson theorem. A good reference for the reader wanting complete proofs.
- **Arnold, V. I.** *Ordinary Differential Equations*, 3rd edition. Springer, 1992. Arnold's geometric approach to ODEs is unique and illuminating. The treatment of flows and the matrix exponential in Chapters 1–5 is particularly elegant. A challenging but rewarding read.
- **Coddington, E. A., and Levinson, N.** *Theory of Ordinary Differential Equations.* McGraw-Hill, 1955. The classic graduate reference for existence, uniqueness, and continuation theorems. Included here for the reader wishing to see the proof of Peano's theorem and other foundational results.

Chapter 4

Poincaré Maps and Recurrence

4.1 Motivation: From Continuous to Discrete

In Chapters 2 and 3, we developed two parallel strands of dynamical systems theory: discrete maps $x_{n+1} = f(x_n)$ and continuous flows $\dot{x} = F(x)$. The discrete case often admits cleaner analysis — fixed points are just solutions to $f(x^*) = x^*$, stability is determined by the derivative $|f'(x^*)|$, and we can iterate directly. Continuous systems in $\mathbb{R}^n$, by contrast, present formidable difficulties: trajectories wind through high-dimensional phase spaces, periodic orbits are harder to detect, and the geometry of the flow can be exceedingly complex.

Henri Poincaré, in his study of the three-body problem in celestial mechanics, introduced an idea of remarkable power: **reduce a continuous-time system to a discrete one by examining where trajectories cross a well-chosen surface**. Instead of tracking a trajectory continuously through $\mathbb{R}^n$, we record only its successive intersections with a codimension-1 surface (a "section"). The resulting discrete map — the *Poincaré map* or *first-return map* — captures the essential dynamics of the original flow in a lower-dimensional setting.

This chapter develops the Poincaré map construction, works through concrete examples, proves Poincaré's recurrence theorem, and briefly introduces Floquet theory for periodic orbits. The recurrence theorem, in particular, will serve as our first encounter with the deep connections between dynamics and measure theory that form the subject of Part II.

4.2 Poincaré Sections and First-Return Maps

4.2.1 Setup and Definition

Consider an autonomous ordinary differential equation in $\mathbb{R}^n$:

$$\dot{x} = F(x), \qquad x \in \mathbb{R}^n,$$

where $F : \mathbb{R}^n \to \mathbb{R}^n$ is smooth (say, C^1). Let $\phi_t(x_0)$ denote the flow: the solution at time t with initial condition x_0, so $\phi_0(x_0) = x_0$ and $\frac{d}{dt}\phi_t(x_0) = F(\phi_t(x_0))$.

Definition 4.1 (Poincaré section)**.** A *Poincaré section* (or *cross-section*) is a codimension-1 smooth submanifold $\Sigma \subset \mathbb{R}^n$ such that the flow is *transverse* to Σ: for every $x \in \Sigma$,

$$F(x) \cdot n(x) \neq 0,$$

where $n(x)$ is a normal vector to Σ at x. In other words, trajectories actually *cross* Σ rather than running tangent to it.

In $\mathbb{R}^3$ (the most common setting for visualization), Σ is a 2-dimensional surface, and the Poincaré map reduces a 3D flow to a 2D discrete dynamical system — a substantial simplification.

Definition 4.2 (First-return map)**.** Given a Poincaré section Σ and a point $x \in \Sigma$, define the *first-return time*

$$\tau(x) = \inf\{t > 0 : \phi_t(x) \in \Sigma\}.$$

If $\tau(x) < \infty$, the *Poincaré map* (or *first-return map*) is

$$P : \Sigma \to \Sigma, \qquad P(x) = \phi_{\tau(x)}(x).$$

The map P sends each point on the section to the next point where the trajectory crosses Σ (in the same direction, by transversality).

Remark 4.1. We require crossing in a *consistent direction.* Since $F(x) \cdot n(x) \neq 0$ on Σ, the sign of this dot product is constant on each connected component of Σ (by continuity), so we can restrict to crossings with a fixed sign — say, $F(x) \cdot n(x) > 0$ — to ensure the orbit crosses Σ in the same direction each time.

4.2.2 Well-Definedness

The definition above hides a nontrivial assumption: *that the flow actually returns to Σ*. This need not happen in general. Several things can go wrong:

1. **The trajectory escapes to infinity.** If the system has unbounded orbits, a trajectory starting on Σ may never return.
2. **The trajectory converges to a fixed point.** A trajectory spiraling into an equilibrium will never cross Σ again (or will take "infinite time" to do so).
3. **The trajectory converges to an invariant set that does not intersect Σ.** For example, the orbit may approach a limit cycle or attractor that misses Σ.

Thus, the domain of P is generally a *subset* of Σ:

$$\mathrm{dom}(P) = \{x \in \Sigma : \tau(x) < \infty\}.$$

In favorable situations — when Σ is chosen in the vicinity of a periodic orbit, or inside a bounded invariant region — the map P is well-defined on an open subset of Σ, and this is sufficient for local analysis.

Proposition 4.1. **If F is C^r $(r \geq 1)$, and if P is well-defined in a neighborhood of $x_0 \in \Sigma$, then P is a C^r map (as a map between submanifolds).**

Proof sketch. sketch.* This follows from the implicit function theorem. The return condition $\phi_t(x) \in \Sigma$ can be written as a smooth equation $g(x,t) = 0$ (where g measures the signed distance from $\phi_t(x)$ to Σ). The transversality condition ensures $\partial g/\partial t \neq 0$, so the implicit function theorem gives $\tau(x)$ as a C^r function of x. Since $\phi_t(x)$ depends smoothly on both x and t, the composition $P(x) = \phi_{\tau(x)}(x)$ is C^r. □

□

4.2.3 Fixed Points and Periodic Orbits

The fundamental correspondence that makes the Poincaré map powerful is:

Theorem 4.4. *A point $x^* \in \Sigma$ is a fixed point of P (i.e., $P(x^*) = x^*$) if and only if the orbit of x^* under the flow ϕ_t is a periodic orbit of period $\tau(x^*)$.*

Proof. * If $P(x^*) = x^*$, then $\phi_{\tau(x^*)}(x^*) = x^*$. Since $\tau(x^*) > 0$, this means x^* lies on a periodic orbit with period $T = \tau(x^*)$. Conversely, if x^* lies on a periodic orbit of period T that intersects Σ transversally, and if T is the minimal period, then $\phi_T(x^*) = x^*$ and $\tau(x^*) = T$, so $P(x^*) = x^*$. □

More generally, a periodic point of P with period k (meaning $P^k(x^*) = x^*$ but $P^j(x^*) \neq x^*$ for $0 < j < k$) corresponds to a periodic orbit of the flow that intersects Σ exactly k times per period.

□

4.2.4 Stability Correspondence

The second key property of the Poincaré map is that it faithfully reflects stability.

Theorem 4.5. *Let γ be a periodic orbit of $\dot{x} = F(x)$ that intersects Σ at x^*. Then γ is an orbitally stable periodic orbit (in the sense of Lyapunov) if and only if x^* is a stable fixed point of P.*

The term "orbitally stable" means that nearby trajectories stay close to the periodic orbit *as a set* (they need not stay in phase with the original trajectory — there is always neutral stability along the direction of the flow, since a small perturbation along the orbit simply shifts the phase).

The linearized stability analysis is also inherited:

Proposition 4.2. **The eigenvalues of $DP(x^*)$ (the derivative of the Poincaré map at a fixed point) determine the orbital stability of the corresponding periodic orbit γ. Specifically:**

*- *If all eigenvalues of $DP(x^*)$ satisfy $|\lambda_i| < 1$, then γ is asymptotically orbitally stable.* - *If some eigenvalue satisfies $|\lambda_i| > 1$, then γ is orbitally unstable.**

Note that $DP(x^*)$ is an $(n-1) \times (n-1)$ matrix (since Σ has dimension $n-1$), so there are $n-1$ eigenvalues. The "missing" eigenvalue compared to the n-dimensional ambient system corresponds to the neutral direction along the flow. We will return to this point in Section 4.7 on Floquet theory.

4.3 Worked Examples

4.3.1 A Simple Limit Cycle: The Radial Example

Consider the planar system in polar coordinates:

$$\dot{r} = r(1-r), \qquad \dot{\theta} = 1.$$

This system has a stable limit cycle at $r = 1$ (the unit circle): for $r < 1$, $\dot{r} > 0$ so trajectories spiral outward; for $r > 1$, $\dot{r} < 0$ so trajectories spiral inward.

Choosing the section. Take Σ to be the positive x-axis, i.e., $\Sigma = \{(r, 0) : r > 0\}$ (in polar coordinates, $\theta = 0$). Since $\dot{\theta} = 1 > 0$, trajectories always cross Σ in the direction of increasing θ, so the transversality condition is satisfied.

Computing the return map. A trajectory starting at $(r_0, 0)$ returns to Σ after one full revolution, at time $\tau(r_0) = 2\pi$ (since $\dot{\theta} = 1$ implies $\theta(t) = t$, and the first return to $\theta = 0$ (mod 2π) is at $t = 2\pi$).

To find $P(r_0)$, we need the value of r at time $t = 2\pi$. The radial equation $\dot{r} = r(1-r)$ is a Bernoulli equation. To solve it, substitute $u = 1/r$:

$$\dot{u} = -\frac{\dot{r}}{r^2} = -\frac{r(1-r)}{r^2} = -\frac{1-r}{r} = -(u-1) = 1-u.$$

This is a linear ODE with solution $u(t) = 1 + (u_0 - 1)e^{-t}$, where $u_0 = 1/r_0$. Therefore:

$$r(t) = \frac{1}{u(t)} = \frac{1}{1 + (1/r_0 - 1)e^{-t}} = \frac{r_0}{r_0 + (1-r_0)e^{-t}}.$$

The Poincaré map is:

$$\boxed{P(r_0) = r(2\pi) = \frac{r_0}{r_0 + (1-r_0)e^{-2\pi}}.}$$

Fixed point. Setting $P(r^*) = r^*$ gives $r^* = r^*/(r^* + (1-r^*)e^{-2\pi})$, which requires $r^* + (1-r^*)e^{-2\pi} = 1$, i.e., $r^*(1-e^{-2\pi}) = 1 - e^{-2\pi}$, so $r^* = 1$. This is exactly the limit cycle.

Stability. Computing:

$$P'(r_0) = \frac{d}{dr_0}\left[\frac{r_0}{r_0 + (1-r_0)e^{-2\pi}}\right].$$

Let $\alpha = e^{-2\pi}$. Then $P(r_0) = r_0/(r_0 + (1-r_0)\alpha) = r_0/(\alpha + r_0(1-\alpha))$. By the quotient rule:

$$P'(r_0) = \frac{\alpha + r_0(1-\alpha) - r_0(1-\alpha)}{(\alpha + r_0(1-\alpha))^2} = \frac{\alpha}{(\alpha + r_0(1-\alpha))^2}.$$

At the fixed point $r^* = 1$:

$$P'(1) = \frac{\alpha}{(\alpha + 1 - \alpha)^2} = \alpha = e^{-2\pi} \approx 0.00187.$$

Since $|P'(1)| = e^{-2\pi} \ll 1$, the fixed point is strongly stable, confirming that the limit cycle is asymptotically orbitally stable. The small value reflects the strong contraction: after one revolution, perturbations in r are reduced by a factor of $e^{-2\pi} \approx 1/535$.

4.3.2 The Forced Van der Pol Oscillator

The Van der Pol oscillator with periodic forcing is:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = A\cos(\omega t),$$

where $\mu > 0$ is the nonlinearity parameter, A is the forcing amplitude, and ω is the forcing frequency. As a first-order system:

$$\dot{x} = y, \qquad \dot{y} = \mu(1 - x^2)y - x + A\cos(\omega t).$$

This is a *nonautonomous* system — the right-hand side depends explicitly on t. The standard technique is to introduce $\theta = \omega t \pmod{2\pi}$ as an additional variable, making the system autonomous in the extended phase space $(x, y, \theta) \in \mathbb{R}^2 \times S^1$:

$$\dot{x} = y, \qquad \dot{y} = \mu(1 - x^2)y - x + A\cos\theta, \qquad \dot{\theta} = \omega.$$

The stroboscopic section. A natural Poincaré section is the *stroboscopic section*:

$$\Sigma = \{(x, y, \theta) : \theta = 0\}.$$

Since $\dot{\theta} = \omega > 0$ everywhere, the flow is transverse to Σ, and every trajectory returns to Σ after time $\tau = 2\pi/\omega$ (one forcing period). The Poincaré map is then:

$$P : \mathbb{R}^2 \to \mathbb{R}^2, \qquad P(x_0, y_0) = (x(2\pi/\omega),\ y(2\pi/\omega)),$$

where $(x(t), y(t))$ is the solution with initial condition (x_0, y_0) at $t = 0$.

For the forced Van der Pol oscillator, we generally cannot write P in closed form. However, the map is perfectly well-defined: given (x_0, y_0), integrate the ODE for one forcing period to obtain $P(x_0, y_0)$.

Interpretation. A fixed point $P(x^*, y^*) = (x^*, y^*)$ corresponds to a periodic solution with the same period as the forcing — the oscillator is *entrained* or *phase-locked* to the drive. A period-k point of P corresponds to a subharmonic response of period $k \cdot 2\pi/\omega$.

For small forcing amplitude A and μ of moderate size, the system typically phase-locks and P has a stable fixed point. As parameters vary, this fixed point can undergo bifurcations — period-doublings, Neimark-Sacker bifurcations — leading to quasiperiodic or chaotic behavior, all visible in the dynamics of P. Numerical iteration of P (i.e., computing the stroboscopic map by numerical integration) is a standard tool for analyzing forced oscillators.

4.3.3 The Lorenz System

The Lorenz equations are:

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = rx - y - xz, \qquad \dot{z} = xy - bz,$$

with the classical parameter values $\sigma = 10$, $b = 8/3$, $r = 28$. The Lorenz attractor is a complicated invariant set in $\mathbb{R}^3$ on which trajectories exhibit chaotic behavior — they alternate

unpredictably between looping around two unstable equilibria.

A Poincaré section for the Lorenz system. A common choice is the plane

$$\Sigma = \{(x, y, z) : z = r - 1\} = \{(x, y, z) : z = 27\},$$

which passes through both nontrivial equilibria (at $z = r - 1$). One restricts to crossings with $\dot{z} < 0$ (downward crossings). The Lorenz attractor intersects this section in a complicated-looking set, but with a striking property.

Lorenz himself discovered (numerically, in his 1963 paper) that if one plots the successive maxima of $z(t)$ — call them z_n and z_{n+1} — the resulting return map is nearly one-dimensional: the data points fall almost perfectly on a curve. More precisely, if one parametrizes the intersection $\Sigma \cap$ (attractor) by a single coordinate (say x), the return map is well-approximated by a one-dimensional map $P : I \to I$ on an interval.

This near-one-dimensional return map has a distinctive shape: it is unimodal (tent-like), with a slope everywhere greater than 1 in absolute value. The fact that $|P'| > 1$ everywhere reflects the stretching that is responsible for the sensitive dependence on initial conditions. The cusp or discontinuity near the peak corresponds to trajectories that pass close to the origin (the unstable equilibrium), where the dynamics is most sensitive.

This dramatic dimensional reduction — from a flow in $\mathbb{R}^3$ to an essentially one-dimensional map — is one of the reasons the Lorenz system is so well-studied. It makes the chaotic dynamics amenable to the techniques of one-dimensional dynamics (symbolic dynamics, kneading theory, etc.), which we will encounter in Chapter 6.

Remark 4.2. The near-one-dimensionality is not exact: the Lorenz return map has a very thin but nonzero "thickness." Proving that the Lorenz system is genuinely chaotic (not merely numerically so) was a major achievement, settled by Tucker (2002) using rigorous computer-assisted methods.

4.4 The Poincaré Recurrence Theorem

We now turn to a foundational result in the theory of dynamical systems — one that connects dynamics to measure theory in a fundamental way.

4.4.1 Setting and Statement

Let $(X, \mathcal{B}, \mu)$ be a measure space with $\mu(X) < \infty$ (a *finite* measure space), and let $T : X \to X$ be a *measure-preserving transformation*, meaning that $\mu(T^{-1}(A)) = \mu(A)$ for every measurable set $A \in \mathcal{B}$.

Recall from Chapter 3 that volume-preserving flows (such as Hamiltonian systems) provide natural examples. If the phase space has finite volume, the time-t map ϕ_t is measure-preserving with respect to the phase space volume (by Liouville's theorem).

Theorem 4.3 (Poincaré Recurrence Theorem)**.** **Let $(X, \mathcal{B}, \mu)$ be a finite measure space and $T : X \to X$ a measure-preserving transformation. Let $A \in \mathcal{B}$ with $\mu(A) > 0$. Then μ-almost every point of A returns to A under iteration of T. That is,**

$$\mu(\{x \in A : T^n(x) \notin A \textit{ for all } n \geq 1\}) = 0.$$

Equivalently: for μ-almost every $x \in A$, there exists $n \geq 1$ such that $T^n(x) \in A$.

4.4.2 Proof

The proof is short and uses only the measure-preserving property and finiteness of the measure. We reproduce it in full.

Proof. Define the set of non-returning* points:

$$B = \{x \in A : T^n(x) \notin A \text{ for all } n \geq 1\}.$$

We wish to show $\mu(B) = 0$.

Step 1. Observe that $B \subset A$ by definition. Also note that if $x \in B$, then $T^n(x) \notin A$ for all $n \geq 1$, which in particular means $T^n(x) \notin B$ for all $n \geq 1$ (since $B \subset A$). Therefore:

$$T^{-n}(B) \cap B = \varnothing \qquad \text{for all } n \geq 1.$$

Step 2. We claim that the sets $B, T^{-1}(B), T^{-2}(B), \ldots$ are *pairwise disjoint*. To see this, suppose for contradiction that $T^{-i}(B) \cap T^{-j}(B) \neq \varnothing$ for some $0 \leq i < j$. Then there exists $x \in X$ with $T^i(x) \in B$ and $T^j(x) \in B$. Since $T^j(x) = T^{j-i}(T^i(x))$ and $T^i(x) \in B$, we have $T^{j-i}(T^i(x)) \in B \subset A$. But this contradicts the fact that $T^i(x) \in B$ (which requires $T^n(T^i(x)) \notin A$ for all $n \geq 1$, and $j - i \geq 1$).

Step 3. Since T is measure-preserving, $\mu(T^{-n}(B)) = \mu(B)$ for all $n \geq 0$. The sets $\{T^{-n}(B)\}_{n=0}^{\infty}$ are pairwise disjoint and all contained in X. By countable additivity of μ:

$$\sum_{n=0}^{\infty} \mu(T^{-n}(B)) = \mu\left(\bigsqcup_{n=0}^{\infty} T^{-n}(B)\right) \leq \mu(X) < \infty.$$

But each term in the sum equals $\mu(B)$, so:

$$\sum_{n=0}^{\infty} \mu(B) \leq \mu(X) < \infty.$$

This is an infinite sum of identical terms. The only way it can converge is if $\mu(B) = 0$. □

Remark on the proof. The argument is beautifully simple. It rests on three ingredients: (1) measure preservation (T does not "lose" measure), (2) finiteness of the total measure (the phase space is bounded in the measure-theoretic sense), and (3) the fact that an infinite sum of a positive constant diverges. The third point is why the theorem fails for infinite measure spaces — there is "room" for the orbit to wander away forever.

4.4.3 A Stronger Form

The theorem as stated guarantees at least one return. In fact, we get infinitely many returns for free.

Corollary 4.8. *Under the hypotheses of Theorem 4.7, for μ-almost every $x \in A$, the orbit of x returns to A infinitely often. That is, the set $\{n \geq 1 : T^n(x) \in A\}$ is infinite for μ-a.e. $x \in A$.*

Proof. * Let $A_1 = \{x \in A : T^n(x) \in A \text{ for some } n \geq 1\}$. By Theorem 4.7, $\mu(A \setminus A_1) = 0$, so $\mu(A_1) = \mu(A)$. Now A_1 is a measurable subset of A with positive measure, and T is still measure-preserving. Apply the recurrence theorem to A_1: μ-a.e. point of A_1 returns to A_1, hence returns to A at least twice. Proceeding inductively, we obtain infinitely many returns for a.e. point. □

□

4.4.4 Physical Significance: The Boltzmann–Zermelo Debate

Poincaré proved his recurrence theorem in 1890, in the context of the three-body problem. Its physical implications sparked one of the great debates in the foundations of statistical mechanics.

The paradox. Consider a box of gas in which all molecules start in one half. The Hamiltonian dynamics governing the gas molecules is measure-preserving (by Liouville's theorem) and the phase space has finite measure (if we impose an energy constraint, the energy surface is compact). Therefore, by Poincaré's theorem, the gas will *eventually return* to a state arbitrarily close to its initial configuration — all molecules in one half of the box.

In 1896, Ernst Zermelo used this fact to argue against Boltzmann's statistical mechanics: if the system recurs, how can entropy increase monotonically (as the second law of thermodynamics demands)?

Boltzmann's response was essentially correct and deeply insightful. He pointed out:

1. **The recurrence time is absurdly large.** For a macroscopic system with $N \sim 10^{23}$ particles, the Poincaré recurrence time is of order e^N — vastly longer than the age of the universe. The theorem guarantees eventual return, but says nothing useful about when.
2. **The second law is statistical.** Entropy increase is overwhelmingly probable, not absolutely certain. Poincaré recurrence is compatible with the second law because the recurrences, while guaranteed in principle, are so rare as to be irrelevant in practice.

This debate clarified the relationship between microscopic reversibility and macroscopic irreversibility and remains relevant in modern discussions of the foundations of statistical mechanics.

4.4.5 Limitations

The Poincaré recurrence theorem is powerful but has important limitations:

1. **Finite measure is essential.** For the flow generated by $\dot{x} = 1$ on $\mathbb{R}$ (with Lebesgue measure), every point escapes to $+\infty$ and never returns. The total measure is infinite, and the theorem does not apply.
2. **Measure preservation is essential.** Dissipative systems (such as $\dot{x} = -x$) contract volumes and do not recur in general. A ball of initial conditions shrinks to a point rather than returning to its initial configuration.

3. **The theorem is purely existential.** It guarantees that return occurs with probability 1, but provides no estimate of *when*. The return time can be — and for typical physical systems, *is* — unimaginably large.
4. **"Almost every" leaves room for exceptions.** There can be a set of measure zero (but possibly uncountable) of non-recurrent points.

4.5 Connections to Ergodic Theory (Preview)

The Poincaré recurrence theorem is the historical and conceptual starting point for ergodic theory. It tells us that measure-preserving systems possess an intrinsic form of "memory" — the system returns, again and again, to configurations it has visited before. Part II of this book will develop this theme systematically. Here we mention two important results that extend the recurrence theorem.

4.5.1 Kac's Lemma

Poincaré's theorem tells us that almost every point of A returns to A, but how long does the return take on average? The answer is given by a striking result due to Mark Kac (1947).

Theorem 4.4 (Kac's Lemma)**.** **Let $(X, \mathcal{B}, \mu)$ be a probability space ($\mu(X) = 1$), let $T : X \to X$ be an ergodic measure-preserving transformation, and let $A \in \mathcal{B}$ with $\mu(A) > 0$. For $x \in A$, define the first-return time**

$$n_A(x) = \min\{n \geq 1 : T^n(x) \in A\}.$$

Then the average return time to A is:

$$\frac{1}{\mu(A)} \int_A n_A(x)\, d\mu(x) = \frac{1}{\mu(A)}.$$

Equivalently, the expected return time (with respect to the conditional measure $\mu|_A/\mu(A)$) is $1/\mu(A)$.

The elegance of this result is hard to overstate. It says that the expected return time to a set depends only on the measure of the set, not on its shape, position, or any other geometric property. A set occupying 1% of the phase space will, on average, be revisited every 100 iterates.

Note the hypothesis of ergodicity, which is stronger than measure preservation. We will define it precisely in Chapter 9. For now, it suffices to say that an ergodic system is one with no nontrivial invariant sets — the dynamics is "indecomposable" in a measure-theoretic sense.

4.5.2 From Recurrence to Ergodic Theorems

The progression from recurrence to ergodic theory follows a natural arc:

- **Poincaré recurrence** (1890): Orbits return to their starting region.
- **Birkhoff's ergodic theorem** (1931): Time averages converge along orbits.
- **Ergodicity**: Time averages equal space averages.

Each statement is strictly stronger than the previous one, and each reveals more about the statistical behavior of dynamical systems. The recurrence theorem is the first hint that measure theory and dynamics are deeply linked — a theme that will occupy us throughout Part II.

4.6 Non-Autonomous Systems and Stroboscopic Maps

Before turning to Floquet theory, we collect some general remarks on Poincaré maps for periodically forced systems, since these arise frequently in applications.

Given a non-autonomous ODE with periodic forcing of period T:

$$\dot{x} = F(x,t), \qquad F(x,t+T) = F(x,t),$$

we can define the *stroboscopic map* $P : \mathbb{R}^n \to \mathbb{R}^n$ by

$$P(x_0) = \phi(T; x_0, 0),$$

where $\phi(t; x_0, t_0)$ denotes the solution at time t with initial condition x_0 at time t_0. This is a Poincaré map with section $\{t = 0\} \subset \mathbb{R}^n \times \mathbb{R}$ in the extended phase space.

The stroboscopic map has several convenient properties:

1. **It is always well-defined** (at least locally), because the return time is exactly T for every point — there is no need to "wait" for the orbit to come back.
2. **Fixed points correspond to T-periodic solutions**, period-k points to subharmonic solutions of period kT.
3. **It preserves the dimension** of the original system: P maps $\mathbb{R}^n$ to $\mathbb{R}^n$, unlike the general Poincaré section which reduces the dimension by one.

The stroboscopic map is the basis for much of the numerical study of forced oscillators, periodically driven systems, and Floquet theory, to which we now turn.

4.7 Floquet Theory (Brief Introduction)

Floquet theory provides the linear algebra underlying the stability analysis of periodic orbits. It connects directly to the derivative of the Poincaré map.

4.7.1 The Variational Equation

Let $\gamma(t)$ be a periodic orbit of $\dot{x} = F(x)$ with period T. To study the stability of γ, we linearize about it. Setting $x(t) = \gamma(t) + \xi(t)$ and keeping only first-order terms in ξ:

$$\dot{\xi} = DF(\gamma(t))\,\xi \equiv A(t)\,\xi,$$

where $A(t) = DF(\gamma(t))$ is the Jacobian of F evaluated along γ. Since γ is T-periodic, $A(t)$ is a T-periodic $n \times n$ matrix. This is the *variational equation* along γ.

4.7.2 The Monodromy Matrix

The fundamental matrix solution $\Phi(t)$ of the variational equation satisfies

$$\dot{\Phi}(t) = A(t)\Phi(t), \qquad \Phi(0) = I.$$

The *monodromy matrix* is the fundamental matrix evaluated after one period:

$$M = \Phi(T).$$

The monodromy matrix propagates perturbations through one full period: if $\xi(0)$ is a small perturbation at time 0, then $\xi(T) = M\xi(0)$ is the perturbation after one period (to first order).

4.7.3 Floquet Multipliers

Definition 4.3. The eigenvalues $\rho_1, \ldots, \rho_n$ of the monodromy matrix M are called the *Floquet multipliers* of the periodic orbit γ.

The Floquet multipliers determine the stability of γ:

- If $|\rho_i| < 1$ for all i (except one — see below), then γ is asymptotically orbitally stable. - If $|\rho_i| > 1$ for some i, then γ is unstable.

Proposition 4.5. *One Floquet multiplier is always equal to 1.*

Proof. * Differentiate the identity $\phi_T(\gamma(0)) = \gamma(0)$ along the orbit. Since $\gamma(t) = \phi_t(\gamma(0))$, the vector $\dot{\gamma}(0) = F(\gamma(0))$ is tangent to γ at the starting point. The variational equation gives:

$$\frac{d}{dt}\left[\Phi(t)F(\gamma(0))\right] = A(t)\Phi(t)F(\gamma(0)).$$

But also $\dot{\gamma}(t) = F(\gamma(t))$ satisfies the variational equation (by differentiating $\dot{\gamma} = F(\gamma)$ with respect to the initial condition along the orbit). Therefore $\Phi(t)F(\gamma(0)) = F(\gamma(t))$, and at $t = T$:

$$M \cdot F(\gamma(0)) = F(\gamma(T)) = F(\gamma(0)).$$

So $F(\gamma(0))$ is an eigenvector of M with eigenvalue 1. □

This multiplier of 1 corresponds to perturbations along the orbit direction, which neither grow nor decay (a phase shift along the orbit persists forever). The remaining $n-1$ multipliers determine the transverse stability.

4.7.4 Connection to the Poincaré Map

The key link between Floquet theory and the Poincaré map is:

Theorem 4.12. *The eigenvalues of $DP(x^*)$ (the derivative of the Poincaré map at the fixed point $x^* = \gamma \cap \Sigma$) are precisely the $n-1$ non-trivial Floquet multipliers (i.e., all Floquet multipliers except the one equal to 1).*

Proof sketch. sketch.* The derivative $DP(x^*)$ describes how the first-return map acts on infinitesimal perturbations in the section Σ. These are perturbations *transverse* to the flow direction. The monodromy matrix M acts on all perturbations, including the tangential one (which

has multiplier 1). The section Σ projects out the tangential component, leaving exactly the other $n-1$ multipliers. A careful computation using the decomposition $\mathbb{R}^n = T_{x^*}\Sigma \oplus \mathrm{span}(F(x^*))$ confirms this. $\square$

This theorem justifies the use of the Poincaré map for stability analysis: the map P automatically factors out the neutral direction along the flow, leaving only the dynamically relevant multipliers.

$\square$

4.7.5 Floquet's Theorem

For completeness, we state the main structural result.

Theorem 4.6 (Floquet)**.** **Let $A(t)$ be a continuous, T-periodic $n \times n$ matrix. Then the fundamental matrix of $\dot{\xi} = A(t)\xi$ can be written in the form**

$$\Phi(t) = Q(t)e^{Rt},$$

where $Q(t)$ is a nonsingular, T-periodic matrix (i.e., $Q(t+T) = Q(t)$), and R is a constant $n \times n$ matrix.

The eigenvalues of R are called the Floquet exponents μ_i, and they relate to the Floquet multipliers by $\rho_i = e^{\mu_i T}$. The significance of Floquet's theorem is that it reduces the study of a periodic-coefficient linear system to a constant-coefficient system (via the coordinate change $\xi = Q(t)\eta$, which transforms $\dot{\xi} = A(t)\xi$ into $\dot{\eta} = R\eta$).

4.8 Summary

The Poincaré map converts the analysis of a continuous flow into the iteration of a discrete map, reducing both the dimension and the conceptual complexity of the problem. Fixed points of the map correspond to periodic orbits of the flow, and stability is faithfully preserved.

The Poincaré recurrence theorem — that measure-preserving systems on finite measure spaces recur almost surely — is the oldest and most basic result connecting dynamics to measure theory. It provides the conceptual foundation for ergodic theory, which we develop in Part II.

Floquet theory provides the linearized version of the Poincaré map construction. The monodromy matrix encodes the linear stability of a periodic orbit, and its nontrivial eigenvalues (the Floquet multipliers) are precisely the eigenvalues of the derivative of the Poincaré map.

Exercises

Exercise 4.1. Consider the system in polar coordinates:

$$\dot{r} = r(1-r)(2-r), \qquad \dot{\theta} = 1.$$

(a) Find all limit cycles.

(b) Using the section $\Sigma = \{\theta = 0\}$, argue (without solving the ODE explicitly) that the Poincaré map has fixed points at $r = 1$ and $r = 2$.

(c) Determine the stability of each limit cycle by analyzing the sign of $\dot{r}$.

(d) What is $P'(1)$ and $P'(2)$? (You need not compute exact values; determine whether each is less than 1 or greater than 1 in absolute value, and the sign.)

Exercise 4.2. (Poincaré recurrence and the doubling map). Let $T : [0, 1) \to [0, 1)$ be the *doubling map* $T(x) = 2x \pmod 1$, equipped with Lebesgue measure.

(a) Verify that T is measure-preserving. (Hint: compute $\mu(T^{-1}([a, b)))$ for intervals $[a, b) \subset [0, 1)$.)

(b) Let $A = [0, 1/4)$. Compute the first few iterates of the point $x_0 = 0.1$ and verify that it returns to A.

(c) Find a point in $[0, 1)$ that *never* returns to $A = [0, 1/4)$. Why does this not contradict the Poincaré recurrence theorem?

Exercise 4.3. (Failure of recurrence for infinite measure). Let $T : \mathbb{R} \to \mathbb{R}$ be the translation $T(x) = x + 1$, equipped with Lebesgue measure.

(a) Verify that T is measure-preserving.

(b) Show that no point of $A = [0, 1)$ ever returns to A under iteration of T.

(c) Explain precisely which hypothesis of the Poincaré recurrence theorem fails.

Exercise 4.4. (Stroboscopic map). Consider the periodically forced linear oscillator:

$$\ddot{x} + \omega_0^2 x = \cos(\omega t).$$

(a) Find the general solution (assume $\omega \neq \omega_0$).

(b) Define the stroboscopic Poincaré map P at the forcing period $T = 2\pi/\omega$, mapping $(x(0), \dot{x}(0))$ to $(x(T), \dot{x}(T))$.

(c) Show that every orbit of P lies on a circle (or a point) in the $(x, \dot{x})$ plane. What does this say about the recurrence of the original system?

(d) What goes wrong at resonance ($\omega = \omega_0$)? Does the Poincaré map still have bounded orbits?

Exercise 4.5. (Floquet multipliers). Consider the Mathieu equation:

$$\ddot{x} + (\delta + \epsilon \cos t)\, x = 0.$$

This is a linear ODE with π-periodic (if we use period 2π, more commonly) or 2π-periodic coefficients.

(a) Write this as a first-order system $\dot{\xi} = A(t)\xi$ with $\xi = (x, \dot{x})^T$. What is $A(t)$?

(b) The monodromy matrix $M = \Phi(2\pi)$ is 2×2. Show that $\det(M) = 1$. (Hint: use Liouville's formula $\det \Phi(t) = \det \Phi(0) \exp(\int_0^t \operatorname{tr} A(s)\, ds)$.)

(c) Given that $\det(M) = 1$, the two Floquet multipliers satisfy $\rho_1 \rho_2 = 1$. What are the possible stability scenarios? (Consider: both real, both complex.)

(d) For $\epsilon = 0$ and $\delta = 1$, compute the monodromy matrix and Floquet multipliers explicitly.

Exercise 4.6. (Kac's lemma — a finite example). Let $X = \{0, 1, 2, 3, 4\}$ with the counting measure normalized to a probability measure ($\mu(\{i\}) = 1/5$). Let T be the cyclic permutation $T(i) = i + 1 \pmod 5$.

(a) Verify that T is measure-preserving and ergodic.

(b) Let $A = \{0, 1\}$. For each point in A, compute the first-return time $n_A(x)$.

(c) Compute the average return time to A and verify Kac's lemma.

Exercise 4.7. (The Lorenz map, numerically). Using a numerical ODE solver of your choice, integrate the Lorenz equations ($\sigma = 10$, $b = 8/3$, $r = 28$) from initial condition $(1, 1, 1)$ for a long time (say $t \in [0, 1000]$, discarding the transient $t \in [0, 50]$).

(a) Record the successive local maxima z_n of the z-coordinate. Plot z_{n+1} vs. z_n. What do you observe?

(b) Estimate the slope of the resulting map. What does the fact that the slope has magnitude greater than 1 imply about the dynamics?

(c) Does the map appear to have a fixed point? If so, at approximately what value of z?

Recommended Reading

1. **Strogatz, S. H.** *Nonlinear Dynamics and Chaos* (2nd ed., Westview Press, 2015). Chapters 7 and 8 give an accessible introduction to limit cycles and the Poincaré map, with many worked examples. An excellent first reference.
2. **Guckenheimer, J. and Holmes, P.** *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields* (Springer, 1983). Chapter 1 covers Poincaré maps, Floquet theory, and their application to bifurcation theory with full mathematical rigor. The standard graduate reference.
3. **Katok, A. and Hasselblatt, B.** *Introduction to the Modern Theory of Dynamical Systems* (Cambridge University Press, 1995). The Poincaré recurrence theorem is proved in Section 4.1, and the connections to ergodic theory are developed throughout Part II. The definitive modern reference for the mathematical theory.
4. **Poincaré, H.** *Les Méthodes Nouvelles de la Mécanique Céleste*, Vols. 1–3 (Gauthier-Villars, 1892–1899). The original source. Volume 1, Chapter 3 introduces the method of sections; the recurrence theorem appears in Volume 1, §8. Difficult but historically essential.
5. **Chicone, C.** *Ordinary Differential Equations with Applications* (2nd ed., Springer, 2006). Chapter 4 contains a careful modern treatment of Floquet theory with complete proofs.
6. **Lorenz, E. N.** "Deterministic Nonperiodic Flow," *Journal of the Atmospheric Sciences* **20** (1963), 130–141. The paper that started it all. The near-one-dimensional return map is described in Section 4.
7. **Kac, M.** "On the Notion of Recurrence in Discrete Stochastic Processes," *Bulletin of the AMS* **53** (1947), 1002–1010. The original paper proving the lemma on expected return times.

Chapter 5

Bifurcations

"The main interest of the theory of bifurcations is to divide the parameter space into regions within which the system is structurally stable, and to study the changes that occur on the boundaries of these regions." — V.I. Arnold, *Geometrical Methods in the Theory of Ordinary Differential Equations*

In the preceding chapters, we studied the qualitative behavior of dynamical systems with *fixed* equations of motion. We classified fixed points, determined their stability, and drew phase portraits. But real systems are never completely isolated: they depend on external conditions — temperature, forcing amplitude, feedback gain, population carrying capacity, laser pump rate. Mathematically, these enter as *parameters*. The central question of this chapter is: **what happens to the qualitative structure of a dynamical system as a parameter varies?**

The answer is that, for most parameter values, nothing dramatic occurs — the phase portrait deforms smoothly but retains its essential topology. At special, isolated parameter values, however, the system undergoes a **bifurcation**: a qualitative change in the dynamics. Fixed points appear, disappear, exchange stability, or give birth to periodic orbits. Understanding these transitions is not merely a mathematical curiosity. Bifurcations govern the onset of oscillations in lasers, the buckling of engineered structures, the extinction thresholds in ecology, and — as we shall see — the route from regular to chaotic dynamics.

5.1 Structural Stability and Bifurcation

5.1.1 Parametric Families of Dynamical Systems

Consider a family of differential equations

$$\dot{x} = f_\mu(x), \quad x \in \mathbb{R}^n, \quad \mu \in \mathbb{R}^p,$$

where μ is a parameter (or vector of parameters). For each fixed value of μ, we have a dynamical system in the sense of Chapter 3. As μ varies, the vector field f_μ changes, and with it the phase portrait.

Definition 5.1 (Bifurcation, informal)**.** A value $\mu = \mu_0$ is called a **bifurcation value** (or **bifurcation point** in parameter space) if the qualitative structure of the phase portrait of $\dot{x} = f_\mu(x)$ changes as μ passes through μ_0.

To make "qualitative structure" precise, we need the concept of topological equivalence.

Definition 5.2 (Topological equivalence)**.** Two autonomous systems $\dot{x} = f(x)$ and $\dot{x} = g(x)$ on $\mathbb{R}^n$ are **topologically equivalent** if there exists a homeomorphism $h : \mathbb{R}^n \to \mathbb{R}^n$ that maps orbits of the first system to orbits of the second, preserving the direction of time.

Note that h need not preserve the parametrization of orbits by time — only the oriented orbit structure. This is intentional: we want to capture the *topology* of the phase portrait (how orbits are arranged), not the precise speed at which trajectories traverse them.

5.1.2 Structural Stability

Definition 5.3 (Structural stability)**.** A system $\dot{x} = f(x)$ is **structurally stable** if every sufficiently small (in an appropriate C^1 topology) perturbation of f yields a topologically equivalent system.

Structurally stable systems are robust: small changes to the right-hand side — whether from modelling uncertainty, parameter drift, or noise — do not alter the qualitative dynamics.

Example 5.1. Consider $\dot{x} = -x$ on $\mathbb{R}$. The origin is a stable fixed point. For any sufficiently small smooth perturbation g, the perturbed system $\dot{x} = -x + g(x)$ still has a unique fixed point near the origin, and it is still stable (by continuity of eigenvalues and the implicit function theorem). The system is structurally stable.

Example 5.2. Consider $\dot{x} = x^2$ on $\mathbb{R}$. The origin is a non-hyperbolic fixed point (the linearization vanishes). The perturbation $\dot{x} = x^2 + \epsilon$ with $\epsilon > 0$ has no fixed points at all, while $\dot{x} = x^2 - \epsilon$ with $\epsilon > 0$ has two fixed points $x = \pm\sqrt{\epsilon}$. The qualitative structure changes under arbitrarily small perturbation. This system is *not* structurally stable.

The following classical result, due to Peixoto (1962) in the two-dimensional case, relates structural stability to the character of fixed points and periodic orbits.

Theorem 5.1 (Sufficient condition for structural stability)**.** **If all fixed points and periodic orbits of a flow on $\mathbb{R}^n$ are hyperbolic, and there are no saddle connections (heteroclinic orbits connecting saddle points), then the system is structurally stable.**

In one dimension the situation is simpler: a system $\dot{x} = f(x)$ on a compact interval is structurally stable if and only if all its fixed points are hyperbolic, i.e., $f'(x^*) \neq 0$ at every fixed point x^*.

The upshot. *Bifurcations occur precisely at the boundary of structural stability. They are associated with the failure of hyperbolicity: an eigenvalue of the linearization at a fixed point crosses the imaginary axis (for flows) or crosses the unit circle (for maps). This is the guiding principle that organizes the entire theory.*

—

5.2 Bifurcations in One Dimension

We begin with scalar equations $\dot{x} = f(x, \mu)$, where $x \in \mathbb{R}$ and $\mu \in \mathbb{R}$. The theory is particularly clean here because the dynamics are constrained: trajectories can only move left or right on the real line, so the only possible invariant sets are fixed points. A bifurcation occurs when fixed points are created, destroyed, or exchange stability.

A fixed point x^* of $\dot{x} = f(x, \mu)$ satisfies $f(x^*, \mu) = 0$. Its stability is determined by the sign of $f_x(x^*, \mu) = \partial f/\partial x|_{(x^*,\mu)}$:

- If $f_x(x^*, \mu) < 0$, the fixed point is **stable**.
- If $f_x(x^*, \mu) > 0$, the fixed point is **unstable**.
- If $f_x(x^*, \mu) = 0$, the fixed point is **non-hyperbolic**, and a bifurcation may occur.

5.2.1 Saddle-Node Bifurcation

The **saddle-node bifurcation** (also called a fold or tangent bifurcation) is the most fundamental mechanism by which fixed points are created or destroyed.

Normal form. The canonical one-dimensional saddle-node bifurcation is given by

$$\dot{x} = \mu - x^2. \tag{5.1}$$

Analysis. Setting $\dot{x} = 0$ gives $x^2 = \mu$, so

$$x^* = \pm\sqrt{\mu}.$$

- For $\mu < 0$: no real fixed points exist. Every trajectory moves to the left (since $\mu - x^2 < 0$ for all x).
- For $\mu = 0$: exactly one fixed point at $x^* = 0$. Since $f_x(0, 0) = -2x|_{x=0} = 0$, it is non-hyperbolic. This is the bifurcation point.
- For $\mu > 0$: two fixed points at $x^* = \pm\sqrt{\mu}$.
- At $x^* = +\sqrt{\mu}$: $f_x = -2\sqrt{\mu} < 0$, so this fixed point is **stable**.
- At $x^* = -\sqrt{\mu}$: $f_x = +2\sqrt{\mu} > 0$, so this fixed point is **unstable**.

Bifurcation diagram. Plotting the fixed points x^* as a function of μ, we obtain a parabola $x^* = \pm\sqrt{\mu}$ opening to the right. The upper branch (stable) is drawn as a solid curve; the lower branch (unstable) as a dashed curve. At $\mu = 0$, the two branches meet and annihilate.

The saddle-node bifurcation has a characteristic *square-root scaling*: near the bifurcation, the distance between the two fixed points grows as $\sim \sqrt{\mu}$. This scaling is universal for saddle-node bifurcations.

Theorem 5.2 (Saddle-node bifurcation)**.** **Consider $\dot{x} = f(x, \mu)$ with f smooth. Suppose that at (x_0, μ_0):**

1. $f(x_0, \mu_0) = 0$ (fixed point exists), 2. $f_x(x_0, \mu_0) = 0$ (non-hyperbolic), 3. $f_\mu(x_0, \mu_0) \neq 0$ (the parameter unfolds the degeneracy), 4. $f_{xx}(x_0, \mu_0) \neq 0$ (quadratic tangency, not higher-order).

**Then in a neighborhood of (x_0, μ_0), the system is locally topologically equivalent to the normal form $\dot{x} = \mu \pm x^2$ (the sign determined by the signs of conditions 3 and 4). In particular,*

*a pair of fixed points — one stable, one unstable — is created or destroyed as μ passes through μ_0.**

Proof sketch. sketch.* By the implicit function theorem, conditions 1–2 imply that the curve of fixed points in the (x, μ)-plane has a fold (a turning point) at (x_0, μ_0). Condition 3 ensures that the curve is not tangent to the x-axis in parameter space, and condition 4 ensures the fold is quadratic, not a cusp or higher-order tangency. A smooth change of coordinates $(x, \mu) \mapsto (X, M)$ brings the equation to the form $\dot{X} = M \pm X^2$ near the bifurcation. □

□

Example 5.3. (Laser threshold). A simplified model for a single-mode laser relates the photon number $n \geq 0$ to the gain G and loss κ:

$$\dot{n} = (G - \kappa)n - \beta n^2 + n_{\text{sp}},$$

where $\beta > 0$ accounts for gain saturation and $n_{\text{sp}} > 0$ represents spontaneous emission (a small noise-like term). Setting $\mu = G - \kappa$ and rescaling, this takes a form similar to $\dot{n} = \mu n - n^2 + \epsilon$. For small ϵ, the system undergoes a saddle-node-like bifurcation: below threshold ($\mu < \mu_c$), the laser output is negligible; above threshold, it jumps to a macroscopic value. The abruptness of this transition (a "kink" rather than a smooth turn-on) is a hallmark of the saddle-node mechanism.

5.2.2 Transcritical Bifurcation

In many physical problems, a fixed point exists for all parameter values but exchanges its stability with another fixed point at the bifurcation. This occurs when the system has a structural reason (e.g., a conserved quantity, a symmetry, or a physical constraint) that forces a fixed point to persist.

Normal form.

$$\dot{x} = \mu x - x^2. \tag{5.2}$$

Analysis. The fixed points are $x^* = 0$ and $x^* = \mu$.

- The linearization at $x^* = 0$ gives $f_x(0, \mu) = \mu$.
- Stable for $\mu < 0$, unstable for $\mu > 0$.
- The linearization at $x^* = \mu$ gives $f_x(\mu, \mu) = \mu - 2\mu = -\mu$.
- Unstable for $\mu < 0$, stable for $\mu > 0$.

At $\mu = 0$, the two fixed points collide (both at $x = 0$) and exchange stability. For $\mu < 0$, the origin is stable and the other fixed point $x = \mu < 0$ is unstable. For $\mu > 0$, the situation reverses. This **exchange of stability** is the defining feature of the transcritical bifurcation.

Bifurcation diagram. Two curves of fixed points cross in the (μ, x)-plane: the horizontal line $x = 0$ and the diagonal $x = \mu$. At $\mu = 0$, they intersect and swap their stability (solid $\leftrightarrow$ dashed).

Theorem 5.3 (Transcritical bifurcation)**.** **Consider $\dot{x} = f(x, \mu)$ with f smooth. Suppose that at (x_0, μ_0):**

1. $f(x_0, \mu_0) = 0$*, 2.* $f_x(x_0, \mu_0) = 0$*, 3.* $f_\mu(x_0, \mu_0) = 0$ **(the fixed point exists for all* μ *near* μ_0*)*, 4.* $f_{x\mu}(x_0, \mu_0) \neq 0$*, 5.* $f_{xx}(x_0, \mu_0) \neq 0$*.*

Then the system is locally topologically equivalent to* $\dot{x} = \mu x \pm x^2$*.

Example 5.4. (Logistic population model). Consider the population model

$$\dot{N} = rN\left(1 - \frac{N}{K}\right) - hN,$$

where r is the intrinsic growth rate, K the carrying capacity, and h a per-capita harvesting rate. The parameter of interest is h. Factoring:

$$\dot{N} = N\big[(r - h) - (r/K)N\big].$$

The fixed points are $N^* = 0$ (extinction) and $N^* = K(1 - h/r)$ (coexistence). These coincide when $h = r$. Setting $\mu = r - h$ and $x = N$, we get $\dot{x} = \mu x - (r/K)x^2$, which is precisely the transcritical normal form (up to rescaling). At $h = r$, the coexistence equilibrium crosses through zero and becomes negative (biologically meaningless), while the extinction state gains stability. The transcritical bifurcation thus represents the harvesting threshold beyond which the population collapses.

5.2.3 Pitchfork Bifurcation

The pitchfork bifurcation occurs in systems with a **symmetry** $x \mapsto -x$. If $f(-x, \mu) = -f(x, \mu)$ for all x, μ, then $x = 0$ is always a fixed point, and any bifurcation from $x = 0$ must respect the symmetry: new fixed points must appear in symmetric pairs.

Supercritical Pitchfork

Normal form.

$$\dot{x} = \mu x - x^3. \tag{5.3}$$

Note the symmetry: $f(-x, \mu) = -\mu x + x^3 = -(\mu x - x^3) = -f(x, \mu)$.

Analysis. Setting $\dot{x} = 0$: $x(\mu - x^2) = 0$, giving $x^* = 0$ and $x^* = \pm\sqrt{\mu}$ (the latter existing only for $\mu > 0$).

Stability:

- $x^* = 0$: $f_x(0, \mu) = \mu$. Stable for $\mu < 0$, unstable for $\mu > 0$.
- $x^* = \pm\sqrt{\mu}$ (for $\mu > 0$): $f_x(\pm\sqrt{\mu}, \mu) = \mu - 3\mu = -2\mu < 0$. **Stable.**

For $\mu < 0$, only the origin exists and is stable. As μ increases through zero, the origin loses stability and two new stable fixed points $\pm\sqrt{\mu}$ are born. The system must "choose" one of the two branches — this is **symmetry-breaking**.

The term *supercritical* indicates that the new (non-trivial) fixed points are stable: the bifurcation produces a "soft" transition.

Bifurcation diagram. A pitchfork shape: the horizontal axis ($x = 0$, switching from solid to dashed at $\mu = 0$) together with the two prongs $x = \pm\sqrt{\mu}$ for $\mu > 0$ (solid, stable).

Subcritical Pitchfork

Normal form.

$$\dot{x} = \mu x + x^3. \tag{5.4}$$

Analysis. Fixed points: $x^* = 0$ and $x^* = \pm\sqrt{-\mu}$ (existing for $\mu < 0$).

Stability:

- $x^* = 0$: $f_x = \mu$. Stable for $\mu < 0$, unstable for $\mu > 0$.
- $x^* = \pm\sqrt{-\mu}$ (for $\mu < 0$): $f_x = \mu + 3(-\mu) = -2\mu > 0$. **Unstable.**

Here the bifurcation is "dangerous": for $\mu < 0$, the origin is stable but is flanked by two unstable fixed points. When μ passes through zero, the origin loses stability and the unstable fixed points have already disappeared — there is no nearby stable state. Trajectories are ejected to large $|x|$ (or to some distant attractor not captured by the local normal form). This is sometimes called a **hard transition** or **catastrophic bifurcation**.

In practice, subcritical pitchforks are often "completed" by adding a stabilizing quintic term: $\dot{x} = \mu x + x^3 - x^5$. This produces a rich bifurcation diagram with hysteresis.

Example 5.5. (Euler buckling of a beam). Consider a thin elastic beam compressed axially by a load P. By symmetry, the straight configuration $\theta = 0$ (where θ measures lateral deflection) is always an equilibrium. For $P < P_c$ (the Euler critical load), the straight state is stable. For $P > P_c$, it becomes unstable, and the beam buckles into one of two symmetric bent configurations $\theta = \pm\theta^*$. The governing equation, after suitable nondimensionalization, takes the form

$$\dot{\theta} = (\mu)\theta - \theta^3 + \text{h.o.t.}, \quad \mu = \frac{P - P_c}{P_c},$$

which is exactly the supercritical pitchfork. The buckling of the beam is a physical realization of symmetry-breaking.

Example 5.6. (Ising-type spin model). In the mean-field Ising model, the magnetization m satisfies the self-consistency equation $m = \tanh(\beta J m + \beta h)$, where $\beta = 1/(k_B T)$, J is the coupling constant, and h is the external field. For $h = 0$ (no external field), expanding tanh for small m:

$$\dot{m} \approx (\beta J - 1)m - \frac{(\beta J)^3}{3} m^3,$$

(interpreting the self-consistency equation as describing relaxation dynamics). Setting $\mu = \beta J - 1$, this is a supercritical pitchfork. The bifurcation at $\mu = 0$ (i.e., $T = T_c = J/k_B$) corresponds to the **ferromagnetic phase transition**: above the critical temperature, the only equilibrium is $m = 0$ (the disordered phase); below it, spontaneous magnetization $m = \pm m^*$ appears.

5.2.4 Summary of One-Dimensional Bifurcations

| Bifurcation | Normal form | Fixed points created/destroyed | Key feature | |—|—|—|—| | Saddle-node | $\dot{x} = \mu - x^2$ | 2 created (or destroyed) | Generic: no symmetry required | | Transcritical | $\dot{x} = \mu x - x^2$ | 0; exchange of stability | Fixed point exists for all μ | | Pitchfork (super) | $\dot{x} = \mu x - x^3$ | 2 created (stable) | Requires $x \mapsto -x$ symmetry | | Pitchfork (sub) | $\dot{x} = \mu x + x^3$ | 2 destroyed (unstable) | Requires $x \mapsto -x$ symmetry |

A fundamental organizing principle: the **saddle-node bifurcation is the only generic one-parameter bifurcation in one dimension**. The transcritical and pitchfork bifurcations occur only when additional structure (a persistent fixed point, a symmetry) constrains the system. Without such constraints, any bifurcation can be perturbed into a saddle-node.

5.3 Normal Form Theory

The analysis of Section 5.2 relied on simple polynomial normal forms. Why should we believe that these capture the behavior of general systems? The answer is provided by **normal form theory**, which guarantees that near a bifurcation, any smooth system can be transformed — by a smooth, parameter-dependent change of coordinates — into one of a small number of canonical forms, up to higher-order terms that do not affect the qualitative behavior.

5.3.1 The Idea

Consider a system $\dot{x} = f(x, \mu)$ with a non-hyperbolic fixed point at (x_0, μ_0). The Taylor expansion of f in $(x - x_0)$ and $(\mu - \mu_0)$ begins with the terms that vanish (by the non-hyperbolicity condition) and continues with quadratic, cubic, and higher-order terms. Not all of these terms are "essential": many can be eliminated by near-identity coordinate changes $x = y + \phi(y, \mu)$.

The terms that *cannot* be eliminated are called **resonant terms**, and the resulting simplified system is the **normal form**. The normal forms we encountered (e.g., $\mu - x^2$ for saddle-node, $\mu x - x^3$ for pitchfork) are precisely these irreducible expressions. The theory guarantees that higher-order terms ($x^3, x^4, \ldots$ in the saddle-node case) can be removed and do not affect the local bifurcation structure.

More formally:

Theorem 5.4 (Normal form reduction, informal)**.** **Let $\dot{x} = f(x, \mu)$ satisfy the non-degeneracy conditions of a given bifurcation theorem (e.g., Theorem 5.2 for saddle-node). Then there exists a smooth, parameter-dependent, near-identity change of coordinates in a neighborhood of the bifurcation point that transforms the system into its normal form plus terms of order higher than those appearing in the normal form. These higher-order terms do not affect the local topological type of the bifurcation.**

For a rigorous treatment, see Guckenheimer and Holmes (1983), Chapter 3, or Kuznetsov (2004), Chapter 2.

5.3.2 Center Manifold Reduction

For higher-dimensional systems, we typically encounter a bifurcation in which *one* eigenvalue (or a pair of complex conjugate eigenvalues) of the linearization crosses the imaginary axis, while the remaining eigenvalues have strictly negative real parts. The **center manifold theorem** allows us to reduce the analysis to a low-dimensional problem.

Theorem 5.5 (Center manifold theorem)**.** **Consider $\dot{x} = f(x)$ with $f(0) = 0$ and $x \in \mathbb{R}^n$. Let the linearization $Df(0)$ have eigenvalues $\lambda_1, \ldots, \lambda_n$. Suppose n_c of these eigenvalues lie on the*

*imaginary axis (the "center" eigenvalues) and $n_s = n - n_c$ have strictly negative real part (the "stable" eigenvalues). Then:**

*1. *There exists a local invariant manifold W^c (the **center manifold**) tangent to the center eigenspace E^c at the origin, of dimension n_c.* 2. *W^c is locally attracting: all nearby trajectories approach W^c exponentially fast.* 3. *The dynamics on W^c determine the local qualitative behavior near the origin. In particular, the stability of the origin for the full system is determined by the stability of the origin for the reduced system on W^c.**

The center manifold is generally not unique (it is C^k for any finite k but typically not analytic), though the Taylor expansion to any finite order is unique — and this is all we need for the normal form analysis.

***The practical recipe.** To analyze a bifurcation in an n-dimensional system:*

1. Compute the linearization at the non-hyperbolic fixed point. 2. Identify the center eigenspace E^c (corresponding to eigenvalues on the imaginary axis). 3. Compute the center manifold W^c as a graph $x_s = h(x_c)$ over the center variables, where h satisfies an invariance equation derived from the original ODE. 4. Restrict the dynamics to W^c, obtaining a low-dimensional (often 1D or 2D) system. 5. Apply normal form theory to the reduced system.

We will see this recipe in action for the Hopf bifurcation in the next section.

—

5.4 Hopf Bifurcation

The one-dimensional bifurcations of Section 5.2 involve the creation, destruction, or destabilization of *fixed points*. In two or more dimensions, a fundamentally new phenomenon becomes possible: a fixed point can lose stability and simultaneously give birth to a **limit cycle** — a periodic orbit. This is the **Hopf bifurcation** (also called the Poincaré–Andronov–Hopf bifurcation).

5.4.1 Setup and Statement

Consider a planar system

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mu), \quad \mathbf{x} \in \mathbb{R}^2, \quad \mu \in \mathbb{R},$$

with a fixed point $\mathbf{x}^*(\mu)$ for μ near zero. Denote the eigenvalues of the Jacobian $Df(\mathbf{x}^*(\mu), \mu)$ by

$$\lambda(\mu) = \alpha(\mu) \pm i\omega(\mu).$$

We assume these eigenvalues are complex conjugate with nonzero imaginary part.

Theorem 5.6 (Hopf bifurcation theorem)**.** *Suppose:*

*1. *At $\mu = 0$, the Jacobian at $\mathbf{x}^*(0)$ has a pair of purely imaginary eigenvalues $\lambda(0) = \pm i\omega_0$ with $\omega_0 > 0$.* 2. *(Transversality) $\alpha'(0) \neq 0$ — the eigenvalues cross the imaginary axis with nonzero speed.* 3. *(Non-degeneracy) The first Lyapunov coefficient ℓ_1 (defined below) satisfies $\ell_1 \neq 0$.**

Then in a neighborhood of $(x^*(0), 0)$ in $\mathbb{R}^2 \times \mathbb{R}$:

*- *If $\ell_1 < 0$ (**supercritical** case): for μ on the side where the fixed point is unstable, a unique stable limit cycle exists, with amplitude growing as $O(\sqrt{|\mu|})$.* - *If $\ell_1 > 0$ (**subcritical** case): for μ on the side where the fixed point is stable, a unique unstable limit cycle exists, with amplitude growing as $O(\sqrt{|\mu|})$.**

In either case, the limit cycle is born from the fixed point at $\mu = 0$.

***The first Lyapunov coefficient** ℓ_1 is a quantity computed from the second- and third-order Taylor coefficients of f at the bifurcation point. Its sign determines whether the nonlinear terms are stabilizing (supercritical) or destabilizing (subcritical). The full formula is somewhat involved; we give it for the case where the system has been put in the form $\dot{z} = (\alpha(\mu) + i\omega(\mu))z + c_1 z|z|^2 + \cdots$ using complex coordinates $z = x_1 + ix_2$:*

$$\ell_1 = \frac{1}{\omega_0} \operatorname{Re}(c_1),$$

where c_1 is the coefficient of the cubic resonant term $z|z|^2$ in the normal form. The computation of c_1 from the original Taylor coefficients is described in detail in Kuznetsov (2004, Section 5.4).

5.4.2 Normal Form for Hopf Bifurcation

In polar coordinates (r, θ), the normal form of the Hopf bifurcation takes the particularly transparent form:

$$\dot{r} = \alpha(\mu)r + \ell_1 r^3 + O(r^5), \tag{5.5}$$

$$\dot{\theta} = \omega(\mu) + O(r^2). \tag{5.6}$$

The angular equation (5.6) simply says that the trajectory rotates with frequency $\approx \omega(\mu)$. The radial equation (5.5) controls the amplitude:

- For the **supercritical** case ($\ell_1 < 0$): when $\alpha > 0$ (fixed point unstable), the equation $\dot{r} = \alpha r + \ell_1 r^3 = 0$ has the nontrivial solution $r^* = \sqrt{-\alpha/\ell_1} = \sqrt{\alpha/|\ell_1|}$. This is a stable limit cycle of radius $\sim \sqrt{\alpha}$.
- For the **subcritical** case ($\ell_1 > 0$): when $\alpha < 0$ (fixed point stable), the equation $\dot{r} = \alpha r + \ell_1 r^3 = 0$ has $r^* = \sqrt{-\alpha/\ell_1} = \sqrt{|\alpha|/\ell_1}$. This is an unstable limit cycle.

5.4.3 Worked Example: A Concrete Hopf Bifurcation

Consider the system

$$\dot{x} = \mu x - y - x(x^2 + y^2), \tag{5.7}$$

$$\dot{y} = x + \mu y - y(x^2 + y^2). \tag{5.8}$$

Step 1: Fixed points. The origin $(0, 0)$ is a fixed point for all μ.

Step 2: Linearization. The Jacobian at the origin is

$$Df(0,0) = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}.$$

The eigenvalues are $\lambda = \mu \pm i$. Thus:

- $\alpha(\mu) = \mu$, $\omega(\mu) = 1$.
- At $\mu = 0$: eigenvalues are $\pm i$ (purely imaginary). Condition 1 is satisfied with $\omega_0 = 1$.
- $\alpha'(0) = 1 \neq 0$. Condition 2 (transversality) is satisfied.

Step 3: Polar coordinates. Let $x = r\cos\theta$, $y = r\sin\theta$. Then

$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r}, \quad \dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2}.$$

Computing $x\dot{x} + y\dot{y}$:

$$\begin{aligned} x\dot{x} + y\dot{y} &= x[\mu x - y - x(x^2+y^2)] + y[x + \mu y - y(x^2+y^2)] \\ &= \mu(x^2+y^2) - (xy - yx) - (x^2+y^2)^2 \\ &= \mu r^2 - r^4. \end{aligned}$$

Therefore $\dot{r} = \mu r - r^3$.

Computing $x\dot{y} - y\dot{x}$:

$$\begin{aligned} x\dot{y} - y\dot{x} &= x[x + \mu y - y(x^2+y^2)] - y[\mu x - y - x(x^2+y^2)] \\ &= x^2 + \mu xy - xy(x^2+y^2) - \mu xy + y^2 + xy(x^2+y^2) \\ &= x^2 + y^2 = r^2. \end{aligned}$$

Therefore $\dot{\theta} = 1$.

Step 4: Analysis of the radial equation. We have the exact (not approximate) equations:

$$\dot{r} = \mu r - r^3, \quad \dot{\theta} = 1.$$

Comparing with (5.5)–(5.6): $\alpha(\mu) = \mu$ and $\ell_1 = -1 < 0$. This is a **supercritical** Hopf bifurcation.

For $\mu \leq 0$: the only non-negative fixed point of $\dot{r} = r(\mu - r^2)$ is $r = 0$, which is stable (since $\mu \leq 0$). The origin is a stable spiral.

For $\mu > 0$: $r = 0$ is unstable. There is a stable fixed point of the radial equation at

$$r^* = \sqrt{\mu}.$$

This corresponds to a **stable limit cycle** of radius $\sqrt{\mu}$ in the (x, y)-plane, traversed with angular velocity $\dot{\theta} = 1$ (period $T = 2\pi$).

Step 5: Summary. At $\mu = 0$, a supercritical Hopf bifurcation occurs: the stable fixed point at the origin becomes an unstable spiral, and a stable limit cycle of radius $\sqrt{\mu}$ and period 2π is born. The amplitude of oscillation grows as the square root of the distance from the

bifurcation — a characteristic signature of the Hopf bifurcation.

This example is special in that the polar form is *exact* (no higher-order corrections). In general systems, one gets additional $O(r^5)$ terms in the radial equation and $O(r^2)$ terms in the angular equation, but these do not change the qualitative picture.

5.4.4 Physical Significance

The Hopf bifurcation is ubiquitous in applications:

- **Electrical circuits:** The onset of oscillation in an LC circuit with nonlinear feedback (e.g., the van der Pol oscillator) is a Hopf bifurcation.
- **Chemical kinetics:** The Belousov–Zhabotinsky reaction exhibits oscillatory behavior via Hopf bifurcation in the Oregonator model.
- **Neuroscience:** The FitzHugh–Nagumo model of neural excitability can undergo Hopf bifurcation, producing repetitive spiking.
- **Fluid dynamics:** The onset of vortex shedding behind a cylinder (at Reynolds number ≈ 47) is a Hopf bifurcation of the steady Navier–Stokes solution.

5.5 Period-Doubling Bifurcation and the Route to Chaos

We now turn to **discrete dynamical systems** (maps) and a bifurcation that has no direct analogue in one-dimensional flows: the **period-doubling** (or flip) bifurcation. This bifurcation is the gateway to one of the most remarkable phenomena in all of mathematics — the **period-doubling cascade** and the onset of chaos.

5.5.1 Period-Doubling Bifurcation for Maps

Consider a one-parameter family of maps $x_{n+1} = g_\mu(x_n)$ on $\mathbb{R}$, with a fixed point $x^*(\mu)$ satisfying $g_\mu(x^*) = x^*$. The stability of x^* is determined by the multiplier $\lambda(\mu) = g'_\mu(x^*)$:

- $|\lambda| < 1$: stable.
- $|\lambda| > 1$: unstable.

In the saddle-node bifurcation for maps, the multiplier leaves the unit circle through $+1$. In the **period-doubling bifurcation**, it leaves through -1.

Theorem 5.7 (Period-doubling bifurcation)**.** **Consider $x_{n+1} = g(x_n, \mu)$ with g smooth. Suppose at $\mu = 0$, the fixed point x^* has multiplier $g'_\mu(x^*) = -1$, and appropriate non-degeneracy conditions hold. Then:**

*1. *The fixed point x^* changes stability as μ passes through 0.* 2. *A period-2 orbit $\{p, q\}$ with $g(p) = q$ and $g(q) = p$ is born (or destroyed) at $\mu = 0$.* 3. If the bifurcation is supercritical, the period-2 orbit is stable when it exists.*

Normal form. *Near the bifurcation, the second iterate $g_\mu^2 = g_\mu \circ g_\mu$ undergoes a pitchfork bifurcation (by the symmetry $g^2(x) - x^*$ inherits from the map structure). The period-2 points are precisely the non-trivial fixed points of g_μ^2.*

***Why $\lambda = -1$?** When the multiplier is -1, successive iterates alternate on opposite sides of the fixed point with equal amplitude: $x_{n+1} - x^* \approx -(x_n - x^*)$. This alternation naturally produces a period-2 pattern when the amplitude is stabilized by nonlinear terms.*

5.5.2 The Logistic Map

The **logistic map** is the single most important example in discrete dynamics:

$$x_{n+1} = f_r(x) = rx(1-x), \quad x \in [0,1], \quad r \in [0,4]. \tag{5.9}$$

Despite its simplicity — it is a quadratic map of the interval — it exhibits the full range of dynamical behavior from stable fixed points through periodic orbits to chaos. We now trace the period-doubling cascade in detail.

Fixed points. Setting $f_r(x) = x$: $rx(1-x) = x$, so $x(r-1-rx) = 0$. The fixed points are

$$x_0^* = 0, \quad x_1^* = 1 - 1/r \quad (r > 1).$$

Stability of $x_0^* = 0$: $f_r'(0) = r$. Stable for $r < 1$, unstable for $r > 1$. At $r = 1$, a transcritical bifurcation occurs.

Stability of $x_1^* = 1 - 1/r$: $f_r'(x_1^*) = r - 2rx_1^* = r - 2(r-1) = 2 - r$. So $|f_r'(x_1^*)| < 1$ iff $-1 < 2 - r < 1$, i.e., $1 < r < 3$. At $r = 3$, the multiplier equals -1 and a period-doubling bifurcation occurs.

The first period-doubling ($r = 3$). For r slightly above 3, the fixed point x_1^* is unstable, and a stable period-2 orbit $\{p, q\}$ appears. These are the solutions of $f_r^2(x) = x$ other than the fixed points. The period-2 orbit is stable for $r \in (3, 1+\sqrt{6})$, where $1 + \sqrt{6} \approx 3.449$.

Let us verify the birth of the period-2 orbit. We need to solve $f_r(f_r(x)) = x$. Define $F(x) = f_r^2(x) = r \cdot f_r(x) \cdot (1 - f_r(x)) = r^2 x(1-x)(1 - rx + rx^2)$. Since the fixed points of f_r are also fixed points of f_r^2, we can factor:

$$f_r^2(x) - x = (f_r(x) - x) \cdot Q(x),$$

where $Q(x)$ is a quadratic. Dividing, one finds that the period-2 points satisfy

$$r^2x^2 - r(r+1)x + (r+1) = 0.$$

By the quadratic formula:

$$x = \frac{r(r+1) \pm \sqrt{r^2(r+1)^2 - 4r^2(r+1)}}{2r^2} = \frac{(r+1) \pm \sqrt{(r+1)(r-3)}}{2r}.$$

These are real and distinct for $r > 3$, confirming the birth of the period-2 orbit at $r = 3$.

The second period-doubling ($r \approx 3.449$). At $r = r_2 = 1 + \sqrt{6} \approx 3.449$, the multiplier of the period-2 orbit (computed as the product $(f_r^2)'(p) = f_r'(p) \cdot f_r'(q)$ by the chain rule) reaches -1. The period-2 orbit loses stability, and a stable period-4 orbit is born.

The cascade continues. This process repeats: each period-2^n orbit loses stability at a parameter value r_n, spawning a stable period-2^{n+1} orbit. The parameter values accumulate:

$$r_1 = 3, \quad r_2 \approx 3.4495, \quad r_3 \approx 3.5441, \quad r_4 \approx 3.5644, \quad \ldots$$

The sequence $\{r_n\}$ converges to a limit

$$r_\infty = \lim_{n\to\infty} r_n \approx 3.5699\ldots$$

Beyond r_∞, the system exhibits chaotic behavior (though interspersed with windows of periodicity, as we discuss in Section 5.6).

5.5.3 Feigenbaum's Universality

In 1978, Mitchell Feigenbaum discovered that the period-doubling cascade possesses remarkable **universal** quantitative properties, independent of the specific map.

Definition 5.4 (Feigenbaum constants)**.** Define the ratios

$$\delta_n = \frac{r_n - r_{n-1}}{r_{n+1} - r_n}.$$

Feigenbaum showed numerically, and later Lanford (1982) proved rigorously for certain cases, that

$$\delta = \lim_{n\to\infty} \delta_n = 4.6692016091029\ldots \tag{5.10}$$

This constant δ is **universal**: it takes the same value for *any* smooth, unimodal map of the interval (a map with a single quadratic maximum).

There is a second universal constant α governing the spatial scaling. At the accumulation point r_∞, the attractor has a self-similar structure. If d_n denotes the distance between the nearest points of the period-2^n orbit near the critical point, then

$$\frac{d_n}{d_{n+1}} \to \alpha = -2.5029078750957\ldots \tag{5.11}$$

The negative sign reflects an alternation in the orientation of the scaled pattern.

Why is this remarkable? Consider the map $x_{n+1} = r\sin(\pi x)$ on $[0, 1]$. This is not the logistic map — it is not even a polynomial. Yet its period-doubling cascade converges with the *same* ratio $\delta \approx 4.669\ldots$ and the *same* spatial scaling $\alpha \approx -2.503\ldots$. The same constants appear for $x_{n+1} = rxe^{-x}$, for $x_{n+1} = r(1 - |2x - 1|^z)$ with $z = 2$, and for any smooth unimodal map with a quadratic maximum. This is a manifestation of **universality** — a concept borrowed from the theory of critical phenomena in statistical physics, where diverse physical systems share identical critical exponents.

The renormalization group explanation. Feigenbaum's universality is explained by a **functional renormalization group** approach. Define the **doubling operator** $\mathcal{T}$ acting on the space of unimodal maps:

$$(\mathcal{T}g)(x) = -\alpha\, g(g(-x/\alpha)),$$

where α is the spatial rescaling factor. The key insight is that at the accumulation point of period-doubling, the map is a **fixed point** of $\mathcal{T}$: there exists a function g^* such that $\mathcal{T}g^* = g^*$. The constant δ is the unstable eigenvalue of the linearization of $\mathcal{T}$ at g^*. Since the renormalization fixed point has only one unstable direction, all unimodal maps with a quadratic maximum are attracted to the same fixed point and hence share the same universal constants.

This is analogous to the universality of critical exponents in statistical mechanics: the "irrelevant" details of the microscopic model (whether it is the logistic map, or $r\sin(\pi x)$, or anything else) are washed away by the renormalization flow, leaving only the universal behavior.

Numerical verification. We can verify the convergence of δ_n for the logistic map:

| n | r_n | $r_n - r_{n-1}$ | δ_n | |—|—|—|—| | 1 | 3.0000000 | — | — | | 2 | 3.4494897 | 0.4494897 | — | | 3 | 3.5440903 | 0.0946006 | 4.7514 | | 4 | 3.5644073 | 0.0203170 | 4.6562 | | 5 | 3.5687594 | 0.0043521 | 4.6684 | | 6 | 3.5696916 | 0.0009322 | 4.6686 | | 7 | 3.5698913 | 0.0001997 | 4.6692 |

The convergence to $\delta \approx 4.6692$ is evident.

—

5.6 Bifurcation Diagrams

5.6.1 Construction

A **bifurcation diagram** for a one-parameter family of maps $x_{n+1} = g_\mu(x_n)$ is a plot of the asymptotic behavior of orbits (the attractor) as a function of μ. The standard numerical procedure is:

1. For each value of μ in a grid:
2. (a) Choose an initial condition x_0 (typically near a critical point).
3. (b) Iterate the map many times (say N_{trans} transient iterations) to approach the attractor.
4. (c) Record the next N_{plot} iterates.
5. Plot all recorded iterates against μ.

This produces a picture that simultaneously reveals fixed points, periodic orbits, and chaotic bands as μ varies.

5.6.2 The Logistic Map Bifurcation Diagram

The bifurcation diagram of the logistic map $f_r(x) = rx(1-x)$ for $r \in [2.5, 4]$ is one of the iconic images of nonlinear science. Its features include:

Region $r \in [2.5, 3)$: A single curve — the stable fixed point $x_1^* = 1 - 1/r$. The attractor is a single point for each r.

Region $r \in [3, 3.449\ldots)$: The fixed point has bifurcated; the attractor consists of 2 points (the period-2 orbit). Two curves are visible, branching from the fixed-point curve.

Region $r \in [3.449\ldots, 3.570\ldots)$: Successive period-doublings produce period-4, period-8, period-16, ... orbits. The diagram shows a cascade of finer and finer branching, accumulating at $r_\infty \approx 3.5699$.

Region $r \in [3.570\ldots, 4]$: **Mostly chaotic behavior, with the attractor consisting of bands of points (indicating sensitive dependence). However, this region is punctuated by windows of periodicity**.

5.6.3 Windows of Periodicity

Within the chaotic regime, there are infinitely many **periodic windows** — intervals of r where a stable periodic orbit suddenly appears. The most prominent is the **period-3 window** near $r \approx 3.83$.

At the left edge of a periodic window, a stable periodic orbit is created via a **saddle-node bifurcation** (of the iterated map). As r increases through the window, this orbit undergoes its own period-doubling cascade, eventually returning to chaos. The self-similarity is striking: each periodic window contains a miniature copy of the entire bifurcation diagram.

The period-3 window deserves special mention because of the theorem of Li and Yorke (1975):

Theorem 5.8 (Li–Yorke)**.** **Let $g : I \to I$ be a continuous map of a compact interval. If g has a point of period 3, then g has points of every period.**

More precisely, Sharkovskii's theorem (1964) provides a complete ordering of the natural numbers:

$$3 \triangleright 5 \triangleright 7 \triangleright \cdots \triangleright 2 \cdot 3 \triangleright 2 \cdot 5 \triangleright \cdots \triangleright 4 \cdot 3 \triangleright \cdots \triangleright 2^3 \triangleright 2^2 \triangleright 2 \triangleright 1$$

such that if g has a periodic point of period m and $m \triangleright k$, then g also has a periodic point of period k. Period 3 is "maximal" in this ordering; its existence forces all others.

5.6.4 Self-Similarity

The bifurcation diagram of the logistic map exhibits **self-similarity** at multiple levels:

1. **In the period-doubling cascade:** Successive bifurcation points approach r_∞ at a rate governed by δ, and the "widths" of the orbit branches scale by α. Zooming into the cascade near r_∞ reveals a structure that looks like a scaled copy of the whole cascade.
2. **In the chaotic regime:** Each periodic window contains its own period-doubling cascade, which is self-similar with the same constants δ and α.
3. **Connection to the Mandelbrot set:** The bifurcation diagram of the logistic map is closely related to a one-dimensional slice through the Mandelbrot set of the quadratic family $z \mapsto z^2 + c$. The "bulbs" of the Mandelbrot set correspond to periodic windows in the logistic map.

5.7 Global Bifurcations

All bifurcations discussed so far are **local**: they can be detected and analyzed by examining the system in a neighborhood of a single fixed point (or periodic orbit). There is another class of bifurcations — **global** — that involve the large-scale rearrangement of invariant sets and cannot be captured by any local analysis.

5.7.1 Homoclinic Bifurcation

Recall from Chapter 3 that a **homoclinic orbit** is a trajectory that is asymptotic to the same fixed point as $t \to +\infty$ and $t \to -\infty$. Equivalently, it lies in the intersection of the stable and unstable manifolds of a saddle point: $W^s(x^*) \cap W^u(x^*)$.

A **homoclinic bifurcation** occurs when a limit cycle grows in amplitude until it collides with a saddle point and becomes a homoclinic orbit. At the bifurcation value, the period of the cycle diverges to infinity (the orbit "stalls" near the saddle point). Beyond the bifurcation, the limit cycle is destroyed.

Example 5.7. Consider a planar system with a saddle point and a limit cycle. As a parameter μ varies, the limit cycle approaches the saddle. At $\mu = \mu_0$, the cycle touches the saddle and forms a homoclinic loop. For $\mu > \mu_0$, the periodic orbit ceases to exist. This is qualitatively different from the Hopf bifurcation: the periodic orbit does not shrink to a point but instead "unzips" at the saddle.

The period T of the limit cycle diverges logarithmically as the homoclinic bifurcation is approached:

$$T(\mu) \sim -\frac{1}{\lambda_s} \ln|\mu - \mu_0| \quad \text{as } \mu \to \mu_0,$$

where λ_s is the stable eigenvalue of the saddle point.

5.7.2 Shilnikov's Theorem

When the saddle point in a homoclinic bifurcation has complex eigenvalues (which requires at least three dimensions), the resulting dynamics can be extraordinarily rich. **Shilnikov's theorem** (1965) states that if a three-dimensional system has a saddle-focus fixed point with eigenvalues $\lambda_s < 0$ and $\lambda_u \pm i\omega$ (with $\lambda_u > 0$) satisfying $|\lambda_s| > \lambda_u$, and if a homoclinic orbit to this saddle-focus exists, then in every neighborhood of the homoclinic orbit there exist infinitely many periodic orbits of arbitrarily long period. This is a global mechanism for chaos.

5.7.3 Crisis

In chaotic systems, a **crisis** is a global bifurcation in which a chaotic attractor is suddenly destroyed, suddenly created, or suddenly changes size as a parameter crosses a critical value.

- **Boundary crisis:** A chaotic attractor collides with an unstable periodic orbit on its basin boundary. At the bifurcation, the attractor is destroyed and almost all trajectories escape to infinity (or to another attractor). In the logistic map, this occurs at $r = 4$: for r slightly above 4, most orbits escape from $[0, 1]$.
- **Interior crisis:** A chaotic attractor collides with an unstable periodic orbit in its interior, causing a sudden expansion in the size of the attractor. In the logistic map, this is visible as the sudden merger of separate chaotic bands.

These phenomena are fundamentally non-local: they involve the interaction between the attractor and invariant sets that may be far away in phase space.

5.8 Connections and Looking Ahead

Bifurcation theory is not merely a catalogue of transitions. It reveals a deep structure:

1. **Universality.** The Feigenbaum constants show that the route to chaos via period-doubling is the same for an enormous class of systems. This is a precursor to the universality we will encounter in ergodic theory (Chapter 12) and has implications for reservoir computing (Chapter 15), where the dynamical regime near the "edge of chaos" is conjectured to optimize computational performance.
2. **From local to global.** Local bifurcations (Sections 5.2–5.4) are amenable to algebraic analysis via normal forms. Global bifurcations (Section 5.7) require topological and geometric methods. The interplay between local and global perspectives is central to the theory of chaos and strange attractors (Chapter 6).
3. **Bifurcations and attractors.** Each bifurcation changes the nature of the attractor: from a fixed point to a periodic orbit (Hopf), from a periodic orbit to a doubled one (period-doubling), from a periodic orbit to chaos (via the cascade), or the sudden destruction of a chaotic attractor (crisis). In Chapter 6, we will study the attractors themselves in detail.
4. **Bifurcations in reservoir computing.** In Part III, we will see that the echo state property of a reservoir is closely related to the stability of its internal dynamics. The spectral radius of the reservoir weight matrix controls the proximity to a bifurcation; tuning it to be near the critical value (the "edge of chaos") is a practical application of the ideas in this chapter.

Exercises

Exercise 5.1. (Saddle-node bifurcation). Consider $\dot{x} = \mu + x - \ln(1 + x)$ for $x > -1$.

(a) Show that $x = 0$ is a fixed point when $\mu = 0$.

(b) Compute $f_x(0, 0)$ and $f_{xx}(0, 0)$. Verify the conditions of Theorem 5.2.

(c) Determine the type of bifurcation and sketch the bifurcation diagram near $\mu = 0$.

(d) Find the fixed points explicitly for small μ by expanding $\ln(1 + x)$ to third order and solving.

—

Exercise 5.2. (Transcritical bifurcation in a chemical reaction). The dimensionless concentration $x \geq 0$ of a chemical species evolves according to

$$\dot{x} = \mu x - x^2 + x^3.$$

(a) Find all fixed points as a function of μ.

(b) Determine the stability of each fixed point.

(c) Identify the bifurcation that occurs at $\mu = 0$. Is it transcritical? Justify using Theorem 5.3.

(d) Is the bifurcation at $\mu = 0$ affected by the cubic term? Discuss.

—

Exercise 5.3. (Subcritical pitchfork with stabilization). Consider

$$\dot{x} = \mu x + x^3 - x^5.$$

(a) Verify that this system has the symmetry $x \mapsto -x$.

(b) Find all fixed points. (Hint: set $u = x^2$ and solve the resulting quadratic in u.)

(c) Determine the stability of each fixed point as a function of μ.

(d) Draw the bifurcation diagram. Show that there is a region of **bistability** (coexistence of stable states) and identify the hysteresis loop.

(e) What happens to a trajectory starting near $x = 0$ if μ is slowly increased from a negative value past zero?

—

Exercise 5.4. (Hopf bifurcation). Consider the system

$$\dot{x} = -y + x(\mu - x^2 - y^2),$$
$$\dot{y} = x + y(\mu - x^2 - y^2).$$

(a) Show that the origin is the only fixed point.

(b) Find the eigenvalues of the Jacobian at the origin as a function of μ.

(c) Convert to polar coordinates and show that $\dot{r} = r(\mu - r^2)$, $\dot{\theta} = 1$.

(d) Determine the bifurcation value, its type (supercritical or subcritical), and the radius and period of the limit cycle for $\mu > 0$.

(e) Now consider the modified system where the cubic term is $+r^3$ instead of $-r^3$ (i.e., $\dot{r} = r(\mu + r^2)$). What changes? Sketch the phase portraits for $\mu < 0$, $\mu = 0$, and $\mu > 0$.

—

Exercise 5.5. (Period-doubling in the logistic map).

(a) For the logistic map $f_r(x) = rx(1-x)$, compute the multiplier $\lambda = (f_r^2)'(p)$ at a period-2 point p, expressed in terms of r. (Use the chain rule: $(f_r^2)'(p) = f_r'(p) \cdot f_r'(q)$ where $q = f_r(p)$.)

(b) Show that the period-2 orbit loses stability when $r = 1 + \sqrt{6}$.

(c) Using the formula for the period-2 points derived in Section 5.5.2, verify numerically that at $r = 1 + \sqrt{6}$, the multiplier of the period-2 orbit equals -1.

(d) Compute numerically (by iterating the map) the first six period-doubling bifurcation values $r_1, \ldots, r_6$. Estimate the Feigenbaum ratio δ_n for $n = 3, 4, 5$.

—

Exercise 5.6. (Center manifold computation). Consider the two-dimensional system

$$\dot{x} = \mu x - xy,$$
$$\dot{y} = -y + x^2.$$

(a) Show that the origin is a fixed point for all μ, and find the eigenvalues of the linearization.

(b) At $\mu = 0$, one eigenvalue is zero (center direction) and one is -1 (stable direction). Write $y = h(x) = ax^2 + bx^3 + \cdots$ for the center manifold and derive the invariance equation by substituting into the system.

(c) Solve for a and b by matching powers of x.

(d) Write the reduced equation on the center manifold: $\dot{x} = \mu x - xh(x)$. What type of bifurcation occurs at $\mu = 0$?

—

Exercise 5.7. (Global bifurcation). Consider the system in polar coordinates:

$$\dot{r} = r(1-r)(r-\mu), \quad \dot{\theta} = 1,$$

where μ is a parameter.

(a) For $0 < \mu < 1$, find all invariant circles (fixed points of the radial equation with $r > 0$) and determine their stability.

(b) What happens at $\mu = 0$? At $\mu = 1$? Classify these bifurcations.

(c) Draw the bifurcation diagram (radii of invariant circles vs. μ) for $\mu \in (-0.5, 1.5)$.

(d) The bifurcation at $\mu = 1$ involves the collision and mutual destruction of two limit cycles. Why can this not be detected by linearization at a fixed point?

—

Exercise 5.8. (Feigenbaum universality). Consider the sine map $x_{n+1} = r\sin(\pi x)$ for $x \in [0, 1]$.

(a) Find the non-trivial fixed point and determine the value of r at which it undergoes a period-doubling bifurcation.

(b) Numerically compute the first five period-doubling bifurcation values $r_1, \ldots, r_5$.

(c) Compute the ratios $\delta_n = (r_n - r_{n-1})/(r_{n+1} - r_n)$ and compare with Feigenbaum's constant $\delta \approx 4.6692$.

(d) Discuss: why does a map with such a different functional form produce the same universal constant?

—

References

- Strogatz, S.H. (2015). *Nonlinear Dynamics and Chaos*, 2nd edition. Westview Press. Chapters 3, 8, 10.
- Guckenheimer, J. and Holmes, P. (1983). *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields.* Springer. Chapters 3, 6.
- Kuznetsov, Y.A. (2004). *Elements of Applied Bifurcation Theory*, 3rd edition. Springer. Chapters 1–5.
- Crawford, J.D. (1991). "Introduction to Bifurcation Theory." *Reviews of Modern Physics*, 63(4), 991–1037.
- Feigenbaum, M.J. (1978). "Quantitative Universality for a Class of Nonlinear Transformations." *Journal of Statistical Physics*, 19(1), 25–52.

- Feigenbaum, M.J. (1979). "The Universal Metric Properties of Nonlinear Transformations." *Journal of Statistical Physics*, 21(6), 669–706.
- Lanford, O.E. (1982). "A Computer-Assisted Proof of the Feigenbaum Conjectures." *Bulletin of the American Mathematical Society*, 6(3), 427–434.
- Peixoto, M.M. (1962). "Structural Stability on Two-Dimensional Manifolds." *Topology*, 1(2), 101–120.
- Li, T.Y. and Yorke, J.A. (1975). "Period Three Implies Chaos." *The American Mathematical Monthly*, 82(10), 985–992.

Recommended Reading

For a first reading, Strogatz (2015) is unmatched in clarity and physical intuition; Chapters 3 (one-dimensional bifurcations), 8 (Hopf bifurcation), and 10 (period-doubling and chaos) parallel our treatment. Crawford (1991) is a masterful review article that covers both local and global bifurcations at a level slightly above this text — it is the natural next step for the motivated reader.

For a deeper mathematical treatment, Guckenheimer and Holmes (1983) remains the standard reference for the interplay between bifurcation theory and dynamical systems. Kuznetsov (2004) is the most comprehensive modern treatment of applied bifurcation theory, including numerical methods and higher-codimension bifurcations.

Feigenbaum's original papers (1978, 1979) are remarkably readable and are recommended as a first encounter with the renormalization group approach to universality. The computer-assisted proof by Lanford (1982) is a landmark in rigorous dynamical systems theory.

For the connections between bifurcation theory and the onset of turbulence, see Ruelle, D. and Takens, F. (1971), "On the Nature of Turbulence," *Communications in Mathematical Physics*, 20, 167–192 — a foundational paper that introduced strange attractors into the study of fluid dynamics.

Chapter 6

Chaos and Attractors

"Chaos is found in greatest abundance wherever order is being sought. It always defeats order, because it is better organized." — Terry Pratchett

In Chapters 2 through 5, we built the machinery of dynamical systems: iteration, fixed points, stability, bifurcations, and the routes by which simple systems can develop complicated behavior. In this chapter, we confront the complicated behavior itself. We give precise mathematical definitions of chaos, develop the tools to measure it (Lyapunov exponents, fractal dimension), study the geometric objects that organize it (strange attractors), and establish a remarkable theorem (Takens' embedding theorem) that will prove essential when we turn to reservoir computing in Part III.

The central tension of this chapter is between *unpredictability* and *structure*. Chaotic systems are deterministic yet practically unpredictable; their trajectories are wildly sensitive to initial conditions yet confined to attractors of intricate, self-similar geometry. Understanding this tension — and making it mathematically precise — is the goal before us.

6.1 What Is Chaos?

The word "chaos" is used loosely in everyday language, but in dynamical systems it has precise mathematical content. Several competing definitions exist in the literature. We begin with the most widely taught.

6.1.1 Devaney's Definition

Let (X, d) be a metric space and $f : X \to X$ a continuous map.

Definition 6.1 (Chaos in the sense of Devaney). The map f is said to be *chaotic on X* if the following three conditions hold:

1. **Sensitive dependence on initial conditions.** There exists $\delta > 0$ such that for every $x \in X$ and every $\epsilon > 0$, there exists $y \in X$ with $d(x, y) < \epsilon$ and $n \in \mathbb{N}$ such that $d(f^n(x), f^n(y)) > \delta$.
2. **Topological transitivity.** For every pair of nonempty open sets $U, V \subseteq X$, there exists $n \in \mathbb{N}$ such that $f^n(U) \cap V \neq \emptyset$.

3. **Dense periodic orbits.** The set of periodic points of f is dense in X.

Condition (1) says that no matter how close two initial conditions are, their orbits eventually separate by at least δ. Condition (2) says that the dynamics cannot be decomposed into two or more invariant pieces — there is a single orbit that comes arbitrarily close to every point. Condition (3) says that periodicity is ubiquitous, even though the generic orbit is aperiodic.

At first glance, these three conditions appear independent. The following remarkable result shows they are not.

Theorem 6.2 (Banks, Brooks, Cairns, Davis, and Stacey, 1992). *Let X be an infinite metric space and $f : X \to X$ a continuous map. If f is topologically transitive and has dense periodic orbits, then f has sensitive dependence on initial conditions.*

In other words, conditions (ii) and (iii) together imply condition (i). Chaos, in Devaney's sense, is "simpler" than it first appears: the sensitive dependence comes for free.

Proof sketch. sketch.* Suppose f is topologically transitive and has dense periodic orbits but does *not* have sensitive dependence. Then for every $\delta > 0$, there exists a point x and a neighborhood U of x such that for all $y \in U$ and all $n \geq 0$, $d(f^n(x), f^n(y)) \leq \delta$. Since periodic points are dense, we can find periodic points $p, q \in U$ with distinct orbits (using the fact that X is infinite and topological transitivity prevents all points in a neighborhood from sharing the same eventual behavior). Choose δ small enough that the orbits of p and q must remain close. But since p and q are periodic, their orbits are finite sets, and topological transitivity forces the existence of points whose orbits wander throughout X. One shows this leads to a contradiction: the finite orbits of nearby periodic points, combined with transitivity, force separation beyond any fixed δ. The full proof, which is not long, can be found in Banks et al. (1992). □

□

Remark 6.1. The converse fails: sensitive dependence on initial conditions alone, or even combined with one of the other conditions, does not imply chaos. Consider the map $f(x) = 2x$ on $\mathbb{R}$: it has sensitive dependence on initial conditions but is neither topologically transitive (orbits escape to infinity or converge to zero depending on sign) nor does it have dense periodic orbits (the origin is the only fixed point, and there are no other periodic points).

6.1.2 Li-Yorke Chaos

An older notion of chaos, predating Devaney's definition, comes from the celebrated paper of Li and Yorke (1975).

Definition 6.3 (Li-Yorke chaos). A continuous map $f : I \to I$ on an interval $I \subseteq \mathbb{R}$ is *Li-Yorke chaotic* if there exists an uncountable set $S \subseteq I$ (called a *scrambled set*) such that for every pair $x, y \in S$ with $x \neq y$:

$$\limsup_{n\to\infty} |f^n(x) - f^n(y)| > 0 \quad \text{and} \quad \liminf_{n\to\infty} |f^n(x) - f^n(y)| = 0.$$

The pair (x, y) is called a *Li-Yorke pair*: their orbits come arbitrarily close together infinitely often, yet also separate by a definite amount infinitely often. This captures a notion of *irregular* or *unpredictable* behavior without requiring dense periodic orbits or topological transitivity.

Theorem 6.4 (Li and Yorke, 1975). *If $f : I \to I$ is a continuous map of a compact interval and f has a periodic point of period 3, then f is Li-Yorke chaotic.*

This is the famous "period three implies chaos" result (the title of their 1975 paper). We state it here without proof; see Devaney (1989, Ch. 1.7) or the original paper.

Comparison. Devaney chaos and Li-Yorke chaos capture different aspects of complicated dynamics:

| | Devaney chaos | Li-Yorke chaos | |—|—|—| | Requires dense periodic orbits | Yes | No | | Requires topological transitivity | Yes | No | | Requires sensitive dependence | Implied | Not directly | | Requires scrambled set | No | Yes | | Applicable to | General metric spaces | Typically interval maps |

It is known that Devaney chaos implies Li-Yorke chaos for interval maps (Huang and Ye, 2002), but the converse does not hold in general. Neither definition is universally accepted as "the" definition of chaos — the appropriate notion depends on the context. In this text, we primarily work with Devaney's definition and the related quantitative tool of Lyapunov exponents.

6.2 Sensitive Dependence on Initial Conditions

We now examine condition (i) in Devaney's definition more carefully, as it is the aspect of chaos most relevant to applications.

6.2.1 Formal Definition

Definition 6.5 (Sensitive dependence on initial conditions). Let (X, d) be a metric space and $f : X \to X$. The map f has *sensitive dependence on initial conditions* if there exists a constant $\delta > 0$ (the *sensitivity constant*) such that for every $x \in X$ and every $\epsilon > 0$, there exist $y \in X$ and $n \in \mathbb{N}$ satisfying:

$$d(x, y) < \epsilon \quad \text{and} \quad d(f^n(x), f^n(y)) > \delta.$$

The quantifier structure matters: δ is *universal* (it does not depend on x or ϵ), while y and n depend on both x and ϵ. In other words, every point has nearby points whose orbits eventually diverge by at least δ, no matter how close "nearby" means.

Example 6.6 (The doubling map). Consider $f : [0, 1) \to [0, 1)$ defined by $f(x) = 2x \pmod 1$. We claim f has sensitive dependence with $\delta = 1/4$. Given any $x \in [0, 1)$ and $\epsilon > 0$, choose n large enough that $2^n\epsilon > 1$. For any y with $|x - y| < \epsilon$ but $y \neq x$, the iterates $f^n(x)$ and $f^n(y)$ differ by $|2^n x - 2^n y| \pmod 1$. Since $2^n|x - y|$ grows without bound, there exists an iterate for which this distance (mod 1) exceeds 1/4. A precise argument uses binary expansions; see Section 6.6.

6.2.2 The Butterfly Effect

The phrase "butterfly effect" is popularly attributed to Edward Lorenz, though his actual 1972 talk was titled "Does the Flap of a Butterfly's Wings in Brazil Set Off a Tornado in Texas?"

The underlying discovery, however, dates to 1961.

Lorenz was running numerical simulations of a simplified atmospheric model (the system we now call the Lorenz equations; see Section 6.4.1). To restart a computation partway through, he entered initial conditions rounded to three decimal places instead of the six his computer stored internally. The resulting trajectory diverged completely from the original within a short time.

This was not a numerical artifact. Lorenz realized that deterministic systems could be *practically unpredictable*: even if the governing equations are known perfectly, unmeasurably small errors in the initial state grow exponentially over time, rendering long-term forecasts impossible. He published this observation in his landmark 1963 paper, "Deterministic Nonperiodic Flow."

Why this matters for prediction. Suppose errors grow as $\|\delta(t)\| \approx \|\delta(0)\| e^{\lambda t}$ for some $\lambda > 0$. If our initial measurement has precision ϵ and we need predictions accurate to tolerance Δ, then the *prediction horizon* is approximately:

$$T_{\text{predict}} \approx \frac{1}{\lambda} \ln \frac{\Delta}{\epsilon}.$$

This grows only *logarithmically* with measurement precision. Improving our instruments by a factor of 1000 buys us only $\frac{\ln 1000}{\lambda} \approx \frac{6.9}{\lambda}$ additional time units of prediction. For weather, with $\lambda \approx 1/\text{day}$, this means roughly one extra week — regardless of how much we improve our measurements. This is the fundamental barrier Lorenz identified.

6.3 Lyapunov Exponents

Sensitive dependence on initial conditions is a qualitative property: either orbits separate or they do not. *Lyapunov exponents* provide a quantitative measure: they tell us the *rate* of exponential divergence (or convergence) of nearby orbits.

6.3.1 Definition for One-Dimensional Maps

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable map and let x_0 be an initial condition with orbit $x_0, x_1 = f(x_0), x_2 = f(x_1), \ldots$

Consider a small perturbation $x_0 + \epsilon$. After one iterate, the perturbed orbit is approximately $f(x_0+\epsilon) \approx f(x_0) + f'(x_0)\epsilon$, so the perturbation has been multiplied by $|f'(x_0)|$. After n iterates, the chain rule gives:

$$f^n(x_0 + \epsilon) - f^n(x_0) \approx \epsilon \prod_{k=0}^{n-1} f'(x_k).$$

We define the *finite-time Lyapunov exponent* as the average logarithmic stretching rate:

$$\lambda_n(x_0) = \frac{1}{n} \sum_{k=0}^{n-1} \ln |f'(x_k)| = \frac{1}{n} \ln \prod_{k=0}^{n-1} |f'(x_k)|.$$

Definition 6.7 (Lyapunov exponent for a 1D map). The *Lyapunov exponent* of f at x_0 is

$$\lambda(x_0) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln |f'(x_k)|,$$

provided the limit exists.

Interpretation. If $\lambda > 0$, nearby orbits diverge exponentially at rate e^λ per iterate on average. If $\lambda < 0$, they converge. If $\lambda = 0$, the behavior is sub-exponential (often power-law). A positive Lyapunov exponent is the hallmark of chaos.

Remark on existence. The limit need not exist for every orbit. However, when f preserves an ergodic measure μ, the Birkhoff ergodic theorem (Chapter 9) guarantees that $\lambda(x_0)$ exists and equals the same value $\lambda = \int \ln |f'|\, d\mu$ for μ-almost every x_0. This connects Lyapunov exponents directly to ergodic theory.

6.3.2 Worked Examples

Example 6.8 (The doubling map). Let $f(x) = 2x \pmod 1$ on $[0, 1)$. Then $f'(x) = 2$ everywhere (where defined), so:

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln |2| = \ln 2 \approx 0.693.$$

The Lyapunov exponent is $\ln 2$ for every orbit. This is consistent with what we already know: the doubling map doubles distances at each step (before the mod operation), and $\ln 2 > 0$ confirms sensitive dependence.

Example 6.9 (The logistic map at $r = 4$). Consider $f(x) = 4x(1 - x)$ on $[0, 1]$. We compute $f'(x) = 4(1 - 2x)$. Rather than trying to evaluate the sum directly along a generic orbit, we exploit a conjugacy.

The change of variables $x = \sin^2(\pi\theta/2)$ (equivalently, $\theta = \frac{2}{\pi} \arcsin(\sqrt{x})$) conjugates f to the *tent map* $g(\theta) = \min(2\theta, 2 - 2\theta)$ on $[0, 1]$, which is in turn semi-conjugate to the doubling map.

More precisely, f preserves the measure $d\mu = \frac{1}{\pi\sqrt{x(1-x)}} dx$ (the arcsine distribution), and with respect to this measure, the system is ergodic. The Lyapunov exponent is:

$$\lambda = \int_0^1 \ln |f'(x)|\, d\mu(x) = \int_0^1 \ln |4(1 - 2x)| \cdot \frac{1}{\pi\sqrt{x(1 - x)}}\, dx.$$

We evaluate this by substituting $x = \sin^2(\pi\theta/2)$, so $dx = \pi \sin(\pi\theta/2) \cos(\pi\theta/2)\, d\theta$ and $\sqrt{x(1 - x)} = \sin(\pi\theta/2) \cos(\pi\theta/2)$. Then:

$$\lambda = \int_0^1 \ln |4 \cos(\pi\theta)|\, d\theta = \int_0^1 \ln 4\, d\theta + \int_0^1 \ln |\cos(\pi\theta)|\, d\theta.$$

The first integral is $\ln 4$. For the second, the classical result (see, e.g., Gradshteyn and Ryzhik, 4.384.1) gives:

$$\int_0^1 \ln |\cos(\pi\theta)|\, d\theta = -\ln 2.$$

Therefore:

$$\lambda = \ln 4 - \ln 2 = \ln 2.$$

The logistic map at $r = 4$ has the same Lyapunov exponent as the doubling map. This is not a coincidence: the two maps are measure-theoretically isomorphic (a fact we will make precise in Chapter 8).

Example 6.10 (The Hénon map). The Hénon map (Section 6.4.3) is a two-dimensional system with two Lyapunov exponents. At the classical parameters $a = 1.4$, $b = 0.3$, numerical computation gives:

$$\lambda_1 \approx 0.42, \quad \lambda_2 \approx -1.62.$$

The positive first exponent indicates chaos, while the strongly negative second exponent reflects the area contraction (the Jacobian determinant is $-b = -0.3$, so $\lambda_1 + \lambda_2 = \ln|b| = \ln 0.3 \approx -1.20$). This is consistent with the numerical values above.

6.3.3 The Lyapunov Spectrum in Higher Dimensions

For a differentiable map $f : \mathbb{R}^d \to \mathbb{R}^d$, nearby points are separated by the Jacobian matrix $Df(x)$. After n iterates starting from x_0, the cumulative effect is captured by the matrix product:

$$M_n(x_0) = Df(x_{n-1}) \cdot Df(x_{n-2}) \cdots Df(x_0),$$

where $x_k = f^k(x_0)$. The Lyapunov exponents are defined as the exponential growth rates of this product. More precisely, they are determined by the singular values of M_n.

Definition 6.11 (Lyapunov spectrum). The *Lyapunov exponents* $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$ are defined by:

$$\lambda_i = \lim_{n\to\infty} \frac{1}{n} \ln \sigma_i(M_n(x_0)),$$

where $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_d$ are the singular values of $M_n(x_0)$, provided these limits exist.

Equivalently, λ_1 is the growth rate of the largest singular value (the rate at which the most-stretched direction stretches), $\lambda_1 + \lambda_2$ is the growth rate of the largest 2D area element, and so on.

For continuous-time systems $\dot{x} = F(x)$, the definition is analogous but with M_n replaced by the fundamental matrix solution of the variational equation $\dot{Y} = DF(x(t))Y$ evaluated at time t, and $1/n$ replaced by $1/t$ as $t \to \infty$.

6.3.4 The Oseledets Multiplicative Ergodic Theorem

The existence of Lyapunov exponents for the examples above relied on specific computations. In general, one needs a deep theorem to guarantee existence and to relate the exponents to the geometry of the dynamics.

Theorem 6.12 (Oseledets, 1968). *Let $f : M \to M$ be a diffeomorphism of a compact Riemannian manifold preserving an ergodic probability measure μ, and suppose $\ln^+ \|Df\| \in$

$L^1(\mu)$. Then there exist numbers $\lambda_1 > \lambda_2 > \cdots > \lambda_s$ (the distinct Lyapunov exponents) and a μ-a.e. defined measurable splitting of the tangent bundle*

$$T_xM = E_1(x) \oplus E_2(x) \oplus \cdots \oplus E_s(x)$$

such that for μ-almost every x and every nonzero $v \in E_i(x)$:

$$\lim_{n\to\infty} \frac{1}{n} \ln \|Df^n(x)v\| = \lambda_i.$$

The subspaces $E_i(x)$ are called the Oseledets subspaces, and $m_i = \dim E_i(x)$ is the multiplicity of λ_i. The collection $\{(\lambda_i, m_i)\}_{i=1}^s$ is the Lyapunov spectrum.

Significance. The Oseledets theorem is the multiplicative analogue of the Birkhoff ergodic theorem (Chapter 9). While Birkhoff's theorem concerns the convergence of *additive* averages $\frac{1}{n}\sum_{k=0}^{n-1} g(f^k(x))$, the Oseledets theorem concerns the convergence of *multiplicative* averages — products of matrices $Df(x_k)$ — which is a fundamentally harder problem because matrices do not commute.

The theorem tells us that:

- The Lyapunov exponents exist almost everywhere (we are not relying on special orbits).
- The tangent space at each point decomposes into directions of definite exponential growth/decay rates.
- These rates are the same for almost every starting point (by ergodicity).

This theorem is foundational for the ergodic theory of smooth dynamical systems. It underpins Pesin theory (the study of nonuniformly hyperbolic systems), which we will encounter in Chapter 12.

6.4 Strange Attractors

6.4.1 Attractors: Formal Definitions

Informally, an attractor is a set toward which typical orbits converge. Making this precise requires some care.

Definition 6.13 (Attractor). Let $f : X \to X$ be a continuous map on a metric space X. A compact set $A \subseteq X$ is an *attractor* if:

1. **Invariance.** $f(A) = A$.
2. **Attracting.** There exists an open neighborhood U of A (called the *basin of attraction*) such that for every $x \in U$, $d(f^n(x), A) \to 0$ as $n \to \infty$.
3. **Topological transitivity (indecomposability).** $f|_A$ is topologically transitive: for every pair of nonempty open sets $V, W \subseteq A$ (in the subspace topology), there exists n with $f^n(V) \cap W \neq \emptyset$.

Condition (3) prevents us from calling the union of two disjoint attractors a single "attractor." It ensures that the attractor is dynamically irreducible.

Definition 6.14 (Basin of attraction). The *basin of attraction* of an attractor A is:

$$\mathcal{B}(A) = \{x \in X : d(f^n(x), A) \to 0 \text{ as } n \to \infty\}.$$

6.4.2 Regular vs. Strange Attractors

Regular attractors are the attractors we encountered in earlier chapters:

- **Fixed points.** A stable fixed point x^* is an attractor consisting of a single point.
- **Limit cycles.** A stable periodic orbit $\{x_0, x_1, \ldots, x_{p-1}\}$ (discrete) or a stable closed orbit γ (continuous) is an attractor.
- **Invariant tori.** In systems with multiple incommensurate frequencies, the attractor can be a torus $\mathbb{T}^k$, supporting quasiperiodic motion.

These attractors have integer topological dimension (0, 1, or 2 respectively) and the dynamics on them is relatively simple (stationary, periodic, or quasiperiodic).

Definition 6.15 (Strange attractor, informal). An attractor is called *strange* if it has fractal structure (non-integer dimension) and supports sensitive dependence on initial conditions.

Remark 6.2. There is no single universally agreed-upon definition. Some authors use "strange" to refer only to the fractal geometry, regardless of the dynamics (so that a "strange nonchaotic attractor" is possible — and such objects do exist; see Grebogi et al., 1984). Others require both fractal geometry and chaos. We adopt the latter convention: for us, a strange attractor is an attractor with fractal structure on which the dynamics is chaotic.

6.4.3 The Lorenz Attractor

The Lorenz system is arguably the most important example in the theory of chaos. It was introduced by the meteorologist Edward Lorenz in 1963 as a drastically simplified model of atmospheric convection, derived from a truncation of the Navier-Stokes equations.

The equations. The Lorenz system is:

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z,$$

where σ, ρ, and β are positive parameters. The classical values are:

$$\sigma = 10, \quad \rho = 28, \quad \beta = 8/3.$$

Here x represents the intensity of convective motion, y represents the temperature difference between ascending and descending currents, and z represents the deviation of the vertical temperature profile from linearity. The parameters σ, ρ, β correspond to the Prandtl number, the Rayleigh number (normalized), and a geometric factor, respectively.

Symmetry and dissipation. The system has the symmetry $(x, y, z) \mapsto (-x, -y, z)$, reflecting the physical equivalence of clockwise and counterclockwise convection. It is also dissipative: the divergence of the vector field is:

$$\nabla \cdot F = \frac{\partial \dot{x}}{\partial x} + \frac{\partial \dot{y}}{\partial y} + \frac{\partial \dot{z}}{\partial z} = -\sigma - 1 - \beta < 0.$$

For the classical parameters, $\nabla \cdot F = -10 - 1 - 8/3 \approx -13.67$. By Liouville's theorem, phase-space volumes contract at rate $e^{-13.67\,t}$, so all trajectories are eventually attracted to a set of zero volume. Yet this set can (and does) have complex geometry.

Equilibria. Setting $\dot{x} = \dot{y} = \dot{z} = 0$:

- For $\rho < 1$: the origin is the only equilibrium and is globally attracting.
- For $\rho > 1$: the origin becomes unstable (a pitchfork bifurcation), and two new equilibria appear at $C^{\pm} = (\pm\sqrt{\beta(\rho-1)}, \pm\sqrt{\beta(\rho-1)}, \rho - 1)$. For the classical parameters, $C^{\pm} = (\pm 6\sqrt{2}, \pm 6\sqrt{2}, 27)$.
- At $\rho \approx 24.74$ (for the standard σ, β), $C^{\pm}$ lose stability via a subcritical Hopf bifurcation. For $\rho = 28$, all three equilibria are unstable, and trajectories wander in a bounded region without settling down.

The geometric structure. The Lorenz attractor has the iconic "butterfly" shape: two wing-like lobes centered near C^+ and C^-. A typical trajectory spirals outward around one equilibrium, then crosses to the other wing and spirals outward around the other equilibrium, switching back and forth in an aperiodic pattern.

The mechanism producing this behavior is *stretching and folding.* Nearby trajectories on one wing separate exponentially (stretching). When they cross to the other wing, the trajectories from different parts of the wing are brought together (folding). This process, repeated infinitely, produces the fractal layered structure of the attractor.

Lyapunov exponents. Numerical computation gives the Lyapunov spectrum for the classical Lorenz attractor:

$$\lambda_1 \approx 0.906, \quad \lambda_2 \approx 0, \quad \lambda_3 \approx -14.57.$$

The positive exponent confirms chaos. The zero exponent corresponds to the flow direction (a general feature of continuous-time autonomous systems — perturbations along the trajectory neither grow nor shrink). The large negative exponent reflects the strong dissipation.

Tucker's theorem. For decades, the existence of the Lorenz attractor as a genuine mathematical object (not merely a numerical artifact) remained an open problem. It was the 14th problem on Smale's list of problems for the 21st century. This was resolved by Warwick Tucker in 1999 (published 2002), using rigorous computer-assisted methods.

Theorem 6.16 (Tucker, 2002). *For the classical parameter values $\sigma = 10$, $\rho = 28$, $\beta = 8/3$, the Lorenz equations have a robust strange attractor. Specifically, the flow admits a "geometric Lorenz attractor" — a compact invariant set that is topologically transitive, supports a unique SRB measure, and has sensitive dependence on initial conditions.*

Tucker's proof uses interval arithmetic to rigorously bound all numerical errors, combined with a careful analysis of the return map to a cross-section of the flow. The key idea is to verify the geometric conditions identified by Afraimovich, Bykov, and Shil'nikov (1977) and by Guckenheimer and Williams (1979) that guarantee the existence of a strange attractor for flows with the qualitative features of the Lorenz system.

6.4.4 The Rössler Attractor

The Rössler system, introduced by Otto Rössler in 1976, provides a simpler example of a strange attractor in three dimensions:

$$\dot{x} = -y - z, \quad \dot{y} = x + ay, \quad \dot{z} = b + z(x - c),$$

with typical parameters $a = 0.2$, $b = 0.2$, $c = 5.7$.

Unlike the Lorenz attractor, which has the symmetry $(x, y, z) \mapsto (-x, -y, z)$ and two "wings," the Rössler attractor has a single band-like structure that twists and folds. The mechanism is simpler to visualize: trajectories spiral outward in the (x, y)-plane, and when x becomes large enough, the z-variable activates and "resets" the trajectory back toward the center.

Depending on the parameter c, the Rössler attractor exhibits different types:

- **Band type** (e.g., $c = 5.7$): the attractor resembles a Möbius-like band with a single fold. The return map to a cross-section is approximately one-dimensional and unimodal, similar to the logistic map.
- **Funnel type** (e.g., $c = 18$): the attractor has a more complex structure with trajectories making varying numbers of loops before reinjection.

The Rössler system is particularly useful for studying the transition from periodic to chaotic behavior via period-doubling cascades (Chapter 5), because the effectively one-dimensional return map makes the analysis more tractable than for the Lorenz system.

Lyapunov exponents for the classical Rössler attractor ($a = b = 0.2$, $c = 5.7$):

$$\lambda_1 \approx 0.07, \quad \lambda_2 \approx 0, \quad \lambda_3 \approx -5.39.$$

6.4.5 The Hénon Map

The Hénon map, introduced by Michel Hénon in 1976 as a simplified model of the Poincaré section of the Lorenz system, is a two-dimensional map:

$$x_{n+1} = 1 - ax_n^2 + y_n, \quad y_{n+1} = bx_n,$$

with classical parameters $a = 1.4$, $b = 0.3$.

Properties. The Jacobian is

$$Df(x, y) = \begin{pmatrix} -2ax & 1 \\ b & 0 \end{pmatrix}$$

with determinant $\det Df = -b$. For $b = 0.3$, each iterate contracts areas by a factor of 0.3. For $b \neq 0$, the map is a diffeomorphism (it is invertible):

$$x_n = y_{n+1}/b, \quad y_n = x_{n+1} - 1 + a(y_{n+1}/b)^2.$$

For $b = 0$, the Hénon map reduces to the one-dimensional quadratic map $x_{n+1} = 1 - ax_n^2$, which is conjugate to the logistic map. Thus the Hénon map can be viewed as a "thickened" version of the logistic map.

The Benedicks-Carleson theorem. The question of whether the numerically observed Hénon attractor is a genuine strange attractor (as opposed to a very long transient, or a stable periodic orbit of very high period) is subtle. For decades it remained an open problem. The breakthrough came with the following result.

Theorem 6.17 (Benedicks and Carleson, 1991). *There exists a set of positive Lebesgue measure in the parameter space (a, b), with $b > 0$ small, such that the Hénon map $f_{a,b}$ has a "genuine" strange attractor: it supports a unique ergodic SRB (Sinai-Ruelle-Bowen) measure with a positive Lyapunov exponent.*

The proof is technically formidable. It extends the methods of Jakobson (1981), who proved the analogous result for the one-dimensional quadratic family, to the two-dimensional setting. The key difficulty is controlling the recurrence of the "critical point" (or rather, the critical region where the map is strongly contracting in one direction), ensuring that the orbit of this region does not return too quickly.

Remark 6.3. Benedicks and Carleson's result applies for b small and positive. Whether it holds at the "classical" value $b = 0.3$ with $a = 1.4$ is a separate, harder question that remains open, though all numerical evidence supports it.

—

6.5 Fractal Dimension

Strange attractors have intricate geometric structure that cannot be captured by integer dimensions. We need a notion of dimension that can take non-integer values.

6.5.1 Box-Counting Dimension

Definition 6.18 (Box-counting dimension). Let $A \subseteq \mathbb{R}^d$ be a bounded set. For $\epsilon > 0$, let $N(\epsilon)$ be the minimum number of d-dimensional cubes (boxes) of side length ϵ needed to cover A. The *box-counting dimension* (also called the *Minkowski dimension* or *capacity dimension*) of A is:

$$\dim_B(A) = \lim_{\epsilon \to 0} \frac{\ln N(\epsilon)}{\ln(1/\epsilon)},$$

provided the limit exists. If the limit does not exist, one defines upper and lower box-counting dimensions using lim sup and lim inf.

Intuition. For a "smooth" set of dimension d, we expect $N(\epsilon) \sim C\epsilon^{-d}$ as $\epsilon \to 0$, so $\frac{\ln N(\epsilon)}{\ln(1/\epsilon)} \to d$. The box-counting dimension generalizes this to sets where the exponent is not an integer.

Example 6.19 (A line segment). A line segment of length L in $\mathbb{R}^2$ requires $N(\epsilon) \sim L/\epsilon$ boxes, so $\dim_B = \frac{\ln(L/\epsilon)}{\ln(1/\epsilon)} \to 1$.

Example 6.20 (A filled square). A square of side L requires $N(\epsilon) \sim (L/\epsilon)^2$ boxes, so $\dim_B = 2$.

Example 6.21 (The middle-thirds Cantor set). The Cantor set C is constructed by iteratively removing the middle third of each interval, starting from $[0, 1]$. At stage n, the set

consists of 2^n intervals of length 3^{-n}. Taking $\epsilon = 3^{-n}$:

$$N(\epsilon) = 2^n, \quad \text{so} \quad \dim_B(C) = \lim_{n\to\infty} \frac{\ln 2^n}{\ln 3^n} = \frac{n \ln 2}{n \ln 3} = \frac{\ln 2}{\ln 3} \approx 0.631.$$

The Cantor set has dimension strictly between 0 and 1: it is "more" than a finite collection of points but "less" than an interval.

6.5.2 Hausdorff Dimension

Box-counting dimension is easy to compute but has some mathematical deficiencies. For instance, a countable dense set has box-counting dimension equal to the dimension of the ambient space, which is arguably the "wrong" answer. The Hausdorff dimension, introduced by Felix Hausdorff in 1918, is the more fundamental notion.

Definition 6.22 (Hausdorff measure and dimension). For $A \subseteq \mathbb{R}^d$ and $s \geq 0$, define the *s-dimensional Hausdorff measure* of A as:

$$\mathcal{H}^s(A) = \lim_{\delta\to 0} \inf \left\{ \sum_{i=1}^{\infty} (\operatorname{diam} U_i)^s : A \subseteq \bigcup_{i=1}^{\infty} U_i,\ \operatorname{diam} U_i \leq \delta \right\}.$$

The infimum is over all countable covers of A by sets of diameter at most δ. The *Hausdorff dimension* of A is:

$$\dim_H(A) = \inf\{s \geq 0 : \mathcal{H}^s(A) = 0\} = \sup\{s \geq 0 : \mathcal{H}^s(A) = \infty\}.$$

Why this is the "right" definition. The Hausdorff dimension has several properties that box-counting dimension lacks:

- $\dim_H(\text{countable set}) = 0$ (correct!).
- $\dim_H$ is countably stable: $\dim_H(\bigcup_{i=1}^{\infty} A_i) = \sup_i \dim_H(A_i)$.
- $\dim_H(A) \leq \dim_B(A)$ always, and equality holds for many "nice" fractal sets (including all self-similar sets satisfying the open set condition).

For the Cantor set, $\dim_H(C) = \ln 2/\ln 3$, agreeing with the box-counting dimension. For self-similar fractals in general, the Hausdorff dimension is given by the solution of Moran's equation.

6.5.3 Information and Correlation Dimensions

For attractors of dynamical systems, we often care not just about the geometry of the set but about how the invariant measure distributes mass on it. This leads to a family of *generalized dimensions.*

Definition 6.23 (Information dimension). Let μ be the natural measure on an attractor. Cover the attractor with boxes of side ϵ and let p_i be the μ-measure of the i-th box. The *information dimension* is:

$$d_1 = \lim_{\epsilon\to 0} \frac{\sum_i p_i \ln p_i}{\ln \epsilon}.$$

The numerator is the Shannon entropy of the partition, so d_1 measures the rate at which information is needed to specify a point on the attractor as the resolution increases.

Definition 6.24 (Correlation dimension). The *correlation dimension* d_2 is defined via the *correlation integral*:

$$C(\epsilon) = \lim_{N\to\infty} \frac{1}{N^2} \#\{(i,j) : \|x_i - x_j\| < \epsilon,\ i \neq j\},$$

where $x_1, x_2, \ldots, x_N$ are points on the attractor (e.g., from a long orbit). Then:

$$d_2 = \lim_{\epsilon\to 0} \frac{\ln C(\epsilon)}{\ln \epsilon}.$$

The correlation dimension was introduced by Grassberger and Procaccia (1983) and is the most commonly used dimension in practice, because it can be estimated efficiently from time series data. One has the general inequality:

$$d_2 \leq d_1 \leq \dim_B .$$

For many "typical" attractors these dimensions are close but not exactly equal.

6.5.4 The Kaplan-Yorke Conjecture

There is a beautiful conjectured relationship between the fractal dimension of an attractor and the Lyapunov exponents of the dynamics on it.

Definition 6.25 (Kaplan-Yorke dimension). Given the ordered Lyapunov exponents $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$, let j be the largest integer such that $\sum_{i=1}^{j} \lambda_i \geq 0$. The *Kaplan-Yorke dimension* (or *Lyapunov dimension*) is:

$$d_{KY} = j + \frac{\sum_{i=1}^{j} \lambda_i}{|\lambda_{j+1}|}.$$

Conjecture 6.26 (Kaplan and Yorke, 1979). *For "typical" attractors, the information dimension d_1 equals the Kaplan-Yorke dimension d_{KY}.*

This conjecture has been proven in certain cases (e.g., for two-dimensional systems by Young, 1982) and is supported by extensive numerical evidence, but remains open in full generality.

Example 6.27 (Kaplan-Yorke dimension of the Lorenz attractor). Using $\lambda_1 \approx 0.906$, $\lambda_2 = 0$, $\lambda_3 \approx -14.57$:

$$j = 2, \quad d_{KY} = 2 + \frac{0.906 + 0}{14.57} \approx 2.06.$$

This is consistent with the numerically estimated fractal dimension of the Lorenz attractor (≈ 2.06). The attractor is a "thin" fractal layer — very nearly a surface, but with a dimension slightly above 2 due to the fractal folding structure.

Example 6.28 (Kaplan-Yorke dimension of the Hénon attractor). Using $\lambda_1 \approx 0.42$, $\lambda_2 \approx -1.62$:

$$j = 1, \quad d_{KY} = 1 + \frac{0.42}{1.62} \approx 1.26.$$

This agrees well with the numerically estimated box-counting dimension of the Hénon attractor (≈ 1.26). The attractor is slightly "thicker" than a curve but much thinner than a surface, consistent with its visual appearance as a system of nearly parallel curves.

6.6 Symbolic Dynamics

Symbolic dynamics provides a powerful combinatorial framework for analyzing chaotic systems. The idea is to replace a continuous dynamical system with a discrete one on a space of symbol sequences, where the analysis becomes more tractable.

6.6.1 Itineraries and Symbol Sequences

Consider a map $f : X \to X$ and a partition of X into finitely many regions $\{P_0, P_1, \ldots, P_{k-1}\}$. Given an initial point $x \in X$, we record which partition element is visited at each time step.

Definition 6.29 (Itinerary). The *itinerary* of x with respect to the partition $\{P_i\}$ is the sequence $s(x) = (s_0, s_1, s_2, \ldots)$ where $s_n \in \{0, 1, \ldots, k-1\}$ and $f^n(x) \in P_{s_n}$.

The itinerary is an element of the *sequence space* $\Sigma_k = \{0, 1, \ldots, k-1\}^{\mathbb{N}_0}$, the set of all one-sided infinite sequences over k symbols. We equip Σ_k with the metric:

$$d(s,t) = \sum_{n=0}^{\infty} \frac{|s_n - t_n|}{k^n},$$

which makes Σ_k a compact metric space. Two sequences are close in this metric if and only if they agree on a long initial segment.

6.6.2 The Shift Map

Definition 6.30 (Shift map). The *(one-sided) shift map* $\sigma : \Sigma_k \to \Sigma_k$ is defined by:

$$\sigma(s_0, s_1, s_2, \ldots) = (s_1, s_2, s_3, \ldots).$$

It simply discards the first symbol and shifts the rest to the left.

Proposition 6.1. **The shift map $\sigma : \Sigma_k \to \Sigma_k$ (for $k \geq 2$) is chaotic in the sense of Devaney.**

Proof. * We verify each condition.

(i) *Dense periodic orbits.* A sequence $s = (s_0, s_1, \ldots, s_{p-1}, s_0, s_1, \ldots, s_{p-1}, \ldots)$ is periodic with period p under σ. Given any sequence $t \in \Sigma_k$ and any $\epsilon > 0$, choose N large enough that $k^{-N} < \epsilon$. Define the periodic sequence s by repeating the block $(t_0, t_1, \ldots, t_{N-1})$. Then $d(s,t) \leq \sum_{n=N}^{\infty} (k-1)/k^n < \epsilon$ (after adjusting constants), so s is within ϵ of t. Hence periodic points are dense.

(ii) *Topological transitivity.* We construct a point whose orbit is dense. Enumerate all finite strings of length 1, then length 2, then length 3, etc., and concatenate them into a single infinite sequence s^*. Then for any open set U in Σ_k, any sequence in U must share a long initial block with some iterate of s^* (since that block appears as a substring of s^*, and shifting brings it to the front). Thus the orbit of s^* is dense.

(iii) *Sensitive dependence on initial conditions.* Take $\delta = 1$ (or any constant less than 1 for the normalized metric). Given any $s \in \Sigma_k$ and $\epsilon > 0$, choose N with $k^{-N} < \epsilon$ and define t by $t_n = s_n$ for $n < N$ and $t_N \neq s_N$. Then $d(s,t) < \epsilon$ but $d(\sigma^N(s), \sigma^N(t)) \geq |s_N - t_N|/k^0 \cdot (\text{leading term}) > 0$. More precisely, since the first symbols of $\sigma^N(s)$ and $\sigma^N(t)$ differ, $d(\sigma^N(s), \sigma^N(t)) \geq 1/k$.

By Theorem 6.2, conditions (ii) and (iii) alone suffice, but we have verified all three directly.

□

6.6.3 Conjugacy with Chaotic Maps

The power of symbolic dynamics lies in the following idea: if we can find a topological conjugacy (or semi-conjugacy) between a chaotic map f and a shift map σ, then the symbolic dynamics of σ completely describes the dynamics of f.

Definition 6.32 (Topological conjugacy). Two dynamical systems $f : X \to X$ and $g : Y \to Y$ are *topologically conjugate* if there exists a homeomorphism $h : X \to Y$ such that $h \circ f = g \circ h$, i.e., the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & X \\ \downarrow h & & \downarrow h \\ Y & \xrightarrow{g} & Y \end{array}$$

If h is merely a continuous surjection (not necessarily a homeomorphism), we say f and g are *semi-conjugate.*

6.6.4 The Doubling Map and Binary Expansions

The doubling map $f(x) = 2x \pmod 1$ on $[0,1)$ provides the cleanest example of the connection to symbolic dynamics.

Every $x \in [0,1)$ has a binary expansion $x = 0.b_1b_2b_3\ldots$ (in base 2), where $b_i \in \{0,1\}$. The action of the doubling map on this expansion is:

$$f(0.b_1b_2b_3\ldots) = 0.b_2b_3b_4\ldots$$

This is exactly the shift map! More precisely, define the *coding map* $h : [0,1) \to \Sigma_2$ by $h(x) = (b_1, b_2, b_3, \ldots)$, the binary expansion of x. Then:

$$h \circ f = \sigma \circ h.$$

The map h is not quite a homeomorphism (binary expansions are not unique for dyadic rationals, e.g., $0.0111\ldots = 0.1000\ldots$), so this is a semi-conjugacy. However, the set of dyadic rationals is countable (hence negligible for most purposes), and the correspondence is otherwise one-to-one.

Consequences. Since the shift map on Σ_2 is chaotic (Proposition 6.31), and semi-conjugacy preserves many dynamical properties, the doubling map is chaotic. More concretely:

- *Dense periodic orbits.* Periodic orbits of the doubling map correspond to eventually repeating binary expansions, i.e., rational numbers with odd denominators. These are dense in $[0, 1)$.
- *Topological transitivity.* The orbit of the point x^* whose binary expansion concatenates all finite binary strings (the Champernowne-like construction) is dense.
- *Sensitive dependence.* Two points that agree in the first n binary digits but differ in the $(n+1)$-st will have f^n-images that differ in the first digit, hence are at least $1/4$ apart.

Example 6.33 (Counting periodic orbits). How many periodic orbits of period n does the doubling map have? A periodic orbit of period n corresponds to a repeating binary block of length n: $x = 0.\overline{b_1 b_2 \dots b_n}$. The number of binary strings of length n is 2^n, and each period-n orbit contains exactly n points, so the number of orbits of *exact* period n is:

$$\frac{1}{n} \sum_{d|n} \mu(n/d)\, 2^d,$$

where μ is the Möbius function (this counts primitive necklaces). For large n, this is approximately $2^n/n$, confirming that the number of periodic orbits grows exponentially.

6.7 Takens' Embedding Theorem

We close this chapter with a theorem that bridges the gap between theory and applications, and which will play a central role in Part III when we discuss reservoir computing.

6.7.1 The Problem

Suppose a deterministic dynamical system $f : M \to M$ on a d-dimensional manifold M produces complicated (possibly chaotic) behavior. In practice, we rarely observe the full state $x \in M$. Instead, we measure a single scalar observable $\phi : M \to \mathbb{R}$ (e.g., one component of the state, or a temperature reading, or a voltage signal). The resulting data is a scalar time series:

$$\phi(x_0),\ \phi(x_1),\ \phi(x_2),\ \dots$$

where $x_n = f^n(x_0)$.

The question. Can we reconstruct the geometry and dynamics of the attractor from this scalar time series alone?

The answer, remarkably, is *yes* — under generic conditions.

6.7.2 Delay Coordinate Embedding

Given the scalar time series $\{y_n\} = \{\phi(f^n(x_0))\}$, we form *delay vectors*:

$$\mathbf{y}_n = (y_n, y_{n+\tau}, y_{n+2\tau}, \dots, y_{n+(m-1)\tau}) \in \mathbb{R}^m,$$

where $\tau \geq 1$ is the *delay* (or *lag*) and m is the *embedding dimension*. The collection of delay vectors $\{\mathbf{y}_n\}$ traces out a set in $\mathbb{R}^m$.

Theorem 6.34 (Takens, 1981). *Let M be a compact manifold of dimension d, let $f : M \to M$ be a C^2 diffeomorphism, and let $\phi : M \to \mathbb{R}$ be a C^2 function. For generic pairs (f, ϕ) (i.e., an open and dense set in the C^2 topology), the delay coordinate map*

$$\Phi_{\phi,f} : M \to \mathbb{R}^{2d+1}, \quad \Phi_{\phi,f}(x) = (\phi(x), \phi(f(x)), \phi(f^2(x)), \ldots, \phi(f^{2d}(x)))$$

is an embedding (i.e., a homeomorphism onto its image whose derivative is injective at every point).

Interpretation. If we take $m = 2d + 1$ delays (with $\tau = 1$, or more generally, with a "generic" delay τ), then the delay coordinate map is one-to-one on M and preserves the topology and differentiable structure. The attractor reconstructed in $\mathbb{R}^{2d+1}$ from the scalar time series is diffeomorphic to the original attractor in M.

The bound $2d + 1$ is analogous to the Whitney embedding theorem, which states that any d-dimensional manifold can be embedded in $\mathbb{R}^{2d+1}$. The extra $+1$ (compared to $2d$) is needed to ensure injectivity generically.

6.7.3 Practical Considerations

Embedding dimension. The theorem requires $m \geq 2d + 1$, but d is the dimension of the ambient manifold M. In practice, what matters is the dimension of the *attractor* $A \subseteq M$, which may be much smaller than d. The question of whether $m \geq 2 \dim_B(A) + 1$ suffices was addressed by Sauer, Yorke, and Casdagli (1991).

Theorem 6.35 (Sauer, Yorke, and Casdagli, 1991). *The delay coordinate map $\Phi_{\phi,f}$ is generically one-to-one on a compact set $A \subseteq M$ provided $m > 2 \dim_B(A)$, where $\dim_B(A)$ is the box-counting dimension of A.*

This "fractal" version of Takens' theorem is essential in practice. For example, the Lorenz attractor has $\dim_B \approx 2.06$, so $m \geq 5$ should suffice for reconstruction, rather than $m \geq 7$ (which would be required if we only knew $M = \mathbb{R}^3$, giving $2 \cdot 3 + 1 = 7$).

Choice of delay τ. The theorem holds for generic τ (or $\tau = 1$), but in practice the choice of τ matters for the quality of the reconstruction. If τ is too small, consecutive delays are highly correlated and the reconstructed attractor is "squashed" along the diagonal. If τ is too large, consecutive delays become effectively independent and the structure is lost. Common heuristics include choosing τ as the first zero of the autocorrelation function, or the first minimum of the mutual information.

6.7.4 Significance for Reservoir Computing

Takens' theorem guarantees that the geometry and topology of a chaotic attractor can be faithfully reconstructed from a single scalar measurement, using only time-delayed copies of that measurement.

This is directly relevant to reservoir computing (Part III). A reservoir computer processes an input time series through a high-dimensional dynamical system (the "reservoir"), producing a high-dimensional state that encodes the input history. If the reservoir dynamics creates a delay

embedding of the input, then the reservoir state faithfully represents the underlying attractor, and a simple linear readout can extract predictions.

More precisely, if the reservoir state at time n depends on the recent input history $(u_n, u_{n-1}, \ldots, u_{n-k})$ in a sufficiently rich way, this is analogous to a delay embedding. The connection between Takens' theorem and the information-processing capacity of reservoirs has been made precise by several authors (see Chapter 16).

This is a deep and somewhat surprising connection: the same theorem that allows physicists to reconstruct attractors from experimental data also explains why reservoir computers can learn to predict chaotic systems.

6.8 Worked Examples

We consolidate the ideas of this chapter with several extended examples.

Example 6.36 (Verifying chaos for the doubling map — complete analysis). We assemble the results obtained throughout this chapter into a complete characterization.

Let $f(x) = 2x \pmod 1$ on $[0, 1)$.

1. *Lyapunov exponent:* $\lambda = \ln 2 > 0$ (Example 6.8).
2. *Sensitive dependence:* with $\delta = 1/4$ (Example 6.6).
3. *Dense periodic orbits:* periodic points are rationals of the form $p/(2^n - 1)$, which are dense.
4. *Topological transitivity:* follows from ergodicity with respect to Lebesgue measure (to be proved in Chapter 9), or directly from the symbolic dynamics (Section 6.6.4).
5. *Fractal dimension of the "attractor":* the map acts on the full interval $[0, 1)$, which has dimension 1. The attractor is the entire interval (for Lebesgue-a.e. initial condition).
6. *Symbolic dynamics:* semi-conjugate to the full shift on two symbols.

The doubling map is the "simplest" chaotic system: it exemplifies every concept in this chapter in its most transparent form.

Example 6.37 (Estimating the correlation dimension from data). Suppose we generate $N = 10{,}000$ iterates of the Hénon map at $a = 1.4$, $b = 0.3$ and wish to estimate the correlation dimension.

Procedure:

1. Discard the first 1000 iterates as transient.
2. For the remaining points $\{(x_i, y_i)\}_{i=1}^{N'}$, compute the correlation integral:
$$C(\epsilon) = \frac{2}{N'(N'-1)} \sum_{i<j} \mathbf{1}[\|(x_i - x_j, y_i - y_j)\| < \epsilon].$$

1. Plot $\ln C(\epsilon)$ versus $\ln \epsilon$ for a range of ϵ.
2. The slope of the linear region gives d_2.

In practice, one finds a slope of approximately 1.21 to 1.26, consistent with the Kaplan-Yorke estimate of ≈ 1.26.

If we instead work with a scalar time series (observing only the x-coordinate) and use delay embedding with $m = 3$ and $\tau = 1$, we form vectors $(x_n, x_{n+1}, x_{n+2}) \in \mathbb{R}^3$ and repeat the Grassberger-Procaccia algorithm. The estimated correlation dimension converges to the same

value ≈ 1.26 (provided $m > 2d_2 \approx 2.52$, which is satisfied since $m = 3$). This is Takens' theorem in action.

Example 6.38 (Symbolic dynamics of the tent map). The tent map $T : [0, 1] \to [0, 1]$ is defined by:

$$T(x) = \begin{cases} 2x & \text{if } x \leq 1/2, \\ 2(1-x) & \text{if } x > 1/2. \end{cases}$$

Partition $[0, 1]$ into $P_0 = [0, 1/2)$ and $P_1 = [1/2, 1]$. The itinerary of x is $s(x) = (s_0, s_1, s_2, \ldots)$ where $s_n = 0$ if $T^n(x) < 1/2$ and $s_n = 1$ if $T^n(x) \geq 1/2$.

Claim: the itinerary map $s : [0, 1] \to \Sigma_2$ is a semi-conjugacy between T and the shift σ.

To see this, note that T maps P_0 linearly onto $[0, 1]$ (stretching by a factor of 2) and P_1 linearly onto $[0, 1]$ (stretching by 2 and reflecting). Thus the symbolic dynamics is equivalent to the full shift on two symbols, confirming that the tent map is chaotic with the same symbolic structure as the doubling map.

The Lyapunov exponent is $\lambda = \ln 2$ (since $|T'(x)| = 2$ wherever T is differentiable), matching the doubling map.

6.9 Summary

This chapter established the mathematical foundations of chaos and strange attractors.

Key ideas:

- *Chaos* (Devaney) requires topological transitivity, dense periodic orbits, and sensitive dependence — but the Banks et al. result shows the first two conditions suffice.
- *Lyapunov exponents* quantify the rate of divergence of nearby orbits. A positive maximal Lyapunov exponent is the signature of chaos. The Oseledets theorem guarantees their existence in the ergodic setting.
- *Strange attractors* are invariant sets with fractal geometry supporting chaotic dynamics. The Lorenz, Rössler, and Hénon attractors are the canonical examples.
- *Fractal dimension* (box-counting, Hausdorff, correlation) quantifies the geometric complexity of attractors. The Kaplan-Yorke conjecture links dimension to Lyapunov exponents.
- *Symbolic dynamics* reduces the study of continuous chaotic systems to combinatorics on symbol sequences.
- *Takens' embedding theorem* guarantees that attractors can be reconstructed from scalar time series, providing the theoretical foundation for data-driven methods including reservoir computing.

Looking ahead. We now have the dynamical systems tools needed to study chaos. But we have repeatedly invoked ergodic-theoretic ideas (invariant measures, the Birkhoff ergodic theorem, ergodicity) without proving them. Part II develops this machinery properly, beginning with measure theory in Chapter 7.

Exercises

Exercise 6.1. Let $f(x) = 3x \pmod 1$ on $[0, 1)$. (a) Compute the Lyapunov exponent. (b) Find the number of fixed points and the number of period-2 orbits. (c) Show that f is semi-conjugate to the one-sided shift on 3 symbols. (d) Conclude that f is chaotic in the sense of Devaney.

Exercise 6.2. Consider the logistic map $f_r(x) = rx(1 - x)$ on $[0, 1]$. (a) Show that f_r has sensitive dependence on initial conditions for $r = 4$ by explicitly constructing, for any $x \in (0, 1)$ and any $\epsilon > 0$, a point y with $|x - y| < \epsilon$ and an iterate n with $|f^n(x) - f^n(y)| > 1/4$. *Hint:* Use the conjugacy to the tent map (or the doubling map) via $x = \sin^2(\pi\theta/2)$. (b) Compute numerically the Lyapunov exponent $\lambda(r)$ as a function of r for $r \in [2.5, 4]$. Where is $\lambda = 0$? Where is $\lambda > 0$? How does this relate to the bifurcation diagram from Chapter 5?

Exercise 6.3. (a) Compute the box-counting dimension of the Sierpinski triangle. *Hint:* At each stage, there are 3^n triangles of side length 2^{-n}. (b) Compute the box-counting dimension of the Koch curve. *Hint:* At each stage, there are 4^n segments of length 3^{-n}. (c) Prove that $\dim_H(A) \leq \dim_B(A)$ for any bounded set $A \subseteq \mathbb{R}^d$.

Exercise 6.4. Consider the map $f(x) = 4x(1 - x)$ on $[0, 1]$ with invariant measure $d\mu = \frac{1}{\pi\sqrt{x(1-x)}}dx$. (a) Verify that μ is a probability measure (i.e., $\mu([0, 1]) = 1$). (b) Verify computationally (by numerical integration or otherwise) that $\int_0^1 \ln|f'(x)|\, d\mu = \ln 2$. (c) The Birkhoff ergodic theorem implies that for μ-a.e. x_0, $\frac{1}{n}\sum_{k=0}^{n-1} \ln|f'(x_k)| \to \ln 2$. Verify this numerically: pick a "random" initial condition, compute 10^6 iterates, and plot the running average of $\ln|f'(x_k)|$.

Exercise 6.5. Consider the Lorenz system with classical parameters. Suppose you observe only the x-coordinate, obtaining a time series $\{x(t_n)\}$. (a) What is the minimum embedding dimension m guaranteed by Takens' theorem (using $d = 3$, the ambient dimension)? (b) What is the minimum embedding dimension suggested by the Sauer-Yorke-Casdagli theorem (using $\dim_B \approx 2.06$)? (c) Explain qualitatively why using too small or too large a delay τ leads to poor reconstructions.

Exercise 6.6. (a) For the shift map on Σ_2, find all periodic orbits of period 4. How many are there? (b) Show that the number of periodic points of period n (not necessarily with minimal period n) of the shift on Σ_k is k^n. (c) Use the Möbius inversion formula to derive the exact number of orbits of *minimal* period n. (d) Compute the *topological entropy* of the shift on Σ_k, defined as $h_{\text{top}} = \lim_{n\to\infty} \frac{1}{n} \ln(\text{number of period-}n \text{ points})$.

Exercise 6.7. A four-dimensional system has Lyapunov exponents $\lambda_1 = 0.5$, $\lambda_2 = 0.1$, $\lambda_3 = -0.3$, $\lambda_4 = -1.8$. (a) Compute the Kaplan-Yorke dimension d_{KY}. (b) What does the sign of $\sum_{i=1}^4 \lambda_i$ tell you about the system? Is it dissipative or conservative? (c) Is the system chaotic? How do you know?

Exercise 6.8. Construct a continuous map $f : [0, 1] \to [0, 1]$ that has sensitive dependence on initial conditions but is *not* topologically transitive. *Hint:* Consider a map with an attracting fixed point and a repelling fixed point, where orbits between them separate before converging.

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References

- Afraimovich, V. S., Bykov, V. V., and Shil'nikov, L. P. (1977). On the origin and structure of the Lorenz attractor. *Dokl. Akad. Nauk SSSR*, 234:336–339.
- Banks, J., Brooks, J., Cairns, G., Davis, G., and Stacey, P. (1992). On Devaney's definition of chaos. *American Mathematical Monthly*, 99(4):332–334.
- Benedicks, M. and Carleson, L. (1991). The dynamics of the Hénon map. *Annals of Mathematics*, 133(1):73–169.
- Devaney, R. L. (1989). *An Introduction to Chaotic Dynamical Systems*. 2nd ed. Addison-Wesley.
- Grassberger, P. and Procaccia, I. (1983). Measuring the strangeness of strange attractors. *Physica D*, 9(1–2):189–208.
- Guckenheimer, J. and Williams, R. F. (1979). Structural stability of Lorenz attractors. *Publications Mathématiques de l'IHÉS*, 50:59–72.
- Hénon, M. (1976). A two-dimensional mapping with a strange attractor. *Communications in Mathematical Physics*, 50:69–77.
- Huang, W. and Ye, X. (2002). Devaney's chaos or 2-scattering implies Li-Yorke's chaos. *Topology and its Applications*, 117:259–272.
- Jakobson, M. V. (1981). Absolutely continuous invariant measures for one-parameter families of one-dimensional maps. *Communications in Mathematical Physics*, 81:39–88.
- Kaplan, J. L. and Yorke, J. A. (1979). Chaotic behavior of multidimensional difference equations. In Peitgen, H.-O. and Walther, H.-O., editors, *Functional Differential Equations and Approximation of Fixed Points*, Lecture Notes in Mathematics 730, pages 204–227. Springer.
- Katok, A. and Hasselblatt, B. (1995). *Introduction to the Modern Theory of Dynamical Systems*. Cambridge University Press.
- Li, T.-Y. and Yorke, J. A. (1975). Period three implies chaos. *American Mathematical Monthly*, 82(10):985–992.
- Lorenz, E. N. (1963). Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20(2):130–141.
- Oseledets, V. I. (1968). A multiplicative ergodic theorem. Lyapunov characteristic numbers for dynamical systems. *Trudy Moskovskogo Matematicheskogo Obshchestva*, 19:179–210.
- Ott, E. (2002). *Chaos in Dynamical Systems*. 2nd ed. Cambridge University Press.
- Rössler, O. E. (1976). An equation for continuous chaos. *Physics Letters A*, 57(5):397–398.
- Sauer, T., Yorke, J. A., and Casdagli, M. (1991). Embedology. *Journal of Statistical Physics*, 65(3–4):579–616.
- Strogatz, S. H. (2015). *Nonlinear Dynamics and Chaos*. 2nd ed. Westview Press.
- Takens, F. (1981). Detecting strange attractors in turbulence. In Rand, D. and Young, L.-S., editors, *Dynamical Systems and Turbulence*, Lecture Notes in Mathematics 898, pages 366–381. Springer.
- Tucker, W. (2002). A rigorous ODE solver and Smale's 14th problem. *Foundations of*

Computational Mathematics, 2(1):53–117.

- Young, L.-S. (1982). Dimension, entropy, and Lyapunov exponents. *Ergodic Theory and Dynamical Systems*, 2:109–124.

Recommended Reading

For a first pass through the ideas of chaos, Strogatz (2015) is excellent — readable, well-motivated, and full of examples, though lighter on proofs. Devaney (1989) provides the standard rigorous introduction to chaos in one-dimensional maps, including a thorough treatment of symbolic dynamics; it is the natural companion to this chapter.

For a deeper treatment, Katok and Hasselblatt (1995) is the definitive graduate reference on dynamical systems and ergodic theory; it is demanding but comprehensive. Ott (2002) strikes a balance between rigor and physical intuition, with excellent coverage of fractal dimensions and Lyapunov exponents.

On the Lorenz attractor specifically, the short survey by Viana (2000), "What's new on Lorenz strange attractors?" (*The Mathematical Intelligencer*, 22(3):6–19), is an accessible overview of the mathematics surrounding Tucker's theorem. Tucker's original paper (2002) is surprisingly readable for a computer-assisted proof.

For Takens' embedding theorem and its applications, the original paper (Takens, 1981) is concise and worth reading. The extension by Sauer, Yorke, and Casdagli (1991) is essential for understanding the practical aspects. Kantz and Schreiber (2003), *Nonlinear Time Series Analysis*, gives an excellent treatment of the practical aspects of attractor reconstruction from data.

For the connections between these ideas and reservoir computing — the subject of Part III — the reader may wish to look ahead to Chapter 16, or consult Lukoševičius and Jaeger (2009), "Reservoir computing approaches to recurrent neural network training" (*Computer Science Review*, 3(3):127–149).

Part II

Ergodic Theory

Chapter 7

Measure Theory Essentials

"The integral is linear — this is the one thing you should remember if you remember nothing else." — Anonymous, but universally agreed upon

This chapter opens Part II of the text. Our goal in the coming chapters is to develop the ergodic theory of dynamical systems: the study of the long-run statistical behavior of orbits. To do this properly, we need a language for assigning sizes to sets, defining averages, and making precise the notion that a property holds for "almost every" initial condition. That language is measure theory.

We do not attempt a complete development of the subject. Instead, we assemble the definitions, theorems, and examples that will be used repeatedly in what follows. The reader who has seen measure theory before may skim this chapter and return to it as needed. The reader encountering it for the first time should work through the definitions carefully and attempt the exercises; the intuition will develop over the next several chapters as we put these tools to work.

7.1 Why Measure Theory?

Consider the doubling map $T : [0, 1) \to [0, 1)$ defined by $T(x) = 2x \pmod 1$. In Chapter 4 we observed that this map exhibits sensitive dependence on initial conditions: nearby orbits diverge exponentially. Yet numerical experiments suggest a remarkable regularity in the *statistical* behavior of orbits. If we histogram the first N iterates of a "typical" starting point x_0, the histogram converges to the uniform distribution on $[0, 1]$ as $N \to \infty$.

To make this precise, we need to answer several questions:

1. What do we mean by the "size" of a subset of $[0, 1]$? (For intervals this is clear, but what about more complicated sets?)
2. What does "typical" mean? We want to exclude a "negligible" set of exceptional initial conditions — but negligible in what sense?
3. How do we define the "average" of a function along an orbit, and in what sense does it converge?

The Riemann integral, familiar from calculus, is insufficient. It cannot integrate the characteristic function $\mathbf{1}_{\mathbb{Q}\cap[0,1]}$ of the rationals in $[0, 1]$, even though the rationals form a "negligible"

set by any reasonable standard. More fundamentally, the space of Riemann-integrable functions is not complete in any useful norm, which makes it ill-suited for limit theorems.

Lebesgue's insight was to partition the *range* of a function rather than its domain: instead of asking "what does f do on each small subinterval?", ask "on what set does f take values in each small range?" This requires us to measure the sizes of potentially complicated subsets of the domain — and that demands a theory of measure.

The payoff is substantial. Lebesgue integration gives us powerful convergence theorems (Section 7.4), complete function spaces (Section 7.5), and a framework in which the fundamental theorems of ergodic theory can be stated and proved.

7.2 Sigma-Algebras and Measurable Spaces

7.2.1 Definition and First Examples

Definition 7.1 (Sigma-algebra)**.** Let X be a nonempty set. A collection $\mathcal{F}$ of subsets of X is a *σ-algebra *(or* σ-field*) on X if:

1. $X \in \mathcal{F}$. 2. If $A \in \mathcal{F}$, then $A^c := X \setminus A \in \mathcal{F}$. (Closure under complementation.) 3. If $A_1, A_2, A_3, \ldots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$. (Closure under countable unions.)

The pair $(X, \mathcal{F})$ is called a *measurable space*, and the elements of $\mathcal{F}$ are called *measurable sets.*

Note that closure under complementation and countable unions implies closure under countable intersections (by De Morgan's law), under finite unions, under finite intersections, and that $\emptyset = X^c \in \mathcal{F}$.

Example 7.1. The following are σ-algebras on a set X:

- **Trivial σ-algebra**: $\mathcal{F} = \{\emptyset, X\}$. This is the smallest σ-algebra on X. - **Discrete σ-algebra**: $\mathcal{F} = 2^X$ (the power set of X). This is the largest σ-algebra on X. - If $A \subseteq X$, then $\mathcal{F} = \{\emptyset, A, A^c, X\}$ is a σ-algebra.

For most of our applications, X will be a topological space (often a metric space), and the relevant σ-algebra is the following.

Definition 7.2 (Borel σ-algebra)**.** Let X be a topological space. The *Borel σ-algebra* $\mathcal{B}(X)$ is the σ-algebra generated by the open subsets of X. Its elements are called *Borel sets.*

We make the notion of "generated by" precise next.

7.2.2 Generated Sigma-Algebras

Proposition 7.1. *Let X be a set and let $\{\mathcal{F}_\alpha\}_{\alpha \in I}$ be any nonempty family of σ-algebras on X. Then $\bigcap_{\alpha \in I} \mathcal{F}_\alpha$ is a σ-algebra on X.*

Proof. * We check the three axioms. (i) $X \in \mathcal{F}_\alpha$ for every α, so $X \in \bigcap \mathcal{F}_\alpha$. (ii) If $A \in \bigcap \mathcal{F}_\alpha$, then $A \in \mathcal{F}_\alpha$ for every α, hence $A^c \in \mathcal{F}_\alpha$ for every α, hence $A^c \in \bigcap \mathcal{F}_\alpha$. (iii) Identical argument for countable unions. □

□

Definition 7.3. Let $\mathcal{C}$ be any collection of subsets of X. The *σ-algebra generated by $\mathcal{C}$*, denoted $\sigma(\mathcal{C})$, is the intersection of all σ-algebras on X that contain $\mathcal{C}$:

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{F} : \mathcal{F} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{C} \subseteq \mathcal{F}\}.$$

This is the smallest σ-algebra containing $\mathcal{C}$. The collection $\mathcal{C}$ is called a *generating family* for $\sigma(\mathcal{C})$.

Thus the Borel σ-algebra $\mathcal{B}(X)$ is $\sigma(\tau)$, where τ is the topology on X. On $\mathbb{R}$, the Borel σ-algebra is also generated by any of the following collections:

- All open intervals (a, b). - All closed intervals $[a, b]$. - All half-open intervals $(-\infty, a]$, for $a \in \mathbb{R}$. - All open intervals with rational endpoints.

The last of these is a countable generating family, which gives some sense of why $\mathcal{B}(\mathbb{R})$, though enormous, is still "manageable." (In fact $|\mathcal{B}(\mathbb{R})| = |\mathbb{R}| = \mathfrak{c}$, while $|2^{\mathbb{R}}| = 2^{\mathfrak{c}} > \mathfrak{c}$, so most subsets of $\mathbb{R}$ are *not* Borel.)

7.2.3 Measurable Functions

Definition 7.4. Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces. A function $f : X \to Y$ is *$(\mathcal{F}, \mathcal{G})$-measurable *(or simply* measurable* when the σ-algebras are understood) if

$$f^{-1}(B) \in \mathcal{F} \quad \text{for every } B \in \mathcal{G}.$$

When $Y = \mathbb{R}$ (or $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$) equipped with its Borel σ-algebra, we call f a *measurable function* on $(X, \mathcal{F})$.

Proposition 7.2. *Let $(X, \mathcal{F})$ be a measurable space. A function $f : X \to \mathbb{R}$ is measurable if and only if $f^{-1}((-\infty, a]) \in \mathcal{F}$ for every $a \in \mathbb{R}$.*

Proof. * The "only if" direction is immediate since $(-\infty, a]$ is a Borel set. For the "if" direction, note that the collection $\{B \in \mathcal{B}(\mathbb{R}) : f^{-1}(B) \in \mathcal{F}\}$ is a σ-algebra (check the axioms) that contains all half-lines $(-\infty, a]$. Since these half-lines generate $\mathcal{B}(\mathbb{R})$, this σ-algebra contains $\mathcal{B}(\mathbb{R})$. □

□

Proposition 7.3. *Let $f, g : X \to \mathbb{R}$ be measurable functions. Then $f + g$, fg, $\max(f, g)$, $\min(f, g)$, and $|f|$ are measurable. If (f_n) is a sequence of measurable functions, then $\sup_n f_n$, $\inf_n f_n$, $\limsup_n f_n$, and $\liminf_n f_n$ are measurable.*

The proof is a sequence of exercises in manipulating preimages and is left to the reader (Exercise 7.1). The crucial point is that measurability is preserved under countable operations — this is the advantage over continuity, which is only preserved under finite operations.

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7.3 Measures

7.3.1 Definition and Basic Properties

Definition 7.5. Let $(X, \mathcal{F})$ be a measurable space. A *measure* on $(X, \mathcal{F})$ is a function $\mu : \mathcal{F} \to [0, +\infty]$ satisfying:

1. $\mu(\emptyset) = 0$. 2. **Countable additivity** (σ-additivity): if $A_1, A_2, \ldots \in \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

The triple $(X, \mathcal{F}, \mu)$ is called a *measure space.*

Definition 7.6. A measure μ is:

- *Finite* if $\mu(X) < \infty$. - A *probability measure* if $\mu(X) = 1$. We then call $(X, \mathcal{F}, \mu)$ a *probability space.* - *σ-finite* if $X = \bigcup_{n=1}^{\infty} A_n$ with $\mu(A_n) < \infty$ for each n.

In ergodic theory, we work almost exclusively with probability measures or finite measures.

The next proposition collects the essential properties that follow from countable additivity. These will be used without comment in later chapters.

Proposition 7.4. *(Basic properties of measures). Let $(X, \mathcal{F}, \mu)$ be a measure space.*

1. ***Monotonicity****: If $A \subseteq B$ and $A, B \in \mathcal{F}$, then $\mu(A) \leq \mu(B)$. 2.* ***Subadditivity****: If $A_1, A_2, \ldots \in \mathcal{F}$, then $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$. 3.* ***Continuity from below****: If $A_1 \subseteq A_2 \subseteq \cdots$ and $A_n \in \mathcal{F}$, then*

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n).$$

4. ***Continuity from above****: If $A_1 \supseteq A_2 \supseteq \cdots$, $A_n \in \mathcal{F}$, and $\mu(A_1) < \infty$, then*

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n).$$

Proof. * (1) Write $B = A \cup (B \setminus A)$ as a disjoint union, so $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$.

(2) Define $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$ for $n \geq 2$. Then the B_n are pairwise disjoint, $\bigcup B_n = \bigcup A_n$, and $B_n \subseteq A_n$, so

$$\mu\left(\bigcup_n A_n\right) = \sum_n \mu(B_n) \leq \sum_n \mu(A_n).$$

(3) Using the same telescoping trick: set $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. These are disjoint with $\bigcup_{k=1}^{n} B_k = A_n$ and $\bigcup_{k=1}^{\infty} B_k = \bigcup_n A_n$. Then

$$\mu\left(\bigcup_n A_n\right) = \sum_{k=1}^{\infty} \mu(B_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \mu(B_k) = \lim_{n\to\infty} \mu(A_n).$$

(4) Apply (3) to the increasing sequence $C_n = A_1 \setminus A_n$:

$$\mu(A_1) - \mu\left(\bigcap_n A_n\right) = \mu\left(\bigcup_n C_n\right) = \lim_n \mu(C_n) = \mu(A_1) - \lim_n \mu(A_n).$$

The finiteness of $\mu(A_1)$ is used to cancel it from both sides. $\square$

The hypothesis $\mu(A_1) < \infty$ in (4) is essential. Without it, the result can fail: let $A_n = [n, \infty)$ with Lebesgue measure. Then $\mu(A_n) = \infty$ for all n, but $\bigcap_n A_n = \emptyset$, so $\mu(\bigcap_n A_n) = 0 \neq \lim_n \mu(A_n)$.

7.3.2 Examples

Example 7.2. (Counting measure). Let X be any set, $\mathcal{F} = 2^X$, and define $\mu(A) = |A|$ (the cardinality of A, with $\mu(A) = \infty$ if A is infinite). This is a measure, called the *counting measure* on X.

Example 7.3. (Dirac measure). Fix a point $x_0 \in X$. The *Dirac measure* (or *point mass*) at x_0 is

$$\delta_{x_0}(A) = \begin{cases} 1 & \text{if } x_0 \in A, \\ 0 & \text{if } x_0 \notin A. \end{cases}$$

This is a probability measure on $(X, \mathcal{F})$ for any σ-algebra $\mathcal{F}$. It models a deterministic outcome: all the "mass" is concentrated at a single point.

Example 7.4. (Lebesgue measure). The *Lebesgue measure* λ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is the unique measure satisfying $\lambda((a, b)) = b - a$ for every bounded open interval. Its construction is sketched in Section 7.3.3 below. The restriction of λ to $[0, 1]$ gives a probability measure on $([0, 1], \mathcal{B}([0, 1]))$.

Lebesgue measure is the formalization of "length" (in $\mathbb{R}$), "area" (in $\mathbb{R}^2$), and "volume" (in $\mathbb{R}^n$). It is the default measure for dynamics on the interval and the torus.

7.3.3 Construction of Lebesgue Measure (Sketch)

The construction proceeds in three steps. We outline it for $[0, 1]$; the construction on $\mathbb{R}$ is analogous.

Step 1: Outer measure. For any $A \subseteq [0, 1]$, define the *Lebesgue outer measure*

$$\lambda^*(A) = \inf\left\{ \sum_{n=1}^{\infty} (b_n - a_n) : A \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n) \right\},$$

where the infimum is over all countable covers of A by open intervals.

The outer measure λ^* is defined on *all* subsets of $[0, 1]$, is monotone, and is countably subadditive: $\lambda^*(\bigcup_n A_n) \leq \sum_n \lambda^*(A_n)$. However, it is *not* countably additive on all subsets.

Step 2: Carathéodory's criterion. Call a set $A \subseteq [0, 1]$ *Lebesgue measurable* if for every $E \subseteq [0, 1]$,

$$\lambda^*(E) = \lambda^*(E \cap A) + \lambda^*(E \cap A^c).$$

Carathéodory's theorem states that the collection $\mathcal{L}$ of Lebesgue measurable sets is a σ-algebra and that λ^* restricted to $\mathcal{L}$ is a complete measure.

Step 3: Verification. One checks that every Borel set is Lebesgue measurable (so $\mathcal{B}([0,1]) \subseteq \mathcal{L}$) and that $\lambda^*((a,b)) = b - a$ for intervals. The restriction $\lambda = \lambda^*|_{\mathcal{L}}$ is *Lebesgue measure.*

The σ-algebra $\mathcal{L}$ is strictly larger than $\mathcal{B}([0,1])$: there exist Lebesgue measurable sets that are not Borel. However, for essentially all purposes in dynamics, we work with the Borel σ-algebra or identify sets that differ by a set of measure zero.

Remark 7.1. Not every subset of $[0,1]$ is Lebesgue measurable. The construction of a non-measurable set (due to Vitali, 1905) requires the axiom of choice: partition $[0,1)$ into equivalence classes under the relation $x \sim y$ iff $x - y \in \mathbb{Q}$, and select one representative from each class. The resulting set cannot be assigned a Lebesgue measure consistent with translation invariance. This is the fundamental reason we need σ-algebras: we must restrict our attention to a well-behaved collection of sets.

7.3.4 Null Sets and "Almost Everywhere"

Definition 7.7. A set $N \in \mathcal{F}$ is a *null set* (or has *measure zero*) if $\mu(N) = 0$. A property is said to hold *μ-almost everywhere* (abbreviated μ-a.e., or simply a.e. when μ is understood) if it holds on $X \setminus N$ for some null set N.

This is the precise meaning of "typical" in dynamics: a property holds for a "typical" orbit if it holds for μ-almost every initial condition.

Example 7.5. The set $\mathbb{Q} \cap [0,1]$ of rationals in $[0,1]$ has Lebesgue measure zero (it is countable, and each singleton has measure zero). Therefore, when we say "almost every orbit of the doubling map equidistributes," we allow for a set of exceptions that includes, but may be larger than, the rationals.

7.3.5 Borel Measures on Metric Spaces

In the setting of dynamics, the state space is typically a metric space (X, d) (often compact), and we work with Borel measures: measures defined on $\mathcal{B}(X)$.

Definition 7.8. A Borel measure μ on a metric space X is:

- *Regular* if for every $B \in \mathcal{B}(X)$,

$$\mu(B) = \inf\{\mu(U) : U \supseteq B,\ U \text{ open}\} = \sup\{\mu(K) : K \subseteq B,\ K \text{ compact}\}.$$

- *Locally finite* if every point has a neighborhood of finite measure.

A locally finite regular Borel measure is called a *Radon measure.* On a compact metric space, every finite Borel measure is Radon. Since we will primarily work with probability measures on compact spaces, we will not need to worry about regularity issues.

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7.4 Lebesgue Integration

7.4.1 Simple Functions

Definition 7.9. Let $(X, \mathcal{F})$ be a measurable space. A *simple function* is a measurable function $\varphi : X \to \mathbb{R}$ that takes only finitely many values. Every simple function can be written as

$$\varphi = \sum_{k=1}^{n} c_k \, \mathbf{1}_{A_k},$$

where $c_1, \ldots, c_n \in \mathbb{R}$ are distinct, $A_k = \varphi^{-1}(\{c_k\}) \in \mathcal{F}$, and $\mathbf{1}_{A_k}$ is the characteristic (indicator) function of A_k. This is called the *standard representation* of φ.

7.4.2 Definition of the Integral

The Lebesgue integral is built in three stages.

Stage 1: Nonnegative simple functions. For a nonnegative simple function $\varphi = \sum_{k=1}^{n} c_k \mathbf{1}_{A_k}$ (standard representation, all $c_k \geq 0$) on a measure space $(X, \mathcal{F}, \mu)$, define

$$\int_X \varphi \, d\mu = \sum_{k=1}^{n} c_k \, \mu(A_k),$$

with the convention $0 \cdot \infty = 0$.

Stage 2: Nonnegative measurable functions. For a nonnegative measurable function $f : X \to [0, +\infty]$, define

$$\int_X f \, d\mu = \sup\left\{ \int_X \varphi \, d\mu : 0 \leq \varphi \leq f, \ \varphi \text{ simple} \right\}.$$

This may equal $+\infty$.

Stage 3: General measurable functions. Write $f = f^+ - f^-$, where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$. If at least one of $\int f^+ \, d\mu$ and $\int f^- \, d\mu$ is finite, define

$$\int_X f \, d\mu = \int_X f^+ \, d\mu - \int_X f^- \, d\mu.$$

If both integrals are finite, f is called *(μ-)integrable *(or* Lebesgue integrable*), and we write $f \in L^1(X, \mathcal{F}, \mu)$ or simply $f \in L^1(\mu)$.

We also write $\int_A f \, d\mu = \int_X f \cdot \mathbf{1}_A \, d\mu$ for integration over a measurable subset A.

Remark 7.2. There is a useful approximation result underlying Stage 2: every nonnegative measurable function is the pointwise limit of an increasing sequence of nonnegative simple functions. This can be constructed explicitly. Given $f \geq 0$ measurable, define

$$\varphi_n(x) = \begin{cases} k \cdot 2^{-n} & \text{if } k \cdot 2^{-n} \leq f(x) < (k+1) \cdot 2^{-n}, \ k = 0, 1, \ldots, n \cdot 2^n - 1, \\ n & \text{if } f(x) \geq n. \end{cases}$$

Then $0 \leq \varphi_1 \leq \varphi_2 \leq \cdots$ and $\varphi_n(x) \to f(x)$ for every x.

7.4.3 Basic Properties

The Lebesgue integral satisfies the following, all of which are used constantly.

Proposition 7.5. *(Properties of the integral). Let $(X, \mathcal{F}, \mu)$ be a measure space and let f, g be measurable functions.*

1. ***Linearity**: If $f, g \in L^1(\mu)$ and $a, b \in \mathbb{R}$, then $af + bg \in L^1(\mu)$ and*

$$\int (af + bg)\, d\mu = a \int f\, d\mu + b \int g\, d\mu.$$

*2. **Monotonicity**: If $f \leq g$ a.e. and both integrals are defined, then $\int f\, d\mu \leq \int g\, d\mu$. 3. **Triangle inequality**: $|\int f\, d\mu| \leq \int |f|\, d\mu$. 4. **Null sets are negligible**: If $\mu(N) = 0$, then $\int_N f\, d\mu = 0$. Equivalently, if $f = g$ a.e., then $\int f\, d\mu = \int g\, d\mu$.*

7.4.4 Convergence Theorems

The three convergence theorems below are the workhorses of measure-theoretic analysis. They tell us when we may interchange limits and integrals — a question that arises repeatedly in ergodic theory.

Theorem 7.6 (Monotone Convergence Theorem; Beppo Levi)**.** *Let (f_n) be a sequence of nonnegative measurable functions on $(X, \mathcal{F}, \mu)$ such that $f_n \leq f_{n+1}$ a.e. for all n. Let $f = \lim_{n\to\infty} f_n$ (pointwise a.e.). Then*

$$\int_X f\, d\mu = \lim_{n\to\infty} \int_X f_n\, d\mu.$$

Proof sketch. sketch.* The inequality $\int f\, d\mu \geq \lim_n \int f_n\, d\mu$ follows from monotonicity (since $f_n \leq f$ implies $\int f_n \leq \int f$). The reverse inequality is the substantive content. Fix a simple function φ with $0 \leq \varphi \leq f$ and fix $\alpha \in (0, 1)$. Define $A_n = \{x : f_n(x) \geq \alpha\varphi(x)\}$. Then $A_1 \subseteq A_2 \subseteq \cdots$ and $\bigcup_n A_n = X$ (a.e., since $f_n \uparrow f \geq \varphi > \alpha\varphi$ wherever $\varphi > 0$). Thus

$$\int_X f_n\, d\mu \geq \int_{A_n} f_n\, d\mu \geq \alpha \int_{A_n} \varphi\, d\mu.$$

By continuity from below applied to the measure $A \mapsto \int_A \varphi\, d\mu$, the right side converges to $\alpha \int_X \varphi\, d\mu$ as $n \to \infty$. Taking $\alpha \uparrow 1$ and then the supremum over all such φ gives $\lim_n \int f_n\, d\mu \geq \int f\, d\mu$. □

Theorem 7.7 (Fatou's Lemma)**.** *Let (f_n) be a sequence of nonnegative measurable functions. Then*

$$\int_X \liminf_{n\to\infty} f_n\, d\mu \leq \liminf_{n\to\infty} \int_X f_n\, d\mu.$$

Proof. * Define $g_n = \inf_{k\geq n} f_k$. Then $g_n \leq f_n$, the sequence (g_n) is nondecreasing, and $g_n \uparrow \liminf_n f_n$. By the Monotone Convergence Theorem,

$$\int \liminf_n f_n\, d\mu = \lim_n \int g_n\, d\mu = \liminf_n \int g_n\, d\mu \leq \liminf_n \int f_n\, d\mu. \quad \square$$

□

Theorem 7.8 (Dominated Convergence Theorem; Lebesgue)**.** *Let (f_n) be a sequence of measurable functions on $(X, \mathcal{F}, \mu)$ such that $f_n \to f$ pointwise a.e. Suppose there exists an integrable function $g \geq 0$ with $|f_n| \leq g$ a.e. for all n. Then f and all f_n are integrable, and*

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Equivalently, $\int |f_n - f| \, d\mu \to 0$.

Proof. * The functions $g + f_n \geq 0$ and $g - f_n \geq 0$ are nonnegative. Applying Fatou's lemma to each:

$$\int (g + f) \, d\mu \leq \liminf_n \int (g + f_n) \, d\mu = \int g \, d\mu + \liminf_n \int f_n \, d\mu,$$

$$\int (g - f) \, d\mu \leq \liminf_n \int (g - f_n) \, d\mu = \int g \, d\mu - \limsup_n \int f_n \, d\mu.$$

Since $\int g \, d\mu < \infty$, we may cancel it. The first inequality gives $\int f \, d\mu \leq \liminf_n \int f_n \, d\mu$, and the second gives $\int f \, d\mu \geq \limsup_n \int f_n \, d\mu$. Together: $\lim_n \int f_n \, d\mu = \int f \, d\mu$.

For the "equivalently" statement, apply the above to $|f_n - f|$, which is dominated by $2g$, and converges to 0 a.e. □

Remark 7.3. The domination hypothesis cannot be dropped. Consider $f_n = n \cdot \mathbf{1}_{(0,1/n)}$ on $[0, 1]$ with Lebesgue measure. Then $f_n \to 0$ pointwise a.e., but $\int f_n \, d\lambda = 1$ for all n.

7.4.5 Comparison with the Riemann Integral

For functions on a bounded interval $[a, b]$:

Theorem 7.26. If $f : [a, b] \to \mathbb{R}$ is Riemann integrable, then f is Lebesgue measurable, Lebesgue integrable, and the two integrals agree:

$$\int_a^b f(x) \, dx \text{ (Riemann)} = \int_{[a,b]} f \, d\lambda \text{ (Lebesgue)}.$$

Moreover, f is Riemann integrable if and only if it is bounded and continuous λ-a.e. (Lebesgue's criterion).

The Lebesgue integral is strictly more general: $\mathbf{1}_{\mathbb{Q}\cap[0,1]}$ is Lebesgue integrable (with integral 0) but not Riemann integrable.

7.5 L^p Spaces

7.5.1 Definitions

Definition 7.10. Let $(X, \mathcal{F}, \mu)$ be a measure space and let $1 \leq p < \infty$. The space $L^p(X, \mathcal{F}, \mu)$ (or $L^p(\mu)$, or simply L^p) consists of all measurable functions $f : X \to \mathbb{R}$ (or $\mathbb{C}$) with

$$\|f\|_p := \left(\int_X |f|^p \, d\mu \right)^{1/p} < \infty.$$

The space $L^\infty(X, \mathcal{F}, \mu)$ consists of all measurable functions that are bounded a.e., with

$$\|f\|_\infty := \inf\{C \geq 0 : |f| \leq C \text{ a.e.}\}$$

(the *essential supremum* of $|f|$).

Convention. Strictly speaking, $\|\cdot\|_p$ is a seminorm: $\|f\|_p = 0$ does not imply $f = 0$, only $f = 0$ a.e. We follow the standard convention of identifying functions that agree almost everywhere, so that elements of L^p are really equivalence classes of functions. With this identification, $\|\cdot\|_p$ is a genuine norm.

The three cases of primary importance for us are:

- **$L^1(\mu)$**: the space of integrable functions. This is where the Birkhoff ergodic theorem (Chapter 9) lives. - **$L^2(\mu)$**: the space of square-integrable functions. This is where the von Neumann mean ergodic theorem lives. - **$L^\infty(\mu)$**: the space of essentially bounded functions. Observables in dynamical systems are often assumed to be in L^∞ or at least L^2.

7.5.2 Key Inequalities

Theorem 7.9 (Hölder's inequality)**.** *Let $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$ (conjugate exponents). If $f \in L^p$ and $g \in L^q$, then $fg \in L^1$ and*

$$\|fg\|_1 \leq \|f\|_p \, \|g\|_q.$$

Theorem 7.10 (Minkowski's inequality)**.** *For $1 \leq p \leq \infty$, if $f, g \in L^p$, then $f + g \in L^p$ and*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

This confirms that $\|\cdot\|_p$ is indeed a norm (the triangle inequality).

7.5.3 Completeness and L^2 as a Hilbert Space

Theorem 7.11 (Riesz-Fischer)**.** *For $1 \leq p \leq \infty$, the space $L^p(X, \mathcal{F}, \mu)$ is a Banach space (a complete normed vector space).*

Proof sketch for L^1. sketch for L^1.* Let (f_n) be a Cauchy sequence in L^1. Pass to a subsequence (f_{n_k}) with $\|f_{n_{k+1}} - f_{n_k}\|_1 < 2^{-k}$. Define $g = \sum_k |f_{n_{k+1}} - f_{n_k}|$; by the Monotone Convergence Theorem, $\|g\|_1 \leq \sum_k 2^{-k} = 1$, so $g < \infty$ a.e. The telescoping series $f_{n_1} + \sum_k (f_{n_{k+1}} - f_{n_k})$ therefore converges absolutely a.e. to some $f \in L^1$. One then verifies $f_n \to f$ in L^1 norm using the Dominated Convergence Theorem. □

The space L^2 has additional structure that makes it especially useful.

□

Definition 7.11. On $L^2(X, \mathcal{F}, \mu)$, define the *inner product*

$$\langle f, g \rangle = \int_X f \cdot \overline{g} \, d\mu.$$

(For real-valued functions, drop the conjugate.) Then $\|f\|_2 = \sqrt{\langle f, f \rangle}$.

By the Riesz-Fischer theorem, L^2 is complete, and hence L^2 is a *Hilbert space*: a complete inner product space.

Why L^2 matters for ergodic theory. Given a measure-preserving transformation $T : X \to X$ on a probability space $(X, \mathcal{F}, \mu)$, the *composition operator* $U_T : L^2(\mu) \to L^2(\mu)$ defined by $U_T f = f \circ T$ is a linear isometry. The mean ergodic theorem (von Neumann, 1932) asserts that the time averages $\frac{1}{N}\sum_{n=0}^{N-1} U_T^n f$ converge in the L^2 norm. The proof uses the Hilbert space structure of L^2 — specifically, the orthogonal decomposition into T-invariant functions and their complement. This is the subject of Chapter 9.

7.5.4 Inclusions Between L^p Spaces

For a probability measure μ (i.e., $\mu(X) = 1$), we have the chain of inclusions

$$L^\infty(\mu) \subseteq \cdots \subseteq L^2(\mu) \subseteq L^1(\mu),$$

and the norm inequalities $\|f\|_1 \leq \|f\|_2 \leq \|f\|_\infty$. This follows from Hölder's inequality with $g = 1$ (which has $\|g\|_q = 1$ on a probability space). These inclusions can *fail* for infinite measures (e.g., $f(x) = 1/x$ is in $L^2((1,\infty), \lambda)$ but not in $L^1((1,\infty), \lambda)$).

Since we work primarily with probability measures in ergodic theory, we have the convenience that bounded functions are in every L^p space.

7.6 Product Measures and Fubini's Theorem

7.6.1 Product Sigma-Algebras and Product Measures

Given measurable spaces $(X, \mathcal{F})$ and $(Y, \mathcal{G})$, the *product σ-algebra* $\mathcal{F} \otimes \mathcal{G}$ on $X \times Y$ is the σ-algebra generated by the *measurable rectangles* $\{A \times B : A \in \mathcal{F}, B \in \mathcal{G}\}$.

Theorem 7.12 (Product measure)**.** *Let $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$ be σ-finite measure spaces. There exists a unique measure $\mu \otimes \nu$ on $(X \times Y, \mathcal{F} \otimes \mathcal{G})$ satisfying*

$$(\mu \otimes \nu)(A \times B) = \mu(A) \cdot \nu(B)$$

for all $A \in \mathcal{F}$, $B \in \mathcal{G}$.

The construction generalizes to finite products and, crucially for symbolic dynamics, to countably infinite products (Section 7.8.3).

7.6.2 Fubini-Tonelli Theorem

Theorem 7.13 (Tonelli)**.** *Let $(X, \mathcal{F}, \mu)$ and $(Y, \mathcal{G}, \nu)$ be σ-finite measure spaces, and let $f : X \times Y \to [0, +\infty]$ be $(\mathcal{F} \otimes \mathcal{G})$-measurable. Then:*

1. For each $x \in X$, the function $y \mapsto f(x, y)$ is $\mathcal{G}$-measurable. 2. The function $x \mapsto \int_Y f(x, y)\, d\nu(y)$ is $\mathcal{F}$-measurable. 3.

$$\int_{X \times Y} f\, d(\mu \otimes \nu) = \int_X \left(\int_Y f(x, y)\, d\nu(y) \right) d\mu(x) = \int_Y \left(\int_X f(x, y)\, d\mu(x) \right) d\nu(y).$$

Theorem 7.14 (Fubini)**.** *Under the same hypotheses as Tonelli's theorem, if $f : X \times Y \to \mathbb{R}$ (or $\mathbb{C}$) is $(\mathcal{F} \otimes \mathcal{G})$-measurable and $\int_{X\times Y} |f|\, d(\mu \otimes \nu) < \infty$, then all three iterated integrals in Tonelli's theorem are finite and equal.*

Remark 7.4. The difference: Tonelli requires $f \geq 0$ (no integrability hypothesis needed); Fubini requires integrability (and then handles sign changes). In practice, one often uses Tonelli first to verify integrability, then Fubini to compute.

We will use product measures and Fubini's theorem in Chapter 11 when working with shift spaces and symbolic dynamics.

—

7.7 Conditional Expectation

Conditional expectation is a more subtle construction that will be needed in Chapter 13 when we develop measure-theoretic entropy. We state the definition and key properties without proof.

Definition and Theorem 7.36 (Conditional expectation). Let $(X, \mathcal{F}, \mu)$ be a probability space, $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra, and $f \in L^1(X, \mathcal{F}, \mu)$. There exists a function $\mathbb{E}[f \mid \mathcal{G}] \in L^1(X, \mathcal{G}, \mu)$, unique up to μ-a.e. equality, such that

$$\int_G \mathbb{E}[f \mid \mathcal{G}]\, d\mu = \int_G f\, d\mu \quad \text{for all } G \in \mathcal{G}.$$

The function $\mathbb{E}[f \mid \mathcal{G}]$ is called the *conditional expectation of f given $\mathcal{G}$*.

The existence proof uses the Radon-Nikodým theorem (the measure $G \mapsto \int_G f\, d\mu$ is absolutely continuous with respect to $\mu|_{\mathcal{G}}$). The proof can be found in Billingsley [3, Section 34] or Halmos [1, Chapter IX].

Key properties:

1. **Linearity**: $\mathbb{E}[af + bg \mid \mathcal{G}] = a\,\mathbb{E}[f \mid \mathcal{G}] + b\,\mathbb{E}[g \mid \mathcal{G}]$ a.e.
2. **Monotonicity**: If $f \leq g$ a.e., then $\mathbb{E}[f \mid \mathcal{G}] \leq \mathbb{E}[g \mid \mathcal{G}]$ a.e.
3. **Tower property**: If $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$, then $\mathbb{E}[\mathbb{E}[f \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[f \mid \mathcal{H}]$ a.e.
4. **Trivial σ-algebra**: $\mathbb{E}[f \mid \{\emptyset, X\}] = \int_X f\, d\mu$ (a constant function).
5. **Full σ-algebra**: $\mathbb{E}[f \mid \mathcal{F}] = f$ a.e.
6. **$\mathcal{G}$-measurable factors**: If g is $\mathcal{G}$-measurable and bounded, then $\mathbb{E}[gf \mid \mathcal{G}] = g\,\mathbb{E}[f \mid \mathcal{G}]$ a.e.

The conditional expectation $\mathbb{E}[f \mid \mathcal{G}]$ can be understood as the "best approximation" of f by a $\mathcal{G}$-measurable function, in the L^2 sense: it is the orthogonal projection of f onto the closed subspace $L^2(X, \mathcal{G}, \mu) \subseteq L^2(X, \mathcal{F}, \mu)$. This geometric viewpoint connects directly to the Hilbert space ideas of Section 7.5.

7.8 Examples from Dynamics

We now introduce the three measures that will appear most frequently in Part II.

7.8.1 Lebesgue Measure on $[0, 1]$

Lebesgue measure λ on $([0, 1], \mathcal{B}([0, 1]))$ is the canonical probability measure for studying maps on the interval. It assigns to each subinterval $[a, b] \subseteq [0, 1]$ its length $b - a$.

Example 7.6. The doubling map $T(x) = 2x \pmod 1$ *preserves* Lebesgue measure: $\lambda(T^{-1}(A)) = \lambda(A)$ for every Borel set $A \subseteq [0, 1]$. To verify this for an interval $[a, b]$:

$$T^{-1}([a, b]) = \left[\frac{a}{2}, \frac{b}{2}\right] \cup \left[\frac{a+1}{2}, \frac{b+1}{2}\right],$$

so $\lambda(T^{-1}([a, b])) = \frac{b-a}{2} + \frac{b-a}{2} = b - a = \lambda([a, b])$. Since intervals generate $\mathcal{B}([0, 1])$, this suffices (by a standard uniqueness argument for measures agreeing on a generating π-system).

Not every map preserves Lebesgue measure. The tent map $T(x) = 1 - |2x - 1|$ does, but the logistic map $T(x) = 4x(1 - x)$ does not (its invariant measure has density $\frac{1}{\pi\sqrt{x(1-x)}}$ with respect to λ).

7.8.2 Haar Measure on the Circle

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ denote the circle (the unit interval with endpoints identified), and let $\mathcal{B}(\mathbb{T})$ be its Borel σ-algebra. *Haar measure* on $\mathbb{T}$ is the unique Borel probability measure m that is invariant under rotations:

$$m(A + \alpha) = m(A) \quad \text{for all } A \in \mathcal{B}(\mathbb{T}),\ \alpha \in \mathbb{T},$$

where $A + \alpha = \{x + \alpha \pmod 1 : x \in A\}$.

Concretely, m coincides with Lebesgue measure on $[0, 1)$ under the identification $\mathbb{T} \cong [0, 1)$. The existence and uniqueness of Haar measure holds for any compact topological group (this is a deep theorem of Haar and von Neumann), but on $\mathbb{T}$ the construction is elementary.

Example 7.7. The rotation $R_\alpha(x) = x + \alpha \pmod 1$ preserves Haar measure for any α. This is immediate from the definition. When α is irrational, R_α is uniquely ergodic: Haar measure is the *only* R_α-invariant Borel probability measure. This will be proved in Chapter 10.

7.8.3 Bernoulli Measure on $\{0, 1\}^{\mathbb{N}}$

Let $\Sigma_2 = \{0, 1\}^{\mathbb{N}}$ be the space of one-sided infinite binary sequences $\omega = (\omega_0, \omega_1, \omega_2, \ldots)$ with each $\omega_i \in \{0, 1\}$. Equip $\{0, 1\}$ with the discrete topology; then Σ_2 with the product topology is a compact metrizable space (it is homeomorphic to the Cantor set).

The Borel σ-algebra $\mathcal{B}(\Sigma_2)$ is generated by the *cylinder sets*

$$[a_0 a_1 \cdots a_{n-1}] = \{\omega \in \Sigma_2 : \omega_0 = a_0, \omega_1 = a_1, \ldots, \omega_{n-1} = a_{n-1}\},$$

where $a_i \in \{0, 1\}$.

Definition 7.12 (Bernoulli measure)**.** Fix $p \in (0, 1)$ and let $q = 1 - p$. The *(p, q)-Bernoulli measure* μ_p on $(\Sigma_2, \mathcal{B}(\Sigma_2))$ is the unique Borel probability measure satisfying

$$\mu_p([a_0 a_1 \cdots a_{n-1}]) = \prod_{i=0}^{n-1} p_i, \quad \text{where } p_i = \begin{cases} p & \text{if } a_i = 1, \\ q & \text{if } a_i = 0. \end{cases}$$

The fair coin measure corresponds to $p = q = 1/2$, in which case every cylinder of length n has measure 2^{-n}.

Formally, μ_p is the countably infinite product measure $\bigotimes_{n=0}^{\infty} \nu$, where ν is the measure on $\{0, 1\}$ with $\nu(\{1\}) = p$ and $\nu(\{0\}) = q$. The existence of such an infinite product measure is guaranteed by the Kolmogorov extension theorem (see Billingsley [3, Theorem 36.1]).

Example 7.8. The *(left) shift map* $\sigma : \Sigma_2 \to \Sigma_2$ defined by $(\sigma(\omega))_n = \omega_{n+1}$ preserves the $(\frac{1}{2}, \frac{1}{2})$-Bernoulli measure. More generally, σ preserves μ_p for every p. Indeed, for a cylinder set,

$$\mu_p(\sigma^{-1}([a_0 \cdots a_{n-1}])) = \mu_p([0a_0 \cdots a_{n-1}]) + \mu_p([1a_0 \cdots a_{n-1}]) = q \prod p_i + p \prod p_i = (p+q) \prod p_i = \mu_p([a_0 \cdots a_{n-1}]$$

The system $(\Sigma_2, \mathcal{B}(\Sigma_2), \mu_p, \sigma)$ is called the *(one-sided) Bernoulli shift.* It is the prototypical example of a mixing dynamical system and will be studied extensively in Chapters 11 and 13.

7.8.4 Connection to the Doubling Map

There is a beautiful connection between the symbolic and interval settings. Define $\pi : \Sigma_2 \to [0, 1]$ by

$$\pi(\omega) = \sum_{n=0}^{\infty} \frac{\omega_n}{2^{n+1}}.$$

This is the binary expansion map. It is surjective, continuous, and satisfies $\pi \circ \sigma = T \circ \pi$, where T is the doubling map. Moreover, the $(\frac{1}{2}, \frac{1}{2})$-Bernoulli measure pushes forward to Lebesgue measure:

$$\mu_{1/2}(\pi^{-1}(A)) = \lambda(A) \quad \text{for every Borel set } A \subseteq [0, 1].$$

This semiconjugacy between (Σ_2, σ) and $([0, 1], T)$ shows that the symbolic and geometric viewpoints are two descriptions of the same dynamical system. We will exploit such connections systematically in Part II.

7.9 Summary

The following table collects the key objects and results for quick reference.

| Concept | Definition/Statement | Where used | |—|—|—| | σ-algebra | Closed under complement and countable unions | Throughout | | Measure | σ-additive set function, $\mu(\emptyset) = 0$ | Throughout | | Lebesgue measure | $\lambda((a, b)) = b - a$; via outer measure construction | Interval maps | | Measurable function | Preimages of Borel sets are measurable | Throughout | | Lebesgue integral | Via simple functions; extends Riemann | Throughout | | Monotone Convergence | $f_n \uparrow f \implies \int f_n \uparrow \int f$ | Ch. 9, 10, 13 | | Dominated Convergence | $f_n \to f$,

$|f_n| \le g \in L^1 \implies \int f_n \to \int f$ | Ch. 9, 10 | | L^2 space | Hilbert space, inner product $\langle f, g\rangle = \int f\bar{g}$ | Mean ergodic thm | | Product measure | $(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$ | Symbolic dynamics | | Conditional expectation | Projection onto sub-σ-algebra | Entropy (Ch. 13) | | Bernoulli measure | Product measure on $\{0,1\}^{\mathbb{N}}$ | Shift spaces |

Exercises

Exercise 7.1. (Measurability is preserved under algebraic operations). Let $(X, \mathcal{F})$ be a measurable space and let $f, g : X \to \mathbb{R}$ be measurable functions.

(a) Prove that $f + g$ is measurable. (*Hint*: $\{f + g < a\} = \bigcup_{r\in\mathbb{Q}}(\{f < r\} \cap \{g < a - r\})$.)

(b) Prove that if (f_n) is a sequence of measurable functions, then $\sup_n f_n$ is measurable. (*Hint*: $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$.)

(c) Deduce that $\limsup_n f_n$ and $\liminf_n f_n$ are measurable.

Exercise 7.2. (Borel-Cantelli lemma). Let $(X, \mathcal{F}, \mu)$ be a measure space and let (A_n) be a sequence of measurable sets.

(a) Prove that if $\sum_{n=1}^{\infty} \mu(A_n) < \infty$, then $\mu(\limsup_n A_n) = 0$, where $\limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$ is the set of points belonging to infinitely many A_n.

(b) Apply this to the doubling map $T(x) = 2x \pmod 1$ on $([0, 1], \lambda)$: show that for λ-almost every x, the orbit $(T^n(x))_{n\ge 0}$ does not visit the interval $(0, 1/n^2)$ infinitely often.

Exercise 7.3. (Approximation in L^p). Let $f \in L^1([0, 1], \lambda)$.

(a) Prove that for every $\varepsilon > 0$, there exists a continuous function $g : [0, 1] \to \mathbb{R}$ with $\|f - g\|_1 < \varepsilon$. (*Hint*: First approximate by simple functions, then approximate indicator functions of intervals by continuous functions.)

(b) Prove that the set of polynomials is dense in $L^2([0, 1], \lambda)$. (*Hint*: By the Weierstrass approximation theorem, polynomials are dense in $C([0, 1])$ in the supremum norm, hence also in L^2 norm on a probability space.)

Exercise 7.4. (Product measures and the Cantor set). Let $C \subset [0, 1]$ be the standard middle-thirds Cantor set.

(a) Express C as a subset of $\{0, 2\}^{\mathbb{N}}$ via ternary expansions. Describe the Borel σ-algebra on C in terms of cylinder sets.

(b) Define the *Cantor measure* μ_C on C as the countably infinite product of the measure ν on $\{0, 2\}$ with $\nu(\{0\}) = \nu(\{2\}) = 1/2$. Show that μ_C is a Borel probability measure on C, and that μ_C assigns measure 2^{-n} to each cylinder of length n.

(c) Show that μ_C is non-atomic (i.e., $\mu_C(\{x\}) = 0$ for every $x \in C$), even though C has Lebesgue measure zero.

Exercise 7.5. (Invariance of Lebesgue measure). Prove that the tent map $T : [0, 1] \to [0, 1]$ defined by

$$T(x) = \begin{cases} 2x & \text{if } 0 \le x \le 1/2, \\ 2 - 2x & \text{if } 1/2 < x \le 1, \end{cases}$$

preserves Lebesgue measure. That is, show $\lambda(T^{-1}(A)) = \lambda(A)$ for every Borel set $A \subseteq [0,1]$. (*Hint*: It suffices to verify this for intervals $[a,b] \subseteq [0,1]$. Compute $T^{-1}([a,b])$ explicitly.)

Exercise 7.6. (Conditional expectation as projection). Let $(X, \mathcal{F}, \mu)$ be a probability space and $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-algebra. Show that if $f \in L^2(X, \mathcal{F}, \mu)$, then $\mathbb{E}[f \mid \mathcal{G}]$ is the unique element of $L^2(X, \mathcal{G}, \mu)$ that minimizes $\|f - g\|_2$ over all $g \in L^2(X, \mathcal{G}, \mu)$. That is, $\mathbb{E}[f \mid \mathcal{G}]$ is the orthogonal projection of f onto the closed subspace $L^2(X, \mathcal{G}, \mu)$.

(*Hint*: Show that $f - \mathbb{E}[f \mid \mathcal{G}]$ is orthogonal to every $g \in L^2(X, \mathcal{G}, \mu)$, using the defining property of conditional expectation.)

Exercise 7.7. (Bernoulli measure and the law of large numbers). Let $\mu = \mu_{1/2}$ be the fair-coin Bernoulli measure on $\Sigma_2 = \{0,1\}^{\mathbb{N}}$, and let σ be the shift map.

(a) Define $f : \Sigma_2 \to \mathbb{R}$ by $f(\omega) = \omega_0$ (the "observation" function that reads the first coordinate). Show that $f \in L^2(\mu)$ and compute $\int f \, d\mu$ and $\|f\|_2$.

(b) Show that the time average $\frac{1}{N}\sum_{n=0}^{N-1} f(\sigma^n(\omega)) = \frac{1}{N}\sum_{n=0}^{N-1} \omega_n$ is the frequency of 1's among the first N entries of ω.

(c) The strong law of large numbers (which we will recover as a consequence of Birkhoff's ergodic theorem in Chapter 9) asserts that this time average converges to $\int f \, d\mu$ for μ-almost every ω. State precisely what this says about the frequency of 1's in a "typical" binary sequence.

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Recommended Reading

The following references provide complete treatments of the material sketched in this chapter.

1. **P. R. Halmos**, *Measure Theory*, Springer, 1950. A classic reference, concise and well-organized. Particularly good on the abstract theory.
2. **W. Rudin**, *Real and Complex Analysis*, 3rd edition, McGraw-Hill, 1987. Chapters 1–3 give an elegant development of measure theory and integration. The treatment of L^p spaces is efficient and authoritative.
3. **P. Billingsley**, *Probability and Measure*, 3rd edition, Wiley, 1995. An excellent bridge between measure theory and probability. Especially recommended for the treatment of product measures, conditional expectation, and the Kolmogorov extension theorem.
4. **R. G. Bartle**, *The Elements of Integration and Lebesgue Measure*, Wiley, 1995. A slim, readable introduction. Good for a first encounter with the subject.
5. **H. L. Royden and P. M. Fitzpatrick**, *Real Analysis*, 4th edition, Pearson, 2010. A standard graduate text with a thorough treatment of measure and integration on $\mathbb{R}^n$.

For the connection to dynamics specifically, the reader may consult:

1. **P. Walters**, *An Introduction to Ergodic Theory*, Springer, 1982. Chapter 1 contains the measure-theoretic prerequisites tailored to ergodic theory.
2. **R. Mañé**, *Ergodic Theory and Differentiable Dynamics*, Springer, 1987. Another treatment of the measure-theoretic foundations with dynamics in mind.

Chapter 8

Measure-Preserving Transformations

In the preceding chapters we studied the qualitative behavior of dynamical systems through the lens of topology: orbit structure, limit sets, and recurrence. We now shift to a quantitative perspective. The central idea is to equip the phase space with a measure and to ask which transformations respect it. This chapter introduces measure-preserving transformations, the principal objects of ergodic theory, and develops the foundational results that will carry us through the rest of the book.

8.1 Definition and Basic Properties

8.1.1 The Definition

Let $(X, \mathcal{B}, \mu)$ be a measure space, where $\mathcal{B}$ is a σ-algebra on X and μ is a measure on $\mathcal{B}$. Unless stated otherwise, we take μ to be a probability measure, so that $\mu(X) = 1$.

Definition 8.1. A measurable map $T : X \to X$ is called **measure-preserving** (with respect to μ) if

$$\mu(T^{-1}(A)) = \mu(A) \quad \text{for all } A \in \mathcal{B}.$$

We also say that μ is **T-invariant**, or that T **preserves** μ. The quadruple $(X, \mathcal{B}, \mu, T)$ is called a **measure-preserving dynamical system**.

Here $T^{-1}(A) = \{x \in X : T(x) \in A\}$ denotes the preimage of A; it is not the image under some inverse function. This is a critical distinction that we must address carefully.

8.1.2 Why the Preimage?

A reader encountering this definition for the first time may wonder: why do we require $\mu(T^{-1}(A)) = \mu(A)$ rather than $\mu(T(A)) = \mu(A)$? There are several compelling reasons.

Measurability. If T is measurable, then $T^{-1}(A) \in \mathcal{B}$ for every $A \in \mathcal{B}$ by definition. However, the forward image $T(A)$ need not be measurable, even when A is. So the condition $\mu(T(A)) = \mu(A)$ may not even make sense.

The preimage preserves set operations. For any collection of sets, we have

$$T^{-1}\left(\bigcup_i A_i\right) = \bigcup_i T^{-1}(A_i), \qquad T^{-1}\left(\bigcap_i A_i\right) = \bigcap_i T^{-1}(A_i), \qquad T^{-1}(A^c) = (T^{-1}(A))^c.$$

These identities mean that the map $A \mapsto T^{-1}(A)$ is a homomorphism of σ-algebras. The forward image $T(A)$ satisfies none of these identities in general. This algebraic compatibility is what makes the preimage formulation natural and powerful.

Non-invertibility. Many important transformations (the doubling map, Bernoulli shifts) are not invertible. The preimage formulation requires no assumption of invertibility. If T happens to be a bijection with measurable inverse, then $\mu(T^{-1}(A)) = \mu(A)$ is equivalent to $\mu(T(A)) = \mu(A)$; but in general, only the preimage formulation is available.

The change-of-variables perspective. The condition $\mu(T^{-1}(A)) = \mu(A)$ is precisely the statement that the pushforward measure $T_*\mu$, defined by $(T_*\mu)(A) = \mu(T^{-1}(A))$, coincides with μ. In other words, μ is a fixed point of the pushforward operation. This is the natural notion of invariance in measure theory.

8.1.3 Equivalent Formulation via Integrals

The definition in terms of sets has a clean reformulation in terms of integrals, which is often more convenient for proofs.

Proposition 8.1. **Let $T : (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ be a measurable map. The following are equivalent:**

*1. *T is measure-preserving: $\mu(T^{-1}(A)) = \mu(A)$ for all $A \in \mathcal{B}$.* 2. *For every $f \in L^1(X, \mathcal{B}, \mu)$,**

$$\int_X f \circ T \, d\mu = \int_X f \, d\mu.$$

*3. *For every bounded measurable $f : X \to \mathbb{R}$,**

$$\int_X f \circ T \, d\mu = \int_X f \, d\mu.$$

Proof. * (1) $\Rightarrow$ (3): We use the standard approximation argument. First, take $f = \mathbf{1}_A$ for some $A \in \mathcal{B}$. Then $f \circ T = \mathbf{1}_{T^{-1}(A)}$, so

$$\int_X f \circ T \, d\mu = \mu(T^{-1}(A)) = \mu(A) = \int_X f \, d\mu.$$

By linearity, the identity extends to simple functions $f = \sum_{i=1}^n c_i \mathbf{1}_{A_i}$. For a general nonnegative bounded measurable f, choose simple functions $0 \le s_n \uparrow f$ pointwise. Then $s_n \circ T \uparrow f \circ T$ pointwise, and by the monotone convergence theorem,

$$\int_X f \circ T \, d\mu = \lim_{n\to\infty} \int_X s_n \circ T \, d\mu = \lim_{n\to\infty} \int_X s_n \, d\mu = \int_X f \, d\mu.$$

For general bounded measurable f, write $f = f^+ - f^-$ and apply the above to each part.

(3) $\Rightarrow$ (2): For nonnegative $f \in L^1$, approximate by bounded measurable functions $f_n = \min(f, n) \uparrow f$ and apply the monotone convergence theorem. For general $f \in L^1$, decompose as

$f = f^+ - f^-$.

(2) $\Rightarrow$ (1): Take $f = \mathbf{1}_A$. □

Remark 8.1. Condition (2) can be read as: "the expectation of any observable is unchanged under the dynamics." This is the statistical mechanics perspective: a measure is invariant if no experiment can detect the passage of time, on average.

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8.2 Examples

We now work through the fundamental examples in detail. Each serves as a prototype for a class of measure-preserving systems.

8.2.1 Rotations on the Circle

Let $X = \mathbb{R}/\mathbb{Z} \cong [0, 1)$ with Lebesgue measure λ and the Borel σ-algebra. Fix $\alpha \in \mathbb{R}$ and define the **rotation** (or **translation**) by

$$T_\alpha(x) = x + \alpha \pmod 1.$$

Proposition 8.2. **The rotation T_α preserves Lebesgue measure λ for every $\alpha \in \mathbb{R}$.**

Proof. * It suffices to verify the invariance condition on intervals $[a, b) \subset [0, 1)$, since these generate the Borel σ-algebra and the resulting measures agree on a generating π-system (by the π-λ theorem).

The preimage $T_\alpha^{-1}([a, b))$ consists of all $x \in [0, 1)$ such that $x + \alpha \pmod 1 \in [a, b)$, which is

$$T_\alpha^{-1}([a, b)) = \{x \in [0, 1) : x + \alpha \pmod 1 \in [a, b)\}.$$

Write $\beta = \alpha \pmod 1 \in [0, 1)$. Then

$$T_\alpha^{-1}([a, b)) = [(a - \beta) \bmod 1,\ (b - \beta) \bmod 1),$$

interpreted cyclically on $[0, 1)$. This set is either a single interval or a union of two intervals (when the subtraction wraps around), but in either case it has total Lebesgue measure $b - a$, since translation modulo 1 is a rigid motion on the circle.

More precisely, if $a - \beta \geq 0$, then $T_\alpha^{-1}([a, b)) = [a - \beta, b - \beta)$, and $\lambda(T_\alpha^{-1}([a, b))) = b - a = \lambda([a, b))$. If $a - \beta < 0$ but $b - \beta \geq 0$, then

$$T_\alpha^{-1}([a, b)) = [0, b - \beta) \cup [1 + a - \beta, 1),$$

which has measure $(b - \beta) + (\beta - a) = b - a$. Similarly for the case $b - \beta < 0$. In every case, $\lambda(T_\alpha^{-1}([a, b))) = b - a = \lambda([a, b))$. □

Remark 8.2. The dynamics of T_α depend critically on whether α is rational or irrational. If $\alpha = p/q$ with $\gcd(p, q) = 1$, then every orbit is periodic with period q. If α is irrational, every orbit is dense in $[0, 1)$ (by Weyl's theorem, which we will meet in Chapter 9). In either case, T_α

preserves Lebesgue measure. However, the ergodic properties differ sharply: T_α is ergodic with respect to Lebesgue measure if and only if α is irrational.

8.2.2 The Doubling Map

Define $T : [0, 1) \to [0, 1)$ by

$$T(x) = 2x \pmod 1 = \begin{cases} 2x & \text{if } 0 \le x < 1/2, \\ 2x - 1 & \text{if } 1/2 \le x < 1. \end{cases}$$

Proposition 8.3. **The doubling map T preserves Lebesgue measure λ.**

Proof. * Again, it suffices to check on intervals. Let $[a, b) \subset [0, 1)$. The preimage $T^{-1}([a, b))$ consists of all $x \in [0, 1)$ with $2x \pmod 1 \in [a, b)$. There are two branches of the preimage:

- From the branch $2x$ (for $x \in [0, 1/2)$): we need $2x \in [a, b)$, so $x \in [a/2, b/2)$.
- From the branch $2x-1$ (for $x \in [1/2, 1)$): we need $2x-1 \in [a, b)$, so $x \in [(a+1)/2, (b+1)/2)$.

These two intervals are disjoint (the first is contained in $[0, 1/2)$ and the second in $[1/2, 1)$), so

$$T^{-1}([a, b)) = \left[\frac{a}{2}, \frac{b}{2}\right) \cup \left[\frac{a+1}{2}, \frac{b+1}{2}\right).$$

Therefore,

$$\lambda(T^{-1}([a, b))) = \frac{b-a}{2} + \frac{b-a}{2} = b - a = \lambda([a, b)).$$

By the π-λ theorem, the measures $\lambda \circ T^{-1}$ and λ agree on all Borel sets. □

Remark 8.3. The doubling map is a 2-to-1 map: every point in $[0, 1)$ has exactly two preimages. Each branch of T^{-1} contracts by a factor of $1/2$, and the two contracted copies together have the same total measure as the original set. This is the mechanism behind measure-preservation for expanding maps more generally.

Remark 8.4. It is instructive to see this in binary. Write $x = 0.x_1x_2x_3\ldots$ in base 2. Then $T(x) = 0.x_2x_3x_4\ldots$, the left shift on the binary expansion. This connects the doubling map to the symbolic dynamics of the Bernoulli shift (Section 8.2.5).

8.2.3 The Baker's Map

The **baker's map** is a transformation of the unit square $[0, 1)^2$ that models the process of kneading dough: stretch horizontally, compress vertically, cut and stack. Define $B : [0, 1)^2 \to [0, 1)^2$ by

$$B(x, y) = \begin{cases} (2x,\ y/2) & \text{if } 0 \le x < 1/2, \\ (2x - 1,\ (y + 1)/2) & \text{if } 1/2 \le x < 1. \end{cases}$$

Geometrically: the left half of the square is stretched horizontally by a factor of 2 and compressed vertically by a factor of $1/2$, producing a rectangle $[0, 1) \times [0, 1/2)$. The right half

undergoes the same stretch and compression, producing $[0,1) \times [0,1/2)$, but is then placed on top of the first piece, occupying $[0,1) \times [1/2,1)$.

Proposition 8.4. **The baker's map B preserves two-dimensional Lebesgue measure λ^2 on $[0,1)^2$.**

Proof. * Let $R = [a_1, b_1) \times [a_2, b_2) \subset [0,1)^2$ be a rectangle. We compute $B^{-1}(R)$.

A point (x, y) maps into R under B if and only if one of the following holds:

Case 1: $0 \le x < 1/2$ and $(2x, y/2) \in R$. This requires $a_1 \le 2x < b_1$ and $a_2 \le y/2 < b_2$, i.e., $x \in [a_1/2, b_1/2)$ and $y \in [2a_2, 2b_2)$. The intersection with $[0,1/2) \times [0,1)$ gives a set of measure

$$\frac{b_1 - a_1}{2} \cdot \min(2b_2, 1) - \frac{b_1 - a_1}{2} \cdot \max(2a_2, 0).$$

Since $a_2, b_2 \in [0,1]$ and $2b_2 \le 2$, we need to be careful about truncation. But if $R \subset [0,1)^2$, then $0 \le a_2 < b_2 \le 1$, so $2a_2 \ge 0$ and $2b_2 \le 2$. The preimage in Case 1 is $[a_1/2, b_1/2) \times ([2a_2, 2b_2) \cap [0,1))$.

Case 2: $1/2 \le x < 1$ and $(2x - 1, (y+1)/2) \in R$. This requires $a_1 \le 2x - 1 < b_1$ and $a_2 \le (y+1)/2 < b_2$, i.e., $x \in [(a_1+1)/2, (b_1+1)/2)$ and $y \in [2a_2 - 1, 2b_2 - 1)$. The preimage in Case 2 is $[(a_1+1)/2, (b_1+1)/2) \times ([2a_2 - 1, 2b_2 - 1) \cap [0,1))$.

We can avoid case analysis by noting that B is a bijection on $[0,1)^2$ (one can verify this directly), and that on each piece B is an affine map. On the left half, B is given by the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}$ with determinant 1. On the right half, B is an affine map with the same linear part $\begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}$ (again determinant 1) and a translation.

Since B is piecewise affine with each piece having Jacobian determinant equal to 1, the change-of-variables formula gives

$$\lambda^2(B^{-1}(R)) = \int_{B^{-1}(R)} 1 \, d\lambda^2 = \int_R \frac{1}{|\det DB|} \, d\lambda^2 = \int_R 1 \, d\lambda^2 = \lambda^2(R)$$

for any measurable R. Here we used that B is a bijection and $|\det DB| = 1$ on each piece. □

8.2.4 Arnold's Cat Map

Let $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ be the two-dimensional torus with Lebesgue measure λ^2 and Borel σ-algebra. Define $T : \mathbb{T}^2 \to \mathbb{T}^2$ by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \pmod 1.$$

This is known as **Arnold's cat map**, named after V. I. Arnold, who illustrated its mixing properties using a picture of a cat on the torus.

Proposition 8.5. **Arnold's cat map preserves Lebesgue measure on $\mathbb{T}^2$.**

Proof. * Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. We note that $\det A = 2 \cdot 1 - 1 \cdot 1 = 1$.

The map T is the composition of the linear map $x \mapsto Ax$ on $\mathbb{R}^2$ with the projection $\pi : \mathbb{R}^2 \to \mathbb{T}^2$. Since A has integer entries, it maps $\mathbb{Z}^2$ to $\mathbb{Z}^2$, so it descends to a well-defined map on $\mathbb{T}^2$.

Since $\det A = 1$, the matrix A is in $\mathrm{SL}(2, \mathbb{Z})$ and hence A^{-1} also has integer entries (specifically, $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$). Therefore T is a bijection on $\mathbb{T}^2$.

For any measurable set $E \subset \mathbb{T}^2$, the preimage is $T^{-1}(E) = A^{-1}E \pmod 1$. The key computation is the change of variables formula for linear maps. On $\mathbb{R}^2$, the linear map $x \mapsto Ax$ transforms Lebesgue measure by the factor $|\det A|$. That is, for any measurable $E \subset \mathbb{R}^2$,

$$\lambda^2(A^{-1}(E)) = \frac{1}{|\det A|}\lambda^2(E) = \lambda^2(E).$$

On the torus, the same formula holds: the fundamental domain $[0, 1)^2$ is mapped by A to a parallelogram of area $|\det A| = 1$. When this parallelogram is folded back onto $\mathbb{T}^2$ by reducing modulo 1, the pieces tile $[0, 1)^2$ exactly, preserving total area. More precisely, for any Borel set $E \subset \mathbb{T}^2$,

$$\lambda^2(T^{-1}(E)) = \lambda^2(A^{-1}E \bmod 1) = \lambda^2(E),$$

where the second equality follows from the fact that $|\det A^{-1}| = 1$. □

Remark 8.5. The same argument applies to any linear toral automorphism $T_A : \mathbb{T}^n \to \mathbb{T}^n$ defined by a matrix $A \in \mathrm{SL}(n, \mathbb{Z})$ (or more generally $A \in \mathrm{GL}(n, \mathbb{Z})$ with $|\det A| = 1$). The condition $\det A = \pm 1$ with integer entries ensures both that T_A is well-defined on the torus and that it preserves Lebesgue measure.

Remark 8.6. Arnold's cat map is a hyperbolic toral automorphism. Its eigenvalues are $\phi = (3+\sqrt{5})/2 > 1$ and $1/\phi = (3-\sqrt{5})/2 < 1$ (both irrational), so it stretches in one eigendirection and contracts in the other, while preserving area. This is the source of its chaotic behavior: nearby points separate exponentially, yet the map is measure-preserving and invertible.

8.2.5 Bernoulli Shifts

Let $\Sigma = \{0, 1, \ldots, k-1\}$ be a finite alphabet and let $\Omega = \Sigma^{\mathbb{N}}$ be the space of one-sided infinite sequences $\omega = (\omega_0, \omega_1, \omega_2, \ldots)$ with $\omega_i \in \Sigma$. Fix a probability vector $\mathbf{p} = (p_0, p_1, \ldots, p_{k-1})$ with $p_i > 0$ and $\sum p_i = 1$. Define the product measure $\mu = \mathbf{p}^{\mathbb{N}}$ on Ω, which assigns to a **cylinder set**

$$[a_0, a_1, \ldots, a_{n-1}] = \{\omega \in \Omega : \omega_0 = a_0, \omega_1 = a_1, \ldots, \omega_{n-1} = a_{n-1}\}$$

the measure

$$\mu([a_0, a_1, \ldots, a_{n-1}]) = p_{a_0} p_{a_1} \cdots p_{a_{n-1}}.$$

The **shift map** $\sigma : \Omega \to \Omega$ is defined by

$$\sigma(\omega_0, \omega_1, \omega_2, \ldots) = (\omega_1, \omega_2, \omega_3, \ldots).$$

The system $(\Omega, \mathcal{B}, \mu, \sigma)$ is called a **(one-sided) Bernoulli shift** and denoted $B(p_0, \ldots, p_{k-1})$.

Proposition 8.6. **The shift map σ preserves the product measure μ.**

Proof. * Cylinder sets generate the σ-algebra $\mathcal{B}$ and form a π-system (closed under finite intersections), so by the π-λ theorem it suffices to check $\mu(\sigma^{-1}(C)) = \mu(C)$ for every cylinder set C.

Let $C = [a_0, a_1, \ldots, a_{n-1}]$. The preimage is

$$\sigma^{-1}(C) = \{\omega \in \Omega : \sigma(\omega) \in C\} = \{\omega \in \Omega : \omega_1 = a_0, \omega_2 = a_1, \ldots, \omega_n = a_{n-1}\}.$$

This is the set of sequences whose entries at positions $1, 2, \ldots, n$ are $a_0, a_1, \ldots, a_{n-1}$, with no constraint on position 0. We can decompose this as a disjoint union over the possible values of ω_0:

$$\sigma^{-1}(C) = \bigsqcup_{j=0}^{k-1} [j, a_0, a_1, \ldots, a_{n-1}].$$

Therefore,

$$\mu(\sigma^{-1}(C)) = \sum_{j=0}^{k-1} p_j \cdot p_{a_0} p_{a_1} \cdots p_{a_{n-1}} = \left(\sum_{j=0}^{k-1} p_j \right) p_{a_0} \cdots p_{a_{n-1}} = 1 \cdot \mu(C) = \mu(C).$$

This completes the proof. □

Remark 8.7. The connection to the doubling map is immediate. The map $x \mapsto 2x \pmod 1$ on $([0, 1), \lambda)$ is measurably conjugate to the Bernoulli shift $B(1/2, 1/2)$ via the binary expansion $x = \sum_{n=1}^{\infty} x_n 2^{-n} \leftrightarrow (x_1, x_2, x_3, \ldots)$. Under this identification, the doubling map becomes the shift on binary sequences, and Lebesgue measure corresponds to the fair coin-tossing measure. This is one of the most illuminating correspondences in ergodic theory.

8.2.6 Hamiltonian Systems and Liouville's Theorem

We briefly indicate how measure-preservation arises naturally in classical mechanics.

A **Hamiltonian system** on $\mathbb{R}^{2n}$ with coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n)$ is governed by Hamilton's equations:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \qquad i = 1, \ldots, n,$$

where $H : \mathbb{R}^{2n} \to \mathbb{R}$ is the Hamiltonian (the energy function). Let φ_t denote the time-t flow of this system.

Theorem 8.15 (Liouville). *The Hamiltonian flow φ_t preserves the standard volume form $\omega = dq_1 \wedge dp_1 \wedge \cdots \wedge dq_n \wedge dp_n$ on phase space. Equivalently, φ_t preserves Lebesgue measure on $\mathbb{R}^{2n}$.*

Proof sketch. sketch.* The rate of change of volume under the flow is governed by the divergence of the vector field $F = (\dot{q}_1, \ldots, \dot{q}_n, \dot{p}_1, \ldots, \dot{p}_n)$. We compute

$$\operatorname{div} F = \sum_{i=1}^{n} \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) = \sum_{i=1}^{n} \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0.$$

By the divergence theorem (or equivalently, by the Jacobi-Liouville formula $\frac{d}{dt} \det D\varphi_t = (\operatorname{div} F) \det D\varphi_t$), the Jacobian determinant $\det D\varphi_t = 1$ for all t, so φ_t preserves volume. □

Remark 8.8. Liouville's theorem is the reason why ergodic theory originated in statistical mechanics. Boltzmann and Gibbs needed to understand the long-time behavior of Hamiltonian systems. The invariance of phase-space volume is the starting point: it allows one to define meaningful statistical ensembles and to apply the recurrence and ergodic theorems we develop in this book.

If the energy surface $\Sigma_E = \{(q, p) : H(q, p) = E\}$ is compact, then the restriction of the flow to Σ_E preserves a natural finite measure (the **microcanonical measure** or **Liouville measure** on the energy surface), making $(\Sigma_E, \mu_E, \varphi_t)$ a measure-preserving dynamical system to which the full machinery of ergodic theory applies.

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8.3 Invariant Measures

8.3.1 The Problem

Given a measurable map $T : X \to X$, we have so far verified that specific measures are preserved. But in practice one often encounters the inverse problem: given T, find all T-invariant measures, or at least show that one exists.

This problem is fundamental because the ergodic-theoretic behavior of T depends heavily on which invariant measure is chosen. The same map can appear orderly under one invariant measure and chaotic under another.

8.3.2 Non-uniqueness of Invariant Measures

A map typically has many invariant measures. This is already visible in simple examples.

Example 8.1. Consider the doubling map $T(x) = 2x \pmod 1$.

- **Lebesgue measure** λ is T-invariant (Proposition 8.6). - **The Dirac measure** δ_0 at the fixed point $x = 0$ is T-invariant, since $T(0) = 0$ and $\delta_0(T^{-1}(A)) = \delta_0(A)$ for all A (the preimage of any set containing 0 contains 0, and vice versa). - More generally, if x_0 is a periodic point with period n (so $T^n(x_0) = x_0$), then the measure $\mu = \frac{1}{n} \sum_{j=0}^{n-1} \delta_{T^j(x_0)}$ is T-invariant. - **Convex combinations**: if μ_1 and μ_2 are T-invariant probability measures, then $\alpha\mu_1 + (1 - \alpha)\mu_2$ is also T-invariant for any $\alpha \in [0, 1]$.

The doubling map has countably many periodic orbits (the rationals with odd denominator), and correspondingly many invariant measures supported on them. In fact, it can be shown that the set of T-invariant probability measures is an infinite-dimensional convex set.

Example 8.2. The identity map $T(x) = x$ on any space $(X, \mathcal{B})$ preserves *every* probability measure on $(X, \mathcal{B})$.

8.3.3 The Krylov-Bogolyubov Theorem

Despite the abundance of invariant measures in many situations, the fundamental existence question is not trivial: does a given map always have at least one invariant measure? In the measurable category the answer is no (consider $T(x) = x + 1$ on $\mathbb{R}$ with Lebesgue measure; no finite invariant measure exists). But in the topological setting, a clean and elegant result guarantees existence.

Theorem 8.19 (Krylov-Bogolyubov). *Let X be a compact metrizable space and $T : X \to X$ a continuous map. Then there exists at least one T-invariant Borel probability measure on X.*

Proof. * Let $\mathcal{M}(X)$ denote the set of all Borel probability measures on X, equipped with the weak-* topology. By the Banach-Alaoglu theorem (or the Riesz representation theorem combined with Alaoglu's theorem), $\mathcal{M}(X)$ is a compact metrizable space in the weak-* topology, since X is compact and metrizable. Explicitly, the weak-* topology is the weakest topology such that $\mu \mapsto \int f \, d\mu$ is continuous for every $f \in C(X)$.

Pick any point $x_0 \in X$ and define the **empirical measures** (Cesaro averages of Dirac masses along the orbit):

$$\mu_n = \frac{1}{n} \sum_{j=0}^{n-1} \delta_{T^j(x_0)}.$$

Each μ_n is a Borel probability measure on X, so $(\mu_n)_{n \geq 1}$ is a sequence in the compact space $\mathcal{M}(X)$. By compactness (sequential, since $\mathcal{M}(X)$ is metrizable), there exists a subsequence $\mu_{n_k} \to \mu$ in the weak-* topology for some $\mu \in \mathcal{M}(X)$.

We claim that μ is T-invariant. To verify this, we must show $\int f \circ T \, d\mu = \int f \, d\mu$ for every $f \in C(X)$, which (by Proposition 8.2) is equivalent to T-invariance.

Compute:

$$\int f \circ T \, d\mu_n = \frac{1}{n} \sum_{j=0}^{n-1} f(T^{j+1}(x_0)) = \frac{1}{n} \sum_{j=1}^{n} f(T^j(x_0)).$$

Therefore,

$$\int f \circ T \, d\mu_n - \int f \, d\mu_n = \frac{1}{n} \left(\sum_{j=1}^{n} f(T^j(x_0)) - \sum_{j=0}^{n-1} f(T^j(x_0)) \right) = \frac{f(T^n(x_0)) - f(x_0)}{n}.$$

Since f is continuous and X is compact, f is bounded: $|f(x)| \leq \|f\|_\infty$ for all x. Hence

$$\left| \int f \circ T \, d\mu_n - \int f \, d\mu_n \right| \leq \frac{2\|f\|_\infty}{n} \to 0 \quad \text{as } n \to \infty.$$

Now take $n = n_k \to \infty$. By weak-* convergence, $\int f \, d\mu_{n_k} \to \int f \, d\mu$. Since $f \circ T$ is also

continuous (as the composition of continuous maps), $\int f \circ T \, d\mu_{n_k} \to \int f \circ T \, d\mu$. The estimate above gives

$$\int f \circ T \, d\mu = \lim_{k\to\infty} \int f \circ T \, d\mu_{n_k} = \lim_{k\to\infty} \int f \, d\mu_{n_k} = \int f \, d\mu.$$

This holds for all $f \in C(X)$, so μ is T-invariant. □

Remark 8.9. The Krylov-Bogolyubov theorem is non-constructive in the sense that it does not specify *which* invariant measure is obtained; different initial points x_0 and different subsequences may yield different limit measures. The theorem tells us the space $\mathcal{M}_T(X)$ of T-invariant probability measures is nonempty. In fact, one can show that $\mathcal{M}_T(X)$ is a nonempty, convex, compact subset of $\mathcal{M}(X)$ in the weak-* topology. Its extreme points are the ergodic measures (a topic for Chapter 9).

Remark 8.10. The proof reveals a useful constructive strategy: to find invariant measures, compute time averages along orbits and take limits. This is precisely the idea behind numerical estimation of invariant measures in applications, including the estimation of reservoir states in echo state networks.

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8.4 Natural and Physical Measures

8.4.1 The Problem of Physical Relevance

The Krylov-Bogolyubov theorem guarantees the existence of invariant measures, and as we saw in Section 8.3.2, there are typically many of them. Not all are equally interesting.

Consider a dissipative system such as the Henon map or the Lorenz attractor. Orbits starting from almost every initial condition (in the sense of Lebesgue measure on the ambient space) are attracted to a set of Lebesgue measure zero — the attractor. On this attractor, there may be uncountably many invariant probability measures, most of them supported on periodic orbits or other dynamically uninteresting subsets. The physically relevant measure is the one that describes the statistics actually observed in experiments or computer simulations.

8.4.2 SRB Measures

The concept of an **SRB measure** (Sinai-Ruelle-Bowen measure) captures the notion of "the physically observable invariant measure."

Definition 8.22 (Informal). Let $T : M \to M$ be a smooth map on a manifold M. A T-invariant probability measure μ is called an **SRB measure** (or a **physical measure**) if there exists a set $B(\mu) \subset M$ of positive Lebesgue measure such that for every continuous observable $\varphi : M \to \mathbb{R}$ and every $x \in B(\mu)$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} \varphi(T^j(x)) = \int \varphi \, d\mu.$$

The set $B(\mu)$ is called the **basin** of μ.

In other words, μ is the measure that governs the time averages for a set of initial conditions that is "visible" to Lebesgue measure. If $B(\mu)$ has full Lebesgue measure in some open set, then μ is the measure that a typical numerical simulation will reveal.

Remark 8.11. For conservative (measure-preserving) systems, the natural invariant measure is often Lebesgue measure itself (or the Liouville measure on an energy surface), and the Birkhoff ergodic theorem (Chapter 9) guarantees that time averages equal space averages for ergodic systems. For dissipative systems, the attractor may have Lebesgue measure zero, and the SRB measure is the correct replacement.

Remark 8.12. The existence of SRB measures is a deep result, proved for Axiom A attractors by Sinai, Ruelle, and Bowen in the 1970s. For more general systems (e.g., the Henon map), the existence of SRB measures remains a major open problem, resolved only in certain parameter ranges.

8.4.3 Connection to Reservoir Computing

In the theory of reservoir computing and echo state networks (which we will develop in Part III of this book), the **echo state property** asserts that the reservoir state $\mathbf{h}_t$ is asymptotically independent of the initial state $\mathbf{h}_0$. When the input is a stationary ergodic process, the reservoir state converges to a unique stationary distribution — an invariant measure for the driven dynamics.

This invariant measure is the analogue of an SRB measure: it describes the statistics observed in practice, regardless of initialization. The Krylov-Bogolyubov theorem guarantees its existence under mild compactness conditions, and the echo state property ensures its uniqueness. This connection between ergodic theory and machine learning is one of the main themes we will develop.

8.5 Recurrence Revisited

8.5.1 Poincare Recurrence

In Chapter 4 we encountered the Poincare recurrence theorem in the topological setting. We now state and prove its measure-theoretic version, which is sharper and more fundamental.

Theorem 8.25 (Poincare Recurrence Theorem). *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving dynamical system with $\mu(X) < \infty$. Let $A \in \mathcal{B}$ with $\mu(A) > 0$. Then μ-almost every point of A returns to A, i.e.,*

$$\mu\left(\{x \in A : T^n(x) \in A \text{ for infinitely many } n \geq 1\}\right) = \mu(A).$$

Proof. * Let $W = \{x \in A : T^n(x) \notin A \text{ for all } n \geq 1\}$ be the set of "wandering" points in A — those that leave A and never return. We will show $\mu(W) = 0$.

Define $W_k = T^{-k}(W)$ for $k \geq 0$, so $W_0 = W$. Note that $W_k = \{x \in X : T^k(x) \in W\}$. We claim that the sets $W_0, W_1, W_2, \ldots$ are pairwise disjoint.

Suppose $x \in W_j \cap W_k$ with $0 \le j < k$. Then $T^j(x) \in W \subset A$ and $T^k(x) \in W \subset A$. But $T^k(x) = T^{k-j}(T^j(x))$, and since $T^j(x) \in W$, we must have $T^m(T^j(x)) \notin A$ for all $m \ge 1$. Yet $k - j \ge 1$ and $T^{k-j}(T^j(x)) = T^k(x) \in A$. Contradiction. So the W_k are pairwise disjoint.

Since T is measure-preserving, $\mu(W_k) = \mu(T^{-k}(W)) = \mu(W)$ for all k. The sets W_k are pairwise disjoint subsets of X, and $\mu(X) < \infty$. Therefore

$$\infty > \mu(X) \ge \mu\left(\bigsqcup_{k=0}^{\infty} W_k\right) = \sum_{k=0}^{\infty} \mu(W_k) = \sum_{k=0}^{\infty} \mu(W).$$

This forces $\mu(W) = 0$. So μ-almost every point of A returns to A at least once.

To get infinitely many returns: let $A_1 = \{x \in A : T^n(x) \in A \text{ for some } n \ge 1\}$. We have shown $\mu(A_1) = \mu(A)$. Note that $A_1 \subset A$ and $A_1 \in \mathcal{B}$. Moreover, T maps returning points to returning points: if $x \in A_1$ and $T^{n_1}(x) \in A$ for some $n_1 \ge 1$, then applying the above argument to $T^{n_1}(x) \in A$ shows that $T^{n_1}(x)$ returns to A again (since $\mu(A) > 0$ and the argument gives almost sure return), and so on. A more precise argument: let $A_\infty = \{x \in A : T^n(x) \in A \text{ for infinitely many } n\}$. Then $A \setminus A_\infty = \{x \in A : T^n(x) \in A \text{ for only finitely many } n\}$. If $x \in A \setminus A_\infty$, let N be the largest n with $T^n(x) \in A$ (possibly $N = 0$). Then $T^N(x) \in W$, so $x \in T^{-N}(W) = W_N$. Hence $A \setminus A_\infty \subset \bigcup_{k=0}^{\infty} W_k$, and $\mu(A \setminus A_\infty) \le \sum_{k=0}^{\infty} \mu(W_k) = 0$. □

8.5.2 Kac's Lemma

Poincare's theorem tells us that almost every point returns. A natural follow-up: *how long* does it take to return? Kac's lemma gives a beautiful answer in the ergodic case.

Setup. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system with $\mu(X) = 1$. For a set $A \in \mathcal{B}$ with $\mu(A) > 0$, the **first return time** to A is the function $\tau_A : A \to \mathbb{N} \cup \{\infty\}$ defined by

$$\tau_A(x) = \inf\{n \ge 1 : T^n(x) \in A\}.$$

By Poincare recurrence, $\tau_A(x) < \infty$ for μ-almost every $x \in A$.

Theorem 8.26 (Kac's Lemma, 1947). *Let $(X, \mathcal{B}, \mu, T)$ be an ergodic measure-preserving system with $\mu(X) = 1$. Let $A \in \mathcal{B}$ with $\mu(A) > 0$. Then the expected return time to A is*

$$\int_A \tau_A \, d\mu = 1.$$

Equivalently, the conditional expected return time is $\mathbb{E}[\tau_A \mid x \in A] = 1/\mu(A)$.

The second formulation follows from the first by dividing by $\mu(A)$:

$$\frac{1}{\mu(A)} \int_A \tau_A \, d\mu = \frac{1}{\mu(A)}.$$

Proof. * For $n \ge 1$, define

$$A_n = \{x \in A : \tau_A(x) = n\} = \{x \in A : T(x) \notin A, T^2(x) \notin A, \ldots, T^{n-1}(x) \notin A, T^n(x) \in A\}.$$

The sets A_n are pairwise disjoint and by Poincare recurrence, $\mu(\bigsqcup_{n=1}^{\infty} A_n) = \mu(A)$ (up to a null set).

Now define, for each $n \geq 1$, the "tower level"

$$E_n = \{x \in X : T^{-1}(x) \in E_{n-1} \setminus A\}$$

More precisely, let us build the Kakutani tower (or Rokhlin tower) over A. For $n \geq 1$ and $0 \leq j \leq n-1$, define

$$A_n^{(j)} = T^j(A_n) \quad \text{for } 0 \leq j \leq n-1.$$

That is, $A_n^{(j)}$ is the set of points that started in A_n and have been iterated j times. By the definition of A_n:

- $A_n^{(0)} = A_n \subset A$.
- $A_n^{(j)} \subset A^c$ for $1 \leq j \leq n-1$ (these points have left A but not yet returned).
- $T(A_n^{(n-1)}) = T^n(A_n) \subset A$ (these points have just returned).

It is more convenient to work with preimage-defined sets to exploit measure-preservation directly. Define

$$B_k = \{x \in A : \tau_A(x) > k\} \quad \text{for } k \geq 0.$$

So $B_0 = A$, and $B_k = \{x \in A : T(x) \notin A, \ldots, T^k(x) \notin A\}$ for $k \geq 1$. Notice that $A_n = B_{n-1} \setminus B_n$ for $n \geq 1$ (with the convention that $\tau_A(x) > k$ means none of $T(x), \ldots, T^k(x)$ are in A).

Now consider the sets $T^j(B_j)$ for $j \geq 0$. We have $T^0(B_0) = A$. For $j \geq 1$:

$$T^j(B_j) = \{T^j(x) : x \in A, T(x) \notin A, \ldots, T^j(x) \notin A\} \subset A^c.$$

We claim that the sets $C_j = T^j(B_j) \setminus A$ for $j \geq 1$, together with A, form a partition of X (up to null sets) when T is ergodic. More precisely, define

$$D_j = \{y \in X \setminus A : \text{the most recent visit to } A \text{ before reaching } y \text{ was exactly } j \text{ steps ago}\}.$$

Formally, for $j \geq 1$, let

$$D_j = \{y \in X : y \notin A,\, T^{-1}(y) \notin A, \ldots, T^{-(j-1)}(y) \notin A,\, T^{-j}(y) \in A\}.$$

But this requires T to be invertible, which we have not assumed. Let us give a cleaner proof. We instead compute $\int_A \tau_A \, d\mu$ directly.

$$\int_A \tau_A \, d\mu = \int_A \sum_{k=1}^{\infty} \mathbf{1}_{\{\tau_A \geq k\}}(x) \, d\mu(x) = \sum_{k=1}^{\infty} \mu(\{x \in A : \tau_A(x) \geq k\}) = \sum_{k=0}^{\infty} \mu(B_k),$$

where $B_k = \{x \in A : \tau_A(x) > k\}$ for $k \geq 0$ (note the shift: $\tau_A \geq k+1$ iff $\tau_A > k$, and $B_0 = A$).

For each $k \geq 0$, B_k consists of points in A whose first k iterates avoid A:

$$B_k = A \cap T^{-1}(A^c) \cap T^{-2}(A^c) \cap \cdots \cap T^{-k}(A^c).$$

Now consider the sets $F_k = T^{-k}(A) \cap T^{-(k-1)}(A^c) \cap \cdots \cap T^{-1}(A^c) \cap A^c$ for $k \geq 1$, and $F_0 = A$. Here F_k (for $k \geq 1$) consists of points not in A that first enter A after exactly k steps, having avoided A for steps $1, \ldots, k-1$ (counted from the point, not from A).

Actually, let us use the cleanest approach. Define the sets:

$$E_j = T^j(B_j) \quad \text{for } j \geq 0.$$

Unpacking: $E_0 = B_0 = A$. For $j \geq 1$,

$$E_j = \{T^j(x) : x \in A,\, T(x) \notin A,\, T^2(x) \notin A, \ldots, T^j(x) \notin A\}.$$

A point $y \in E_j$ (with $j \geq 1$) has the property that $y \notin A$ (since $T^j(x) \notin A$ is part of the condition defining B_j). Moreover, y has a preimage $x = T^{-j}(y)$ in A such that the orbit from x avoids A for j steps.

Claim: *The sets $E_0, E_1, E_2, \ldots$ are pairwise disjoint (modulo null sets), and if T is ergodic, $\bigsqcup_{j=0}^{\infty} E_j = X$ (modulo null sets).*

Proof of disjointness. of disjointness.* The set $E_0 = A$ is disjoint from E_j for $j \geq 1$ since $E_j \subset A^c$. For $j \neq k$ with $j, k \geq 1$, the disjointness is more subtle and actually the sets E_j need not be disjoint in general (since T may not be injective). Instead, we use a different decomposition.

Let us restart the proof with the standard approach from Petersen.

Proof of Kac's Lemma (standard argument). of Kac's Lemma (standard argument).** Define $\widetilde{A}_j$ for $j \geq 0$ as follows. Let

$$\widetilde{A}_0 = A, \qquad \widetilde{A}_j = T^{-1}(\widetilde{A}_{j-1}) \setminus A \quad \text{for } j \geq 1.$$

Unpacking this inductive definition:

- $\widetilde{A}_0 = A$.
- $\widetilde{A}_1 = T^{-1}(A) \setminus A = \{x \notin A : T(x) \in A\}$ — points outside A that enter A in one step.
- $\widetilde{A}_2 = T^{-1}(\widetilde{A}_1) \setminus A = \{x \notin A : T(x) \in \widetilde{A}_1\} = \{x \notin A : T(x) \notin A,\, T^2(x) \in A\}$ — points outside A that first enter A in exactly two steps.
- In general, $\widetilde{A}_j = \{x \notin A : T(x) \notin A, \ldots, T^{j-1}(x) \notin A, T^j(x) \in A\}$ for $j \geq 1$.

So $\widetilde{A}_j$ (for $j \geq 1$) is the set of points in A^c whose first hitting time of A is exactly j.

Claim 1: *The sets $\widetilde{A}_0, \widetilde{A}_1, \widetilde{A}_2, \ldots$ are pairwise disjoint.*

This is clear: $\widetilde{A}_0 = A$, while $\widetilde{A}_j \subset A^c$ for $j \geq 1$. For $j, k \geq 1$ with $j \neq k$, the first hitting times to A differ, so the sets are disjoint.

Claim 2: *$\mu\Big(\bigsqcup_{j=0}^{\infty} \widetilde{A}_j\Big) = 1$ when T is ergodic.*

Let $W = X \setminus \bigsqcup_{j=0}^{\infty} \widetilde{A}_j$. This is the set of points in A^c that *never* enter A:

$$W = \{x \in A^c : T^n(x) \notin A \text{ for all } n \geq 1\} = \{x \in X : T^n(x) \notin A \text{ for all } n \geq 0\}.$$

Note that $T^{-1}(W) \subset W$: if $T(x) \in W$ then $T^n(T(x)) \notin A$ for all $n \geq 0$, and $T(x) \notin A$ (since $T(x) \in W \subset A^c$), so $T^n(x) \notin A$ for all $n \geq 1$. If also $x \notin A$, then $x \in W$. If $x \in A$, then $x \notin W$. So $T^{-1}(W) \subset W \cup A$.

Actually, more carefully: $W = \bigcap_{n=0}^{\infty} T^{-n}(A^c)$. Therefore $T^{-1}(W) = \bigcap_{n=0}^{\infty} T^{-(n+1)}(A^c) = \bigcap_{n=1}^{\infty} T^{-n}(A^c) \supset W$ (since $W = A^c \cap \bigcap_{n=1}^{\infty} T^{-n}(A^c) \subset \bigcap_{n=1}^{\infty} T^{-n}(A^c) = T^{-1}(W)$).

So $W \subset T^{-1}(W)$. By measure-preservation, $\mu(T^{-1}(W)) = \mu(W)$, and since $W \subset T^{-1}(W)$, we get $\mu(T^{-1}(W) \setminus W) = 0$.

Now $T^{-1}(W) \setminus W \subset A$ (the only points in $T^{-1}(W)$ that could fail to be in W are those in A, since $T^{-1}(W) \subset \bigcap_{n=1}^{\infty} T^{-n}(A^c)$ and $W = A^c \cap \bigcap_{n=1}^{\infty} T^{-n}(A^c)$, so $T^{-1}(W) \setminus W \subset A$).

Consider the T-invariant set $\widetilde{W} = \bigcap_{n=0}^{\infty} T^{-n}(A^c)$. This equals W as defined above. We have $T^{-1}(\widetilde{W}) = \bigcap_{n=1}^{\infty} T^{-n}(A^c) \supset \widetilde{W}$, but $T^{-1}(\widetilde{W})$ is not generally equal to $\widetilde{W}$.

Let us instead consider $\widehat{W} = \bigcap_{n \in \mathbb{Z}_{\geq 0}} T^{-n}(A^c)$, which is the same as W. This set satisfies $T^{-1}(W) \supset W$. Now define $W^* = \bigcup_{n=0}^{\infty} T^{-n}(W)$. Then $T^{-1}(W^*) = \bigcup_{n=1}^{\infty} T^{-n}(W) \subset W^*$, and $W^* \subset T^{-1}(W^*)$ (taking $n = 0$), so $T^{-1}(W^*) = W^*$: the set W^* is strictly T-invariant.

Since $W \subset A^c$ and $T^{-n}(W) \subset T^{-n}(A^c)$, the set W^* is disjoint from A: if $x \in W^* \cap A$, then $x \in T^{-n}(W)$ for some n, so $T^n(x) \in W \subset A^c$. But $x \in A$, so $T^n(x) \notin A$ for all $n \geq 0$ would be needed for $T^n(x) \in W$, which contradicts $x \in A$ when $n = 0$. Wait, $T^n(x) \in W$ means $T^m(T^n(x)) \notin A$ for all $m \geq 0$, which is compatible with $x \in A$ (as long as $n \geq 1$). So the issue is that W^* could contain points of A.

Let us reconsider. The set W^* is T-invariant and $W^* \cap A = \emptyset$ would imply $\mu(W^*) = 0$ by ergodicity (since A has positive measure). But we need to verify $W^* \cap A = \emptyset$.

If $x \in A \cap W^*$, then $T^n(x) \in W$ for some $n \geq 0$. If $n = 0$, then $x \in W \subset A^c$, contradicting $x \in A$. If $n \geq 1$, then $T^n(x) \in W$, meaning $T^m(T^n(x)) \notin A$ for all $m \geq 0$, i.e., $T^k(x) \notin A$ for all $k \geq n$. But $x \in A$, so the orbit visits A at time 0 and never again after time $n - 1$. Now, is $x \in A$ with only finitely many visits to A possible under an ergodic measure? Actually, by Poincare recurrence, μ-almost every $x \in A$ returns to A infinitely often, so $\mu(A \cap W^*) = 0$. Since $T^{-1}(W^*) = W^*$ and T is ergodic, $\mu(W^*) = 0$ or $\mu(W^*) = 1$. But $\mu(A \cap W^*) = 0$ and $\mu(A) > 0$, so $\mu(W^*) \neq 1$, hence $\mu(W^*) = 0$. In particular, $\mu(W) = 0$, proving Claim 2.

Claim 3: $^*\mu(\widetilde{A}_j) = \mu(B_j)$ for all $j \geq 0$, where $B_j = \{x \in A : \tau_A(x) > j\}$.*

We prove this via the relation $T^{-1}(\widetilde{A}_j) = \widetilde{A}_{j+1} \sqcup A_{j+1}$ for $j \geq 0$, where $A_{j+1} = \{x \in A : \tau_A(x) = j + 1\}$.

Indeed, $x \in T^{-1}(\widetilde{A}_j)$ means $T(x) \in \widetilde{A}_j$. If $j = 0$: $T(x) \in A$. Then either $x \in A$ (so $\tau_A(x) = 1$ if $T(x) \in A$, meaning $x \in A_1$) or $x \notin A$ (so $x \in \widetilde{A}_1$). So $T^{-1}(\widetilde{A}_0) = T^{-1}(A) = \widetilde{A}_1 \sqcup A_1$ (where $A_1 = \{x \in A : T(x) \in A\}$).

For general $j \geq 1$: $T(x) \in \widetilde{A}_j$ means $T(x) \notin A$ and $T^{j+1}(x) \in A$ and $T^k(x) \notin A$ for $1 \leq k \leq j$. If $x \in A$, then $\tau_A(x) = j + 1$, so $x \in A_{j+1}$. If $x \notin A$, then $x \notin A$, $T(x) \notin A, \ldots, T^j(x) \notin A, T^{j+1}(x) \in A$, so $x \in \widetilde{A}_{j+1}$.

Hence $T^{-1}(\widetilde{A}_j) = \widetilde{A}_{j+1} \sqcup A_{j+1}$, and by measure-preservation:

$$\mu(\widetilde{A}_j) = \mu(T^{-1}(\widetilde{A}_j)) = \mu(\widetilde{A}_{j+1}) + \mu(A_{j+1}). \tag{$*$}$$

Also, $B_j = \bigsqcup_{n=j+1}^{\infty} A_n$ (points in A with return time $> j$), so $B_j = B_{j+1} \sqcup A_{j+1}$, giving

$$\mu(B_j) = \mu(B_{j+1}) + \mu(A_{j+1}). \tag{$**$}$$

From $(*)$ and $(**)$, $\mu(\widetilde{A}_j) - \mu(B_j) = \mu(\widetilde{A}_{j+1}) - \mu(B_{j+1})$ for all $j \geq 0$. Since $\widetilde{A}_0 = A = B_0$, we get $\mu(\widetilde{A}_j) = \mu(B_j)$ for all $j \geq 0$ by induction.

Conclusion. Using Claims 2 and 3:

$$\int_A \tau_A \, d\mu = \sum_{j=0}^{\infty} \mu(B_j) = \sum_{j=0}^{\infty} \mu(\widetilde{A}_j) = \mu\left(\bigsqcup_{j=0}^{\infty} \widetilde{A}_j\right) = \mu(X) = 1.$$

Dividing by $\mu(A)$, the conditional expected return time is $1/\mu(A)$. □

□

□

Remark 8.13. Kac's lemma has a beautiful intuitive content: the smaller the target set A, the longer it takes to return. A set occupying a fraction p of the space requires, on average, $1/p$ iterates to revisit. This is consistent with a "random" exploration of the space, but the result holds for deterministic, measure-preserving maps — provided they are ergodic.

Remark 8.14. The ergodicity hypothesis is essential. Without it, the result fails: consider $X = [0, 1]$, $T = \mathrm{id}$, $A = [0, 1/2]$. Then $\tau_A(x) = 1$ for all $x \in A$ (since $T(x) = x \in A$), and $\int_A \tau_A \, d\mu = \mu(A) = 1/2 \neq 1$. The problem is that T is not ergodic: the set A is invariant with $0 < \mu(A) < 1$. Actually wait — $\tau_A(x) = 1$ for $x \in A$, so $\int_A \tau_A \, d\mu = \int_A 1 \, d\mu = 1/2$. And indeed $1/\mu(A) = 2 \neq 1/\mu(A) \cdot \mu(A)$... Let me reconsider. For the identity, $\tau_A(x) = 1$ since $T^1(x) = x \in A$, so $\int_A \tau_A \, d\mu = 1 \cdot \mu(A) = 1/2$. Kac says this should be 1 if T is ergodic; but the identity is not ergodic, and indeed $1/2 \neq 1$.

Remark 8.15. In the non-ergodic case, one still has the weaker statement $\int_A \tau_A \, d\mu \leq 1$ (with equality holding if and only if $\bigsqcup_{j=0}^{\infty} \widetilde{A}_j$ has full measure, which is guaranteed by ergodicity but may fail otherwise).

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8.6 Summary

This chapter introduced the central objects of ergodic theory:

1. **Measure-preserving transformations**: maps that leave a measure invariant, formulated via preimages ($\mu(T^{-1}(A)) = \mu(A)$) or equivalently via integrals ($\int f \circ T = \int f$).
2. **Examples**: We verified measure-preservation for circle rotations (rigid translations), the doubling map (an expanding map with two branches), the baker's map (piecewise affine with unit Jacobian), Arnold's cat map (a hyperbolic toral automorphism with det = 1), Bernoulli shifts (shifts on product spaces), and Hamiltonian flows (via Liouville's theorem and div = 0).
3. **Invariant measures**: A given map typically has many invariant measures. The Krylov-Bogolyubov theorem guarantees existence for continuous maps on compact spaces, using

weak-* compactness of the space of probability measures and Cesaro averaging along orbits.

4. **Physical measures**: Not all invariant measures are physically relevant. SRB measures govern the time averages observed for typical initial conditions and connect to the echo state property in reservoir computing.
5. **Recurrence**: The Poincare recurrence theorem (almost every point returns) and Kac's lemma (the expected return time to a set A in an ergodic system is $1/\mu(A)$).

In Chapter 9, we will develop the concept of ergodicity in depth and prove the Birkhoff ergodic theorem, which provides the bridge between time averages and space averages.

Exercises

Exercise 8.1. tripling map** $T(x) = 3x \pmod 1$. (a) Show that T preserves Lebesgue measure. (b) Find all fixed points of T. For each fixed point x_0, verify that δ_{x_0} is a T-invariant measure. (c) Show that T is measurably conjugate to the one-sided Bernoulli shift $B(1/3, 1/3, 1/3)$.

Exercise 8.2. Let $T_\alpha : \mathbb{T} \to \mathbb{T}$ be the rotation $T_\alpha(x) = x + \alpha \pmod 1$. (a) Show that when $\alpha = p/q$ with $\gcd(p, q) = 1$, the measure $\mu = \frac{1}{q}\sum_{j=0}^{q-1} \delta_{j/q}$ is T_α-invariant. (b) Find all T_α-invariant measures when $\alpha = 1/2$. (*Hint:* every convex combination of the measure from (a) and Lebesgue measure works, but are there others?) (c) Show that when α is irrational, T_α is **uniquely ergodic**: the only T_α-invariant Borel probability measure is Lebesgue measure. (*Hint:* use the Fourier characterization. If μ is T_α-invariant, then $\hat{\mu}(n) = e^{2\pi i n\alpha}\hat{\mu}(n)$ for all $n \in \mathbb{Z}$.)

Exercise 8.3. tent map**:

$$T(x) = \begin{cases} 2x & \text{if } 0 \le x < 1/2, \\ 2 - 2x & \text{if } 1/2 \le x < 1. \end{cases}$$

(a) Show that T preserves Lebesgue measure on $[0, 1)$. (b) Show that T^2 (the second iterate) is conjugate to the doubling map via a suitable homeomorphism. (*Hint:* consider the map $h(x) = \frac{2}{\pi}\arcsin(\sqrt{x})$, or work out T^2 directly.)

Exercise 8.4. The **Gauss map** $G : (0, 1] \to [0, 1)$ is defined by $G(x) = \{1/x\}$ (the fractional part of $1/x$). (a) Show that the **Gauss measure** $d\mu(x) = \frac{1}{\log 2} \cdot \frac{1}{1+x}\, dx$ is G-invariant. (*Hint:* check the invariance on intervals $[a, b] \subset (0, 1]$ by computing $G^{-1}([a, b])$ and summing the resulting series.) (b) Is Lebesgue measure G-invariant? Justify your answer.

Exercise 8.5. Let $T_1 : (X_1, \mu_1) \to (X_1, \mu_1)$ be measure-preserving, and let $\phi : X_1 \to \mathbb{R}/\mathbb{Z}$ be measurable. Define $T : X_1 \times \mathbb{R}/\mathbb{Z} \to X_1 \times \mathbb{R}/\mathbb{Z}$ by

$$T(x, y) = (T_1(x),\, y + \phi(x) \pmod 1).$$

Show that T preserves the product measure $\mu_1 \times \lambda$, where λ is Lebesgue measure on $\mathbb{R}/\mathbb{Z}$.

Exercise 8.6. Consider the doubling map on $[0, 1)$ with Lebesgue measure (which is ergodic; you may assume this). (a) Let $A = [0, 1/4)$. Compute the expected return time $\mathbb{E}[\tau_A]$ using Kac's lemma. (b) Let $A = [0, 1/4)$. Compute $\mathbb{E}[\tau_A]$ directly by finding the return time function $\tau_A(x)$ for each $x \in A$ and integrating, as follows. Show that for $x \in [0, 1/8)$, $\tau_A(x) = 1$; for $x \in [1/8, 1/4)$, determine $\tau_A(x)$ by tracing the orbit and verify that the integral gives $1/\mu(A) = 4$ for the conditional expectation. (c) Let $A = [1/3, 2/3)$. What does Kac's lemma predict for $\mathbb{E}[\tau_A \mid x \in A]$? (*Note:* the detailed verification is more involved here.)

Exercise 8.7. Consider the doubling map $T(x) = 2x \pmod 1$. (a) Show that $x = 0$ and $x = 1/3$ are fixed points of T and T^2 respectively, and that $x = 1/7, 2/7, 4/7$ is a periodic orbit of period 3. Construct explicit invariant probability measures supported on each of these orbits. (b) Show that the set $\mathcal{M}_T$ of all T-invariant Borel probability measures on $[0, 1)$ is convex: if $\mu, \nu \in \mathcal{M}_T$ and $0 \leq \alpha \leq 1$, then $\alpha\mu + (1-\alpha)\nu \in \mathcal{M}_T$. (c) Show that $\mathcal{M}_T$ is closed in the weak-* topology.

Exercise 8.8. State and prove the analogue of the Krylov-Bogolyubov theorem for continuous flows $\varphi_t : X \to X$ on a compact metrizable space X. (*Hint:* replace $\frac{1}{n}\sum_{j=0}^{n-1} \delta_{T^j(x_0)}$ with $\frac{1}{T}\int_0^T \delta_{\varphi_t(x_0)}\, dt$ and adapt the proof.)

—

Recommended Reading

- **P. Walters**, *An Introduction to Ergodic Theory*, Springer, 1982. Chapter 1 covers measure-preserving transformations and their basic properties. This is one of the most accessible introductions to the subject.
- **K. Petersen**, *Ergodic Theory*, Cambridge University Press, 1983. Chapters 2-3 provide a careful treatment of measure-preserving systems, the Poincare recurrence theorem, and the Krylov-Bogolyubov theorem, with many examples.
- **R. Mane**, *Ergodic Theory and Differentiable Dynamics*, Springer, 1987. A more advanced treatment with emphasis on smooth dynamics. The discussion of SRB measures and physical measures in Chapter III is particularly relevant to Section 8.4.
- **A. Katok and B. Hasselblatt**, *Introduction to the Modern Theory of Dynamical Systems*, Cambridge University Press, 1995. The definitive reference for the interaction between dynamics and ergodic theory. Part II develops the measure-theoretic foundations in great depth.
- **M. Viana and K. Oliveira**, *Foundations of Ergodic Theory*, Cambridge University Press, 2016. A modern textbook with an excellent treatment of invariant measures, SRB measures, and their role in dynamical systems.

For the original work on SRB measures, the foundational papers are:

- **Ya. G. Sinai**, *Gibbs measures in ergodic theory*, Russian Mathematical Surveys 27 (1972), 21–69.
- **D. Ruelle**, *A measure associated with Axiom A attractors*, American Journal of Mathematics 98 (1976), 619–654.

- **R. Bowen and D. Ruelle**, *The ergodic theory of Axiom A flows*, Inventiones Mathematicae 29 (1975), 181–202.

Chapter 9

Ergodicity and the Ergodic Theorems

9.1 The Fundamental Question

We arrive at the central question of ergodic theory — arguably one of the most important questions in all of mathematical physics.

Suppose $(X, \mathcal{B}, \mu)$ is a probability space and $T : X \to X$ is a measure-preserving transformation. Given a measurable function $f : X \to \mathbb{R}$ (an "observable"), we can form its **time average** along the orbit of a point x:

$$\bar{f}_n(x) = \frac{1}{n} \sum_{k=0}^{n-1} f(T^k x).$$

We can also form the **space average** (or **ensemble average**) of f:

$$\langle f \rangle = \int_X f \, d\mu.$$

The fundamental question is:

> **When does the time average converge to the space average?** That is, when does
>
> $$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f(T^k x) = \int_X f \, d\mu$$
>
> hold for μ-almost every $x \in X$?

This question has its origins in statistical mechanics. Ludwig Boltzmann, in the late 19th century, proposed what became known as the **ergodic hypothesis**: that a physical system, over sufficient time, visits all accessible states compatible with its energy. If this were true, one could replace the practically impossible computation of a time average (requiring observation of a system for infinite time) with the more tractable computation of a space average (an integral over phase space). We will return to the physical interpretation in Section 9.8.

The mathematical resolution of this question came in the early 1930s, in two landmark theorems:

- **John von Neumann's Mean Ergodic Theorem** (1932): the time averages converge in the L^2 norm.
- **George David Birkhoff's Pointwise Ergodic Theorem** (1931): the time averages converge pointwise almost everywhere.

The answer to *when* the limit equals $\int f\,d\mu$ (rather than some other invariant function) turns out to be precisely when the system is **ergodic** — a property we will define and study in detail.

Throughout this chapter, $(X, \mathcal{B}, \mu)$ denotes a probability space (so $\mu(X) = 1$), and $T : X \to X$ is a measure-preserving transformation (so $\mu(T^{-1}A) = \mu(A)$ for all $A \in \mathcal{B}$), unless stated otherwise.

9.2 The Koopman Operator

Before stating the ergodic theorems, we introduce an operator that translates the dynamics of T on the space X into linear algebra on function spaces. This perspective, due to B.O. Koopman (1931), is fundamental.

Definition 9.1 (Koopman operator)**.** Let $T : X \to X$ be a measurable transformation. The **Koopman operator** $U_T : L^2(X, \mu) \to L^2(X, \mu)$ is defined by

$$U_T f = f \circ T.$$

That is, $(U_T f)(x) = f(T(x))$ for all $x \in X$.

When T is measure-preserving, the Koopman operator has several important properties.

Proposition 9.1. *If T is measure-preserving, then U_T is a linear isometry on $L^2(X, \mu)$. That is, $\|U_T f\|_2 = \|f\|_2$ for all $f \in L^2$.*

Proof. * We compute directly:

$$\|U_T f\|_2^2 = \int_X |f(Tx)|^2\,d\mu(x) = \int_X |f|^2\,d(T_*\mu) = \int_X |f|^2\,d\mu = \|f\|_2^2,$$

where the second equality uses the change-of-variables formula and the third uses the fact that $T_*\mu = \mu$ (i.e., T preserves μ). □

In particular, U_T is a bounded operator with $\|U_T\| = 1$. Note that U_T is an isometry but not necessarily unitary — it is unitary if and only if T is invertible (mod μ). Observe also that

$$U_T^n f = f \circ T^n,$$

so the time average can be written as

$$\bar{f}_n = \frac{1}{n} \sum_{k=0}^{n-1} U_T^k f.$$

This reformulation is the key to von Neumann's approach: the question of convergence of time averages becomes a question about the behavior of the **Cesàro averages** of iterates of a bounded linear operator on a Hilbert space.

9.3 Von Neumann's Mean Ergodic Theorem

We now state and prove von Neumann's theorem. The proof is elegant and relies only on the Hilbert space structure of L^2.

Notation. For a measure-preserving transformation T, define the subspace of **T-invariant functions**:

$$\mathcal{I} = \{f \in L^2(X, \mu) : U_T f = f\} = \{f \in L^2 : f \circ T = f \text{ a.e.}\}.$$

This is a closed subspace of L^2 (as the kernel of the bounded operator $U_T - I$). Let $P : L^2 \to \mathcal{I}$ denote the orthogonal projection onto $\mathcal{I}$.

Theorem 9.2 (Von Neumann's Mean Ergodic Theorem, 1932)**.** *Let $(X, \mathcal{B}, \mu)$ be a probability space, $T : X \to X$ measure-preserving, and $f \in L^2(X, \mu)$. Then*

$$\frac{1}{n} \sum_{k=0}^{n-1} U_T^k f \xrightarrow{L^2} Pf \quad \text{as } n \to \infty,$$

where P is the orthogonal projection onto the subspace $\mathcal{I}$ of T-invariant functions.

Proof. * The idea is to decompose L^2 into two subspaces on which the convergence is easy to check separately.

Step 1: Orthogonal decomposition. We claim that

$$L^2(X, \mu) = \mathcal{I} \oplus \overline{\mathcal{C}},$$

where

$$\mathcal{C} = \{g - U_T g : g \in L^2(X, \mu)\}$$

is the space of **coboundaries**, and $\overline{\mathcal{C}}$ is its closure in L^2.

To establish this, it suffices to show that $\mathcal{I} = \mathcal{C}^\perp$ (since $L^2 = \mathcal{C}^\perp \oplus \overline{\mathcal{C}}$ for any subspace $\mathcal{C}$).

First, suppose $f \in \mathcal{I}$, so $U_T f = f$. For any $g \in L^2$,

$$\langle f, g - U_T g \rangle = \langle f, g \rangle - \langle f, U_T g \rangle.$$

Now we use the fact that U_T is an isometry with adjoint U_T^* satisfying $\langle U_T^* f, g \rangle = \langle f, U_T g \rangle$. Since T is measure-preserving, $U_T^* U_T = I$ (on L^2), and one can check that $U_T^* f = f$ whenever $U_T f = f$. (Indeed, $\|U_T f - f\| = 0$ implies $\langle U_T f, f \rangle = \|f\|^2$, and then $\|f - U_T^* f\|^2 = \|f\|^2 - 2\mathrm{Re}\langle f, U_T^* f \rangle + \|U_T^* f\|^2 = \|f\|^2 - 2\mathrm{Re}\langle U_T f, f \rangle + \|U_T^* f\|^2$. Since $U_T^* U_T = I$, we have $\|U_T^* f\|^2 \leq \|f\|^2$ (as U_T^* is a contraction), and $\mathrm{Re}\langle U_T f, f \rangle = \|f\|^2$, giving $\|f - U_T^* f\|^2 \leq 0$.)

So $U_T^* f = f$, and therefore $\langle f, U_T g \rangle = \langle U_T^* f, g \rangle = \langle f, g \rangle$. Thus $\langle f, g - U_T g \rangle = 0$. Since g was arbitrary, $f \perp \mathcal{C}$.

Conversely, suppose $f \perp \mathcal{C}$. Then for every $g \in L^2$, $\langle f, g - U_T g \rangle = 0$, i.e., $\langle f, g \rangle = \langle f, U_T g \rangle = \langle U_T^* f, g \rangle$. Since this holds for all g, we get $U_T^* f = f$. Then $\|U_T f - f\|^2 = \|U_T f\|^2 - 2\mathrm{Re}\langle U_T f, f \rangle + \|f\|^2 = \|f\|^2 - 2\mathrm{Re}\langle f, U_T^* f \rangle + \|f\|^2 = 2\|f\|^2 - 2\|f\|^2 = 0$ (using $\|U_T f\| = \|f\|$ and $U_T^* f = f$). So $U_T f = f$, i.e., $f \in \mathcal{I}$.

This establishes $\mathcal{I} = \mathcal{C}^\perp$, and therefore $L^2 = \mathcal{I} \oplus \overline{\mathcal{C}}$.

Step 2: Convergence on each subspace. Write $f = f_{\text{inv}} + h$, where $f_{\text{inv}} = Pf \in \mathcal{I}$ and $h \in \overline{\mathcal{C}}$. Then

$$\frac{1}{n}\sum_{k=0}^{n-1} U_T^k f = \frac{1}{n}\sum_{k=0}^{n-1} U_T^k f_{\text{inv}} + \frac{1}{n}\sum_{k=0}^{n-1} U_T^k h = f_{\text{inv}} + \frac{1}{n}\sum_{k=0}^{n-1} U_T^k h,$$

since $U_T^k f_{\text{inv}} = f_{\text{inv}}$ for all k (because f_{inv} is T-invariant). So it suffices to show that $\frac{1}{n}\sum_{k=0}^{n-1} U_T^k h \to 0$ in L^2 for all $h \in \overline{\mathcal{C}}$.

Case (a): $h = g - U_T g$ is an exact coboundary. Then

$$\sum_{k=0}^{n-1} U_T^k h = \sum_{k=0}^{n-1} (U_T^k g - U_T^{k+1} g) = g - U_T^n g,$$

a telescoping sum. Therefore

$$\left\| \frac{1}{n}\sum_{k=0}^{n-1} U_T^k h \right\|_2 = \frac{1}{n}\|g - U_T^n g\|_2 \le \frac{2\|g\|_2}{n} \to 0.$$

Case (b): $h \in \overline{\mathcal{C}}$ is a general element. Given $\varepsilon > 0$, choose a coboundary $h_0 = g_0 - U_T g_0$ with $\|h - h_0\|_2 < \varepsilon$. Since U_T is an isometry, $\|\frac{1}{n}\sum_{k=0}^{n-1} U_T^k\| \le 1$ (as an operator on L^2). Therefore

$$\left\| \frac{1}{n}\sum_{k=0}^{n-1} U_T^k h \right\|_2 \le \left\| \frac{1}{n}\sum_{k=0}^{n-1} U_T^k h_0 \right\|_2 + \left\| \frac{1}{n}\sum_{k=0}^{n-1} U_T^k (h - h_0) \right\|_2 \le \frac{2\|g_0\|_2}{n} + \varepsilon.$$

For n large enough, the first term is less than ε, so the whole expression is less than 2ε. Since ε was arbitrary, $\frac{1}{n}\sum_{k=0}^{n-1} U_T^k h \to 0$ in L^2. □

Remark 9.1. Von Neumann's theorem is really a theorem in functional analysis: it holds for any isometry on a Hilbert space. The dynamics enter only through the identification of $\mathcal{I}$ with the invariant functions. This level of generality is both a strength and a weakness — the theorem tells us that time averages converge *in norm*, but it says nothing about convergence at individual points. For that, we need Birkhoff's deeper result.

9.4 The Maximal Ergodic Theorem

Before proving Birkhoff's theorem, we need a crucial technical tool. The maximal ergodic theorem controls the size of the set where partial time averages are large. The proof we give is due to A. Garsia (1965) and is notable for its brevity and elegance.

Notation. For $f \in L^1(X, \mu)$ and $n \ge 1$, define the partial sums

$$S_n f(x) = \sum_{k=0}^{n-1} f(T^k x), \qquad S_0 f(x) = 0,$$

and the **maximal function**

$$M_N f(x) = \max_{1 \le n \le N} S_n f(x).$$

Theorem 9.3 (Maximal Ergodic Theorem)**.** *Let T be a measure-preserving transformation on $(X, \mathcal{B}, \mu)$ and let $f \in L^1(X, \mu)$. For each $N \geq 1$, define*

$$E_N = \{x \in X : M_N f(x) > 0\} = \left\{x \in X : \max_{1 \leq n \leq N} S_n f(x) > 0\right\}.$$

Then

$$\int_{E_N} f \, d\mu \geq 0.$$

Proof (Garsia). (Garsia).* Fix $N \geq 1$. Define $f^* = M_N f = \max_{1 \leq n \leq N} S_n f$. Note that $E_N = \{f^* > 0\}$ and $f^* \geq S_n f$ for $1 \leq n \leq N$, with $f^* \geq S_1 f = f$.

We claim:

$$f \geq f^* - f^* \circ T. \tag{$*$}$$

To prove $(*)$, note that for each n with $1 \leq n \leq N$,

$$S_n f(x) = f(x) + S_{n-1} f(Tx).$$

For $n \geq 2$, $S_{n-1} f(Tx) \leq M_N f(Tx) = f^*(Tx)$ (since $n - 1 \leq N - 1 < N$ and $M_N f$ takes the maximum over indices up to N; more precisely, $S_{n-1} f(Tx) = \sum_{k=0}^{n-2} f(T^{k+1}x) \leq \max_{1 \leq j \leq N} S_j f(Tx) = f^*(Tx)$). Also $S_1 f(x) = f(x) = f(x) + 0 \leq f(x) + f^*(Tx)$ (since $f^* \geq 0$ on E_N and $f^* \geq S_1 f = f$ everywhere; actually we need to be slightly more careful).

Let us redo this cleanly. For $1 \leq n \leq N$:

- If $n = 1$: $S_1 f(x) = f(x) \leq f(x) + f^*(Tx)$ since $f^*(Tx) \geq 0$ when $Tx \in E_N$, and $f^*(Tx) \geq S_0 f(Tx) = 0$... but f^* might be negative.

Let us use a cleaner version of Garsia's argument. Define

$$f^*(x) = \max(0, S_1 f(x), S_2 f(x), \ldots, S_N f(x)) = \max_{0 \leq n \leq N} S_n f(x) \geq 0.$$

Now $f^* \geq 0$ everywhere, and $E_N = \{f^* > 0\}$. For $1 \leq n \leq N$:

$$S_n f(x) = f(x) + S_{n-1} f(Tx) \leq f(x) + f^*(Tx),$$

since $S_{n-1} f(Tx) \leq \max_{0 \leq j \leq N} S_j f(Tx) = f^*(Tx)$ (note $n-1$ ranges from 0 to $N-1$, and f^* takes the max over 0 to N, so indeed $S_{n-1} f(Tx) \leq f^*(Tx)$). Also $S_0 f(x) = 0 \leq f(x) + f^*(Tx)$ is not necessarily true, but we do not need it: we only need

$$f^*(x) = \max_{0 \leq n \leq N} S_n f(x) \leq \max\big(0,\ f(x) + f^*(Tx)\big) \leq f(x) + f^*(Tx),$$

where the last inequality uses the fact that $f^*(Tx) \geq 0$. (The first inequality holds because each $S_n f(x) \leq f(x) + f^*(Tx)$ for $n \geq 1$, and $S_0 f(x) = 0$.)

Wait — to be fully precise: if $f^*(x) = 0$, the inequality $f^*(x) \leq f(x) + f^*(Tx)$ might fail if $f(x) + f^*(Tx) < 0$. But $f^*(Tx) \geq 0$, so $f(x) + f^*(Tx) \geq f(x)$, and we would need $f(x) \geq 0$, which need not hold. However, if $f^*(x) = 0$, then $x \notin E_N$, and we only integrate over E_N anyway.

So: on E_N, we have $f^*(x) > 0$, which means $f^*(x) = \max_{1\le n\le N} S_n f(x)$, and for each such n, $S_n f(x) = f(x) + S_{n-1}f(Tx) \le f(x) + f^*(Tx)$. Therefore

$$f^*(x) \le f(x) + f^*(Tx) \qquad \text{for all } x \in E_N.$$

Rearranging: $f(x) \ge f^*(x) - f^*(Tx)$ for $x \in E_N$. Since $f^* \ge 0$ everywhere and $f^*(x) > 0$ on E_N:

$$\int_{E_N} f\,d\mu \ge \int_{E_N} (f^* - f^* \circ T)\,d\mu = \int_{E_N} f^*\,d\mu - \int_{E_N} f^*(Tx)\,d\mu.$$

Now, $\int_{E_N} f^*\,d\mu = \int_X f^*\,d\mu$ since $f^* = 0$ on $X \setminus E_N$ (by definition, $f^* \ge 0$ and $f^* = 0$ outside E_N). Also, $\int_{E_N} f^*(Tx)\,d\mu \le \int_X f^*(Tx)\,d\mu = \int_X f^*\,d\mu$, where the equality uses the fact that T is measure-preserving. Therefore

$$\int_{E_N} f\,d\mu \ge \int_X f^*\,d\mu - \int_X f^*\,d\mu = 0. \quad \square$$

$\square$

Remark 9.2. The maximal ergodic theorem is sometimes stated in the following equivalent form. Define $E^* = \{x : \sup_{n\ge1} S_n f(x) > 0\}$. Since $E^* = \bigcup_{N=1}^{\infty} E_N$ and $E_1 \subseteq E_2 \subseteq \cdots$, continuity of the measure gives $\int_{E^*} f\,d\mu = \lim_{N\to\infty} \int_{E_N} f\,d\mu \ge 0$.

More precisely, for any $\alpha \in \mathbb{R}$, applying the theorem to $f - \alpha$ (which is still in L^1), we obtain the following useful corollary.

Corollary 9.4 (Maximal Ergodic Inequality)**.** *Let $f \in L^1(X,\mu)$ and $\alpha \in \mathbb{R}$. Define*

$$E_\alpha = \left\{ x \in X : \sup_{n\ge1} \frac{1}{n} S_n f(x) > \alpha \right\}.$$

Then $\int_{E_\alpha} f\,d\mu \ge \alpha \cdot \mu(E_\alpha)$.

Proof. * Apply Remark 9.6 to the function $g = f - \alpha$. Note that $S_n g = S_n f - n\alpha$, so $\sup_n \frac{1}{n} S_n g(x) > 0$ if and only if $\sup_n \frac{1}{n} S_n f(x) > \alpha$. Thus $\{x : \sup_n S_n g(x) > 0\} = E_\alpha$, and $\int_{E_\alpha} g\,d\mu \ge 0$ gives $\int_{E_\alpha} f\,d\mu \ge \alpha \cdot \mu(E_\alpha)$. $\square$

$\square$

9.5 Birkhoff's Pointwise Ergodic Theorem

We now prove the most important theorem of ergodic theory.

Theorem 9.5 (Birkhoff's Pointwise Ergodic Theorem, 1931)**.** *Let $(X, \mathcal{B}, \mu)$ be a probability space, $T : X \to X$ a measure-preserving transformation, and $f \in L^1(X,\mu)$. Then:*

(i) *The limit*

$$\bar{f}(x) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f(T^k x)$$

exists for μ-almost every $x \in X$.

(ii) *The limit function $\bar{f}$ is T-invariant: $\bar{f}(Tx) = \bar{f}(x)$ a.e.*
(iii) *$\bar{f} \in L^1(X, \mu)$ and $\int_X \bar{f}\, d\mu = \int_X f\, d\mu$.*

Proof. *

Part (ii) is immediate once the limit exists: if $\bar{f}(x) = \lim_n \frac{1}{n}\sum_{k=0}^{n-1} f(T^k x)$, then

$$\bar{f}(Tx) = \lim_n \frac{1}{n}\sum_{k=0}^{n-1} f(T^{k+1}x) = \lim_n \frac{1}{n}\sum_{k=1}^{n} f(T^k x) = \lim_n \frac{1}{n}\sum_{k=0}^{n-1} f(T^k x) = \bar{f}(x),$$

where the third equality uses the fact that $\frac{1}{n}(S_n f(x) - f(x) + f(T^n x)) \to \bar{f}(x)$ since $\frac{f(x)}{n} \to 0$ and $\frac{f(T^n x)}{n} \to 0$ (the latter requires a small argument, but follows from the fact that $f \in L^1$ implies $\frac{f(T^n x)}{n} \to 0$ a.e. — this is a consequence of the almost-everywhere convergence itself, or can be seen from the Borel-Cantelli lemma; for now we note that the shift of indices does not affect the Cesàro limit when it exists).

Part (i): The limit exists a.e. This is the hard part. We use the maximal ergodic theorem.

Define the upper and lower time averages:

$$\overline{f}(x) = \limsup_{n\to\infty} \frac{1}{n} S_n f(x), \qquad \underline{f}(x) = \liminf_{n\to\infty} \frac{1}{n} S_n f(x).$$

Clearly $\underline{f}(x) \le \overline{f}(x)$ for all x. The limit $\bar{f}(x)$ exists if and only if $\underline{f}(x) = \overline{f}(x)$. We will show that $\underline{f}(x) = \overline{f}(x)$ for μ-a.e. x.

For rational numbers $\alpha < \beta$, define

$$A_{\alpha,\beta} = \{x \in X : \underline{f}(x) < \alpha < \beta < \overline{f}(x)\}.$$

Then

$$\{x : \underline{f}(x) < \overline{f}(x)\} = \bigcup_{\alpha<\beta,\ \alpha,\beta\in\mathbb{Q}} A_{\alpha,\beta}.$$

It suffices to show that $\mu(A_{\alpha,\beta}) = 0$ for every pair of rationals $\alpha < \beta$.

The sets $A_{\alpha,\beta}$ are T-invariant: if $x \in A_{\alpha,\beta}$, then $\overline{f}(Tx) = \overline{f}(x)$ and $\underline{f}(Tx) = \underline{f}(x)$ (by the same index-shifting argument as in Part (ii)), so $Tx \in A_{\alpha,\beta}$.

Now fix $\alpha < \beta$ in $\mathbb{Q}$. We apply Corollary 9.7 twice.

Applying the maximal inequality to get an upper bound. On the set $A_{\alpha,\beta}$, we have $\overline{f}(x) > \beta$, so $A_{\alpha,\beta} \subseteq E_\beta = \{x : \sup_n \frac{1}{n} S_n f(x) > \beta\}$. But we want to work on $A_{\alpha,\beta}$ itself. Since $A_{\alpha,\beta}$ is T-invariant, we can restrict our attention to it.

Let $g = f - \beta$. On $A_{\alpha,\beta}$, $\overline{f}(x) > \beta$ means $\limsup_n \frac{1}{n} S_n g(x) > 0$, so $\sup_n \frac{1}{n} S_n g(x) > 0$, so $A_{\alpha,\beta} \subseteq \{x : \sup_n \frac{1}{n} S_n g(x) > 0\} = E_0(g)$.

By the maximal ergodic inequality (Corollary 9.7 with $\alpha = 0$ applied to g):

$$\int_{E_0(g)} g\, d\mu \ge 0,$$

and since $A_{\alpha,\beta} \subseteq E_0(g)$... but this is not quite enough, because $E_0(g)$ might be larger than $A_{\alpha,\beta}$.

Let us use a cleaner approach. We work directly with Corollary 9.7 applied to the restriction of the system to the invariant set $A_{\alpha,\beta}$.

Since $A_{\alpha,\beta}$ is T-invariant and measurable, T restricts to a measure-preserving transformation on $A_{\alpha,\beta}$ (with the restricted measure — or more precisely, $T^{-1}(A_{\alpha,\beta}) = A_{\alpha,\beta}$ since the set is T-invariant).

On $A_{\alpha,\beta}$, we have $\overline{f}(x) > \beta$, so $\sup_n \frac{1}{n} S_n f(x) > \beta$. Apply Corollary 9.7 to f restricted to $A_{\alpha,\beta}$ with the threshold β: the set $\{x \in A_{\alpha,\beta} : \sup_n \frac{1}{n} S_n f(x) > \beta\} = A_{\alpha,\beta}$ (since every point in $A_{\alpha,\beta}$ has $\overline{f}(x) > \beta$). Thus:

$$\int_{A_{\alpha,\beta}} f \, d\mu \geq \beta \cdot \mu(A_{\alpha,\beta}). \tag{1}$$

Similarly, on $A_{\alpha,\beta}$, $\underline{f}(x) < \alpha$, so $\liminf_n \frac{1}{n} S_n f(x) < \alpha$, which means $\limsup_n \frac{1}{n} S_n(-f)(x) > -\alpha$, hence $\sup_n \frac{1}{n} S_n(-f)(x) > -\alpha$. Applying Corollary 9.7 to $-f$ on $A_{\alpha,\beta}$ with threshold $-\alpha$:

$$\int_{A_{\alpha,\beta}} (-f) \, d\mu \geq -\alpha \cdot \mu(A_{\alpha,\beta}),$$

i.e.,

$$\int_{A_{\alpha,\beta}} f \, d\mu \leq \alpha \cdot \mu(A_{\alpha,\beta}). \tag{2}$$

Combining (1) and (2):

$$\beta \cdot \mu(A_{\alpha,\beta}) \leq \int_{A_{\alpha,\beta}} f \, d\mu \leq \alpha \cdot \mu(A_{\alpha,\beta}).$$

Since $\beta > \alpha$, this implies $(\beta - \alpha)\mu(A_{\alpha,\beta}) \leq 0$, and since $\beta - \alpha > 0$, we conclude $\mu(A_{\alpha,\beta}) = 0$.

Since this holds for all rational $\alpha < \beta$, $\mu(\{x : \underline{f}(x) < \overline{f}(x)\}) = 0$, so the limit $\bar{f}(x) = \lim_n \frac{1}{n} S_n f(x)$ exists μ-a.e.

Part (iii): $\bar{f} \in L^1$ and $\int \bar{f} = \int f$. By Fatou's lemma applied to the non-negative functions $\frac{1}{n}|S_n f|$:

$$\int_X |\bar{f}| \, d\mu \leq \liminf_{n\to\infty} \frac{1}{n} \int_X |S_n f| \, d\mu \leq \liminf_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \int_X |f \circ T^k| \, d\mu = \|f\|_1,$$

where we used the fact that $\|f \circ T^k\|_1 = \|f\|_1$ (since T is measure-preserving). So $\bar{f} \in L^1$ and $\|\bar{f}\|_1 \leq \|f\|_1$.

For the equality of integrals, note that $|\frac{1}{n} S_n f(x)| \leq \frac{1}{n} \sum_{k=0}^{n-1} |f(T^k x)|$, and the right-hand side converges a.e. to $\overline{|f|}(x) \in L^1$ by Part (i) applied to $|f|$. So $\{\frac{1}{n} S_n f\}$ is dominated by an L^1 function (eventually, and can be handled carefully). More directly, by the T-invariance of μ:

$$\int_X \frac{1}{n} S_n f \, d\mu = \frac{1}{n} \sum_{k=0}^{n-1} \int_X f \circ T^k \, d\mu = \frac{1}{n} \sum_{k=0}^{n-1} \int_X f \, d\mu = \int_X f \, d\mu$$

for every n. Now, define $g_n = \frac{1}{n} S_n f$. We have $g_n \to \bar{f}$ a.e. and $|g_n| \leq \frac{1}{n} \sum_{k=0}^{n-1} |f| \circ T^k$. We need to apply dominated convergence. Define $h_n = \frac{1}{n} \sum_{k=0}^{n-1} |f| \circ T^k$, which converges a.e. to $\overline{|f|}$. We have $|g_n| \leq h_n$, but h_n is not a fixed dominating function.

Instead, we use a truncation argument. For $M > 0$, let $f_M = \max(-M, \min(f, M))$ be the truncation of f at level M. Then $|f_M| \leq M$, so $|\frac{1}{n} S_n f_M| \leq M$, and by the dominated convergence theorem:

$$\int_X \bar{f}_M \, d\mu = \lim_{n\to\infty} \int_X \frac{1}{n} S_n f_M \, d\mu = \int_X f_M \, d\mu.$$

Now let $M \to \infty$. Since $f_M \to f$ in L^1 and $\|\bar{f}_M - \bar{f}\|_1 \leq \|f_M - f\|_1 \to 0$ (by the bound $\|\bar{g}\|_1 \leq \|g\|_1$ from above, applied to $g = f_M - f$), we get $\int \bar{f}_M \to \int \bar{f}$ and $\int f_M \to \int f$, so $\int \bar{f} = \int f$. □

Remark 9.3. The relationship between the two ergodic theorems is worth noting. Von Neumann's theorem gives L^2 convergence, which is weaker than pointwise a.e. convergence in general (the two are not directly comparable, but L^2 convergence implies convergence in measure). Birkhoff's theorem gives the stronger pointwise convergence, but requires a harder proof. Historically, Birkhoff proved his theorem slightly before von Neumann published, though von Neumann had communicated his result earlier — leading to some priority disputes. Both theorems are fundamental, and in many applications the L^2 version suffices.

9.6 Ergodicity

We now define the central concept of this chapter — and arguably of the entire subject.

Definition 9.2. A measure-preserving transformation $T : (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ is **ergodic** if every T-invariant measurable set has measure 0 or 1. That is, if $A \in \mathcal{B}$ and $T^{-1}(A) = A$, then $\mu(A) = 0$ or $\mu(A) = 1$.

Informally, ergodicity means the system cannot be decomposed into two non-trivial invariant pieces. The dynamics is "indecomposable" from the measure-theoretic point of view.

Remark 9.4. Some authors use the weaker condition $\mu(T^{-1}A \triangle A) = 0 \Rightarrow \mu(A) \in \{0, 1\}$ (invariance mod μ). For most purposes these are equivalent. We will use strict invariance $T^{-1}A = A$ in the definition, noting that the equivalence with the "mod μ" version is straightforward when the system is defined on a complete measure space.

The following theorem gives several equivalent characterizations of ergodicity. This is a result of fundamental importance — it tells us precisely when time averages equal space averages.

Theorem 9.6 (Equivalent characterizations of ergodicity)**.** *Let $T : (X, \mathcal{B}, \mu) \to (X, \mathcal{B}, \mu)$ be a measure-preserving transformation on a probability space. The following are equivalent:*

(a) *T is ergodic: every T-invariant set $A \in \mathcal{B}$ (i.e., $T^{-1}A = A$) satisfies $\mu(A) = 0$ or $\mu(A) = 1$.*

(b) *Every T-invariant function in $L^1(X, \mu)$ is constant a.e. That is, if $f \in L^1$ and $f \circ T = f$ a.e., then f is constant a.e.*

(c) *For every $f \in L^1(X, \mu)$:*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} f(T^k x) = \int_X f \, d\mu \qquad \text{for } \mu\text{-a.e. } x.$$

(d) *For all measurable sets $A, B \in \mathcal{B}$:*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mu(T^{-k}A \cap B) = \mu(A)\mu(B).$$

Proof. *

(a) $\Rightarrow$ (b). Suppose T is ergodic and $f \circ T = f$ a.e. with $f \in L^1$. We first handle the case f real-valued. For any $c \in \mathbb{R}$, the set $A_c = \{x : f(x) \leq c\}$ satisfies

$$T^{-1}(A_c) = \{x : f(Tx) \leq c\} = \{x : f(x) \leq c\} = A_c \quad \text{(a.e.)}.$$

More precisely, $\mu(T^{-1}(A_c) \triangle A_c) = 0$, so by adjusting on a null set (or using the "mod μ" version of ergodicity), $\mu(A_c) \in \{0, 1\}$. Define

$$c^* = \inf\{c \in \mathbb{R} : \mu(A_c) = 1\}.$$

For $c > c^*$, $\mu(A_c) = 1$, so $f \leq c$ a.e. For $c < c^*$, $\mu(A_c) = 0$, so $f > c$ a.e. Taking a sequence $c_n \downarrow c^*$, we get $f \leq c^*$ a.e., and taking $c_n \uparrow c^*$, we get $f \geq c^*$ a.e. Therefore $f = c^*$ a.e. (The value c^* is finite since $f \in L^1$.)

For complex-valued f, apply the above to the real and imaginary parts.

(b) $\Rightarrow$ (a). Suppose every T-invariant L^1 function is constant a.e. Let A be a T-invariant set. Then $f = \mathbf{1}_A$ is a T-invariant L^1 function (since $\mathbf{1}_A \circ T = \mathbf{1}_{T^{-1}A} = \mathbf{1}_A$). By (b), $\mathbf{1}_A$ is constant a.e., so $\mathbf{1}_A = 0$ a.e. or $\mathbf{1}_A = 1$ a.e., i.e., $\mu(A) = 0$ or $\mu(A) = 1$.

(b) $\Rightarrow$ (c). By Birkhoff's theorem (Theorem 9.8), the time average $\bar{f}(x) = \lim_n \frac{1}{n} \sum_{k=0}^{n-1} f(T^k x)$ exists a.e., is T-invariant, and satisfies $\int \bar{f} = \int f$. By (b), $\bar{f}$ is constant a.e. Since $\int \bar{f} = \int f$, this constant must equal $\int f \, d\mu$.

(c) $\Rightarrow$ (b). Suppose $f \in L^1$ and $f \circ T = f$ a.e. Then $\frac{1}{n} \sum_{k=0}^{n-1} f(T^k x) = f(x)$ a.e. for all n. By (c), this equals $\int f \, d\mu$ a.e. So $f(x) = \int f \, d\mu$ for a.e. x.

(c) $\Rightarrow$ (d). Apply (c) to $f = \mathbf{1}_A$, where $A \in \mathcal{B}$:

$$\frac{1}{n} \sum_{k=0}^{n-1} \mathbf{1}_A(T^k x) \to \mu(A) \qquad \text{a.e.}$$

Multiply both sides by $\mathbf{1}_B(x)$ and integrate:

$$\frac{1}{n} \sum_{k=0}^{n-1} \int_X \mathbf{1}_A(T^k x) \mathbf{1}_B(x) \, d\mu(x) \to \mu(A) \cdot \mu(B).$$

The left side equals $\frac{1}{n} \sum_{k=0}^{n-1} \mu(T^{-k}A \cap B)$ (since $\mathbf{1}_A(T^k x)\mathbf{1}_B(x) = \mathbf{1}_{T^{-k}A \cap B}(x)$). The convergence of the integral is justified by dominated convergence (the integrands are bounded by 1). This gives (d).

(d) $\Rightarrow$ (a). Let A be T-invariant, i.e., $T^{-1}A = A$. Apply (d) with $B = A$:

$$\frac{1}{n} \sum_{k=0}^{n-1} \mu(T^{-k}A \cap A) = \frac{1}{n} \sum_{k=0}^{n-1} \mu(A \cap A) = \mu(A) \to \mu(A)\mu(A) = \mu(A)^2.$$

So $\mu(A) = \mu(A)^2$, which implies $\mu(A) = 0$ or $\mu(A) = 1$. $\square$

Remark 9.5. The equivalence (a) $\Leftrightarrow$ (c) is the key takeaway of this chapter. It says:

> *The time average of any observable equals its space average (for a.e. initial condition) if and only if the system is ergodic.*

This is the precise mathematical content of Boltzmann's ergodic hypothesis (in its modern formulation).

Condition (d) is often called **Cesàro mixing** or **ergodicity in the sense of mixing on average**. It says that, on average, the sets $T^{-k}A$ become independent of any fixed set B. This is weaker than (strong) mixing, which requires $\mu(T^{-n}A \cap B) \to \mu(A)\mu(B)$ without the Cesàro averaging.

9.7 Examples of Ergodic Systems

We now prove ergodicity for several fundamental examples introduced in Chapter 8.

9.7.1 Irrational Rotation

Let $X = \mathbb{R}/\mathbb{Z} \cong [0, 1)$ with Lebesgue measure μ, and let $T_\alpha(x) = x+\alpha \pmod 1$, where $\alpha \in \mathbb{R}\backslash\mathbb{Q}$.

Theorem 9.14. The irrational rotation T_α is ergodic with respect to Lebesgue measure.

Proof. * We use Fourier analysis. We show that characterization (b) holds: every T_α-invariant L^2 function is constant a.e. (Since $L^2 \subset L^1$ on a probability space, this suffices.)

Let $f \in L^2(\mathbb{R}/\mathbb{Z})$ satisfy $f \circ T_\alpha = f$ a.e. Expand f in a Fourier series:

$$f(x) = \sum_{n\in\mathbb{Z}} \hat{f}(n)e^{2\pi inx}, \qquad \hat{f}(n) = \int_0^1 f(x)e^{-2\pi inx}\,dx.$$

The condition $f(x + \alpha) = f(x)$ a.e. gives, at the level of Fourier coefficients:

$$\hat{f}(n)e^{2\pi in\alpha} = \hat{f}(n) \qquad \text{for all } n \in \mathbb{Z}.$$

This is because $(f \circ T_\alpha)\hat{}(n) = e^{2\pi in\alpha}\hat{f}(n)$.

For $n \neq 0$, $e^{2\pi in\alpha} \neq 1$ (since α is irrational and $n \neq 0$, we have $n\alpha \notin \mathbb{Z}$). Therefore $\hat{f}(n) = 0$ for all $n \neq 0$.

So $f(x) = \hat{f}(0) = \int_0^1 f\,dx$ a.e. That is, f is constant a.e. □

Remark 9.6. This proof breaks down completely when $\alpha = p/q$ is rational. In that case, $e^{2\pi in\alpha} = 1$ whenever $q \mid n$, so there exist non-constant invariant functions — for instance, $f(x) = e^{2\pi iqx}$. We will see below that rational rotations are indeed not ergodic.

9.7.2 The Doubling Map

Let $X = [0, 1)$ with Lebesgue measure μ, and $T(x) = 2x \pmod 1$.

Theorem 9.16. The doubling map is ergodic with respect to Lebesgue measure.

Proof. * We again use Fourier analysis. Suppose $f \in L^2([0, 1))$ is T-invariant: $f(2x \bmod 1) = f(x)$ a.e.

We compute the Fourier coefficients of $f \circ T$:

$$(f \circ T)\hat{}(n) = \int_0^1 f(2x \bmod 1)e^{-2\pi inx}\,dx.$$

Split the integral into $[0, 1/2)$ and $[1/2, 1)$. On $[0, 1/2)$, $2x \bmod 1 = 2x$; substituting $u = 2x$:

$$\int_0^{1/2} f(2x)e^{-2\pi inx}\,dx = \frac{1}{2}\int_0^1 f(u)e^{-\pi inu}\,du.$$

On $[1/2, 1)$, $2x \bmod 1 = 2x - 1$; substituting $u = 2x - 1$:

$$\int_{1/2}^1 f(2x-1)e^{-2\pi inx}\,dx = \frac{1}{2}\int_0^1 f(u)e^{-\pi in(u+1)}\,du = \frac{e^{-\pi in}}{2}\int_0^1 f(u)e^{-\pi inu}\,du.$$

Adding:

$$(f \circ T)\hat{}(n) = \frac{1 + e^{-\pi in}}{2}\int_0^1 f(u)e^{-\pi inu}\,du.$$

Now $\frac{1+e^{-\pi in}}{2} = \begin{cases} 1 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$.

So for n even, say $n = 2m$:

$$(f \circ T)\hat{}(2m) = \int_0^1 f(u)e^{-2\pi imu}\,du = \hat{f}(m).$$

And for n odd: $(f \circ T)\hat{}(n) = 0$.

The condition $f \circ T = f$ a.e. means $(f \circ T)\hat{}(n) = \hat{f}(n)$ for all n. This gives:

- For n odd: $\hat{f}(n) = 0$.
- For n even, $n = 2m$: $\hat{f}(2m) = \hat{f}(m)$.

The second relation, applied repeatedly, gives $\hat{f}(m) = \hat{f}(2m) = \hat{f}(4m) = \hat{f}(8m) = \cdots = \hat{f}(2^k m)$ for all $k \geq 0$.

If $m \neq 0$, then $|2^k m| \to \infty$, but $\hat{f}(n) \to 0$ as $|n| \to \infty$ by the Riemann-Lebesgue lemma (since $f \in L^1$, and in particular $f \in L^2$ implies $\sum |\hat{f}(n)|^2 < \infty$, hence $\hat{f}(n) \to 0$). Therefore $\hat{f}(m) = 0$ for all $m \neq 0$.

We also need to handle the case where m is even: write $m = 2^j m'$ with m' odd. Then $\hat{f}(m) = \hat{f}(m')$ (by iterating $\hat{f}(2m) = \hat{f}(m)$ a total of j times). But m' is odd, so $\hat{f}(m') = 0$. Hence $\hat{f}(m) = 0$ for all $m \neq 0$.

So $f(x) = \hat{f}(0)$ a.e., and T is ergodic. □

9.7.3 Bernoulli Shifts

Let $\Sigma = \{0, 1, \ldots, k-1\}^{\mathbb{N}}$ be the one-sided shift space on k symbols (with the product σ-algebra), equipped with the product measure $\mu = p^{\mathbb{N}}$ where $p = (p_0, p_1, \ldots, p_{k-1})$ is a probability vector ($p_i > 0$, $\sum p_i = 1$). The **shift map** $\sigma : \Sigma \to \Sigma$ is defined by $(\sigma(\omega))_n = \omega_{n+1}$.

Recall that the **cylinder sets**

$$[a_0, a_1, \ldots, a_{m-1}] = \{\omega \in \Sigma : \omega_0 = a_0, \omega_1 = a_1, \ldots, \omega_{m-1} = a_{m-1}\}$$

form a semi-algebra generating $\mathcal{B}$, and $\mu([a_0, \ldots, a_{m-1}]) = p_{a_0} p_{a_1} \cdots p_{a_{m-1}}$.

Theorem 9.17. The Bernoulli shift (Σ, μ, σ) is ergodic.

Proof. * We verify characterization (d) of ergodicity for cylinder sets, which suffices since cylinders generate the σ-algebra (and both sides of (d) extend by standard measure-theoretic approximation arguments).

Let $A = [a_0, \ldots, a_{m-1}]$ and $B = [b_0, \ldots, b_{\ell-1}]$ be cylinder sets, specifying coordinates $0, \ldots, m-1$ and $0, \ldots, \ell-1$ respectively. Note that

$$\sigma^{-k}(A) = \{\omega : \omega_k = a_0, \omega_{k+1} = a_1, \ldots, \omega_{k+m-1} = a_{m-1}\},$$

which specifies coordinates $k, k+1, \ldots, k+m-1$.

For $k \geq \ell$, the sets $\sigma^{-k}(A)$ and B specify disjoint sets of coordinates, so by independence of coordinates under the product measure:

$$\mu(\sigma^{-k}A \cap B) = \mu(\sigma^{-k}A) \cdot \mu(B) = \mu(A) \cdot \mu(B).$$

So for $n > \ell$:

$$\frac{1}{n}\sum_{k=0}^{n-1} \mu(\sigma^{-k}A \cap B) = \frac{1}{n}\sum_{k=0}^{\ell-1} \mu(\sigma^{-k}A \cap B) + \frac{1}{n}\sum_{k=\ell}^{n-1} \mu(A)\mu(B).$$

The first sum contributes at most $\ell/n \to 0$, and the second sum equals $\frac{n-\ell}{n}\mu(A)\mu(B) \to \mu(A)\mu(B)$.

Therefore $\frac{1}{n}\sum_{k=0}^{n-1} \mu(\sigma^{-k}A \cap B) \to \mu(A)\mu(B)$.

To extend from cylinders to arbitrary measurable sets: let $A, B \in \mathcal{B}$. Given $\varepsilon > 0$, approximate A and B by finite unions of cylinders A' and B' with $\mu(A \triangle A') < \varepsilon$ and $\mu(B \triangle B') < \varepsilon$. Since

$$|\mu(\sigma^{-k}A \cap B) - \mu(\sigma^{-k}A' \cap B')| \leq \mu(A \triangle A') + \mu(B \triangle B') < 2\varepsilon$$

(using the fact that $\mu(\sigma^{-k}(A \triangle A')) = \mu(A \triangle A')$ since σ preserves μ), and (d) holds for A', B', a routine 3ε-argument shows (d) holds for A, B. □

Remark 9.7. In fact, the Bernoulli shift is **mixing** (not just ergodic): $\mu(\sigma^{-n}A \cap B) \to \mu(A)\mu(B)$ as $n \to \infty$, without the need for Cesàro averaging. This is stronger than what we needed. We will discuss mixing in Chapter 10.

9.7.4 Non-Example: Rational Rotation

Let $T_{p/q}(x) = x + p/q \pmod 1$ where p/q is a rational number in lowest terms with $q \geq 2$.

Proposition 9.7. *The rational rotation $T_{p/q}$ is NOT ergodic with respect to Lebesgue measure.*

Proof. * We exhibit a $T_{p/q}$-invariant set of intermediate measure. Consider the set

$$A = \bigcup_{j=0}^{q-1} \left[\frac{j}{q}, \frac{j}{q} + \frac{1}{2q}\right).$$

This is a union of the "first halves" of the intervals $[j/q, (j+1)/q)$ for $j = 0, \ldots, q-1$. We have $\mu(A) = q \cdot \frac{1}{2q} = \frac{1}{2}$.

Since $T_{p/q}$ permutes the intervals $[j/q, (j+1)/q)$ (it maps the j-th interval to the $(j + p \bmod q)$-th interval), and A consists of the first half of each such interval, A is $T_{p/q}$-invariant. So $\mu(A) = 1/2 \notin \{0, 1\}$, and $T_{p/q}$ is not ergodic.

Alternatively: $f(x) = e^{2\pi i q x}$ satisfies $f(x + p/q) = e^{2\pi i q(x+p/q)} = e^{2\pi i q x} e^{2\pi i p} = f(x)$, so f is a non-constant $T_{p/q}$-invariant function. □

Remark 9.8. Geometrically, rational rotation partitions the circle into finitely many orbits, each of length q. The dynamics on each orbit is periodic. One can decompose the system into q invariant subsets (or, thinking continuously, into invariant "strips"), showing the system is far from indecomposable.

9.8 Unique Ergodicity

So far, Birkhoff's theorem tells us that time averages converge *almost everywhere*. But "almost everywhere" depends on the measure, and there may be exceptional points. In some situations, we can do better.

Definition 9.3. A continuous map $T : X \to X$ on a compact metrizable space X is **uniquely ergodic** if there is exactly one T-invariant Borel probability measure on X.

Theorem 9.8 (Unique ergodicity and uniform convergence)**.** *Let $T : X \to X$ be a continuous map on a compact metrizable space X, and let μ be a T-invariant Borel probability measure. The following are equivalent:*

(a) *T is uniquely ergodic (with unique invariant measure μ).*

(b) *For every continuous function $f \in C(X)$:*

$$\frac{1}{n} \sum_{k=0}^{n-1} f(T^k x) \to \int_X f \, d\mu \qquad \textit{uniformly in } x \in X.$$

(c) *For every continuous function $f \in C(X)$:*

$$\frac{1}{n} \sum_{k=0}^{n-1} f(T^k x) \to \int_X f \, d\mu \qquad \textit{for every } x \in X.$$

Note the strength of this result: convergence holds for every point, not just almost every point. There is no exceptional set.

Proof. *

(a) ⇒ (b). Suppose for contradiction that the convergence is not uniform for some $f \in C(X)$. Then there exist $\varepsilon > 0$, a sequence $n_j \to \infty$, and points $x_j \in X$ such that

$$\left| \frac{1}{n_j} \sum_{k=0}^{n_j-1} f(T^k x_j) - \int f \, d\mu \right| \geq \varepsilon. \tag{†}$$

Define the probability measures $\nu_j = \frac{1}{n_j}\sum_{k=0}^{n_j-1}\delta_{T^k x_j}$ (empirical measures along orbits). By the Banach-Alaoglu theorem (or equivalently, Prokhorov's theorem, since X is compact), the sequence $\{\nu_j\}$ has a weak-$*$ convergent subsequence $\nu_{j_i} \rightharpoonup \nu$ for some Borel probability measure ν on X.

We claim ν is T-invariant. For any $g \in C(X)$:

$$\int g\, d(T_*\nu_j) = \int g \circ T\, d\nu_j = \frac{1}{n_j}\sum_{k=0}^{n_j-1} g(T^{k+1}x_j) = \frac{1}{n_j}\sum_{k=1}^{n_j} g(T^k x_j).$$

The difference between this and $\int g\, d\nu_j$ is at most $\frac{2\|g\|_\infty}{n_j} \to 0$. So $T_*\nu_j$ and ν_j have the same weak-$*$ limit, hence $T_*\nu = \nu$.

By unique ergodicity, $\nu = \mu$. But then $\int f\, d\nu_j \to \int f\, d\nu = \int f\, d\mu$, i.e., $\frac{1}{n_{j_i}}\sum_{k=0}^{n_{j_i}-1} f(T^k x_{j_i}) \to \int f\, d\mu$, contradicting (†).

(b) $\Rightarrow$ (c). Trivial (uniform convergence implies pointwise convergence).

(c) $\Rightarrow$ (a). Suppose ν is any T-invariant probability measure. For any $f \in C(X)$, by (c), $\frac{1}{n}\sum_{k=0}^{n-1} f(T^k x) \to \int f\, d\mu$ for every x. Integrating against ν and using T-invariance of ν:

$$\int f\, d\nu = \int \frac{1}{n}\sum_{k=0}^{n-1} f(T^k x)\, d\nu(x) \to \int \left(\int f\, d\mu\right) d\nu = \int f\, d\mu.$$

(The exchange of limit and integral is justified by dominated convergence since $|f| \le \|f\|_\infty$.) Since $\int f\, d\nu = \int f\, d\mu$ for all $f \in C(X)$, the Riesz representation theorem gives $\nu = \mu$. □

9.8.1 Irrational Rotations Are Uniquely Ergodic

Theorem 9.23. The irrational rotation $T_\alpha : x \mapsto x + \alpha \pmod 1$ on $\mathbb{R}/\mathbb{Z}$ is uniquely ergodic, with Lebesgue measure as its unique invariant measure.

Proof. * Let ν be any T_α-invariant Borel probability measure on $\mathbb{R}/\mathbb{Z}$. We show ν is Lebesgue measure by computing its Fourier-Stieltjes coefficients.

For $n \in \mathbb{Z}$, define

$$\hat{\nu}(n) = \int_0^1 e^{-2\pi i n x}\, d\nu(x).$$

Since ν is T_α-invariant, $(T_\alpha)_*\nu = \nu$, so

$$\hat{\nu}(n) = \int_0^1 e^{-2\pi i n x}\, d\nu(x) = \int_0^1 e^{-2\pi i n(x+\alpha)}\, d\nu(x) = e^{-2\pi i n\alpha}\hat{\nu}(n).$$

For $n \neq 0$, $e^{-2\pi i n\alpha} \neq 1$ (since α is irrational), so $\hat{\nu}(n) = 0$.

For $n = 0$, $\hat{\nu}(0) = \nu(\mathbb{R}/\mathbb{Z}) = 1$.

Since $\hat{\nu}(n) = \hat{\lambda}(n)$ for all n (where λ is Lebesgue measure: $\hat{\lambda}(0) = 1$ and $\hat{\lambda}(n) = 0$ for $n \neq 0$), we conclude $\nu = \lambda$ by the uniqueness theorem for Fourier-Stieltjes coefficients of measures on $\mathbb{R}/\mathbb{Z}$. □

9.8.2 Weyl's Equidistribution Theorem

As a beautiful application of unique ergodicity, we obtain a classical result in number theory.

Theorem 9.9 (Weyl's Equidistribution Theorem, 1916)**.** *If α is irrational, then the sequence $\{n\alpha\}_{n=0}^{\infty}$ (where $\{y\} = y - \lfloor y \rfloor$ denotes the fractional part) is* ***equidistributed modulo 1****. That is, for every interval $[a, b] \subseteq [0, 1]$:*

$$\lim_{N\to\infty} \frac{1}{N} \#\{0 \le n < N : \{n\alpha\} \in [a, b]\} = b - a.$$

Proof. * Apply Theorem 9.22(b) to $T_\alpha(x) = x + \alpha \pmod 1$, which is uniquely ergodic with invariant measure $\mu = \lambda$ (Lebesgue measure), and to $x_0 = 0$. For $f = \mathbf{1}_{[a,b]}$ ... but f is not continuous. However, for continuous f, Theorem 9.22 gives

$$\frac{1}{N} \sum_{n=0}^{N-1} f(\{n\alpha\}) = \frac{1}{N} \sum_{n=0}^{N-1} f(T_\alpha^n(0)) \to \int_0^1 f(x)\, dx$$

uniformly in the starting point (and in particular for $x_0 = 0$).

To handle indicator functions, approximate $\mathbf{1}_{[a,b]}$ from above and below by continuous functions: for any $\varepsilon > 0$, there exist continuous $g_\varepsilon \le \mathbf{1}_{[a,b]} \le h_\varepsilon$ with $\int (h_\varepsilon - g_\varepsilon) < \varepsilon$. Then

$$\int g_\varepsilon \le \liminf_N \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{1}_{[a,b]}(\{n\alpha\}) \le \limsup_N \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{1}_{[a,b]}(\{n\alpha\}) \le \int h_\varepsilon,$$

and both $\int g_\varepsilon$ and $\int h_\varepsilon$ tend to $b - a$ as $\varepsilon \to 0$. □

Remark 9.9. Weyl originally proved this using exponential sum estimates. The ergodic-theoretic proof given here, while requiring more machinery, is arguably more conceptual: equidistribution is simply a consequence of the dynamics being uniquely ergodic.

9.9 Physical Interpretation and Significance

9.9.1 The Ergodic Hypothesis

The word "ergodic" was coined by Boltzmann (from the Greek *ergon* = work and *hodos* = path) in the context of statistical mechanics. The setup is as follows.

Consider a classical mechanical system of N particles in a box. The state of the system is described by a point x in a $6N$-dimensional phase space Γ (three position coordinates and three momentum coordinates per particle). The system evolves according to Hamilton's equations, generating a flow $\phi_t : \Gamma \to \Gamma$.

Liouville's theorem (from Hamiltonian mechanics) states that this flow preserves the Lebesgue measure on Γ (or more precisely, the measure induced by the symplectic structure). Since the total energy $H(x) = E$ is conserved, the system actually moves on the energy surface $\Sigma_E = \{x \in \Gamma : H(x) = E\}$, and the flow preserves a natural measure on Σ_E (the microcanonical measure).

Boltzmann's ergodic hypothesis (in its modern formulation) asserts:

The flow ϕ_t on Σ_E is ergodic with respect to the microcanonical measure.

If this holds, then for any observable f (temperature, pressure, etc.), the time average

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T f(\phi_t(x))\, dt$$

equals the ensemble average $\int_{\Sigma_E} f\, d\mu$ for a.e. initial condition x. This is precisely Birkhoff's theorem combined with ergodicity.

9.9.2 Why This Matters

The practical significance is immense:

1. **Time averages are what we measure.** An experimentalist measuring the pressure of a gas is effectively computing a time average (averaged over the measurement time, which is enormously long compared to molecular timescales).
2. **Ensemble averages are what we can compute.** Statistical mechanics gives us tools (partition functions, etc.) to compute integrals over phase space.
3. **Ergodicity bridges the gap.** If the system is ergodic, these two quantities agree, providing the theoretical foundation for statistical mechanics.

9.9.3 Limitations

The ergodic hypothesis, despite its foundational importance, has significant limitations:

1. **Not all systems are ergodic.** The KAM theorem (Kolmogorov-Arnold-Moser) shows that many Hamiltonian systems with few degrees of freedom are *not* ergodic: they possess invariant tori that partition phase space into invariant regions of positive measure.
2. **Ergodicity may be too slow.** Even when a system is ergodic, the convergence of time averages to space averages may be so slow as to be physically irrelevant. The system may have multiple timescales, and on observable timescales, different regions of phase space may be effectively disconnected (a phenomenon related to *metastability*).
3. **Hard systems.** Proving ergodicity for realistic physical systems is extremely difficult. The ergodicity of hard sphere gases was a major open problem for decades. Sinai proved the case of two disks on a torus (1970), and Ya. Sinai and N. Chernov, along with others, made progress on more general cases, but a complete proof for N hard spheres remains incomplete for large N.
4. **Beyond ergodicity.** For many applications in statistical mechanics, one needs stronger properties than ergodicity — such as mixing, or quantitative estimates on rates of mixing (decay of correlations). We will explore mixing in Chapter 10.

Despite these caveats, the ergodic theorems remain cornerstones of both pure mathematics and mathematical physics. They provide the conceptual framework within which we understand the relationship between microscopic dynamics and macroscopic observables.

9.10 Summary

Let us collect the main ideas of this chapter.

1. **Von Neumann's theorem** establishes L^2-convergence of time averages to the projection onto the invariant subspace. The proof is a clean application of Hilbert space geometry.

2. **Birkhoff's theorem** establishes pointwise a.e. convergence of time averages to a T-invariant function $\bar{f}$, with $\int \bar{f} = \int f$. The proof uses the maximal ergodic theorem.
3. **Ergodicity** is the condition under which the invariant function $\bar{f}$ is forced to be the constant $\int f\, d\mu$. It is equivalent to: no non-trivial invariant sets, no non-constant invariant functions, and the Cesàro mixing condition.
4. We verified ergodicity for **irrational rotations** (Fourier analysis), the **doubling map** (Fourier analysis + Riemann-Lebesgue), and **Bernoulli shifts** (independence of coordinates).
5. **Unique ergodicity** is a topological strengthening: it guarantees convergence for *every* point, not just almost every point. Irrational rotations are uniquely ergodic, yielding Weyl's equidistribution theorem.
6. The physical significance: ergodicity justifies the replacement of time averages by ensemble averages, the foundational step of statistical mechanics.

In Chapter 10, we will study stronger forms of "mixing" — strong mixing and weak mixing — which refine the ergodic hierarchy and have deep connections to spectral theory.

Exercises

Exercise 9.1. Let T be a measure-preserving transformation on $(X, \mathcal{B}, \mu)$. Show that T is ergodic if and only if: for every $f \in L^2(X, \mu)$ with $f \circ T = f$ a.e., f is constant a.e. (That is, characterization (b) of Theorem 9.12 holds with L^2 in place of L^1.)

Exercise 9.2. Let T_α be an irrational rotation on $\mathbb{R}/\mathbb{Z}$. Using the ergodic theorem (not Weyl's theorem directly), prove that for any Riemann-integrable function $f : [0, 1] \to \mathbb{R}$:

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} f(\{n\alpha\}) = \int_0^1 f(x)\, dx.$$

Exercise 9.3. (Ergodicity of a product). Let $T_1 : (X_1, \mu_1) \to (X_1, \mu_1)$ and $T_2 : (X_2, \mu_2) \to (X_2, \mu_2)$ be ergodic measure-preserving transformations. Define $T = T_1 \times T_2 : X_1 \times X_2 \to X_1 \times X_2$ by $T(x_1, x_2) = (T_1 x_1, T_2 x_2)$, preserving $\mu_1 \times \mu_2$.

(a) Show by example that T need not be ergodic. (*Hint*: consider $T_1 = T_2 =$ irrational rotation by α. What invariant sets does $T_1 \times T_2$ have?)

(b) Show that if T_1 is ergodic and T_2 is weak mixing (defined as: $T_2 \times T_2$ is ergodic), then $T_1 \times T_2$ is ergodic. You may assume results about weak mixing from Chapter 10 if needed, or prove it directly.

Exercise 9.4. (A non-ergodic system). Let $T : [0, 1] \to [0, 1]$ be the "tent map" restricted to $[0, 1/2]$: $T(x) = 2x$ for $x \in [0, 1/4]$ and $T(x) = 1 - 2x$ for $x \in [1/4, 1/2]$. Note T maps $[0, 1/2]$ to itself, and also $[1/2, 1]$ to itself (define T appropriately on $[1/2, 1]$). Show that T is not ergodic with respect to Lebesgue measure on $[0, 1]$.

Exercise 9.5. (Maximal inequality and weak type bound). Using the maximal ergodic theorem, prove the following: for $f \in L^1(X, \mu)$, T measure-preserving, and $\lambda > 0$:

$$\mu\left(\left\{x : \sup_{n\geq 1}\frac{1}{n}\left|\sum_{k=0}^{n-1} f(T^k x)\right| > \lambda\right\}\right) \leq \frac{3}{\lambda}\|f\|_1.$$

(*Hint*: Apply Corollary 9.7 to $f-\lambda/3$ and to $-f-\lambda/3$, then handle the region where partial sums oscillate.)

Exercise 9.6. (Computing time averages). Let $T(x) = 2x \pmod 1$ on $[0,1)$ with Lebesgue measure.

(a) Compute $\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1}\cos(2\pi T^k(x))$ for a.e. x.

(b) Compute $\lim_{n\to\infty}\frac{1}{n}\sum_{k=0}^{n-1}\mathbf{1}_{[0,1/3)}(T^k(x))$ for a.e. x.

(c) For $x = 1/3$, verify your answer to (b) by directly analyzing the orbit of $1/3$ under the doubling map.

Exercise 9.7. minimal** if every orbit is dense: $\overline{\{T^n x : n \geq 0\}} = X$ for all $x \in X$.

(a) Show that if T is uniquely ergodic and μ has full support (i.e., $\mu(U) > 0$ for every nonempty open U), then T is minimal.

(b) Give an example of a minimal homeomorphism that is not uniquely ergodic. (*This is harder — you may describe the construction without full details. See Furstenberg's example on the torus.*)

(c) Show that irrational rotations on $\mathbb{R}/\mathbb{Z}$ are minimal.

Exercise 9.8. (Ergodicity via invariant functions — a direct approach). Let $T(x) = 2x \pmod 1$ on $[0,1)$. Give an alternative proof of ergodicity that does not use Fourier analysis, using the following outline:

(a) Show that if A is a T-invariant measurable set, then for every dyadic interval $I = [j/2^n, (j+1)/2^n)$, $\mu(A \cap I) = \mu(A)\cdot\mu(I)$. (*Hint*: What does T^{-n} do to dyadic intervals?)

(b) Conclude that $\mu(A \cap J) = \mu(A)\cdot\mu(J)$ for every interval $J \subseteq [0,1)$.

(c) Deduce that $\mu(A) = 0$ or 1.

Exercise 9.9. (Convergence of ergodic averages for the doubling map). Let $T(x) = 2x \pmod 1$ and consider $f(x) = x$.

(a) Compute $\int_0^1 f\,dx$. What does the ergodic theorem predict about $\frac{1}{n}\sum_{k=0}^{n-1} T^k(x)$ for a.e. x?

(b) Write a careful argument (or numerical computation) to verify the prediction for $x = \sqrt{2}-1$, at least numerically.

(c) Find a specific x for which the time average does NOT converge to $\int f\,d\mu$. Why does this not contradict the ergodic theorem?

References

- G.D. Birkhoff. "Proof of the ergodic theorem." *Proceedings of the National Academy of Sciences*, 17(12):656–660, 1931.
- J. von Neumann. "Proof of the quasi-ergodic hypothesis." *Proceedings of the National Academy of Sciences*, 18(1):70–82, 1932.

- A.M. Garsia. "A simple proof of E. Hopf's maximal ergodic theorem." *Journal of Mathematics and Mechanics*, 14(3):381–382, 1965.
- P. Walters. *An Introduction to Ergodic Theory.* Graduate Texts in Mathematics, vol. 79. Springer-Verlag, 1982.
- K. Petersen. *Ergodic Theory.* Cambridge Studies in Advanced Mathematics, vol. 2. Cambridge University Press, 1983.
- M. Einsiedler and T. Ward. *Ergodic Theory: with a view towards Number Theory.* Graduate Texts in Mathematics, vol. 259. Springer, 2011.
- P.R. Halmos. *Lectures on Ergodic Theory.* Mathematical Society of Japan, 1956. Reprinted by Chelsea, 1960.

Recommended Reading

For a first reading of the ergodic theorems, **Walters** (Chapters 1 and 2) is clear and well-organized. **Petersen** gives a more thorough treatment with excellent motivation. For readers interested in the number-theoretic applications (Weyl's theorem and beyond), **Einsiedler and Ward** is outstanding. **Halmos** is a classic — short, elegant, and still worth reading for its clarity of exposition, though some notation is dated. For the original papers, Birkhoff (1931) and von Neumann (1932) are historically significant and surprisingly readable.

For the physical context and the ergodic hypothesis in statistical mechanics, the survey by Gallavotti and others in *Statistical Mechanics* (Springer, various editions) provides an accessible discussion, as does the treatment in Arnol'd and Avez, *Ergodic Problems of Classical Mechanics* (1968).

Chapter 10

Mixing and Spectral Theory

In Chapter 9 we established ergodicity as the fundamental notion of indecomposability for a measure-preserving system: time averages equal space averages for every integrable observable. Ergodicity, however, is a relatively weak property. It tells us that orbits visit every set of positive measure, but says nothing about how quickly correlations between events at different times decay. In this chapter we introduce two stronger notions — *weak mixing* and *strong mixing* — that quantify the rate at which a system "forgets" its past. We then develop the spectral theory of the Koopman operator, which provides an elegant functional-analytic framework in which these dynamical properties correspond to properties of the spectrum. The interplay between dynamics, measure theory, and operator theory that emerges is one of the most beautiful aspects of ergodic theory, and it will also prove essential when we later analyze the fading-memory properties of reservoir computers.

Throughout this chapter, $(X, \mathcal{B}, \mu, T)$ denotes a measure-preserving system: $(X, \mathcal{B}, \mu)$ is a probability space and $T : X \to X$ is a measure-preserving transformation ($\mu(T^{-1}A) = \mu(A)$ for all $A \in \mathcal{B}$). We assume the probability space is a standard Borel space (a complete separable metric space equipped with its Borel σ-algebra), which covers all examples of interest. We write $\langle f, g\rangle = \int_X f\bar{g}\, d\mu$ for the inner product on $L^2(X, \mu)$.

10.1 Strong Mixing

We begin with the stronger notion, as it is the most intuitive.

Definition 10.1 (Strong mixing). A measure-preserving transformation T is *strongly mixing* (or simply *mixing*) if for all measurable sets $A, B \in \mathcal{B}$,

$$\lim_{n\to\infty} \mu(T^{-n}A \cap B) = \mu(A)\mu(B). \tag{10.1}$$

The interpretation is immediate and compelling. Think of B as the event "the system is currently in the region B" and $T^{-n}A$ as the event "after n steps, the system will be in A." Strong mixing says that for large n, the probability of being in B now and in A after n steps is approximately the product of the individual probabilities — the events become asymptotically independent. The system progressively "forgets" its initial condition.

Remark 10.1. Equation (10.1) can be rewritten in functional-analytic terms. For characteristic functions $f = \mathbf{1}_A$ and $g = \mathbf{1}_B$, condition (10.1) reads $\langle f \circ T^n, g\rangle \to \langle f, 1\rangle\langle 1, g\rangle$. By linearity and density, this is equivalent to requiring that for all $f, g \in L^2(X, \mu)$,

$$\lim_{n\to\infty} \int_X (f \circ T^n) \cdot \bar{g}\, d\mu = \int_X f\, d\mu \cdot \int_X \bar{g}\, d\mu. \tag{10.2}$$

We will revisit this formulation in Section 10.5 when we discuss the Koopman operator.

10.2 Weak Mixing

Strong mixing demands that $\mu(T^{-n}A \cap B) - \mu(A)\mu(B) \to 0$ for *every* $n \to \infty$. Weak mixing relaxes this: we only require convergence to zero in the Cesàro sense (i.e., on average).

Definition 10.3 (Weak mixing). A measure-preserving transformation T is *weakly mixing* if for all measurable sets $A, B \in \mathcal{B}$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \left|\mu(T^{-k}A \cap B) - \mu(A)\mu(B)\right| = 0. \tag{10.3}$$

Since convergence of a sequence of non-negative reals implies Cesàro convergence to the same limit, strong mixing immediately implies weak mixing. The converse is false; we will exhibit a counterexample in Section 10.3.

The following characterization of weak mixing is one of the foundational results in ergodic theory.

Theorem 10.4. The following are equivalent for a measure-preserving system $(X, \mathcal{B}, \mu, T)$:

1. T is weakly mixing.
2. For all $A, B \in \mathcal{B}$,
$$\frac{1}{n} \sum_{k=0}^{n-1} \left|\mu(T^{-k}A \cap B) - \mu(A)\mu(B)\right|^2 \to 0.$$
1. The product transformation $T \times T$ is ergodic on $(X \times X, \mathcal{B} \otimes \mathcal{B}, \mu \times \mu)$.
2. For every ergodic measure-preserving system $(Y, \mathcal{C}, \nu, S)$, the product $T \times S$ is ergodic on $X \times Y$.

Proof. * We prove the equivalence of (1) and (3); the equivalences with (2) and (4) can be found in Walters [Wal82, Theorem 1.24] and Petersen [Pet89, §2.5].

(1) $\Rightarrow$ (3): Suppose T is weakly mixing. To show $T \times T$ is ergodic on $X \times X$, it suffices (by the L^2 characterization of ergodicity from Chapter 9) to show that measurable rectangles satisfy the ergodic condition. Let $A_1, A_2, B_1, B_2 \in \mathcal{B}$. We need to show

$$\frac{1}{n} \sum_{k=0}^{n-1} (\mu \times \mu)\big((T \times T)^{-k}(A_1 \times A_2) \cap (B_1 \times B_2)\big) \to (\mu \times \mu)(A_1 \times A_2) \cdot (\mu \times \mu)(B_1 \times B_2).$$

The left side equals

$$\frac{1}{n} \sum_{k=0}^{n-1} \mu(T^{-k}A_1 \cap B_1) \cdot \mu(T^{-k}A_2 \cap B_2).$$

Define $a_k = \mu(T^{-k}A_1 \cap B_1) - \mu(A_1)\mu(B_1)$ and $b_k = \mu(T^{-k}A_2 \cap B_2) - \mu(A_2)\mu(B_2)$. Then

$$\mu(T^{-k}A_1 \cap B_1)\mu(T^{-k}A_2 \cap B_2) = \big(\mu(A_1)\mu(B_1) + a_k\big)\big(\mu(A_2)\mu(B_2) + b_k\big).$$

Expanding, the Cesàro average becomes

$$\mu(A_1)\mu(B_1)\mu(A_2)\mu(B_2) + \mu(A_2)\mu(B_2) \cdot \frac{1}{n}\sum a_k + \mu(A_1)\mu(B_1) \cdot \frac{1}{n}\sum b_k + \frac{1}{n}\sum a_k b_k.$$

The first two correction terms tend to zero because weak mixing implies, a fortiori, that $\frac{1}{n}\sum a_k \to 0$ (since $\frac{1}{n}\sum |a_k| \to 0$). For the last term, use the Cauchy–Schwarz inequality:

$$\left|\frac{1}{n}\sum_{k=0}^{n-1} a_k b_k\right| \le \left(\frac{1}{n}\sum |a_k|^2\right)^{1/2} \left(\frac{1}{n}\sum |b_k|^2\right)^{1/2}.$$

Since $|a_k| \le 1$, we have $|a_k|^2 \le |a_k|$, so $\frac{1}{n}\sum |a_k|^2 \le \frac{1}{n}\sum |a_k| \to 0$ by weak mixing (and similarly for b_k). Therefore the last term also vanishes, and $T \times T$ is ergodic.

(3) $\Rightarrow$ (1): Suppose $T \times T$ is ergodic. By von Neumann's mean ergodic theorem applied to $T \times T$, for any $F \in L^2(X \times X)$,

$$\frac{1}{n}\sum_{k=0}^{n-1} F \circ (T \times T)^k \to \int F \, d(\mu \times \mu)$$

in $L^2(X \times X)$. Now take $F(x, y) = \mathbf{1}_A(x) \cdot \mathbf{1}_B(y)$ for measurable sets A, B and evaluate $\langle \frac{1}{n}\sum F \circ (T \times T)^k, F\rangle$ on $X \times X$. The left side gives

$$\frac{1}{n}\sum_{k=0}^{n-1} \mu(T^{-k}A \cap A) \cdot \mu(T^{-k}B \cap B),$$

which converges to $\mu(A)^2\mu(B)^2$. Applying this machinery more carefully (see Petersen [Pet89, Theorem 2.11] for the complete argument via the van der Corput lemma), one deduces the full Cesàro convergence in (10.3). □

10.3 The Mixing Hierarchy and Examples

Proposition 10.5 (Mixing hierarchy). For a measure-preserving system, we have the implications

$$\text{strong mixing} \implies \text{weak mixing} \implies \text{ergodic},$$

and neither implication can be reversed.

Proof. Strong mixing implies weak mixing:* If $|\mu(T^{-k}A \cap B) - \mu(A)\mu(B)| \to 0$, then the Cesàro averages of these non-negative terms also tend to zero.

Weak mixing implies ergodic: If T is weakly mixing, then (10.3) holds in particular for $A = B$:

$$\frac{1}{n}\sum_{k=0}^{n-1} |\mu(T^{-k}A \cap A) - \mu(A)^2| \to 0.$$

Hence $\frac{1}{n}\sum_{k=0}^{n-1}\mu(T^{-k}A \cap A) \to \mu(A)^2$. But by the mean ergodic theorem, this Cesàro limit always exists and equals $\mu(A)^2$ if and only if T is ergodic (for all A). More directly: if $T^{-1}A = A$, then each term in the sum is $|\mu(A) - \mu(A)^2|$, so $\mu(A)(1 - \mu(A)) = 0$, giving $\mu(A) \in \{0, 1\}$.

No converse holds: Irrational rotations are ergodic but not weakly mixing (Example 10.8). Chacon's transformation is weakly mixing but not strongly mixing (Example 10.9). □

We now examine the key examples in detail.

10.3.1 Bernoulli Shifts Are Strongly Mixing

Let $\Sigma = \{0, 1, \ldots, r-1\}^{\mathbb{Z}}$ (or $\{0, \ldots, r-1\}^{\mathbb{N}}$) be the full shift space with the product measure $\mu = p^{\mathbb{Z}}$ where $p = (p_0, \ldots, p_{r-1})$ is a probability vector. The shift map σ acts by $(\sigma x)_n = x_{n+1}$.

Theorem 10.6. Every Bernoulli shift is strongly mixing.

Proof. It suffices to verify (10.1) for cylinder sets*, since these generate the σ-algebra and form a $\cap$-stable generating class (an application of the Dynkin π-λ theorem extends the result to all measurable sets).

A cylinder set has the form

$$C = \{x \in \Sigma : x_{i_1} = a_1, \ldots, x_{i_m} = a_m\}$$

for some finite collection of coordinates $i_1 < \cdots < i_m$ and symbols $a_1, \ldots, a_m$. Let A and B be two cylinder sets, say A depends on coordinates in $\{i_1, \ldots, i_m\}$ and B depends on coordinates in $\{j_1, \ldots, j_\ell\}$. Then $\sigma^{-n}A$ depends on coordinates $\{i_1 + n, \ldots, i_m + n\}$.

For n large enough (specifically, $n > \max\{j_1, \ldots, j_\ell\} - \min\{i_1, \ldots, i_m\}$, or more precisely, once the coordinate sets are disjoint), the sets $\sigma^{-n}A$ and B depend on *disjoint* collections of coordinates. By the independence built into the product measure,

$$\mu(\sigma^{-n}A \cap B) = \mu(\sigma^{-n}A) \cdot \mu(B) = \mu(A) \cdot \mu(B).$$

Thus for each pair of cylinder sets, $\mu(\sigma^{-n}A \cap B) = \mu(A)\mu(B)$ for all sufficiently large n. In particular, (10.1) holds. □

10.3.2 Hyperbolic Toral Automorphisms Are Strongly Mixing

Let $A \in \mathrm{GL}(d, \mathbb{Z})$ with $|\det A| = 1$. Then A induces a measure-preserving automorphism $T_A : \mathbb{T}^d \to \mathbb{T}^d$ of the d-dimensional torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ equipped with Lebesgue measure μ. The classic example is Arnold's cat map:

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

which has eigenvalues $\frac{3\pm\sqrt{5}}{2}$, both irrational and off the unit circle.

Theorem 10.7. If $A \in \mathrm{GL}(d, \mathbb{Z})$ is *hyperbolic* — that is, no eigenvalue of A has modulus 1 — then T_A is strongly mixing.

Proof. * We use Fourier analysis on $\mathbb{T}^d$. The characters $e_{\mathbf{m}}(x) = e^{2\pi i\langle \mathbf{m},x\rangle}$ for $\mathbf{m} \in \mathbb{Z}^d$ form an orthonormal basis of $L^2(\mathbb{T}^d)$. It suffices to verify (10.2) for $f = e_{\mathbf{m}}$ and $g = e_{\mathbf{n}}$.

We compute:

$$\langle e_{\mathbf{m}} \circ T_A^n, e_{\mathbf{n}}\rangle = \int_{\mathbb{T}^d} e^{2\pi i\langle \mathbf{m},A^n x\rangle} e^{-2\pi i\langle \mathbf{n},x\rangle}\, d\mu(x) = \int_{\mathbb{T}^d} e^{2\pi i\langle (A^T)^n\mathbf{m}-\mathbf{n},x\rangle}\, d\mu(x),$$

where A^T is the transpose. By orthogonality of characters, this integral equals

$$\begin{cases} 1 & \text{if } (A^T)^n\mathbf{m} = \mathbf{n}, \\ 0 & \text{if } (A^T)^n\mathbf{m} \neq \mathbf{n}. \end{cases}$$

Case 1: $\mathbf{m} = \mathbf{0}$ or $\mathbf{n} = \mathbf{0}$. If $\mathbf{m} = \mathbf{0}$, then $\langle e_{\mathbf{0}} \circ T_A^n, e_{\mathbf{n}}\rangle = \langle 1, e_{\mathbf{n}}\rangle = \delta_{\mathbf{n},\mathbf{0}}$, and $\int e_{\mathbf{0}}\, d\mu \cdot \int \bar{e}_{\mathbf{n}}\, d\mu = \delta_{\mathbf{n},\mathbf{0}}$. So (10.2) holds trivially.

Case 2: $\mathbf{m} \neq \mathbf{0}$. Since A is hyperbolic, all eigenvalues of A^T lie off the unit circle. For any $\mathbf{m} \neq \mathbf{0}$, we have $\|(A^T)^n\mathbf{m}\| \to \infty$ as $n \to \infty$. (Indeed, decompose $\mathbf{m}$ in the basis of generalized eigenvectors of A^T: at least one component lies along an eigenspace with eigenvalue of modulus > 1, and since $\mathbf{m} \neq \mathbf{0}$ and A^T has no eigenvalue of modulus 1, the expanding directions dominate.) Therefore, for any fixed $\mathbf{n} \in \mathbb{Z}^d$, the equation $(A^T)^n\mathbf{m} = \mathbf{n}$ can hold for at most finitely many values of n. Hence for all sufficiently large n,

$$\langle e_{\mathbf{m}} \circ T_A^n, e_{\mathbf{n}}\rangle = 0 = \int e_{\mathbf{m}}\, d\mu \cdot \int \bar{e}_{\mathbf{n}}\, d\mu,$$

since $\int e_{\mathbf{m}}\, d\mu = 0$ when $\mathbf{m} \neq \mathbf{0}$.

Since (10.2) holds for all pairs of characters and these span $L^2(\mathbb{T}^d)$, an approximation argument (using the fact that (10.2) for a dense set in L^2 implies (10.2) for all $f, g \in L^2$, which follows from the uniform boundedness of the operators U_T^n) shows that T_A is strongly mixing. □

Remark 10.2. The key input was that the hyperbolicity of A ensures $\|(A^T)^n\mathbf{m}\| \to \infty$ for $\mathbf{m} \neq \mathbf{0}$. When A has an eigenvalue on the unit circle, the argument breaks down, and indeed the system may fail to be mixing.

10.3.3 Irrational Rotations Are Ergodic but Not Weakly Mixing

Let $X = \mathbb{T}^1 = \mathbb{R}/\mathbb{Z}$ with Lebesgue measure, and let $T_\alpha(x) = x + \alpha \pmod 1$ for irrational α.

Theorem 10.8. The irrational rotation T_α is ergodic but not weakly mixing.

Proof. * Ergodicity was proved in Chapter 9 using Fourier analysis: if $f \circ T_\alpha = f$ a.e. with $f \in L^2$, expanding $f = \sum \hat{f}(n)e^{2\pi inx}$ gives $\hat{f}(n)e^{2\pi in\alpha} = \hat{f}(n)$ for all n, so $\hat{f}(n) = 0$ for $n \neq 0$ (since α is irrational), hence f is a.e. constant.

To show T_α is *not* weakly mixing, we exhibit a non-constant eigenfunction. The function $f(x) = e^{2\pi ix}$ satisfies

$$f(T_\alpha x) = e^{2\pi i(x+\alpha)} = e^{2\pi i\alpha} \cdot e^{2\pi ix} = e^{2\pi i\alpha} f(x).$$

So f is an eigenfunction of the Koopman operator U_{T_α} with eigenvalue $\lambda = e^{2\pi i\alpha} \neq 1$.

We will show in Theorem 10.16 that weak mixing is equivalent to having no eigenvalues other than 1. Since T_α has eigenvalue $e^{2\pi i\alpha} \neq 1$, it is not weakly mixing. □

Remark 10.3. In fact, the full spectrum of U_{T_α} consists of all the eigenvalues $\{e^{2\pi in\alpha} : n \in \mathbb{Z}\}$, each with multiplicity one, with eigenfunctions $e_n(x) = e^{2\pi inx}$. The spectrum is pure point (i.e., L^2 has an orthonormal basis of eigenfunctions). This is the spectral opposite of weak mixing.

10.3.4 Chacon's Transformation: Weakly Mixing but Not Strongly Mixing

Example 10.9 (Chacon's transformation). This is a measure-preserving transformation of $[0, 1]$ constructed by a *cutting-and-stacking* procedure. At stage 0, we start with the interval $[0, 1)$. At each subsequent stage, we cut the current column into three equal pieces, add a *spacer* interval on top of the middle piece, and then stack them: the left piece on the bottom, the middle piece (with spacer) in the middle, the right piece on top. The transformation T maps each point to the point directly above it in the column; spacers are new intervals drawn from a supply of measure.

After infinitely many stages, the construction defines a measure-preserving transformation T on a space of total measure $\frac{3}{2}$ (which can be normalized to a probability space).

Chacon [Cha69] proved that this transformation is weakly mixing. The proof proceeds by verifying that T has no non-constant eigenfunctions (which, by the spectral characterization in Theorem 10.16, is equivalent to weak mixing). The argument uses the detailed combinatorial structure of the cutting-and-stacking construction to show that any eigenfunction must be constant on longer and longer columns, hence constant a.e.

On the other hand, Chacon also showed that T is *not* strongly mixing. The idea is to find sets A for which $\mu(T^{-n}A \cap A)$ does not converge to $\mu(A)^2$ along a specific subsequence. Let A be a level in one of the columns. Along the subsequence $n_k = 3^k$ (the height of the columns at stage k), a non-trivial fraction of A maps back into A in a structured way that prevents the correlations from decaying. More precisely, one can show that $\limsup_n \mu(T^{-n}A \cap A) > \mu(A)^2$ along the subsequence of column heights.

A full proof can be found in Petersen [Pet89, §3.5] or in the original paper of Chacon [Cha69]. This example is of great importance: it demonstrates that the mixing hierarchy is strict.

10.4 The Koopman Operator

The passage from dynamics on the phase space to dynamics on the function space is one of the great organizing ideas of ergodic theory. It replaces a possibly nonlinear transformation T with a *linear* operator.

Definition 10.10 (Koopman operator). Given a measure-preserving transformation $T : X \to X$, the *Koopman operator* $U_T : L^2(X, \mu) \to L^2(X, \mu)$ is defined by

$$U_T f = f \circ T. \tag{10.4}$$

Proposition 10.1. *The Koopman operator has the following properties:*

1. U_T is a linear isometry: $\|U_T f\|_2 = \|f\|_2$ for all $f \in L^2$. 2. U_T is a positive operator: $f \geq 0 \implies U_T f \geq 0$. 3. $U_T \mathbf{1} = \mathbf{1}$ (where $\mathbf{1}$ denotes the constant function 1). 4. If T is invertible (a.e.), then U_T is unitary, with $U_T^* = U_T^{-1} = U_{T^{-1}}$. 5. $U_{T^n} = U_T^n$ for all $n \geq 1$.

Proof. * (1) Since T is measure-preserving:

$$\|U_T f\|_2^2 = \int_X |f(Tx)|^2 \, d\mu(x) = \int_X |f(x)|^2 \, d\mu(x) = \|f\|_2^2,$$

where we used the change-of-variables formula $\int g \circ T \, d\mu = \int g \, d\mu$.

(2)–(3) are immediate.

(4) If T is invertible, $U_{T^{-1}}$ is also a well-defined isometry. We check $U_T U_{T^{-1}} = U_{T^{-1}} U_T = I$: indeed, $U_T U_{T^{-1}} f = f \circ T^{-1} \circ T = f$ a.e. Moreover, for all $f, g \in L^2$:

$$\langle U_T f, g \rangle = \int (f \circ T) \bar{g} \, d\mu = \int f \cdot (\bar{g} \circ T^{-1}) \, d\mu = \langle f, U_{T^{-1}} g \rangle,$$

so $U_T^* = U_{T^{-1}} = U_T^{-1}$, confirming unitarity.

(5) is immediate: $U_T^n f = f \circ T^n = U_{T^n} f$. □

When T is not invertible, U_T is still an isometry, hence $U_T^* U_T = I$, but in general $U_T U_T^* \neq I$ — that is, U_T is a *non-surjective isometry*. In this case one works with the Koopman operator as an isometry rather than a unitary operator, and the spectral theory is slightly more delicate. We will primarily consider the invertible case.

10.5 Spectral Characterization of Mixing Properties

The central results of this chapter connect the dynamical notions of ergodicity, weak mixing, and strong mixing to spectral properties of the Koopman operator U_T. These connections allow us to bring the powerful machinery of functional analysis — in particular, the spectral theorem for unitary operators — to bear on dynamical questions.

10.5.1 The Spectral Theorem for Unitary Operators

We state the result we need; proofs can be found in Reed and Simon [RS80] or Halmos [Hal51].

Theorem 10.12 (Spectral theorem for unitary operators). Let U be a unitary operator on a separable Hilbert space H. For each $f \in H$, there exists a unique finite positive Borel measure σ_f on the unit circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, called the *spectral measure* of f, such that

$$\langle U^n f, f \rangle = \int_{S^1} z^n \, d\sigma_f(z) \quad \text{for all } n \in \mathbb{Z}. \tag{10.5}$$

Moreover, $\sigma_f(S^1) = \|f\|^2$.

More generally, for each pair $f, g \in H$ there exists a complex Borel measure $\sigma_{f,g}$ on S^1 such that

$$\langle U^n f, g \rangle = \int_{S^1} z^n \, d\sigma_{f,g}(z) \quad \text{for all } n \in \mathbb{Z}. \tag{10.6}$$

Every spectral measure can be decomposed (Lebesgue decomposition) into three mutually singular parts: $\sigma_f = \sigma_f^{(pp)} + \sigma_f^{(ac)} + \sigma_f^{(sc)}$, where $\sigma_f^{(pp)}$ is *pure point* (a sum of atoms), $\sigma_f^{(ac)}$ is

absolutely continuous with respect to Lebesgue measure on S^1, and $\sigma_f^{(sc)}$ is *singular continuous* (singular with respect to Lebesgue measure but has no atoms).

Definition 10.1. The *maximal spectral type* of U is a measure class (equivalence class of measures under mutual absolute continuity) $[\sigma]$ on S^1 such that $\sigma_f \ll \sigma$ for every $f \in H$, and σ is minimal with this property.

10.5.2 Spectral Characterizations

We now state and prove the main spectral characterizations.

Theorem 10.14 (Spectral characterization of ergodicity). T is ergodic if and only if $\lambda = 1$ is a simple eigenvalue of U_T (i.e., $\dim \ker(U_T - I) = 1$, spanned by the constant functions).

Proof. * ($\Rightarrow$) If T is ergodic and $U_T f = f$, then $f \circ T = f$ a.e., so f is T-invariant. By ergodicity, f is constant a.e. Thus the eigenspace for eigenvalue 1 is $\mathbb{C} \cdot \mathbf{1}$.

($\Leftarrow$) If $\lambda = 1$ is simple, then every T-invariant function f (i.e., $f \circ T = f$ a.e.) belongs to $\mathbb{C} \cdot \mathbf{1}$, so f is constant a.e. This is precisely ergodicity. □

Lemma 10.15. Suppose $U_T f = \lambda f$ for some $f \in L^2 \setminus \{0\}$ and $\lambda \in \mathbb{C}$. Then $|\lambda| = 1$ and $|f|$ is T-invariant a.e. If in addition T is ergodic, then $|f|$ is constant a.e.

Proof. * Since U_T is an isometry, $\|f\| = \|U_T f\| = |\lambda| \|f\|$, so $|\lambda| = 1$. Moreover, $|f| \circ T = |f \circ T| = |\lambda f| = |f|$ a.e. □

Theorem 10.16 (Spectral characterization of weak mixing). T is weakly mixing if and only if the only eigenvalue of U_T is $\lambda = 1$ (with eigenspace $\mathbb{C} \cdot \mathbf{1}$). Equivalently, U_T restricted to $L^2_0 = \{f \in L^2 : \int f \, d\mu = 0\}$ has *continuous spectrum* (no eigenvalues).

Proof. * ($\Rightarrow$) Suppose T is weakly mixing and $U_T f = \lambda f$ for some $f \neq 0$. Then for any measurable B,

$$\mu(T^{-k} A \cap B) - \mu(A)\mu(B)$$

with A chosen to involve f leads to a contradiction unless $\lambda = 1$. More precisely, let $g \in L^2$. Then

$$\langle U_T^k f, g \rangle = \lambda^k \langle f, g \rangle.$$

Now, weak mixing (in its L^2 formulation) gives

$$\frac{1}{n} \sum_{k=0}^{n-1} \left| \langle U_T^k f, g \rangle - \langle f, \mathbf{1} \rangle \langle \mathbf{1}, g \rangle \right| \to 0.$$

Taking $g = f$ and noting $\langle U_T^k f, f \rangle = \lambda^k \|f\|^2$, we get

$$\frac{1}{n} \sum_{k=0}^{n-1} \left| \lambda^k \|f\|^2 - \langle f, \mathbf{1} \rangle \langle \mathbf{1}, f \rangle \right| \to 0.$$

If $\lambda \neq 1$, set $c = \langle f, \mathbf{1} \rangle \langle \mathbf{1}, f \rangle / \|f\|^2$. The above says $\frac{1}{n} \sum |\lambda^k - c| \to 0$. Since $|\lambda| = 1$ and $\lambda \neq 1$, the sequence λ^k is either periodic (if λ is a root of unity) or equidistributed on S^1 (if not). In

either case, it does not Cesàro-converge to any single complex number in modulus unless $\|f\|^2$ conspires to make everything vanish — but let us be more careful.

If λ is a root of unity, say $\lambda^p = 1$, then $\frac{1}{n}\sum_{k=0}^{n-1}|\lambda^k - c| \to \frac{1}{p}\sum_{j=0}^{p-1}|\lambda^j - c|$. For this to be zero, we need $c = \lambda^j$ for every j, which is impossible for $p > 1$. Contradiction.

If λ is not a root of unity, then $\{\lambda^k\}$ is equidistributed on S^1 by Weyl's theorem, so $\frac{1}{n}\sum|\lambda^k - c| \to \int_{S^1}|z-c|\,dm(z) > 0$ for any $c \in \mathbb{C}$. Contradiction.

Therefore $\lambda = 1$ and f is constant.

($\Leftarrow$) Suppose U_T has no eigenvalues on L^2_0. Let $f \in L^2_0$. The spectral measure σ_f has no atoms (since atoms of σ_f correspond to eigenvalues of U_T). By (10.5),

$$\langle U_T^k f, f\rangle = \int_{S^1} z^k\,d\sigma_f(z).$$

We need to show $\frac{1}{n}\sum_{k=0}^{n-1}|\langle U_T^k f, f\rangle| \to 0$. By the Cauchy–Schwarz inequality and Parseval-type reasoning,

$$\frac{1}{n}\sum_{k=0}^{n-1}|\langle U_T^k f, f\rangle|^2 = \frac{1}{n}\sum_{k=0}^{n-1}\left|\int_{S^1} z^k\,d\sigma_f(z)\right|^2 = \int_{S^1}\int_{S^1}\frac{1}{n}\sum_{k=0}^{n-1}(z\bar{w})^k\,d\sigma_f(z)\,d\sigma_f(w).$$

The inner sum is $\frac{1}{n}\sum_{k=0}^{n-1}(z\bar{w})^k$, which converges to $\mathbf{1}_{\{z=w\}}$ pointwise (it equals 1 when $z = w$ and converges to 0 when $z \neq w$). By bounded convergence,

$$\frac{1}{n}\sum_{k=0}^{n-1}|\langle U_T^k f, f\rangle|^2 \to \int_{S^1}\int_{S^1}\mathbf{1}_{\{z=w\}}\,d\sigma_f(z)\,d\sigma_f(w) = \sum_{\lambda\in S^1}\sigma_f(\{\lambda\})^2.$$

Since σ_f has no atoms, this is 0. Thus $\frac{1}{n}\sum|\langle U_T^k f, f\rangle|^2 \to 0$, which implies $\frac{1}{n}\sum|\langle U_T^k f, f\rangle| \to 0$ (since Cesàro convergence of $|a_k|^2$ to 0 implies Cesàro convergence of $|a_k|$ to 0 for bounded sequences). A polarization argument extends this to all pairs $f, g \in L^2_0$, giving weak mixing. □

Theorem 10.17 (Spectral characterization of strong mixing). T is strongly mixing if and only if for all $f, g \in L^2(X,\mu)$,

$$\langle U_T^n f, g\rangle \to \langle f, \mathbf{1}\rangle\langle \mathbf{1}, g\rangle \quad \text{as } n \to \infty. \tag{10.7}$$

Equivalently, T is strongly mixing if and only if for every $f \in L^2_0$, the spectral measure σ_f is a *Rajchman measure* (i.e., its Fourier coefficients tend to zero: $\hat{\sigma}_f(n) \to 0$).

Proof. * Condition (10.7) is simply the restatement of (10.2) and is thus equivalent to strong mixing by definition (and the approximation argument in Remark 10.2).

For the spectral reformulation: restricting to L^2_0, condition (10.7) becomes $\langle U_T^n f, g\rangle \to 0$ for all $f, g \in L^2_0$. In particular, taking $g = f$:

$$\hat{\sigma}_f(n) = \int_{S^1} z^n\,d\sigma_f(z) = \langle U_T^n f, f\rangle \to 0.$$

This is precisely the condition that σ_f is a Rajchman measure. Conversely, if all spectral measures are Rajchman, then by polarization $\langle U_T^n f, g\rangle \to 0$ for all $f, g \in L^2_0$. □

Remark 10.4. The spectral characterizations reveal a clean hierarchy:

| Property | Spectral condition on L^2_0 | |—|—| | Ergodic | 1 is not an eigenvalue of $U_T\|_{L^2_0}$ | | Weakly mixing | $U_T\|_{L^2_0}$ has no eigenvalues (continuous spectrum) | | Strongly mixing | $\hat{\sigma}_f(n) \to 0$ for all $f \in L^2_0$ (Rajchman spectral measures) |

Note that having no eigenvalues (continuous spectrum) does not force the Fourier coefficients of the spectral measures to decay. For instance, singular continuous measures need not be Rajchman. This is the spectral reason why weak mixing does not imply strong mixing.

10.6 Decay of Correlations

The notion of mixing has a natural quantitative refinement: how *fast* do correlations decay?

Definition 10.19 (Correlation function). For observables $f, g \in L^2(X, \mu)$, the *correlation function* (or *correlation coefficient*) is

$$C_n(f, g) = \int_X (f \circ T^n) \cdot \bar{g}\, d\mu - \int_X f\, d\mu \cdot \int_X \bar{g}\, d\mu = \langle U_T^n f, g\rangle - \langle f, \mathbf{1}\rangle\langle \mathbf{1}, g\rangle. \tag{10.8}$$

Mixing is the statement that $C_n(f, g) \to 0$ for all $f, g \in L^2$. The *rate* of decay is of great importance in applications.

Definition 10.20 (Exponential mixing). A system is *exponentially mixing* for a class of observables $\mathcal{F}$ if there exist constants $C > 0$ and $0 < \rho < 1$ such that for all $f, g \in \mathcal{F}$,

$$|C_n(f, g)| \leq C\|f\|_{\mathcal{F}}\|g\|_{\mathcal{F}}\, \rho^n, \tag{10.9}$$

where $\|\cdot\|_{\mathcal{F}}$ is an appropriate norm (typically a Hölder or Sobolev norm, stronger than the L^2 norm).

Remark 10.5. The class $\mathcal{F}$ matters. No system can be exponentially mixing for *all* L^2 functions, because L^2 functions can be arbitrarily rough. One typically works with Hölder continuous functions, functions of bounded variation, or smooth functions.

Example 10.22 (Exponential mixing for hyperbolic systems). The Arnold cat map and, more generally, Anosov diffeomorphisms are exponentially mixing for Hölder continuous observables. This was proved by Sinai [Sin72] and later refined by Bowen and Ruelle using symbolic dynamics and transfer operators. For the cat map T_A on $\mathbb{T}^2$ with matrix A having eigenvalues λ_1, λ_2 with $|\lambda_1| > 1 > |\lambda_2|$, the rate of exponential decay is governed by $|\lambda_2|$ (or equivalently $|\lambda_1|^{-1}$).

Example 10.23 (Polynomial mixing). Some systems mix, but only at a polynomial rate: $|C_n(f, g)| = O(n^{-\alpha})$ for some $\alpha > 0$. This is typical of systems with "intermittent" behavior, such as the Manneville–Pomeau map $T(x) = x + x^{1+\alpha} \pmod 1$ for $\alpha \in (0, 1)$, where correlations of Hölder observables decay as $O(n^{1-1/\alpha})$.

10.6.1 Connection to Statistical Mechanics and Reservoir Computing

The decay of correlations is central to statistical mechanics, where it governs the approach to equilibrium and the validity of the central limit theorem for dynamical systems. If $C_n(f, f) =$

$O(\rho^n)$ (exponential decay), then for a large class of observables, the time averages $\frac{1}{n}\sum_{k=0}^{n-1} f \circ T^k$ satisfy a central limit theorem: properly normalized, they converge in distribution to a Gaussian.

For reservoir computing, the decay of correlations is directly connected to the *fading memory property.* A reservoir with state dynamics driven by a mixing dynamical system will "forget" its initial state at a rate governed by the correlation decay. We will make this connection precise in Chapter 14, where we show that exponential mixing of the reservoir dynamics ensures that the echo state property holds and that the reservoir's output depends primarily on recent inputs.

10.7 Multiple Mixing and Rokhlin's Problem

Definition 10.24 (k-mixing). A measure-preserving transformation T is *k-mixing *(or* mixing of order k*) if for all measurable sets $A_0, A_1, \ldots, A_k$,

$$\mu\Big(A_0 \cap T^{-n_1}A_1 \cap T^{-(n_1+n_2)}A_2 \cap \cdots \cap T^{-(n_1+\cdots+n_k)}A_k\Big) \to \mu(A_0)\mu(A_1)\cdots\mu(A_k)$$

as $\min(n_1, n_2, \ldots, n_k) \to \infty$.

Note that 1-mixing is ordinary (strong) mixing, and k-mixing implies $(k-1)$-mixing.

Rokhlin's Problem (1949). Does 2-mixing imply 3-mixing?

This is one of the oldest and most famous open problems in ergodic theory. Despite more than seven decades of work, it remains unresolved in full generality.

Partial results:

1. **Bernoulli shifts** are mixing of all orders, by the same independence argument used in Theorem 10.6.
2. **Hyperbolic toral automorphisms** are mixing of all orders. This follows from higher-order versions of the Fourier-analytic argument in Theorem 10.7.
3. **Theorem 10.25 (Host, 1991 [Hos91]).** If T is mixing and has *singular spectrum* (i.e., the maximal spectral type of $U_T|_{L^2_0}$ is singular with respect to Lebesgue measure on S^1), then T is mixing of all orders.

Host's theorem is a remarkable result: it covers a very large class of mixing transformations, since "most" mixing transformations (in various senses) have singular spectrum. In particular, all known examples of mixing rank-one transformations have singular spectrum, so they are mixing of all orders by Host's theorem.

1. **Kalikow (1984) [Kal84]** proved that mixing rank-one transformations are mixing of all orders, predating Host's more general result.
2. **Ryzhikov (1993)** extended Host's theorem in various directions.

No counterexample to Rokhlin's question is known. Many experts believe that 2-mixing does in fact imply mixing of all orders, but a proof for the general case remains elusive.

10.8 Summary

We summarize the main results of this chapter.

The dynamical properties of a measure-preserving system arrange in a strict hierarchy:

$$\text{strong mixing} \implies \text{weak mixing} \implies \text{ergodic},$$

with neither implication reversible. These properties correspond precisely to spectral properties of the Koopman operator U_T:

- **Ergodicity** $\iff$ 1 is a simple eigenvalue of U_T.
- **Weak mixing** $\iff$ $U_T|_{L_0^2}$ has no eigenvalues (purely continuous spectrum).
- **Strong mixing** $\iff$ $\langle U_T^n f, g\rangle \to \langle f, \mathbf{1}\rangle\langle \mathbf{1}, g\rangle$ for all f, g (Rajchman spectral measures).

The correlation function $C_n(f, g)$ quantifies how quickly a mixing system forgets its past, with exponential mixing being characteristic of hyperbolic dynamics and directly relevant to the fading memory properties of reservoir computers.

Exercises

Exercise 10.1. Let T be a measure-preserving transformation on $(X, \mathcal{B}, \mu)$. Show that T is strongly mixing if and only if for all $f, g \in L^2(X, \mu)$,

$$\int_X (f \circ T^n) \cdot g \, d\mu \to \int_X f \, d\mu \cdot \int_X g \, d\mu.$$

(*Hint:* Start with characteristic functions and use an approximation argument.)

Exercise 10.2. Let $\sigma : \{0, 1\}^{\mathbb{N}} \to \{0, 1\}^{\mathbb{N}}$ be the one-sided shift with the $(1/2, 1/2)$ Bernoulli measure. Find the eigenvalues and eigenfunctions of the Koopman operator U_σ on L^2, or prove that the only eigenvalue is 1. Is the one-sided shift weakly mixing?

Exercise 10.3. Let T be a measure-preserving transformation. Show directly (without using the spectral characterization) that if $T \times T$ is ergodic, then T is ergodic. Then show by example that the converse fails: exhibit a transformation that is ergodic but for which $T \times T$ is not ergodic. (*Hint:* For the counterexample, consider an irrational rotation.)

Exercise 10.4. Show that T is strongly mixing if and only if $T^{-n}A \to \mu(A)$ in the *weak topology* on measurable sets, in the following sense: for every $B \in \mathcal{B}$, $\mu(T^{-n}A \cap B) \to \mu(A)\mu(B)$. Interpret this as saying that $T^{-n}A$ becomes "uniformly spread out" over X.

Exercise 10.5. Let T_α be an irrational rotation of $\mathbb{T}^1$. For $f(x) = e^{2\pi i m x}$ with $m \in \mathbb{Z}$, compute the spectral measure σ_f explicitly. Verify that it is a pure point measure and identify its support.

Exercise 10.6. Consider the doubling map $T(x) = 2x \pmod 1$ on $[0, 1)$ with Lebesgue measure.

(a) Show that T is strongly mixing. (*Hint:* Show that T is isomorphic to the one-sided $(1/2, 1/2)$ Bernoulli shift via the binary expansion map.)

(b) For $f(x) = e^{2\pi i x}$ and $g(x) = e^{2\pi i x}$, compute $C_n(f, g)$ explicitly and determine its rate of decay.

(c) Show that for Lipschitz observables f, g, the doubling map is exponentially mixing: there exists $0 < \rho < 1$ such that $|C_n(f, g)| \leq C \cdot \mathrm{Lip}(f) \cdot \|g\|_\infty \cdot \rho^n$. What is the optimal ρ?

Exercise 10.7. Suppose T is weakly mixing and S is an arbitrary ergodic measure-preserving transformation. Prove that $T \times S$ is ergodic. (*Hint:* Use the characterization from Theorem 10.4(4), or prove it from the spectral characterization.)

Exercise 10.8. Let T be a measure-preserving transformation of a compact metric space X that is an *isometry*: $d(Tx, Ty) = d(x, y)$ for all x, y. Show that T is never strongly mixing (unless X is a single point). (*Hint:* Consider small balls.)

Recommended Reading

The following references provide thorough treatments of the material in this chapter:

- **Walters, P.** *An Introduction to Ergodic Theory*, Springer, 1982 [Wal82]. Chapter 1, Sections 5–7 cover mixing and spectral theory with full proofs. An excellent first reference.
- **Petersen, K.** *Ergodic Theory*, Cambridge University Press, 1989 [Pet89]. Chapters 2–3 treat mixing and spectral theory in depth. Section 3.5 contains a complete treatment of Chacon's transformation.
- **Cornfeld, I.P., Fomin, S.V., Sinai, Ya.G.** *Ergodic Theory*, Springer, 1982 [CFS82]. A comprehensive reference with extensive coverage of spectral theory, multiple mixing, and entropy.
- **Halmos, P.R.** *Lectures on Ergodic Theory*, AMS Chelsea, 1956 (reprinted 2006) [Hal56]. A classic concise treatment. The spectral approach to ergodic theory is developed beautifully.
- **Nadkarni, M.G.** *Spectral Theory of Dynamical Systems*, Birkhäuser, 1998 [Nad98]. Focuses specifically on the spectral theory of measure-preserving transformations. Contains detailed treatments of spectral types, multiplicity theory, and their dynamical implications. Particularly recommended for the material in Sections 10.5–10.7.
- **Host, B.** "Mixing of all orders and pairwise independent joinings of systems without singular part," *Israel Journal of Mathematics*, 76 (1991), 289–298 [Hos91]. The original paper proving Theorem 10.25.
- **Chacon, R.V.** "Weakly mixing transformations which are not strongly mixing," *Proceedings of the AMS*, 22 (1969), 559–562 [Cha69]. The original construction of a weakly mixing, non-strongly mixing transformation.

Chapter 11

Entropy

11.1 Introduction

How unpredictable is a dynamical system? If we observe a measure-preserving transformation T acting on a probability space, how much new information does each application of T reveal? These are the questions that the theory of entropy answers.

The concept of entropy enters dynamical systems from two directions. From information theory, Shannon (1948) introduced entropy as a measure of uncertainty in a random variable. From the dynamical systems side, Kolmogorov (1958) and Sinai (1959) adapted Shannon's ideas to define a numerical invariant of measure-preserving transformations—the *measure-theoretic entropy* (or *Kolmogorov–Sinai entropy*). This invariant measures the asymptotic rate at which a dynamical system generates information.

Entropy has proven to be one of the most powerful tools in ergodic theory. It provides a way to distinguish systems that are otherwise difficult to tell apart; it connects measure-theoretic and topological dynamics through the variational principle; and it relates to Lyapunov exponents through Pesin's formula, tying together themes from Chapters 6 and 10.

We begin with Shannon's information-theoretic foundations, build up to the Kolmogorov–Sinai entropy of a measure-preserving transformation, and then explore topological entropy and the deep theorems that connect these ideas.

Throughout this chapter, log denotes the natural logarithm unless otherwise stated. Some authors use $\log_2$; the choice of base amounts to a choice of units (nats vs. bits) and does not affect any of the structural results.

11.2 Shannon Entropy

11.2.1 Motivation from Information Theory

Consider a random variable X taking values in a finite set $\{x_1, x_2, \ldots, x_n\}$ with probabilities $p_1, p_2, \ldots, p_n$ respectively, where $p_i \geq 0$ and $\sum_{i=1}^{n} p_i = 1$. We seek a function $H(X)$ that quantifies the *uncertainty* or *information content* of X.

What properties should such a function have?

1. **Continuity**: H should depend continuously on the probabilities $p_1, \ldots, p_n$.

2. **Maximality at uniformity**: When all outcomes are equally likely ($p_i = 1/n$ for all i), uncertainty should be maximal.
3. **Additivity for independent events**: If X and Y are independent random variables, then $H(X, Y) = H(X) + H(Y)$.

Shannon proved that, up to a positive multiplicative constant, there is a unique function satisfying these axioms.

Definition 11.1 (Shannon entropy)**.** The *Shannon entropy* of a discrete random variable X with probability distribution $(p_1, \ldots, p_n)$ is

$$H(X) = -\sum_{i=1}^{n} p_i \log p_i,$$

where we adopt the convention $0 \log 0 = 0$ (justified by $\lim_{p \to 0^+} p \log p = 0$).

The quantity $-\log p_i$ is called the *information content* (or *surprisal*) of the outcome x_i. Rare events carry more information; certain events carry none. The entropy $H(X)$ is the expected information content:

$$H(X) = \mathbb{E}[-\log p_X] = \sum_{i=1}^{n} p_i(-\log p_i).$$

Example 11.1. Consider a coin flip with $P(\text{heads}) = p$ and $P(\text{tails}) = 1 - p$. The entropy is

$$H(p) = -p \log p - (1 - p) \log(1 - p).$$

This function equals 0 at $p = 0$ and $p = 1$ (deterministic outcomes), and achieves its maximum of $\log 2$ at $p = 1/2$ (fair coin).

11.2.2 Basic Properties

Proposition 11.1. *(Properties of Shannon entropy). Let X be a random variable taking n values with probabilities $(p_1, \ldots, p_n)$. Then:*

(i) $H(X) \geq 0$, with equality if and only if some $p_i = 1$.

(ii) $H(X) \leq \log n$, with equality if and only if $p_i = 1/n$ for all i.

(iii) H is a concave function of the probability vector $(p_1, \ldots, p_n)$.

Proof. *

(i) Since $0 \leq p_i \leq 1$, we have $\log p_i \leq 0$, so $-p_i \log p_i \geq 0$ for each i. Thus $H(X) \geq 0$. We have $H(X) = 0$ if and only if $-p_i \log p_i = 0$ for every i, which occurs if and only if each p_i is either 0 or 1. Since the p_i sum to 1, exactly one must equal 1.

(ii) We use the inequality $\log x \leq x - 1$ for $x > 0$, with equality iff $x = 1$. For each i with $p_i > 0$:

$$\log \frac{1/n}{p_i} \leq \frac{1/n}{p_i} - 1.$$

Multiplying by p_i and summing:

$$\sum_{i=1}^{n} p_i \log \frac{1/n}{p_i} \leq \sum_{i=1}^{n} p_i \left(\frac{1/n}{p_i} - 1 \right) = \sum_{i=1}^{n} \frac{1}{n} - \sum_{i=1}^{n} p_i = 1 - 1 = 0.$$

Expanding the left side:

$$\sum_{i=1}^{n} p_i \log(1/n) - \sum_{i=1}^{n} p_i \log p_i \leq 0,$$

which gives $-\log n + H(X) \leq 0$, i.e., $H(X) \leq \log n$. Equality holds iff $1/(np_i) = 1$ for all i, i.e., $p_i = 1/n$.

(iii) Let $\varphi(x) = -x \log x$ for $x > 0$ and $\varphi(0) = 0$. Then $\varphi''(x) = -1/x < 0$ for $x > 0$, so φ is strictly concave. Since $H(p_1, \ldots, p_n) = \sum_{i=1}^{n} \varphi(p_i)$ is a sum of concave functions of the p_i, and each p_i is a linear (hence concave) function of the probability vector, H is concave. □

11.2.3 Entropy of a Partition

We now translate these ideas into the language of measure theory, which is the natural setting for ergodic theory.

Definition 11.2 (Measurable partition)**.** Let $(X, \mathcal{B}, \mu)$ be a probability space. A *finite measurable partition* of X is a finite collection $\alpha = \{A_1, A_2, \ldots, A_n\}$ of pairwise disjoint measurable sets with $\bigcup_{i=1}^{n} A_i = X$.

A partition α represents a finite-resolution observation of the system: we can determine which atom A_i a point x belongs to, but nothing finer. The entropy of α quantifies how much information this observation provides.

Definition 11.3 (Entropy of a partition)**.** The *entropy* of a finite measurable partition $\alpha = \{A_1, \ldots, A_n\}$ with respect to a probability measure μ is

$$H(\alpha) = H_\mu(\alpha) = -\sum_{A \in \alpha} \mu(A) \log \mu(A).$$

This is simply the Shannon entropy of the probability distribution $(\mu(A_1), \ldots, \mu(A_n))$.

Definition 11.4 (Refinement and join)**.** Let α and β be finite measurable partitions.

- We say α *refines* β (written $\alpha \succeq \beta$ or $\beta \preceq \alpha$) if every atom of β is a union of atoms of α. - The *join* (or *common refinement*) $\alpha \vee \beta$ is the partition whose atoms are all nonempty sets of the form $A \cap B$ with $A \in \alpha$ and $B \in \beta$.

If $\alpha \succeq \beta$, then α is at least as fine as β: observing through α provides at least as much information as observing through β.

Proposition 11.2. *If $\alpha \succeq \beta$, then $H(\alpha) \geq H(\beta)$.*

Proof. * Each atom B_j of β is a disjoint union of atoms of α: say $B_j = \bigcup_{i \in I_j} A_i$, where the index sets I_j partition $\{1, \ldots, n\}$. Within each B_j, the concavity of $\varphi(x) = -x \log x$ gives (by Jensen's inequality applied to the identity):

$$-\mu(B_j)\log\mu(B_j) = \varphi\left(\sum_{i\in I_j}\mu(A_i)\right) \le \sum_{i\in I_j}\varphi(\mu(A_i)) = -\sum_{i\in I_j}\mu(A_i)\log\mu(A_i),$$

where we have used the fact that φ is concave and φ of a sum $\ge$ the same sum when each summand is non-negative (this actually requires the *superadditivity* of φ on non-negative reals, which is the correct direction here, or one can argue via the grouping axiom for entropy). Summing over j:

$$H(\beta) = \sum_j \varphi(\mu(B_j)) \le \sum_j \sum_{i\in I_j} \varphi(\mu(A_i)) = H(\alpha). \quad \blacksquare$$

□

11.3 Conditional Entropy and Joint Entropy

11.3.1 Conditional Entropy

Definition 11.5 (Conditional entropy)**.** Let $\alpha = \{A_1, \ldots, A_m\}$ and $\beta = \{B_1, \ldots, B_n\}$ be finite measurable partitions of $(X, \mathcal{B}, \mu)$. The *conditional entropy* of α given β is

$$H(\alpha \mid \beta) = -\sum_{j=1}^{n}\sum_{i=1}^{m} \mu(A_i \cap B_j)\log\frac{\mu(A_i \cap B_j)}{\mu(B_j)},$$

where terms with $\mu(B_j) = 0$ are omitted. Equivalently:

$$H(\alpha \mid \beta) = \sum_{j=1}^{n} \mu(B_j)\, H(\alpha \mid B_j),$$

where $H(\alpha \mid B_j) = -\sum_{i=1}^{m} \frac{\mu(A_i\cap B_j)}{\mu(B_j)} \log \frac{\mu(A_i\cap B_j)}{\mu(B_j)}$ is the entropy of α computed with respect to the conditional measure $\mu(\cdot \mid B_j)$.

The conditional entropy measures how much additional information α provides once β is already known.

11.3.2 The Chain Rule

Theorem 11.3 (Chain rule)**.** *For finite measurable partitions α and β:*

$$H(\alpha \vee \beta) = H(\beta) + H(\alpha \mid \beta).$$

Proof. * We compute directly:

$$H(\alpha \vee \beta) = -\sum_{i,j} \mu(A_i \cap B_j)\log\mu(A_i \cap B_j).$$

Writing $\mu(A_i \cap B_j) = \mu(B_j) \cdot \frac{\mu(A_i\cap B_j)}{\mu(B_j)}$ and using $\log(ab) = \log a + \log b$:

$$H(\alpha \vee \beta) = -\sum_{i,j} \mu(A_i \cap B_j) \left[\log \mu(B_j) + \log \frac{\mu(A_i \cap B_j)}{\mu(B_j)} \right].$$

Splitting the sum:

$$= -\sum_j \log \mu(B_j) \sum_i \mu(A_i \cap B_j) \; - \; \sum_{i,j} \mu(A_i \cap B_j) \log \frac{\mu(A_i \cap B_j)}{\mu(B_j)}.$$

Since $\sum_i \mu(A_i \cap B_j) = \mu(B_j)$, the first term is $-\sum_j \mu(B_j) \log \mu(B_j) = H(\beta)$. The second term is $H(\alpha \mid \beta)$ by definition. □

By symmetry of the join ($\alpha \vee \beta = \beta \vee \alpha$), we also have

$$H(\alpha \vee \beta) = H(\alpha) + H(\beta \mid \alpha).$$

Corollary 11.4 (Conditioning reduces entropy)**.**

$$H(\alpha \mid \beta) \leq H(\alpha),$$

with equality if and only if α and β are independent (i.e., $\mu(A_i \cap B_j) = \mu(A_i)\mu(B_j)$ for all i, j).

Proof. * We have $H(\alpha \vee \beta) \leq H(\alpha) + H(\beta)$ (subadditivity, which follows from the log-sum inequality or from the non-negativity of mutual information). Combining with the chain rule:

$$H(\beta) + H(\alpha \mid \beta) = H(\alpha \vee \beta) \leq H(\alpha) + H(\beta),$$

so $H(\alpha \mid \beta) \leq H(\alpha)$.

For the subadditivity $H(\alpha \vee \beta) \leq H(\alpha) + H(\beta)$, one applies the inequality $\log x \leq x - 1$ as follows. Let $p_{ij} = \mu(A_i \cap B_j)$, $a_i = \mu(A_i)$, $b_j = \mu(B_j)$. Then:

$$H(\alpha) + H(\beta) - H(\alpha \vee \beta) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{a_i b_j}.$$

Wait—that is $-\sum_{i,j} p_{ij} \log \frac{a_i b_j}{p_{ij}}$. Applying $\log x \leq x - 1$ with $x = \frac{a_i b_j}{p_{ij}}$:

$$-\sum_{i,j} p_{ij} \log \frac{a_i b_j}{p_{ij}} \geq -\sum_{i,j} p_{ij} \left(\frac{a_i b_j}{p_{ij}} - 1 \right) = -\sum_{i,j} a_i b_j + \sum_{i,j} p_{ij} = -1 + 1 = 0.$$

So $H(\alpha) + H(\beta) - H(\alpha \vee \beta) \geq 0$. Equality holds iff $a_i b_j / p_{ij} = 1$ for all i, j, i.e., α and β are independent. □

11.3.3 Further Properties

The following properties will be needed in the construction of measure-theoretic entropy.

Proposition 11.5. *(i) $H(\alpha \mid \beta) \geq 0$.*

(ii) If $\beta \preceq \gamma$ (i.e., γ refines β), then $H(\alpha \mid \gamma) \leq H(\alpha \mid \beta)$.

(iii) For any partitions α, β, γ: $H(\alpha \vee \beta \mid \gamma) \leq H(\alpha \mid \gamma) + H(\beta \mid \gamma)$.

(iv) (Chain rule, general form): $H(\alpha \vee \beta \mid \gamma) = H(\alpha \mid \gamma) + H(\beta \mid \alpha \vee \gamma)$.

Part (ii) says that conditioning on more information can only reduce (or maintain) uncertainty. Part (iii) is conditional subadditivity. We omit the proofs, which follow the same techniques as above.

—

11.4 Measure-Theoretic Entropy (Kolmogorov–Sinai Entropy)

We now come to the central definition of this chapter. The idea is to measure how fast a dynamical system generates information by tracking how the complexity of iterated partitions grows.

11.4.1 Setup

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving dynamical system: $T\colon X \to X$ is measurable, μ is a probability measure, and $\mu(T^{-1}B) = \mu(B)$ for all $B \in \mathcal{B}$.

Given a finite partition α, define its *iterates* under T:

$$T^{-k}\alpha = \{T^{-k}A : A \in \alpha\}.$$

Since T is measure-preserving, $\mu(T^{-k}A) = \mu(A)$, so $H(T^{-k}\alpha) = H(\alpha)$.

The join $\bigvee_{k=0}^{n-1} T^{-k}\alpha$ is the partition whose atoms are sets of the form

$$A_{i_0} \cap T^{-1}A_{i_1} \cap \cdots \cap T^{-(n-1)}A_{i_{n-1}}, \qquad A_{i_j} \in \alpha.$$

A point x and a point y lie in the same atom of this partition if and only if x and y visit the same sequence of atoms of α under the first n iterates of T: that is, $T^k(x)$ and $T^k(y)$ belong to the same atom of α for $k = 0, 1, \ldots, n-1$. Thus, $\bigvee_{k=0}^{n-1} T^{-k}\alpha$ encodes the trajectory of a point up to time $n-1$, as seen through the lens of α.

The entropy $H\left(\bigvee_{k=0}^{n-1} T^{-k}\alpha\right)$ measures the total uncertainty in an n-step trajectory. The entropy *rate* is the per-step contribution:

11.4.2 Definition and Existence of the Limit

Theorem 11.12. Let T be a measure-preserving transformation on $(X, \mathcal{B}, \mu)$ and let α be a finite measurable partition. Define

$$H_n = H\left(\bigvee_{k=0}^{n-1} T^{-k}\alpha\right).$$

Then:

(i) The sequence (H_n) is *subadditive*: $H_{m+n} \leq H_m + H_n$ for all $m, n \geq 1$.

(ii) The limit

$$h_\mu(T, \alpha) = \lim_{n\to\infty} \frac{H_n}{n}$$

exists and equals $\inf_{n\geq 1} \frac{H_n}{n}$.

Proof. *

(i) Write $\bigvee_{k=0}^{m+n-1} T^{-k}\alpha = \bigvee_{k=0}^{m-1} T^{-k}\alpha \ \vee \ \bigvee_{k=m}^{m+n-1} T^{-k}\alpha$. By subadditivity of entropy (Corollary 11.10 and the discussion preceding it):

$$H_{m+n} \leq H\left(\bigvee_{k=0}^{m-1} T^{-k}\alpha\right) + H\left(\bigvee_{k=m}^{m+n-1} T^{-k}\alpha\right) = H_m + H\left(\bigvee_{k=0}^{n-1} T^{-(k+m)}\alpha\right).$$

Now, since T is measure-preserving, the map T^{-m} preserves the measure of every set, so:

$$H\left(\bigvee_{k=0}^{n-1} T^{-(k+m)}\alpha\right) = H\left(T^{-m}\bigvee_{k=0}^{n-1} T^{-k}\alpha\right) = H\left(\bigvee_{k=0}^{n-1} T^{-k}\alpha\right) = H_n.$$

Therefore $H_{m+n} \leq H_m + H_n$.

(ii) This is a direct application of Fekete's lemma.

□

Lemma 11.6 (Fekete)**.** *If $(a_n)_{n\geq 1}$ is a subadditive sequence of real numbers (i.e., $a_{m+n} \leq a_m + a_n$), then $\lim_{n\to\infty} a_n/n$ exists (in $[-\infty, \infty)$) and equals $\inf_{n\geq 1} a_n/n$.*

Proof of Fekete's lemma. of Fekete's lemma.* Fix $m \geq 1$. For any n, write $n = qm + r$ with $0 \leq r < m$. By subadditivity applied repeatedly, $a_n \leq q \cdot a_m + a_r$. Dividing by $n = qm + r$:

$$\frac{a_n}{n} \leq \frac{q \cdot a_m}{qm + r} + \frac{a_r}{n}.$$

As $n \to \infty$, $q/n \to 1/m$ and $a_r/n \to 0$ (since $0 \leq r < m$ and a_r is bounded). Therefore:

$$\limsup_{n\to\infty} \frac{a_n}{n} \leq \frac{a_m}{m}.$$

Since this holds for every m:

$$\limsup_{n\to\infty} \frac{a_n}{n} \leq \inf_{m\geq 1} \frac{a_m}{m} \leq \liminf_{n\to\infty} \frac{a_n}{n}.$$

Therefore the limit exists and equals the infimum. □

Since $H_n \geq 0$ (entropy is non-negative) and (H_n) is subadditive, Fekete's lemma gives $h_\mu(T, \alpha) = \lim_n H_n/n = \inf_n H_n/n \geq 0$. Moreover, $h_\mu(T, \alpha) \leq H_1/1 = H(\alpha)$.

Remark 11.1. There is an alternative formula:

$$h_\mu(T, \alpha) = \lim_{n\to\infty} H\left(\alpha \,\middle|\, \bigvee_{k=1}^{n} T^{-k}\alpha\right).$$

This expresses the entropy rate as the new information gained by observing the present, given the entire past. The limit is a decreasing limit (conditioning on more of the past can only decrease uncertainty), so it exists by monotonicity. We will not prove the equivalence here but refer to Walters (1982), Theorem 4.12.

11.4.3 The Entropy of a Transformation

Definition 11.6 (Kolmogorov–Sinai entropy)**.** The *measure-theoretic entropy* (or *metric entropy*, or *Kolmogorov–Sinai entropy*) of a measure-preserving transformation T is

$$h_\mu(T) = \sup_\alpha h_\mu(T, \alpha),$$

where the supremum is taken over all finite measurable partitions α of X.

Interpretation. The entropy $h_\mu(T)$ is the maximal rate of information creation per unit time, optimized over all possible finite-resolution observations of the system. A system with $h_\mu(T) = 0$ is, in a precise sense, deterministic at the information-theoretic level: the past determines the future (up to measure zero). A system with $h_\mu(T) > 0$ is genuinely unpredictable—new information is created at every step.

11.4.4 Basic Properties of Metric Entropy

Proposition 11.7. *(i)* $h_\mu(T) \geq 0$.

(ii) $h_\mu(T^n) = n \cdot h_\mu(T)$ *for all* $n \geq 1$.

(iii) If T *is invertible,* $h_\mu(T^{-1}) = h_\mu(T)$.

(iv) If (X_1, μ_1, T_1) *and* (X_2, μ_2, T_2) *are measure-theoretically isomorphic, then* $h_{\mu_1}(T_1) = h_{\mu_2}(T_2)$.

Proof sketch. sketch.* (i) is immediate. For (ii), observe that $\bigvee_{k=0}^{n-1}(T^n)^{-k}\alpha = \bigvee_{k=0}^{n-1} T^{-nk}\alpha$, and a careful computation shows $h_\mu(T^n, \alpha) = h_\mu(T, \bigvee_{k=0}^{n-1} T^{-k}\alpha)$; taking the supremum over α yields the result. Part (iii) follows from (ii) and the observation that $h_\mu(\mathrm{id}) = 0$. Part (iv) follows because an isomorphism establishes a bijection between partitions preserving all relevant quantities. See Walters (1982), Chapter 4, for complete proofs. ■

Property (iv) is the reason entropy is so important: it is a *conjugacy invariant*. If two systems have different entropies, they cannot be isomorphic.

□

11.5 Computing Entropy: The Generator Theorem

The supremum in the definition of $h_\mu(T)$ is taken over all finite partitions—an enormous collection. The Kolmogorov–Sinai theorem tells us that this supremum is achieved by any *generating partition*, reducing the computation of entropy to a single partition.

11.5.1 Generating Partitions

Definition 11.7 (Generating partition)**.** A finite measurable partition α is a *generator* (or *generating partition*) for a measure-preserving transformation T if the σ-algebra generated by the iterated partitions $\{T^{-k}\alpha : k \geq 0\}$ is the full σ-algebra $\mathcal{B}$ (up to sets of measure zero). That is:

$$\bigvee_{k=0}^{\infty} T^{-k}\alpha \doteq \mathcal{B},$$

where $\doteq$ means equality modulo μ-null sets. If T is invertible, we may also require the *two-sided* generating condition:

$$\bigvee_{k=-\infty}^{\infty} T^{-k}\alpha \doteq \mathcal{B}.$$

Informally, α is a generator if, by observing the trajectory of a point through the atoms of α for all time, one can (in principle) determine the point exactly.

Example 11.2. For the full shift σ on $\Sigma_k = \{1, 2, \ldots, k\}^{\mathbb{Z}}$, the partition $\alpha = \{[i] : i = 1, \ldots, k\}$, where $[i] = \{x \in \Sigma_k : x_0 = i\}$ is the cylinder set at position 0, is a generating partition. Indeed, knowing $x_0, x_1, x_2, \ldots$ (by applying $\sigma^{-k}\alpha$ for all $k \geq 0$) determines the entire forward sequence; for the two-sided shift, knowing x_n for all $n \in \mathbb{Z}$ determines x exactly.

Example 11.3. For the doubling map $T(x) = 2x \pmod 1$ on $[0, 1)$ with Lebesgue measure, the partition $\alpha = \{[0, 1/2), [1/2, 1)\}$ is a generator. The atom containing $T^k(x)$ tells us the $(k+1)$-th digit in the binary expansion of x; knowing all digits determines x (up to the countable set of dyadic rationals, which has measure zero).

11.5.2 The Kolmogorov–Sinai Theorem

Theorem 11.8 (Kolmogorov–Sinai)**.** *Let T be a measure-preserving transformation on $(X, \mathcal{B}, \mu)$ and let α be a generating partition (one-sided for non-invertible T, two-sided for invertible T). Then:*

$$h_\mu(T) = h_\mu(T, \alpha).$$

This theorem is the workhorse for computing entropy: instead of taking a supremum over all partitions, we need only find a single generator and compute $h_\mu(T, \alpha)$.

Proof sketch. sketch.* The key idea is that for any finite partition β, if α is a generator, then the partitions $\bigvee_{k=0}^{n-1} T^{-k}\alpha$ eventually approximate β arbitrarily well (in the sense that β can be approximated by a partition measurable with respect to $\bigvee_{k=0}^{n-1} T^{-k}\alpha$). One shows that for any $\epsilon > 0$ and any finite partition β, if n is large enough, then $h_\mu(T, \beta) \leq h_\mu(T, \alpha) + \epsilon$. Taking $\epsilon \to 0$ and the supremum over β gives $h_\mu(T) \leq h_\mu(T, \alpha)$. The reverse inequality $h_\mu(T) \geq h_\mu(T, \alpha)$ holds by definition. See Petersen (1983), Theorem 4.17, or Walters (1982), Theorem 4.18, for the complete proof. ■

□

11.6 Examples

We now compute entropy explicitly for several important dynamical systems.

11.6.1 Bernoulli Shifts

The fair coin: $B(1/2, 1/2)$. Consider the Bernoulli shift σ on $\{0,1\}^{\mathbb{Z}}$ (or $\{0,1\}^{\mathbb{N}}$) where each coordinate independently takes the value 0 or 1 with equal probability 1/2. The partition $\alpha = \{[0], [1]\}$ into cylinder sets at position 0 is a generator (Example 11.18).

We compute $h_\mu(\sigma, \alpha)$ directly. The join $\bigvee_{k=0}^{n-1} \sigma^{-k}\alpha$ partitions the space according to the first n coordinates $(x_0, x_1, \ldots, x_{n-1})$. There are 2^n atoms, each of measure $(1/2)^n = 2^{-n}$. Therefore:

$$H\left(\bigvee_{k=0}^{n-1} \sigma^{-k}\alpha\right) = -\sum_{w\in\{0,1\}^n} 2^{-n}\log 2^{-n} = -2^n \cdot 2^{-n} \cdot (-n\log 2) = n\log 2.$$

Thus:

$$h_\mu(\sigma) = h_\mu(\sigma, \alpha) = \lim_{n\to\infty} \frac{n\log 2}{n} = \log 2.$$

General Bernoulli shift $B(p_1, \ldots, p_k)$. Now consider the Bernoulli shift on $\{1, 2, \ldots, k\}^{\mathbb{Z}}$ where the i-th symbol appears with probability p_i. Again, $\alpha = \{[1], [2], \ldots, [k]\}$ is a generating partition. Each atom of $\bigvee_{k=0}^{n-1} \sigma^{-k}\alpha$ corresponds to a word $w = (w_0, w_1, \ldots, w_{n-1}) \in \{1, \ldots, k\}^n$ with measure $p_{w_0}p_{w_1}\cdots p_{w_{n-1}}$.

$$H\left(\bigvee_{j=0}^{n-1} \sigma^{-j}\alpha\right) = -\sum_{w\in\{1,\ldots,k\}^n} \left(\prod_{j=0}^{n-1} p_{w_j}\right) \log\left(\prod_{j=0}^{n-1} p_{w_j}\right).$$

Since the coordinates are independent:

$$= -\sum_w \prod_j p_{w_j} \sum_{j=0}^{n-1} \log p_{w_j} = -\sum_{j=0}^{n-1} \sum_w \prod_\ell p_{w_\ell} \cdot \log p_{w_j}.$$

For each fixed j, summing over w with $w_j = i$ gives $p_i \log p_i$ (the other coordinates sum to 1 independently). So:

$$= -\sum_{j=0}^{n-1} \sum_{i=1}^{k} p_i \log p_i = -n \sum_{i=1}^{k} p_i \log p_i = n \cdot H(\alpha).$$

Therefore:

$$\boxed{h_\mu(\sigma) = -\sum_{i=1}^{k} p_i \log p_i.}$$

The entropy of a Bernoulli shift is exactly the Shannon entropy of its weight vector.

11.6.2 Irrational Rotations

Proposition 11.9. *Let $R_\theta \colon \mathbb{T} \to \mathbb{T}$ be the rotation $x \mapsto x + \theta \pmod 1$ on the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, equipped with Lebesgue measure λ. Then $h_\lambda(R_\theta) = 0$.*

Proof. * Let α be any finite partition of $\mathbb{T}$ into intervals (or any finite measurable partition). We will show $h_\lambda(R_\theta, \alpha) = 0$. It suffices to show this for interval partitions, since every finite

partition can be approximated by one.

Let $\alpha = \{A_1, \ldots, A_m\}$ where each A_i is an interval (or a finite union of intervals). The partition $\alpha_n = \bigvee_{k=0}^{n-1} R_\theta^{-k}\alpha$ has at most mn atoms (each application of R_θ^{-1} introduces at most m new boundary points, so after n steps there are at most mn intervals). In general, for a partition into intervals, $|\alpha_n| \leq mn$.

By the bound $H(\alpha_n) \leq \log |\alpha_n|$:

$$h_\lambda(R_\theta, \alpha) = \lim_{n\to\infty} \frac{H(\alpha_n)}{n} \leq \lim_{n\to\infty} \frac{\log(mn)}{n} = 0.$$

Since α was arbitrary, $h_\lambda(R_\theta) = \sup_\alpha h_\lambda(R_\theta, \alpha) = 0$. □

Remark 11.2. More generally, *every isometry* of a compact metric space has zero topological entropy (and hence zero metric entropy for every invariant measure). Zero entropy does not mean the system is uninteresting—irrational rotations are uniquely ergodic and exhibit equidistribution—but it does mean that the system is, in an information-theoretic sense, deterministic: the past determines the future.

11.6.3 Arnold's Cat Map

The *Arnold cat map* is the automorphism $T_A \colon \mathbb{T}^2 \to \mathbb{T}^2$ induced by the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

acting on $\mathbb{R}^2/\mathbb{Z}^2$, preserving Lebesgue measure λ on $\mathbb{T}^2$.

The eigenvalues of A are

$$\lambda_{1,2} = \frac{3 \pm \sqrt{5}}{2}.$$

Thus $\lambda_1 = \frac{3+\sqrt{5}}{2} > 1$ and $\lambda_2 = \frac{3-\sqrt{5}}{2} = 1/\lambda_1 \in (0, 1)$. The Lyapunov exponents are $\log \lambda_1 > 0$ and $\log \lambda_2 < 0$.

Proposition 11.10. *The entropy of Arnold's cat map with respect to Lebesgue measure is*

$$h_\lambda(T_A) = \log\left(\frac{3 + \sqrt{5}}{2}\right).$$

Proof. * This can be computed directly using the theory of toral automorphisms. More generally, for a hyperbolic toral automorphism induced by a matrix $A \in \mathrm{GL}(d, \mathbb{Z})$, the entropy with respect to Lebesgue measure is

$$h_\lambda(T_A) = \sum_{|\lambda_i|>1} \log |\lambda_i|,$$

where the sum is over eigenvalues λ_i of A with absolute value greater than 1 (counted with multiplicity).

One approach is via Pesin's formula (Section 11.9), which applies because Lebesgue measure is the SRB measure for toral automorphisms. The positive Lyapunov exponent is $\log \lambda_1 = \log \frac{3+\sqrt{5}}{2}$, and Pesin's formula gives

$$h_\lambda(T_A) = \log \frac{3+\sqrt{5}}{2}.$$

Alternatively, one can construct a Markov partition for T_A and use the Kolmogorov–Sinai theorem. See Katok and Hasselblatt (1995), Section 3.2, for details. □

Numerically, $h_\lambda(T_A) = \log\left(\frac{3+\sqrt{5}}{2}\right) \approx 0.9624$.

11.6.4 The Doubling Map

The *doubling map* is $T\colon [0,1) \to [0,1)$ defined by $T(x) = 2x \pmod 1$, preserving Lebesgue measure.

Proposition 11.11. $h_\lambda(T) = \log 2$.

Proof. * The partition $\alpha = \{[0,1/2), [1/2,1)\}$ is a generator (Example 11.19). The join $\bigvee_{k=0}^{n-1} T^{-k}\alpha$ partitions $[0,1)$ into 2^n equal intervals of the form $[j/2^n, (j+1)/2^n)$, $j = 0, 1, \ldots, 2^n - 1$. Each has Lebesgue measure 2^{-n}. Therefore:

$$H\left(\bigvee_{k=0}^{n-1} T^{-k}\alpha\right) = -2^n \cdot 2^{-n} \log 2^{-n} = n \log 2.$$

So $h_\lambda(T, \alpha) = \log 2$, and by the Kolmogorov–Sinai theorem, $h_\lambda(T) = \log 2$. □

Remark 11.3. The doubling map is measurably isomorphic to the one-sided Bernoulli shift $B(1/2, 1/2)$. The isomorphism is given by the binary expansion map: $x \mapsto (x_1, x_2, x_3, \ldots)$ where $x = \sum x_i 2^{-i}$. That both systems have entropy $\log 2$ is consistent with the fact that entropy is a conjugacy invariant.

11.6.5 Entropy as a Conjugacy Invariant

Theorem 11.26. Measure-theoretic entropy is an invariant of measure-theoretic isomorphism. That is, if $\varphi\colon (X_1, \mu_1) \to (X_2, \mu_2)$ is a measure-preserving isomorphism with $\varphi \circ T_1 = T_2 \circ \varphi$ (a.e.), then $h_{\mu_1}(T_1) = h_{\mu_2}(T_2)$.

This is Proposition 11.16(iv). As a consequence:

Corollary 11.27. The Bernoulli shifts $B(1/2, 1/2)$ and $B(1/3, 1/3, 1/3)$ are *not* isomorphic.

Proof. * $h(B(1/2, 1/2)) = \log 2 \approx 0.693$ and $h(B(1/3, 1/3, 1/3)) = \log 3 \approx 1.099$. Since $\log 2 \neq \log 3$, the systems cannot be isomorphic. ■

Before entropy was available, distinguishing these two systems was an open problem. Both are ergodic, mixing, and have similar qualitative behavior. Entropy resolved the question definitively.

□

11.7 Topological Entropy

We now turn to an entropy concept that lives in the topological category rather than the measure-theoretic one. Topological entropy, introduced by Adler, Konheim, and McAndrew (1965) and reformulated by Bowen (1971) and Dinaburg (1970), measures the complexity of a continuous map without reference to any invariant measure.

11.7.1 Definition via Separated Sets

Let (X, d) be a compact metric space and $T\colon X \to X$ a continuous map.

Definition 11.8 (Bowen metric)**.** For $n \geq 1$, the *Bowen metric* (or *dynamical metric*) is

$$d_n(x, y) = \max_{0 \leq k \leq n-1} d(T^k x, T^k y).$$

Two points are close in the Bowen metric d_n if their first n iterates remain close.

Definition 11.9 (Separated and spanning sets)**.** Let $\epsilon > 0$ and $n \geq 1$.

- A set $E \subset X$ is *(n, ϵ)-separated* if for every pair $x \neq y$ in E, $d_n(x, y) > \epsilon$: the orbits of any two distinct points in E differ by more than ϵ within the first n steps. - A set $F \subset X$ is *(n, ϵ)-spanning* if for every $x \in X$, there exists $y \in F$ with $d_n(x, y) \leq \epsilon$: the orbits of points in F approximate every orbit to within ϵ.

Let $s_n(\epsilon, T)$ denote the maximal cardinality of an (n, ϵ)-separated set, and $r_n(\epsilon, T)$ the minimal cardinality of an (n, ϵ)-spanning set. Since X is compact, both are finite.

Lemma 11.30. $r_n(\epsilon, T) \leq s_n(\epsilon, T) \leq r_n(\epsilon/2, T)$.

Proof. * For the first inequality: a maximal (n, ϵ)-separated set is automatically (n, ϵ)-spanning (if some x were not within d_n-distance ϵ of any point in E, we could add x to E, contradicting maximality). For the second: if F is $(n, \epsilon/2)$-spanning, then any (n, ϵ)-separated set E has $|E| \leq |F|$, because the map assigning to each $e \in E$ a nearest point in F (within d_n-distance $\epsilon/2$) must be injective. ■

□

Definition 11.10 (Topological entropy)**.** The *topological entropy* of T is

$$h_{\text{top}}(T) = \lim_{\epsilon \to 0} \limsup_{n \to \infty} \frac{1}{n} \log s_n(\epsilon, T).$$

By Lemma 11.30, this equals $\lim_{\epsilon \to 0} \limsup_{n \to \infty} \frac{1}{n} \log r_n(\epsilon, T)$.

Remark 11.4. As ϵ decreases, $s_n(\epsilon, T)$ can only increase, so $\frac{1}{n} \log s_n(\epsilon, T)$ is non-decreasing in $1/\epsilon$. Therefore the outer limit $\lim_{\epsilon \to 0}$ exists (as a limit in $[0, \infty]$). In many cases the lim sup is actually a limit.

11.7.2 Definition via Open Covers

There is an equivalent definition using open covers, which was the original approach. Let $\mathcal{U}$ be a finite open cover of X. Define $N(\mathcal{U})$ to be the minimal cardinality of a subcover, and for two covers $\mathcal{U}, \mathcal{V}$, define the join $\mathcal{U} \vee \mathcal{V} = \{U \cap V : U \in \mathcal{U}, V \in \mathcal{V}\}$.

$$h_{\text{top}}(T) = \sup_{\mathcal{U}} \lim_{n\to\infty} \frac{1}{n} \log N\left(\bigvee_{k=0}^{n-1} T^{-k}\mathcal{U}\right).$$

The limit exists by a subadditivity argument analogous to that in Theorem 11.12.

11.7.3 Examples

Example 11.4. (Full shift on k symbols). Let $\Sigma_k = \{1, \ldots, k\}^{\mathbb{Z}}$ with the product topology (metrized by, e.g., $d(x, y) = 2^{-\min\{|n| : x_n \neq y_n\}}$) and let σ be the shift map. For any $\epsilon < 1$ small enough that the 0-th coordinate is determined, an (n, ϵ)-separated set can distinguish the coordinates $0, 1, \ldots, n-1$, so $s_n(\epsilon, \sigma) = k^n$. Therefore:

$$h_{\text{top}}(\sigma) = \lim_{n\to\infty} \frac{1}{n} \log k^n = \log k.$$

Example 11.5. (Topological entropy of rotations). For an irrational rotation R_θ on the circle, R_θ is an isometry: $d(R_\theta x, R_\theta y) = d(x, y)$. Therefore $d_n(x, y) = d(x, y)$ for all n, and $s_n(\epsilon, R_\theta) = s_1(\epsilon, R_\theta)$ is independent of n. Thus:

$$h_{\text{top}}(R_\theta) = \lim_{n\to\infty} \frac{1}{n} \log s_1(\epsilon, R_\theta) = 0.$$

This is consistent with the measure-theoretic result (Proposition 11.21).

Example 11.6. (Doubling map). The doubling map $T(x) = 2x \pmod 1$ on $[0, 1)$ has topological entropy $h_{\text{top}}(T) = \log 2$. To see this, note that T is semiconjugate to the one-sided shift on $\{0, 1\}^{\mathbb{N}}$ via binary expansion, and the shift on two symbols has topological entropy $\log 2$.

—

11.8 The Variational Principle

The variational principle is the bridge between measure-theoretic and topological entropy.

Theorem 11.12 (Variational principle)**.** *Let $T\colon X \to X$ be a continuous map on a compact metrizable space. Then:*

$$h_{\text{top}}(T) = \sup_{\mu \in \mathcal{M}_T(X)} h_\mu(T),$$

where $\mathcal{M}_T(X)$ denotes the set of all T-invariant Borel probability measures on X.

This remarkable theorem, proved independently by Goodwyn (1969, the $\geq$ direction—actually Goodwyn proved $h_\mu(T) \leq h_{\text{top}}(T)$ for all μ), Dinaburg (1970), and Goodman (1971, the $\leq$ direction), says that topological entropy is the supremum of metric entropies over all invariant measures.

Proof outline. outline.* We sketch the two directions.

Direction 1: $h_\mu(T) \le h_{\text{top}}(T)$ for every $\mu \in \mathcal{M}_T(X)$.

Let α be a finite partition and $\epsilon > 0$ such that every atom of α has diameter less than ϵ (possible by refining α). Then the atoms of $\bigvee_{k=0}^{n-1} T^{-k}\alpha$ are contained in d_n-balls of radius ϵ. An (n, ϵ)-separated set E can intersect each atom of $\bigvee_{k=0}^{n-1} T^{-k}\alpha$ in at most one point (since any two points in the same atom have d_n-distance $\le \epsilon$, but separated points have d_n-distance $> \epsilon$). Wait—actually this requires ϵ to be the diameter of atoms. Let us be more precise.

Take α such that every atom of α has diameter less than ϵ. Then the atoms of $\alpha_n = \bigvee_{k=0}^{n-1} T^{-k}\alpha$ have diameter less than ϵ in the d_n metric. The number of atoms of α_n with positive measure is at most $|\alpha_n|$, and $|\alpha_n|$ is bounded below by $s_n(\epsilon, T)$... actually, the correct argument uses the fact that $H(\alpha_n) \le \log |\alpha_n|$ and that $|\alpha_n|$ is related to spanning/separated sets. More carefully: the number of nonempty atoms of α_n is at least $r_n(\epsilon, T)$ (since these atoms form an (n, ϵ)-spanning cover in some sense). The complete argument requires the Misiurewicz proof; see Walters (1982), Theorem 8.6.

Direction 2: $h_{\text{top}}(T) \le \sup_\mu h_\mu(T)$.

This is the deeper direction. For each n and ϵ, let E_n be a maximal (n, ϵ)-separated set. Define an atomic measure $\nu_n = \frac{1}{|E_n|} \sum_{x \in E_n} \delta_x$, and let $\mu_n = \frac{1}{n} \sum_{k=0}^{n-1} (T^k)_* \nu_n$ be its time average. Take a weak-* limit point μ; it is T-invariant. A careful analysis shows $h_\mu(T) \ge \limsup_n \frac{1}{n} \log |E_n| - \delta(\epsilon)$ where $\delta(\epsilon) \to 0$ as $\epsilon \to 0$. See Walters (1982), Theorem 8.6, for the full proof. □

Definition 11.11 (Measure of maximal entropy)**.** An invariant measure μ is called a *measure of maximal entropy* (or *MME*) if $h_\mu(T) = h_{\text{top}}(T)$. Such a measure always exists when X is compact and T is continuous (by upper semicontinuity of the entropy map and compactness of $\mathcal{M}_T(X)$ in the weak-* topology).

Example 11.7. For the full shift on $\{1, \dots, k\}^{\mathbb{Z}}$, the measure of maximal entropy is the uniform Bernoulli measure $(1/k, \dots, 1/k)$. Indeed, $h_{\text{top}}(\sigma) = \log k$ and the entropy of $B(1/k, \dots, 1/k)$ is $-k \cdot \frac{1}{k} \log \frac{1}{k} = \log k$.

—

11.9 Pesin's Formula (The Entropy Formula)

Pesin's formula connects entropy to Lyapunov exponents, providing a bridge between the information-theoretic and geometric descriptions of chaos. Recall from Chapter 6 that the Lyapunov exponents $\lambda_1(x) \ge \lambda_2(x) \ge \cdots \ge \lambda_d(x)$ measure the asymptotic rates of expansion and contraction along different directions.

11.9.1 The Ruelle Inequality

Theorem 11.13 (Ruelle inequality, 1978)**.** *Let M be a compact smooth Riemannian manifold and $T\colon M \to M$ a C^1 diffeomorphism preserving a Borel probability measure μ. Then:*

$$h_\mu(T) \le \int_M \sum_{\lambda_i(x) > 0} \lambda_i(x)\, d\mu(x),$$

where $\lambda_1(x) \geq \cdots \geq \lambda_d(x)$ are the Lyapunov exponents at x (counted with multiplicity).

The Ruelle inequality holds for all T-invariant probability measures μ. The right side is the average exponential rate at which T stretches volume in the unstable directions—an upper bound on how fast information can be created.

Proof idea. idea.* The key is that entropy measures distinguishability of orbits, and Lyapunov exponents control how fast nearby orbits separate. A ball of radius ϵ in the Bowen metric d_n has volume roughly $\epsilon^d \cdot \prod_i e^{-n\lambda_i^+(x)}$ near x (where λ_i^+ denotes the positive part). The number of such balls needed to cover M is roughly $\prod_i e^{n\lambda_i^+(x)}$, giving the bound. The rigorous proof uses the theory of Lyapunov charts and Pesin's theory of non-uniform hyperbolicity. See Katok and Hasselblatt (1995), Supplement S.3.

□

11.9.2 Pesin's Formula

Theorem 11.14 (Pesin's entropy formula)**.** *Let M be a compact smooth Riemannian manifold and $T\colon M \to M$ a $C^{1+\alpha}$ diffeomorphism (i.e., C^1 with α-Hölder continuous derivative for some $\alpha > 0$). If μ is an SRB measure (Sinai–Ruelle–Bowen measure)—in particular, if μ is absolutely continuous with respect to the Riemannian volume on the unstable manifolds—then equality holds in the Ruelle inequality:*

$$h_\mu(T) = \int_M \sum_{\lambda_i(x)>0} \lambda_i(x)\, d\mu(x).$$

This was proved by Pesin (1977) for the case when μ is equivalent to the Riemannian volume (Lebesgue measure on the manifold), and extended by Ledrappier and Strelcyn (1982) and Ledrappier and Young (1985) to the SRB case.

Remark 11.5. The converse direction was established by Ledrappier and Young (1985): if $h_\mu(T) = \int \sum_{\lambda_i>0} \lambda_i\, d\mu$, then μ has absolutely continuous conditional measures on unstable manifolds—i.e., μ is an SRB measure.

11.9.3 Examples Revisited

Arnold's cat map. For T_A with $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, the Lyapunov exponents with respect to Lebesgue measure are constant: $\lambda_1 = \log \frac{3+\sqrt{5}}{2} > 0$ and $\lambda_2 = \log \frac{3-\sqrt{5}}{2} < 0$. Lebesgue measure is the SRB measure (it is smooth, hence absolutely continuous). Pesin's formula gives:

$$h_\lambda(T_A) = \lambda_1 = \log \frac{3+\sqrt{5}}{2}.$$

This confirms the result of Proposition 11.23.

Doubling map. The doubling map $T(x) = 2x \pmod 1$ is not invertible, but there is a version of Pesin's formula for endomorphisms. The single Lyapunov exponent is $\lambda = \log 2$ (since $T'(x) = 2$ everywhere). Lebesgue measure is the natural measure, and:

$$h_\lambda(T) = \log 2,$$

consistent with Proposition 11.24.

11.9.4 Significance

Pesin's formula is a deep result that ties together three fundamental quantities:

1. **Entropy** (information theory / ergodic theory),
2. **Lyapunov exponents** (dynamical instability / chaos),
3. **SRB measures** (physically observable measures).

It tells us that for "physically relevant" measures, the entropy is entirely accounted for by the expansion rates in the unstable directions. No information is wasted, so to speak: the Ruelle inequality becomes an equality. Systems where entropy is strictly less than $\int \sum \lambda_i^+ \, d\mu$ are, in a sense, not in equilibrium—the measure μ fails to "see" all the complexity that the dynamics is capable of producing.

11.10 Ornstein's Theorem

We conclude with one of the deepest and most celebrated results in ergodic theory.

11.10.1 The Classification Problem

After Kolmogorov introduced entropy in 1958, the immediate question was: *Is entropy a complete invariant for Bernoulli shifts?* That is, if two Bernoulli shifts have the same entropy, must they be isomorphic?

Entropy can *distinguish* non-isomorphic Bernoulli shifts (Corollary 11.27), but it was not clear whether it was a strong enough invariant to *classify* them. For example, $B(1/2, 1/2)$ and $B(1/4, 1/4, 1/4, 1/4)$ both have entropy $\log 2$. Are they isomorphic?

11.10.2 The Theorem

Theorem 11.15 (Ornstein, 1970)**.** *Two Bernoulli shifts are measurably isomorphic if and only if they have the same entropy.*

In other words, entropy is a complete invariant for the class of Bernoulli shifts: it classifies them up to measure-theoretic isomorphism.

Corollary 11.43. $B(1/2, 1/2) \cong B(1/4, 1/4, 1/4, 1/4)$, since both have entropy $\log 2$.

This is a remarkable and non-obvious statement. These two shifts have different alphabet sizes (2 vs. 4) and different probability vectors, yet they are measure-theoretically the same system. The isomorphism is highly non-trivial and cannot be described by any simple formula.

11.10.3 Significance and Methods

The proof of Ornstein's theorem introduced a powerful new technique: the theory of $*\bar{d}$-distance *(or* finitary coding*) between processes. Ornstein defined a metric on the space of stationary processes and showed that Bernoulli shifts with the same entropy can be approximated and matched via this metric. The key technical tool is the* Ornstein copying lemma*.

Ornstein's work opened a rich program of classifying dynamical systems. Several important classes of systems were subsequently shown to be Bernoulli (and hence classified by entropy):

- Geodesic flows on negatively curved manifolds (Ornstein and Weiss, 1973).
- Hyperbolic toral automorphisms (Katznelson, 1971).
- Mixing Markov chains (Friedman and Ornstein, 1970).
- The Gauss map $x \mapsto \{1/x\}$ on $(0, 1]$ with the Gauss measure (a result that follows from its being a natural extension of a one-sided Bernoulli process).

The full proof of Ornstein's theorem is beyond our scope (it requires substantial machinery from the theory of joinings and the $\bar{d}$-metric), but we refer the interested reader to Ornstein (1974), Shields (1973), or the excellent exposition in Petersen (1983), Chapter 6.

11.11 Summary

| System | Entropy | |——|———| | Bernoulli shift $B(p_1, \ldots, p_k)$ | $-\sum p_i \log p_i$ | | Irrational rotation R_θ | 0 | | Arnold's cat map | $\log \frac{3+\sqrt{5}}{2}$ | | Doubling map $x \mapsto 2x \pmod 1$ | $\log 2$ | | Full shift on k symbols (topological) | $\log k$ |

The theory of entropy provides a hierarchy of results:

1. **Shannon entropy** quantifies the uncertainty in a single observation (partition).
2. **Kolmogorov–Sinai entropy** measures the rate of information production of a measure-preserving system.
3. **Topological entropy** measures the complexity of a continuous map without reference to a measure.
4. **The variational principle** connects (2) and (3): $h_{\text{top}}(T) = \sup_\mu h_\mu(T)$.
5. **Pesin's formula** connects entropy to Lyapunov exponents for SRB measures.
6. **Ornstein's theorem** shows that entropy completely classifies Bernoulli shifts.

Exercises

Exercise 11.1. Compute the Shannon entropy $H(p, 1-p)$ for $p \in [0, 1]$ and verify that it is maximized at $p = 1/2$ with value $\log 2$. Sketch the graph.

Exercise 11.2. Let $\alpha = \{A_1, A_2, A_3\}$ be a partition of a probability space with $\mu(A_1) = 1/2$, $\mu(A_2) = 1/3$, $\mu(A_3) = 1/6$. Compute $H(\alpha)$. Compare this to $\log 3$ and explain why $H(\alpha) < \log 3$.

Exercise 11.3. (Subadditivity in action). Let T be a measure-preserving transformation and α a finite partition. Define $H_n = H\left(\bigvee_{k=0}^{n-1} T^{-k}\alpha\right)$. Show by direct computation that for the Bernoulli shift $B(p, 1-p)$ with the generating partition $\alpha = \{[0], [1]\}$, the sequence H_n/n is constant (i.e., $H_n = nH(\alpha)$ for all n). Why does this happen? (Hint: independence.)

Exercise 11.4. (Entropy of a Markov chain). Consider the Markov shift on $\{0, 1\}^{\mathbb{N}}$ with transition matrix $P = \left(\begin{smallmatrix} 1/2 & 1/2 \\ 1/3 & 2/3 \end{smallmatrix}\right)$ and stationary distribution $\pi = (2/5, 3/5)$. Show that the measure-theoretic entropy of the shift with respect to the Markov measure is

$$h = -\sum_i \pi_i \sum_j P_{ij} \log P_{ij}.$$

Compute this value numerically. Is it larger or smaller than the entropy of the Bernoulli shift $B(2/5, 3/5)$? Explain why.

Exercise 11.5. Let $T\colon \mathbb{T}^2 \to \mathbb{T}^2$ be the toral automorphism given by the matrix $A = \left(\begin{smallmatrix} 3 & 2 \\ 1 & 1 \end{smallmatrix}\right)$. Compute the Lyapunov exponents and, using Pesin's formula, determine $h_\lambda(T)$.

Exercise 11.6. (Topological entropy of a subshift). Let $\Sigma \subset \{0,1\}^{\mathbb{Z}}$ be the subshift of finite type defined by forbidding the word 11 (i.e., no two consecutive 1's). Show that the number of admissible words of length n is the Fibonacci number F_{n+2}, and conclude that

$$h_{\text{top}}(\sigma|_\Sigma) = \log\left(\frac{1+\sqrt{5}}{2}\right).$$

Exercise 11.7. (Variational principle application). The full shift on $\{0,1\}^{\mathbb{Z}}$ has topological entropy $\log 2$. The Bernoulli measure $B(p, 1-p)$ has metric entropy $-p\log p - (1-p)\log(1-p)$. Verify that this is maximized over $p \in [0,1]$ at $p = 1/2$ with value $\log 2$, confirming the variational principle in this case.

Exercise 11.8. Prove that if T is an isometry of a compact metric space (X, d), then $h_{\text{top}}(T) = 0$. (Hint: $d_n = d$ for all n.)

Exercise 11.9. (Entropy and factors). Let $\pi\colon (X, \mu, T) \to (Y, \nu, S)$ be a factor map (i.e., π is measure-preserving and $\pi \circ T = S \circ \pi$ a.e.). Show that $h_\nu(S) \le h_\mu(T)$. (Hint: if β is a partition of Y, then $\pi^{-1}\beta$ is a partition of X with $H_\mu(\pi^{-1}\beta) = H_\nu(\beta)$.)

—

Recommended Reading

- **Walters, P.** *An Introduction to Ergodic Theory.* Springer, 1982. — Chapter 4 (entropy) and Chapter 8 (topological entropy and the variational principle) are the standard graduate-level reference.
- **Petersen, K.** *Ergodic Theory.* Cambridge University Press, 1983. — Chapters 4–6 cover entropy, generators, and Ornstein theory with careful proofs.
- **Downarowicz, T.** *Entropy in Dynamical Systems.* Cambridge University Press, 2011. — A modern and comprehensive treatment of both measure-theoretic and topological entropy.
- **Katok, A. and Hasselblatt, B.** *Introduction to the Modern Theory of Dynamical Systems.* Cambridge University Press, 1995. — Chapter 4 covers entropy; the supplements discuss Pesin theory and Lyapunov exponents.
- **Shannon, C. E.** "A Mathematical Theory of Communication." *Bell System Technical Journal*, 27(3):379–423, 1948. — The founding paper of information theory.

- **Kolmogorov, A. N.** "A New Metric Invariant of Transitive Dynamical Systems and Automorphisms of Lebesgue Spaces." *Doklady Akademii Nauk SSSR*, 119(5):861–864, 1958. — Where measure-theoretic entropy was introduced.
- **Ornstein, D. S.** "Bernoulli Shifts with the Same Entropy are Isomorphic." *Advances in Mathematics*, 4(3):337–352, 1970. — The landmark classification theorem.
- **Ornstein, D. S.** *Ergodic Theory, Randomness, and Dynamical Systems.* Yale University Press, 1974. — Ornstein's own exposition of his theory.
- **Brin, M. and Stuck, G.** *Introduction to Dynamical Systems.* Cambridge University Press, 2002. — An accessible introduction, covering entropy in Chapters 4–5 at a level suitable for undergraduates.

In the next chapter, we will explore how the ideas of ergodic theory and dynamical systems connect to reservoir computing, where the dynamics of a high-dimensional system are harnessed for computation.

Chapter 12

Connections — From Ergodic Theory Back to Dynamics

In Part I, we studied dynamical systems: their orbits, stability, bifurcations, and chaos. In Part II, we built a measure-theoretic framework — invariant measures, ergodicity, mixing, entropy — for understanding the statistical behavior of those systems. The two parts have developed in parallel, with many of the same examples recurring, but we have not yet brought them together in a systematic way.

This chapter closes the loop. We return to the dynamical systems of Part I, now armed with the ergodic-theoretic tools of Part II, and show how the two theories interlock. The central theme is that **hyperbolic** dynamical systems — those exhibiting a clean splitting of directions into expanding and contracting — are the systems for which the ergodic theory is richest and most complete. For these systems, there exist distinguished invariant measures (the SRB measures) that govern the statistical behavior of Lebesgue-typical orbits, and the statistical properties of the dynamics — decay of correlations, central limit theorems, large deviations — rival those of genuinely random processes.

We also organize the various notions of "randomness" for deterministic systems into the **ergodic hierarchy**, survey a striking application of ergodic theory to number theory (the Gauss map and continued fractions), and look ahead to Part III, where we will see that reservoir computers are themselves dynamical systems whose computational properties are governed by the very concepts we have developed.

12.1 Ergodic Theory of Hyperbolic Systems

12.1.1 What Is Hyperbolicity?

Throughout Part I, we observed that chaos tends to arise when a dynamical system simultaneously stretches and folds regions of phase space. Stretching amplifies small differences between nearby orbits (sensitive dependence), while folding keeps orbits confined to a bounded region. **Hyperbolicity** is the mathematical formalization of this stretching-and-contracting structure.

Informal picture. A diffeomorphism $T : M \to M$ on a smooth manifold M is hyperbolic on an invariant set Λ if at every point $x \in \Lambda$, the tangent space T_xM splits into two complementary

subspaces: one along which DT contracts (the **stable direction**) and one along which DT expands (the **unstable direction**). The key requirement is that this splitting is **uniform** — the rates of contraction and expansion are bounded away from zero and from infinity, uniformly over all points in Λ.

Definition 12.1 (Uniformly Hyperbolic Invariant Set)**.** Let M be a smooth Riemannian manifold, and let $T : M \to M$ be a C^1 diffeomorphism. A compact T-invariant set $\Lambda \subset M$ (meaning $T(\Lambda) = \Lambda$) is called **uniformly hyperbolic** (or simply **hyperbolic**) if there exist:

1. a continuous splitting of the tangent bundle over Λ:

$$T_xM = E^s(x) \oplus E^u(x), \quad \text{for all } x \in \Lambda,$$

2. constants $C > 0$ and $0 < \lambda < 1$,

such that for all $x \in \Lambda$ and all $n \geq 0$:

- $DT^n(x)\, v \in E^s(T^n x)$ for all $v \in E^s(x)$, and $\|DT^n(x)\, v\| \leq C\lambda^n \|v\|$ (contraction on E^s),
- $DT^{-n}(x)\, v \in E^u(T^{-n}x)$ for all $v \in E^u(x)$, and $\|DT^{-n}(x)\, v\| \leq C\lambda^n \|v\|$ (contraction on E^u under T^{-1}, i.e., expansion under T).

The subspace $E^s(x)$ is the **stable subspace** and $E^u(x)$ is the **unstable subspace** at x. The splitting $E^s \oplus E^u$ is called the **hyperbolic splitting**.

Remark 12.1. By choosing an **adapted metric** (a standard technique), one can always arrange that $C = 1$ in Definition 12.1, so that the contraction and expansion are immediate at each step rather than merely asymptotic. This simplifies many arguments.

The condition of uniform hyperbolicity is strong: it requires that every tangent direction at every point of Λ is either contracted or expanded, with uniform bounds. There is no room for "neutral" directions where the derivative neither contracts nor expands.

12.1.2 Anosov Diffeomorphisms

The most important special case of uniform hyperbolicity arises when the hyperbolic set Λ is the entire manifold M.

Definition 12.2 (Anosov Diffeomorphism)**.** A C^1 diffeomorphism $T : M \to M$ on a compact Riemannian manifold M is called an **Anosov diffeomorphism** if the entire manifold M is a uniformly hyperbolic set for T.

The term honors Dmitri Anosov, who in the 1960s carried out a systematic study of these systems and proved their structural stability.

Example 12.1. (Arnold's Cat Map). The most important and best-understood example of an Anosov diffeomorphism is the **hyperbolic toral automorphism** introduced by V. I. Arnold. Consider the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Since $\det(A) = 1$ and all entries are integers, A defines a map $T_A : \mathbb{T}^2 \to \mathbb{T}^2$ on the 2-torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ by

$$T_A(x, y) = A \begin{pmatrix} x \\ y \end{pmatrix} \pmod 1 = (2x + y,\, x + y) \pmod 1.$$

Because $\det(A) = 1$, the map T_A is invertible (with A^{-1} also having integer entries) and preserves Lebesgue measure on $\mathbb{T}^2$.

The eigenvalues of A are

$$\lambda_u = \frac{3 + \sqrt{5}}{2} \approx 2.618, \qquad \lambda_s = \frac{3 - \sqrt{5}}{2} \approx 0.382.$$

Note that $\lambda_u > 1 > \lambda_s > 0$ and $\lambda_u \lambda_s = 1$ (since $\det A = 1$). The corresponding eigenvectors

$$v_u = \begin{pmatrix} 1 \\ (\sqrt{5} - 1)/2 \end{pmatrix}, \qquad v_s = \begin{pmatrix} 1 \\ -(1 + \sqrt{5})/2 \end{pmatrix}$$

define the unstable and stable directions at every point. Since A is a linear map, the hyperbolic splitting is constant: $E^u(x) = \mathrm{span}(v_u)$ and $E^s(x) = \mathrm{span}(v_s)$ for all $x \in \mathbb{T}^2$.

The map T_A stretches in the direction of v_u by a factor of $\lambda_u \approx 2.618$ and contracts in the direction of v_s by a factor of $\lambda_s \approx 0.382$, at every point. This is manifestly uniformly hyperbolic with $C = 1$ and $\lambda = \lambda_s < 1$.

We showed in Chapters 8 and 10 that the cat map preserves Lebesgue measure and is ergodic and mixing with respect to it. In fact, much more is true: it is Bernoulli (Section 12.4). The cat map is the paradigmatic example of a hyperbolic system, and we will use it repeatedly in what follows.

Why "cat map"? Arnold illustrated the action of this map by showing what it does to a picture of a cat drawn on the torus. One application of T_A stretches, shears, and wraps the image back onto the torus, making the cat unrecognizable. After many iterations, the image appears to be uniformly smeared across the torus — a visual demonstration of mixing. (Remarkably, because the map is invertible and the torus is compact, Poincare recurrence guarantees that the cat image will eventually return arbitrarily close to its original form, but the recurrence time is astronomically large.)

Structural stability. A fundamental property of Anosov diffeomorphisms is that they are **structurally stable**: any C^1-small perturbation of an Anosov diffeomorphism is topologically conjugate to the original. The hyperbolic structure persists. This was proved by Anosov in 1967, building on earlier work of Smale. Structural stability means that the qualitative dynamics — the orbit structure, the mixing properties, the entropy — are robust. Small perturbations change the details of individual orbits but not the statistical behavior.

12.1.3 Smale's Horseshoe

Anosov diffeomorphisms are "globally" hyperbolic — the hyperbolicity pervades the entire manifold. In many systems, hyperbolicity is present only on a particular invariant set, often a fractal one. The cleanest example of this phenomenon is **Smale's horseshoe**.

Construction. Consider a map T of the plane that acts on a unit square $S = [0, 1]^2$ as follows:

1. **Stretch** S vertically by a factor $\mu > 2$ and compress horizontally by a factor $1/\mu$.
2. **Fold** the resulting thin, tall strip into a horseshoe shape and lay it back across S.

The result is that $T(S) \cap S$ consists of two vertical strips, and $T^{-1}(S) \cap S$ consists of two horizontal strips. Let

$$\Lambda = \bigcap_{n=-\infty}^{\infty} T^n(S).$$

This is the set of points that remain in S under all forward and backward iterates of T. It is a compact, T-invariant set, and it has a remarkable structure.

Theorem 12.1 (Smale)**.** *The invariant set Λ of the horseshoe map is:*

1. A uniformly hyperbolic set for T. 2. Homeomorphic to a Cantor set. 3. Topologically conjugate to the ***full two-sided shift*** *on two symbols: there exists a homeomorphism $h : \Lambda \to \{0,1\}^{\mathbb{Z}}$ such that $h \circ T = \sigma \circ h$, where σ is the shift map.*

Proof sketch. sketch.* The conjugacy is constructed by **symbolic dynamics**. Each point $x \in \Lambda$ is assigned a bi-infinite sequence $(\ldots, a_{-1}, a_0, a_1, \ldots) \in \{0,1\}^{\mathbb{Z}}$ according to the rule: $a_n = 0$ if $T^n(x)$ lies in the left vertical strip, and $a_n = 1$ if $T^n(x)$ lies in the right vertical strip. The hyperbolicity guarantees that distinct sequences correspond to distinct points (the stable and unstable manifolds of distinct symbol sequences intersect transversally in a unique point), and the coding map h is a homeomorphism that intertwines T with the shift. □

Significance. Through the conjugacy with the full shift, the horseshoe inherits all the dynamical complexity of symbolic dynamics:

- The number of periodic orbits of period n is 2^n.
- The topological entropy is $\log 2$.
- The dynamics on Λ are topologically mixing.
- There exist dense orbits, and the periodic orbits are dense in Λ.

Smale's horseshoe is important not because it is a contrived example, but because horseshoes are **ubiquitous**. Whenever a diffeomorphism has a transverse homoclinic point — a point that lies on both the stable and unstable manifolds of a hyperbolic fixed point, with these manifolds intersecting transversally — the Smale-Birkhoff homoclinic theorem guarantees that a horseshoe is present in the dynamics. Transverse homoclinic points arise in countless systems, from the forced Duffing oscillator to the restricted three-body problem. The horseshoe is the fundamental mechanism through which chaos enters smooth dynamics.

□

12.1.4 Axiom A Systems

Anosov diffeomorphisms (where the entire manifold is hyperbolic) and Smale's horseshoe (where a Cantor set is hyperbolic) are both special cases of a more general framework introduced by Smale.

Definition 12.3 (Non-wandering Set)**.** A point $x \in M$ is **non-wandering** for $T : M \to M$ if for every open neighborhood U of x, there exists $n \geq 1$ such that $T^n(U) \cap U \neq \emptyset$. The set of all non-wandering points is denoted $\Omega(T)$.

The non-wandering set is the part of phase space where the dynamics are "recurrent" in a topological sense. It contains all fixed points, all periodic orbits, and all limit sets.

Definition 12.4 (Axiom A)**.** A C^1 diffeomorphism $T : M \to M$ satisfies **Axiom A** (or is an **Axiom A diffeomorphism**) if:

1. The non-wandering set $\Omega(T)$ is a uniformly hyperbolic set. 2. The periodic orbits of T are dense in $\Omega(T)$.

The **Smale spectral decomposition theorem** states that if T satisfies Axiom A, then $\Omega(T)$ can be uniquely decomposed into finitely many disjoint, closed, T-invariant sets $\Omega_1, \ldots, \Omega_k$, called **basic sets**, on each of which T is topologically transitive.

The Axiom A framework was the setting in which Sinai, Ruelle, and Bowen developed the thermodynamic formalism and proved the existence of the measures that now bear their names.

12.2 SRB Measures Revisited

12.2.1 The Problem of Physical Measures

In Chapter 8, we introduced the idea that not all invariant measures are equally relevant for understanding the observable behavior of a dynamical system. A dissipative system — one that contracts volume in phase space — cannot preserve Lebesgue measure. Orbits typically converge to an attractor of lower dimension, and the relevant invariant measure is supported on this attractor. But attractors typically carry infinitely many invariant measures. Which one describes the statistics that a physical observer (or a computer simulation) would see?

The answer, for hyperbolic attractors, is the **SRB measure**, named after Yakov Sinai, David Ruelle, and Rufus Bowen.

12.2.2 Definition and Key Properties

Definition 12.5 (SRB Measure)**.** Let $T : M \to M$ be a $C^{1+\alpha}$ diffeomorphism, and let μ be a T-invariant Borel probability measure on M. The measure μ is called an **SRB measure** (or **Sinai-Ruelle-Bowen measure**) if it has the following property: for μ-a.e. x, the conditional measures of μ on the local unstable manifolds $W^u_{\text{loc}}(x)$ are absolutely continuous with respect to the Riemannian volume on those manifolds.

This definition requires some unpacking.

Local stable and unstable manifolds. At each point x of a hyperbolic set, there exist smooth embedded disks

$$W^s_{\text{loc}}(x) = \{y \in M : d(T^n(x), T^n(y)) \leq \epsilon \text{ for all } n \geq 0, \text{ and } d(T^n(x), T^n(y)) \to 0\},$$

$$W^u_{\text{loc}}(x) = \{y \in M : d(T^{-n}(x), T^{-n}(y)) \leq \epsilon \text{ for all } n \geq 0, \text{ and } d(T^{-n}(x), T^{-n}(y)) \to 0\},$$

tangent to $E^s(x)$ and $E^u(x)$ respectively. The **stable manifold theorem** (one of the foundational results in hyperbolic dynamics) guarantees their existence and smoothness. Points on

$W^s_{\text{loc}}(x)$ converge to x under forward iteration; points on $W^u_{\text{loc}}(x)$ converge to x under backward iteration.

Absolute continuity along unstable manifolds. The condition in Definition 12.7 says that the SRB measure, when "sliced" along unstable manifolds, gives something that looks like a smooth density on each such slice. Along stable manifolds, by contrast, the measure may be singular — concentrated on a fractal. This asymmetry reflects the dynamical asymmetry: the unstable direction is the one in which information is generated (positive Lyapunov exponents), and the SRB measure is smooth in precisely those directions.

Why this is the "physical" measure. The fundamental property of SRB measures is the following:

Theorem 12.2 (Physical Property of SRB Measures)**.** *Let Λ be a uniformly hyperbolic attractor for a $C^{1+\alpha}$ diffeomorphism T, and let μ be the SRB measure on Λ. Then for Lebesgue-almost every initial condition x in the basin of attraction $\mathcal{B}(\Lambda)$,*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \varphi(T^k(x)) = \int \varphi \, d\mu$$

for every continuous observable $\varphi : M \to \mathbb{R}$.

In words: time averages computed along typical orbits converge to space averages with respect to the SRB measure. Here "typical" means for a set of initial conditions of full Lebesgue measure in the basin of attraction — not full measure with respect to μ (which lives on the attractor, a set of Lebesgue measure zero) but full Lebesgue measure in the ambient space.

This is what makes the SRB measure physically relevant. When you run a numerical simulation starting from a "random" initial condition and compute time averages of some observable, the numbers you get converge to integrals against the SRB measure — regardless of which initial condition you happen to choose (with probability one).

12.2.3 Existence Theorems

The existence of SRB measures for uniformly hyperbolic attractors is one of the landmark results of the field.

Theorem 12.3 (Sinai-Ruelle-Bowen, 1970s)**.** *Let $T : M \to M$ be a $C^{1+\alpha}$ diffeomorphism satisfying Axiom A, and let Λ be a topologically transitive attractor (a basic set that is an attractor). Then Λ supports a unique SRB measure μ. Moreover:*

1. μ is ergodic. 2. μ has absolutely continuous conditional measures on unstable manifolds. 3. For Lebesgue-a.e. x in the basin of attraction of Λ, the time averages along the orbit of x converge to integrals against μ. 4. μ is the unique equilibrium state for the potential $\varphi^u(x) = -\log|\det DT|_{E^u(x)}|$ (i.e., μ maximizes $h_\mu(T) + \int \varphi^u \, d\mu$ over all T-invariant probability measures).

Historical note. *The construction proceeded in stages. Sinai (1968, 1972) constructed what are now called SRB measures for Anosov diffeomorphisms using the technique of* ***Markov partitions*** *— partitions of the manifold into sets with a product structure (stable $\times$ unstable) that the dynamics maps cleanly from one partition element to another, analogous to the states*

of a Markov chain. Bowen (1970, 1975) extended Markov partitions to general Axiom A basic sets. Ruelle (1976) placed the entire theory in the framework of ***thermodynamic formalism****, drawing on ideas from statistical mechanics: invariant measures are analogues of Gibbs states, and the SRB measure corresponds to the equilibrium state for a particular potential function (part 4 of the theorem).*

The construction via Markov partitions and the thermodynamic formalism is the subject of Bowen's monograph Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms, which remains one of the most elegant treatments of the subject.

Proof idea for the cat map. idea for the cat map.* For the Arnold cat map $T_A : \mathbb{T}^2 \to \mathbb{T}^2$, the SRB measure is simply Lebesgue measure. This is because T_A preserves Lebesgue measure (being a linear map with $|\det A| = 1$), and Lebesgue measure already has absolutely continuous conditional measures on unstable manifolds — in fact, it has absolutely continuous conditional measures on *all* smooth curves, being a smooth measure. This is the "easy" case. For dissipative systems (like the Henon attractor or the solenoid), the SRB measure is genuinely singular with respect to Lebesgue measure on the ambient space. □

□

12.2.4 Beyond Uniform Hyperbolicity

Uniform hyperbolicity is a powerful hypothesis, but it is also restrictive. Many of the most important examples in dynamics — the Henon map for "generic" parameters, the Lorenz attractor, billiards, geodesic flows on manifolds with some zero curvature — fail to be uniformly hyperbolic. Several frameworks have been developed to extend the theory.

Non-uniform hyperbolicity (Pesin theory). The key idea, due to Yakov Pesin (1976, 1977), is to replace the *uniform* bounds in Definition 12.1 with *pointwise asymptotic* conditions. A measure-preserving diffeomorphism T is **non-uniformly hyperbolic** (with respect to an invariant measure μ) if the Lyapunov exponents (Chapter 6, Chapter 11) are nonzero μ-almost everywhere: for μ-a.e. x, every Lyapunov exponent $\lambda_i(x) \neq 0$.

Recall from the multiplicative ergodic theorem (Oseledets' theorem) that the Lyapunov exponents $\lambda_1(x) \geq \lambda_2(x) \geq \cdots \geq \lambda_d(x)$ are defined for μ-a.e. x and are constant on ergodic components. Non-uniform hyperbolicity means that none of these exponents is zero — every direction is either expanding or contracting, but the rates may vary from point to point and the "hyperbolic estimates" hold only asymptotically, not uniformly.

Pesin showed that, under these weaker hypotheses, much of the theory of stable and unstable manifolds carries over: at μ-a.e. point, there exist local stable and unstable manifolds tangent to the stable and unstable Oseledets subspaces. However, the manifolds vary only measurably (not continuously) with the base point, and their sizes can be arbitrarily small. This makes the theory technically more difficult.

Partial hyperbolicity. A diffeomorphism is **partially hyperbolic** if the tangent bundle splits as $TM = E^s \oplus E^c \oplus E^u$, where E^s is uniformly contracted, E^u is uniformly expanded, and E^c is a **center direction** that may be mildly expanding, mildly contracting, or neutral, but at a rate strictly between those of E^s and E^u. Many physically relevant systems — time-one maps

of Anosov flows, frame flows, certain skew products — are partially hyperbolic. The ergodic theory of partially hyperbolic systems is an active area of research.

Challenges. Without uniform hyperbolicity, the existence and uniqueness of SRB measures becomes a much harder problem. Benedicks and Young (1993) proved the existence of SRB measures for the Henon map for a positive-measure set of parameters, a landmark result that required delicate analysis of the non-uniform expansion. For the Lorenz attractor, the existence of an SRB measure was established by Tucker (2002) as part of his computer-assisted proof of the geometric Lorenz model, and separately by Araujo, Pacifico, Pujals, and Viana (2009) in a broader setting.

The general question — for which smooth dynamical systems do SRB measures exist? — remains one of the central open problems in smooth ergodic theory. Viana has conjectured that any diffeomorphism with all Lyapunov exponents nonzero (with respect to a natural reference measure) should admit an SRB measure. This is not yet proved in full generality.

12.3 Statistical Properties of Chaotic Systems

One of the most striking features of hyperbolic dynamics is that deterministic systems can exhibit statistical behavior indistinguishable from that of random processes. In this section, we make this precise.

12.3.1 Decay of Correlations

Let $T : M \to M$ be a measure-preserving transformation with invariant probability measure μ. In Chapter 10, we defined strong mixing by the condition that $\mu(T^{-n}A \cap B) \to \mu(A)\mu(B)$ as $n \to \infty$ for all measurable sets A, B. The **correlation function** quantifies the rate at which this convergence occurs.

Definition 12.6 (Correlation Function)**.** For observables $\varphi, \psi \in L^2(\mu)$, the **correlation function** (or **correlation coefficient**) at lag n is

$$C_n(\varphi, \psi) = \int \varphi \circ T^n \cdot \psi \, d\mu - \int \varphi \, d\mu \int \psi \, d\mu.$$

The system (T, μ) is strongly mixing if and only if $C_n(\varphi, \psi) \to 0$ as $n \to \infty$ for all $\varphi, \psi \in L^2(\mu)$ (Proposition 10.5). But for statistical applications, the *rate* of convergence matters enormously.

Theorem 12.4 (Exponential Decay of Correlations for Anosov Systems)**.** *Let $T : M \to M$ be a C^2 Anosov diffeomorphism with SRB measure μ. Then there exist constants $C > 0$ and $0 < \tau < 1$ such that for all Holder continuous observables $\varphi, \psi : M \to \mathbb{R}$,*

$$|C_n(\varphi, \psi)| \leq C \|\varphi\|_\alpha \|\psi\|_\alpha \, \tau^n,$$

where $\| \cdot \|_\alpha$ denotes the Holder norm with exponent $\alpha > 0$.

In words: correlations decay exponentially fast. The system forgets its past at an exponential rate. This is the deterministic analogue of the exponential mixing property of Markov chains.

Proof sketch. sketch.* The proof goes through the construction of a **Markov partition** and the associated **symbolic dynamics**. The Markov partition codes the dynamics of T by a subshift of finite type (an extension of the coding we described for the horseshoe in Section 12.1.3, but now for the entire manifold). The invariant measure μ becomes a Gibbs measure for a Holder continuous potential on the symbolic space. For such Gibbs measures, exponential decay of correlations follows from the spectral properties of the **Ruelle transfer operator** (also called the Ruelle-Perron-Frobenius operator): this operator has a simple leading eigenvalue at 1 (corresponding to the invariant measure) and the rest of its spectrum is contained in a disk of radius $\tau < 1$. The spectral gap translates directly into exponential decay of correlations. For the full argument, see Bowen [1975] or Katok and Hasselblatt [1995, Chapter 20]. □

Beyond exponential decay. For systems with weaker hyperbolicity, the decay of correlations can be slower. Intermittent maps — such as the Manneville-Pomeau map $T(x) = x + x^{1+\alpha} \pmod 1$ for $\alpha > 0$ — have an indifferent fixed point at 0 (with $T'(0) = 1$) that traps orbits for long stretches, destroying exponential mixing. For these systems, Liverani, Saussol, and Vaienti (1999) proved that correlations decay polynomially:

$$|C_n(\varphi, \psi)| = O(n^{1-1/\alpha}).$$

The transition from exponential to polynomial decay of correlations is intimately related to the transition from uniform to non-uniform hyperbolicity, and it has profound consequences for the statistical limit theorems we discuss next.

□

12.3.2 The Central Limit Theorem for Dynamical Systems

Recall the classical Central Limit Theorem (CLT) from probability theory: if $X_1, X_2, \ldots$ are i.i.d. random variables with mean μ and variance $\sigma^2 > 0$, then

$$\frac{1}{\sqrt{n}} \sum_{k=1}^{n} (X_k - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Now consider a deterministic dynamical system (T, μ) and an observable $\varphi : M \to \mathbb{R}$ with $\int \varphi \, d\mu = 0$ (centered). The sequence $\varphi, \varphi \circ T, \varphi \circ T^2, \ldots$ is a stationary process (since μ is T-invariant), but the "random variables" $\varphi \circ T^k$ are *not* independent — they are deterministically linked. Nevertheless, for hyperbolic systems, a CLT holds.

Theorem 12.5 (Central Limit Theorem for Anosov Diffeomorphisms)**.** *Let $T : M \to M$ be a C^2 Anosov diffeomorphism with SRB measure μ, and let $\varphi : M \to \mathbb{R}$ be Holder continuous with $\int \varphi \, d\mu = 0$. Then either:*

*1. $\varphi = u \circ T - u$ for some measurable u (i.e., φ is a **coboundary**), in which case $\sum_{k=0}^{n-1} \varphi(T^k x) = u(T^n x) - u(x)$ is bounded along orbits and $\sigma^2 = 0$; or 2. There exists $\sigma^2 > 0$ such that for μ-a.e. x,*

$$\frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \varphi(T^k x) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

where the convergence is in distribution with respect to μ.

The variance is given by the ***Green-Kubo formula****:*

$$\sigma^2 = \int \varphi^2 \, d\mu + 2 \sum_{n=1}^{\infty} \int \varphi \cdot (\varphi \circ T^n) \, d\mu = \sum_{n=-\infty}^{\infty} C_n(\varphi, \varphi).$$

The series converges absolutely because of the exponential decay of correlations.

Proof idea. idea.* The proof proceeds by approximating the deterministic process $\varphi \circ T^k$ by a sequence of nearly independent random variables. One standard approach uses the **martingale method**: construct a filtration adapted to the dynamics and decompose φ into a martingale difference sequence plus a coboundary term. The CLT for martingale differences then yields the result. The exponential decay of correlations is the key input that makes the approximation work. An alternative approach goes through the spectral theory of the transfer operator: the CLT is deduced from the analytic properties of the **characteristic function** $n \mapsto \int e^{itS_n\varphi/\sqrt{n}} \, d\mu$ using perturbation theory for the transfer operator. See Liverani [1996] or the treatment in Viana and Oliveira [2016]. □

What makes this remarkable. The system is entirely deterministic. Given the initial condition x, the entire orbit $x, Tx, T^2x, \ldots$ is determined. There is no randomness whatsoever. Yet the fluctuations of the time average

$$\bar{\varphi}_n(x) = \frac{1}{n} \sum_{k=0}^{n-1} \varphi(T^k x)$$

around the space average $\int \varphi \, d\mu$ are governed by a Gaussian distribution, just as if the observations $\varphi(T^k x)$ were drawn from a random process. Deterministic chaos produces statistical regularity.

□

Example 12.2. For the doubling map $T(x) = 2x \pmod 1$ on $[0, 1)$ with Lebesgue measure, the observable $\varphi(x) = \cos(2\pi x)$ satisfies $\int \varphi \, d\mu = 0$. Since the doubling map is isomorphic to the Bernoulli shift $(\frac{1}{2}, \frac{1}{2})$, the sequence $\varphi(T^k x)$ is essentially a function of i.i.d. coin flips (the binary digits of x), and the CLT holds with a variance that can be computed explicitly.

12.3.3 Almost Sure Invariance Principle

The CLT describes the distribution of $S_n\varphi = \sum_{k=0}^{n-1} \varphi(T^k x)$ after normalization by $\sqrt{n}$. A much stronger result, the **almost sure invariance principle** (ASIP), asserts that the entire process $\{S_n\varphi\}_{n \geq 0}$ can be approximated, on a suitably enlarged probability space, by a Brownian motion $B(t)$:

$$S_n\varphi = B(n\sigma^2) + o(n^{1/2}) \quad \text{a.s.}$$

The ASIP was proved for Anosov diffeomorphisms by Denker and Philipp (1984) and has been extended to many other hyperbolic and partially hyperbolic systems. It implies the CLT,

the law of the iterated logarithm, and essentially all of the "classical" limit theorems of probability theory, in one stroke. The dynamical system behaves, at the level of fluctuations, as a Brownian motion.

12.3.4 Large Deviations

While the ergodic theorem tells us that $\bar{\varphi}_n(x) \to \int \varphi \, d\mu$ for a.e. x, and the CLT describes the typical fluctuations of order $1/\sqrt{n}$, one may ask: how fast does $\mu(\{x : |\bar{\varphi}_n(x) - \int \varphi \, d\mu| > \epsilon\})$ decay as $n \to \infty$ for fixed $\epsilon > 0$?

For hyperbolic systems, the answer is **exponential decay**:

Theorem 12.6 (Large Deviations)**.** *Under the hypotheses of Theorem 12.12, for each $\epsilon > 0$, there exists $c(\epsilon) > 0$ such that*

$$\mu\left(\left\{x : \left|\frac{1}{n}\sum_{k=0}^{n-1}\varphi(T^k x) - \int \varphi \, d\mu\right| > \epsilon\right\}\right) \leq e^{-c(\epsilon)n}$$

for all sufficiently large n.

This is the dynamical analogue of Cramer's theorem in probability. The rate function $c(\epsilon)$ is related to the pressure function of the thermodynamic formalism. Large deviation principles for hyperbolic systems were established by Orey and Pelikan (1988), Kifer (1990), and Young (1990), among others.

12.4 The Ergodic Hierarchy

Throughout Part II, we have encountered several notions expressing increasing degrees of "randomness" for a deterministic measure-preserving system. In this section, we organize these notions into a strict hierarchy.

12.4.1 The Hierarchy

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system on a probability space. We consider the following properties, listed from strongest to weakest.

Bernoulli. The system is **(measure-theoretically) isomorphic to a Bernoulli shift**. Recall from Chapter 10 that the Bernoulli shift $B(p_1, \ldots, p_k)$ is the shift map on $\{1, \ldots, k\}^{\mathbb{Z}}$ with the product measure $\prod_{n=-\infty}^{\infty}(p_1, \ldots, p_k)$. Being Bernoulli means there exists a measure-preserving isomorphism between $(X, \mathcal{B}, \mu, T)$ and some Bernoulli shift. By Ornstein's theorem (1970), two Bernoulli shifts are isomorphic if and only if they have the same entropy. So the Bernoulli property, together with the entropy value, completely classifies the system up to measure-theoretic isomorphism.

Kolmogorov (K-mixing). The system has the **Kolmogorov property** (or is a **K-system**, or **K-automorphism**) if there exists a sub-σ-algebra $\mathcal{K} \subset \mathcal{B}$ such that:

1. $T^{-1}\mathcal{K} \subset \mathcal{K}$ (the σ-algebra increases under T),
2. $\bigvee_{n=0}^{\infty} T^{-n}\mathcal{K} = \mathcal{B}$ (the increasing σ-algebras generate $\mathcal{B}$),
3. $\bigcap_{n=0}^{\infty} T^{n}\mathcal{K} = \{\emptyset, X\}$ (the decreasing σ-algebras are trivial).

The K-property is equivalent to having **completely positive entropy**: every non-trivial factor of the system has positive entropy ($h_\mu(T, \mathcal{P}) > 0$ for every non-trivial finite partition $\mathcal{P}$). The K-property implies that the remote past and the remote future are asymptotically independent.

Strong mixing. The system is **strongly mixing** if for all measurable sets $A, B \in \mathcal{B}$,

$$\lim_{n\to\infty} \mu(T^{-n}A \cap B) = \mu(A)\mu(B).$$

This was our Definition 10.3 in Chapter 10.

Weak mixing. The system is **weakly mixing** if for all measurable sets $A, B \in \mathcal{B}$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \Big|\mu(T^{-k}A \cap B) - \mu(A)\mu(B)\Big| = 0.$$

Equivalently (by the spectral characterization of Chapter 10), T is weakly mixing if and only if the only eigenvalue of the associated Koopman operator U_T on $L^2(\mu)$ is 1, with eigenspace consisting only of the constants.

Ergodic. The system is **ergodic** if every T-invariant measurable set has measure 0 or 1. Equivalently (Birkhoff's theorem), time averages equal space averages for every L^1 observable.

The hierarchy is:

$$\textbf{Bernoulli} \implies \textbf{K-mixing} \implies \textbf{Strong mixing} \implies \textbf{Weak mixing} \implies \textbf{Ergodic}$$

None of these implications reverses.

12.4.2 The Implications and Their Strictness

Bernoulli $\implies$ K-mixing. A Bernoulli shift has the K-property: one can take $\mathcal{K}$ to be the σ-algebra generated by coordinates $\{x_n : n \geq 0\}$, and the three conditions are easily verified.

K-mixing $\implies$ Strong mixing. If (T, μ) is a K-system, then for any $A \in \mathcal{B}$, the martingale convergence theorem applied to the decreasing sequence $T^n\mathcal{K}$ shows that $\mu(A \mid T^n\mathcal{K}) \to \mu(A)$ in L^2 as $n \to \infty$. For any $B \in \mathcal{K}$, this yields $\mu(T^{-n}A \cap B) \to \mu(A)\mu(B)$, and the general case follows from the generating property.

Strong mixing $\implies$ Weak mixing. Immediate: if $a_n \to 0$, then $\frac{1}{n}\sum_{k=0}^{n-1} |a_k| \to 0$.

Weak mixing $\implies$ Ergodic. If A is T-invariant, then $\mu(T^{-k}A \cap A) = \mu(A)$ for all k, so $|\mu(A)^2 - \mu(A)| = 0$, giving $\mu(A) \in \{0, 1\}$.

Strictness. Each implication is strict — there exist systems at each level that do not satisfy the next stronger property:

- *Ergodic but not weakly mixing:* Irrational rotation R_α on the circle. It is ergodic (Chapter 9), but the Koopman operator has eigenvalues $e^{2\pi i n\alpha}$ for all $n \in \mathbb{Z}$, so it is not weakly mixing.
- *Weakly mixing but not strongly mixing:* Such systems exist but are harder to construct explicitly. Chacon's transformation (a rank-one construction from the 1960s) is the standard example: it is weakly mixing but not strongly mixing.

- *Strongly mixing but not K:* The horocycle flow on a surface of constant negative curvature (considered at appropriate times) provides examples.
- *K but not Bernoulli:* The first example was given by Ornstein and Shields (1973), who constructed a K-automorphism that is not isomorphic to any Bernoulli shift.

12.4.3 Where the Standard Examples Sit

The following table summarizes the position of the main examples from this textbook in the ergodic hierarchy.

| **System** | **Bernoulli** | **K** | **Strong Mix.** | **Weak Mix.** | **Ergodic** | |—|—|—|—|—|—| | Irrational rotation R_α on S^1 | No | No | No | No | Yes | | Doubling map $x \mapsto 2x \pmod 1$ (Lebesgue) | Yes | Yes | Yes | Yes | Yes | | Baker's map (Lebesgue) | Yes | Yes | Yes | Yes | Yes | | Arnold's cat map (Lebesgue) | Yes | Yes | Yes | Yes | Yes | | Full shift on k symbols (product measure) | Yes | Yes | Yes | Yes | Yes | | Gauss map (Gauss measure, see Sec. 12.5) | Yes | Yes | Yes | Yes | Yes | | Geodesic flow on neg. curved surface | Yes | Yes | Yes | Yes | Yes |

Remark 12.2. The irrational rotation stands out as the only example in the table that is ergodic but not mixing. This reflects a general phenomenon: systems with any hyperbolicity tend to be high in the hierarchy, while systems with only "toral" or "rotational" dynamics tend to be merely ergodic. The cat map and the geodesic flow on negatively curved surfaces are Bernoulli — a deep result due to Ornstein and Weiss (1973) for the former, and Ornstein and Weiss (1973) building on Sinai's work for the latter.

That Arnold's cat map — a linear map on a torus, defined by a 2×2 integer matrix — is Bernoulli is remarkable. It means the deterministic, algebraically defined dynamics of the cat map are measure-theoretically indistinguishable from an infinite sequence of independent coin flips (with appropriate weights). The algebraic simplicity of the map gives no hint of this extreme stochastic behavior.

12.5 Ergodic Theory and Number Theory: The Gauss Map

One of the most beautiful applications of ergodic theory lies in the theory of continued fractions. This is a case where the abstract machinery of invariant measures and ergodicity delivers concrete, quantitative information about the digits of real numbers — information that would be extremely difficult to obtain by other means.

12.5.1 Continued Fractions and the Gauss Map

Every irrational number $x \in (0, 1)$ has a unique (infinite) **continued fraction expansion**

$$x = \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cdots}}} = [a_1, a_2, a_3, \ldots],$$

where the **partial quotients** (or **digits**) $a_1, a_2, a_3, \ldots$ are positive integers. The digits are generated by the **Gauss map** $T : (0,1) \to (0,1)$ defined by

$$T(x) = \left\{\frac{1}{x}\right\} = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor,$$

where $\{y\} = y - \lfloor y \rfloor$ denotes the fractional part. The n-th digit is

$$a_n(x) = \left\lfloor \frac{1}{T^{n-1}(x)} \right\rfloor.$$

In other words, the continued fraction expansion of x is generated by iterating the Gauss map and recording $\lfloor 1/T^{n-1}(x) \rfloor$ at each step. The continued fraction algorithm is a dynamical system, and the digits are the symbolic dynamics.

Example 12.3. Let us trace the first few steps for $x = \pi - 3 = 0.14159265\ldots$ (the fractional part of π):

- $a_1 = \lfloor 1/0.14159\ldots \rfloor = \lfloor 7.0625\ldots \rfloor = 7$, and $T(x) = 1/0.14159\ldots - 7 = 0.0625\ldots$ - $a_2 = \lfloor 1/0.0625\ldots \rfloor = 15$, and $T^2(x) = 1/0.0625\ldots - 15 = 0.9965\ldots$ - $a_3 = \lfloor 1/0.9965\ldots \rfloor = 1$, and so on.

Indeed, $\pi = 3 + [7, 15, 1, 292, 1, 1, 1, 2, \ldots]$. The large digit $a_4 = 292$ reflects the fact that the convergent $355/113$ is an exceptionally good rational approximation to π.

12.5.2 The Gauss Measure

Lebesgue measure on $(0,1)$ is *not* invariant under T (the Gauss map is not Lebesgue-measure-preserving). However, Gauss himself discovered the correct invariant measure:

Proposition 12.7. *The probability measure μ_G on $(0,1)$ defined by*

$$d\mu_G = \frac{1}{\ln 2} \cdot \frac{dx}{1+x}$$

is T-invariant.

Proof. * We must show that $\mu_G(T^{-1}(A)) = \mu_G(A)$ for every measurable set $A \subset (0,1)$. It suffices to check this for intervals $A = (a,b) \subset (0,1)$.

For each positive integer k, the Gauss map sends the interval $\left(\frac{1}{k+1}, \frac{1}{k}\right)$ monotonically onto $(0,1)$ via $T(x) = 1/x - k$. Specifically, $T(x) \in (a,b)$ when $x \in \left(\frac{1}{k+b}, \frac{1}{k+a}\right)$. Therefore,

$$T^{-1}((a,b)) = \bigcup_{k=1}^{\infty} \left(\frac{1}{k+b}, \frac{1}{k+a}\right),$$

and

$$\mu_G(T^{-1}((a,b))) = \frac{1}{\ln 2} \sum_{k=1}^{\infty} \int_{1/(k+b)}^{1/(k+a)} \frac{dx}{1+x} = \frac{1}{\ln 2} \sum_{k=1}^{\infty} \left[\ln\left(1 + \frac{1}{k+a}\right) - \ln\left(1 + \frac{1}{k+b}\right)\right].$$

Writing $\ln(1 + \frac{1}{k+a}) = \ln \frac{k+1+a}{k+a}$, this becomes

$$\frac{1}{\ln 2}\sum_{k=1}^{\infty}\left[\ln(k+1+a)-\ln(k+a)-\ln(k+1+b)+\ln(k+b)\right].$$

This is a telescoping sum. The partial sum from $k=1$ to $k=N$ is

$$\frac{1}{\ln 2}\left[\ln(N+1+a)-\ln(1+a)-\ln(N+1+b)+\ln(1+b)\right].$$

As $N\to\infty$, $\ln(N+1+a)-\ln(N+1+b)=\ln\frac{N+1+a}{N+1+b}\to\ln 1=0$. Therefore,

$$\mu_G(T^{-1}((a,b)))=\frac{1}{\ln 2}\left[\ln(1+b)-\ln(1+a)\right]=\frac{1}{\ln 2}\int_a^b\frac{dx}{1+x}=\mu_G((a,b)).$$

This completes the proof. □

Remark 12.3. The factor $\frac{1}{\ln 2}$ is the normalizing constant that makes μ_G a probability measure: $\int_0^1\frac{dx}{1+x}=\ln 2$.

12.5.3 Ergodicity and Its Consequences

Theorem 12.16. The Gauss map $T:(0,1)\to(0,1)$ is ergodic with respect to the Gauss measure μ_G.

Proof sketch. sketch.* The standard proof uses the fact that the Gauss map is **expanding on average** (the derivative $|T'(x)|=1/x^2\geq 1$ everywhere, with equality only at $x=1$) and has good distortion properties. More precisely, one shows that T is **exact**: for any measurable set A with $\mu_G(A)>0$, the sets $T^n(A)$ eventually have measure close to 1. Exactness implies the K-property, which implies ergodicity. The key estimate is that the Gauss map has **bounded distortion** on the intervals $I_k=(1/(k+1),1/k)$ — the ratio $|T'(x)|/|T'(y)|$ is bounded uniformly in k for $x,y\in I_k$ — which is a consequence of the Koebe distortion principle for Mobius transformations. For details, see Einsiedler and Ward [2011, Chapter 3] or Khinchin [1964]. □

Now we harvest the consequences. By Birkhoff's ergodic theorem (Chapter 9), for any $\varphi\in L^1(\mu_G)$ and μ_G-a.e. $x\in(0,1)$,

$$\lim_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1}\varphi(T^jx)=\int_0^1\varphi\,d\mu_G.$$

Since μ_G is absolutely continuous with respect to Lebesgue measure (and vice versa on $(0,1)$), "μ_G-a.e." is the same as "Lebesgue-a.e." in this context.

Application 1: Frequency of digits.

For a positive integer k, define the indicator function $\varphi_k(x)=\mathbf{1}_{(1/(k+1),1/k]}(x)$, which equals 1 if $a_1(x)=k$ and 0 otherwise. The frequency of the digit k in the continued fraction expansion of x is

$$\mathrm{freq}(k,x)=\lim_{n\to\infty}\frac{1}{n}\#\{1\leq j\leq n: a_j(x)=k\}.$$

By the ergodic theorem, this limit exists for a.e. x and equals

$$\mathrm{freq}(k,x) = \int_0^1 \varphi_k \, d\mu_G = \frac{1}{\ln 2}\int_{1/(k+1)}^{1/k} \frac{dx}{1+x} = \frac{1}{\ln 2}\ln\frac{k^2+2k+1}{k^2+2k} = \log_2 \frac{(k+1)^2}{k(k+2)}.$$

□

Theorem 12.8 (Gauss-Kuzmin)**.** *For Lebesgue-almost every $x \in (0,1)$, the digit k appears in the continued fraction expansion of x with frequency*

$$\mathit{freq}(k,x) = \log_2 \frac{(k+1)^2}{k(k+2)}.$$

Some numerical values:

/ k / freq(k) / /—/—/ / 1 / $\log_2(4/3) \approx 0.41504$ / / 2 / $\log_2(9/8) \approx 0.16993$ / / 3 / $\log_2(16/15) \approx 0.09311$ / / 4 / $\log_2(25/24) \approx 0.05889$ / / 5 / $\log_2(36/35) \approx 0.04064$ /

The digit 1 appears about 41.5% of the time, the digit 2 about 17%, and higher digits appear with rapidly decreasing frequency. This is a universal law: it holds for almost every real number, with no exceptions of positive measure.

Application 2: The Khinchin-Levy theorem.

By the ergodic theorem applied to $\varphi(x) = \ln a_1(x) = \ln\lfloor 1/x \rfloor$,

$$\lim_{n\to\infty} \frac{1}{n}\sum_{j=1}^{n} \ln a_j(x) = \int_0^1 \ln\lfloor 1/x \rfloor \, d\mu_G(x)$$

for a.e. x. Evaluating the integral (by summing over the intervals I_k):

$$\int_0^1 \ln\lfloor 1/x \rfloor \, d\mu_G = \frac{1}{\ln 2}\sum_{k=1}^{\infty} \ln k \int_{1/(k+1)}^{1/k} \frac{dx}{1+x} = \frac{1}{\ln 2}\sum_{k=1}^{\infty} \ln k \cdot \ln\frac{(k+1)^2}{k(k+2)}.$$

*Exponentiating, we obtain **Khinchin's constant**:*

$$\lim_{n\to\infty} (a_1(x)a_2(x)\cdots a_n(x))^{1/n} = \prod_{k=1}^{\infty}\left[\frac{(k+1)^2}{k(k+2)}\right]^{\log_2 k/\ln 2} \equiv K_0 \approx 2.6854520\ldots$$

for Lebesgue-almost every x. This is a remarkable result: the geometric mean of the continued fraction digits of almost every real number converges to a universal constant. The constant K_0 is known to many decimal places but is not known to be irrational — though it almost certainly is.

Application 3: Growth rate of denominators (Levy's constant).

*Let p_n/q_n denote the n-th convergent of the continued fraction expansion of x. The **Levy constant** describes the exponential growth rate of the denominators:*

$$\lim_{n\to\infty} \frac{1}{n}\ln q_n = \frac{\pi^2}{12\ln 2} \approx 1.1865691\ldots$$

for a.e. x. This too is a consequence of the ergodic theorem applied to the function $\varphi(x) = -\ln x$ (since the recursion for continued fraction convergents gives $\ln q_n \approx -\sum_{j=0}^{n-1} \ln T^j(x)$ for large n).

These results illustrate the extraordinary power of ergodic theory in number theory: a single abstract theorem (Birkhoff) combined with a single invariant measure (Gauss) yields a family of precise, quantitative statements about the digits of almost every real number.

12.6 Summary and Bridge to Part III

12.6.1 What We Have Built

Over the course of Part II, we have constructed a theory that describes the **statistical behavior of deterministic dynamical systems**. Let us take stock of the main tools and results.

Invariant measures (Chapter 8). Every continuous map on a compact space has at least one invariant measure (Krylov-Bogolyubov). But not all invariant measures are created equal. The **SRB measures** (Section 12.2) are the ones that describe the statistics of Lebesgue-typical orbits — the statistics you would observe in a numerical simulation.

Ergodicity and the ergodic theorems (Chapter 9). Birkhoff's theorem guarantees that time averages converge to space averages for almost every initial condition. Ergodicity is the condition under which the space average is the same for all orbits, yielding a unique statistical description of the long-term behavior.

Mixing and spectral theory (Chapter 10). Mixing is a stronger form of statistical independence that describes how rapidly the system "forgets" its initial state. The Koopman operator provides a spectral characterization: ergodicity corresponds to the eigenvalue 1 being simple, and mixing corresponds to the absence of other eigenvalues (in the weak mixing case) or the vanishing of matrix coefficients (in the strong mixing case).

Entropy (Chapter 11). Kolmogorov-Sinai entropy quantifies the rate of information production — how much uncertainty is generated per unit time. The variational principle connects measure-theoretic entropy to topological entropy, and Pesin's formula relates entropy to Lyapunov exponents.

Hyperbolicity and statistical properties (this chapter). For hyperbolic systems, the theory reaches its most complete form: SRB measures exist and are unique, correlations decay exponentially, and the full battery of probabilistic limit theorems (CLT, large deviations, ASIP) holds. The ergodic hierarchy — Bernoulli, K, strongly mixing, weakly mixing, ergodic — provides a taxonomy of the "degree of randomness" of deterministic systems.

12.6.2 The Bridge to Reservoir Computing

In Part III, we turn to **reservoir computing** — a computational paradigm in which a high-dimensional dynamical system (the "reservoir") is used as a computational substrate. The fundamental idea is simple: drive a nonlinear dynamical system with an input signal, observe the resulting high-dimensional state, and train a linear readout to extract the desired computation. The reservoir is not trained; only the readout is.

Why should this work? And what does it have to do with everything we have developed in Parts I and II? Here is a preview of the connections we will develop.

The echo state property and stability. The **echo state property** (ESP) asserts that the reservoir's state is asymptotically determined by the input history alone, independent of

the initial condition. This is the reservoir analogue of a contraction: driven by the same input, all initial conditions converge to the same trajectory. In our language, the ESP is related to the reservoir having all Lyapunov exponents negative (when considered as a driven system) — a form of uniform stability. We will see in Chapter 14 that the spectral radius condition for echo state networks is a practical sufficient condition for this kind of contraction.

Information processing and entropy. A reservoir's computational power is related to its ability to separate distinct inputs into distinct states — a form of **sensitivity**. But too much sensitivity (chaos, positive Lyapunov exponents) destroys the echo state property. The most computationally powerful reservoirs operate near the **edge of chaos**: they are sensitive enough to distinguish inputs but stable enough to be driven. This tension has a natural formulation in terms of Lyapunov exponents and, through Pesin's formula, in terms of entropy. We will explore this in Chapter 16.

Generalization and ergodic averages. When a reservoir computer is tested on new data, it must **generalize** — produce correct outputs for inputs it has not seen during training. If the input signal is generated by an ergodic process, then training on a sufficiently long time series amounts to computing ergodic averages, and generalization follows from the ergodic theorem: the training data is statistically representative of the test data. The theory of mixing and decay of correlations controls how quickly these averages converge — and hence how much training data is needed.

Fading memory and mixing. The **fading memory property** — the requirement that the reservoir's current state depends more strongly on recent inputs than on distant ones — is the reservoir computing analogue of the decay of correlations. Systems with exponential decay of correlations have exponential fading memory, while systems with slower mixing have longer memory but worse generalization properties.

Takens' embedding and state reconstruction. Takens' embedding theorem (previewed in Chapter 6) shows that the state of a dynamical system can be reconstructed from observations of a single scalar observable, provided enough time-delayed copies are used. Reservoir computing can be understood as a high-dimensional, nonlinear version of Takens embedding: the reservoir's state space provides a rich set of "observables" from which a linear readout can reconstruct any desired function of the input history.

These connections are not merely analogies. They are mathematical relationships that we will develop rigorously in Part III. The ergodic theory of Parts I and II provides the conceptual and technical foundation for understanding when and why reservoir computers work, what their limitations are, and how to design them optimally.

With this, we close Part II. We have traveled from the abstract axioms of measure theory, through the deep theorems of Birkhoff, von Neumann, Sinai, Ruelle, and Bowen, to a point where the deterministic dynamics of Part I are understood in statistical terms as rich and precise as those of probability theory. Part III will show that this understanding is not merely retrospective — it is predictive, guiding the design and analysis of a new class of computational systems.

Exercises

Exercise 12.1. (Hyperbolicity of the Cat Map). Consider the Arnold cat map $T_A : \mathbb{T}^2 \to \mathbb{T}^2$ with $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

(a) Compute the eigenvalues λ_s, λ_u and eigenvectors v_s, v_u of A. Verify that $\lambda_s \lambda_u = 1$ and that $0 < \lambda_s < 1 < \lambda_u$.

(b) Show that $\|A^n v\| \leq \lambda_s^n \|v\|$ for all $v \in E^s = \text{span}(v_s)$ and $\|A^{-n} v\| \leq \lambda_s^n \|v\|$ for all $v \in E^u = \text{span}(v_u)$.

(c) Show that the Lyapunov exponents of T_A with respect to Lebesgue measure are $\ln \lambda_u$ and $\ln \lambda_s = -\ln \lambda_u$. What is the Kolmogorov-Sinai entropy? (Use Pesin's formula.)

(d) The cat map has periodic points: show that T_A has exactly $|\det(A^n - I)|$ periodic points of period dividing n. Compute this for $n = 1, 2, 3$.

—

Exercise 12.2. (Symbolic Dynamics of the Horseshoe). Let $\sigma : \{0,1\}^{\mathbb{Z}} \to \{0,1\}^{\mathbb{Z}}$ be the full two-sided shift on two symbols.

(a) Show that the number of periodic orbits of σ with minimal period exactly n is $\frac{1}{n} \sum_{d|n} \mu(n/d)\, 2^d$, where μ is the Mobius function. Compute this for $n = 1, 2, 3, 4$.

(b) Prove that the topological entropy of σ is $\log 2$.

(c) Show that σ has a dense orbit. (Hint: construct a sequence that contains every finite binary word as a substring.)

—

Exercise 12.3. (Gauss Map Computations). Let $T(x) = \{1/x\}$ be the Gauss map on $(0, 1)$.

(a) Compute the continued fraction expansion of $\sqrt{2} - 1$. (Show that $T(\sqrt{2} - 1) = \sqrt{2} - 1$, so $\sqrt{2} - 1 = [2, 2, 2, \ldots]$.)

(b) Verify the invariance of the Gauss measure for the specific interval $A = (0, 1/2)$: show that $\mu_G(T^{-1}(A)) = \mu_G(A)$.

(c) Using Theorem 12.17, compute the probability that a randomly chosen real number has $a_1 = 1$ and $a_2 = 1$ in its continued fraction expansion. (Hint: this is $\mu_G(T^{-1}(\{a_1 = 1\}) \cap T^{-2}(\{a_2 = 1\}))$, but by the ergodic theorem and the mixing property, it is approximately $\text{freq}(1)^2$ for large separations. For adjacent digits, compute directly: the event $\{a_1 = 1, a_2 = 1\}$ corresponds to $x \in (1/3, 1/2)$.)

(d) What fraction of real numbers have continued fraction expansions where all digits are ≤ 2? (This is μ_G-measure zero. Explain why the ergodic theorem does not apply to answer this question directly, and relate the answer to the theory of badly approximable numbers.)

—

Exercise 12.4. (Decay of Correlations and the CLT). Consider the doubling map $T(x) = 2x \pmod 1$ on $[0, 1)$ with Lebesgue measure, and the observable $\varphi(x) = \cos(2\pi x)$.

(a) Show that $\int_0^1 \varphi \, dx = 0$.

(b) Compute $C_n(\varphi, \varphi) = \int_0^1 \varphi(T^n x)\varphi(x)\, dx$ for $n \geq 1$. (Hint: $\varphi(T^n x) = \cos(2\pi \cdot 2^n x)$, and use orthogonality of trigonometric functions.)

(c) Using (b), compute the variance $\sigma^2 = \sum_{n=-\infty}^{\infty} C_n(\varphi, \varphi)$ appearing in the CLT.

(d) State what the CLT (Theorem 12.12) says about the distribution of $\frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\cos(2\pi\cdot 2^k x)$ for Lebesgue-typical x.

—

Exercise 12.5. (The Ergodic Hierarchy). For each of the following systems, determine where it sits in the ergodic hierarchy (Bernoulli, K, strongly mixing, weakly mixing, ergodic, or none of these). Justify your answers.

(a) The rotation $R_\alpha(\theta) = \theta + \alpha \pmod 1$ on $[0,1)$ with Lebesgue measure, where α is irrational.

(b) The identity map $T(x) = x$ on $[0,1)$ with Lebesgue measure.

(c) The tent map $T(x) = 1 - |2x - 1|$ on $[0,1]$ with Lebesgue measure.

(d) The Bernoulli shift σ on $\{1,2,3\}^{\mathbb{Z}}$ with the product measure $(\frac{1}{3},\frac{1}{3},\frac{1}{3})^{\mathbb{Z}}$.

(e) Let T be the Arnold cat map on $\mathbb{T}^2$ and consider the product system $S = R_\alpha \times T$ on $S^1 \times \mathbb{T}^2$ with the product of Lebesgue measures, where α is irrational. (Hint: consider how mixing properties behave under products.)

—

Exercise 12.6. (SRB Measures and Time Averages). Consider the solenoid map: let D be the solid torus $S^1 \times \overline{D}^2$ (where $\overline{D}^2$ is the closed unit disk in $\mathbb{R}^2$), and define $T : D \to D$ by

$$T(\theta, z) = (2\theta \pmod 1,\ \tfrac{1}{4}z + \tfrac{1}{2}e^{2\pi i\theta}),$$

where we identify $\overline{D}^2 \cong \{z \in \mathbb{C} : |z| \le 1\}$.

(a) Show that T maps D into its interior, so the **attractor** $\Lambda = \bigcap_{n=0}^{\infty} T^n(D)$ is non-empty.

(b) Show that T is not volume-preserving (it contracts volume by a factor of 1/8 at each step). Explain why Lebesgue measure on D cannot be T-invariant.

(c) The attractor Λ has a complicated (locally Cantor set $\times$ interval) structure. Explain in qualitative terms why the SRB measure on Λ should be smooth in the "S^1" (unstable) direction and singular in the "disk" (stable) direction.

—

Exercise 12.7. (Khinchin's Constant, Numerically). Write a computer program (or carry out a hand computation for a specific number) to estimate Khinchin's constant $K_0 \approx 2.6854\ldots$ as follows.

(a) Compute the first $N = 1000$ continued fraction digits of a "random" number (e.g., $\sqrt{3}-1$, or $e-2$, or a number produced by a random number generator).

(b) Compute the geometric mean $(a_1 a_2 \cdots a_N)^{1/N}$ and compare it to K_0.

(c) The convergence is known to be slow (logarithmic). Approximately how large must N be for the geometric mean to agree with K_0 to 2 decimal places? Relate this to the rate of decay of correlations for the Gauss map.

References

- D. V. Anosov, "Geodesic flows on closed Riemannian manifolds with negative curvature," *Proceedings of the Steklov Institute of Mathematics*, 90, 1967.

- R. Bowen, *Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms*, Lecture Notes in Mathematics 470, Springer-Verlag, 1975. Second revised edition, edited by J.-R. Chazottes, 2008.
- M. Einsiedler and T. Ward, *Ergodic Theory: With a View Towards Number Theory*, Graduate Texts in Mathematics 259, Springer, 2011.
- A. Katok and B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*, Encyclopedia of Mathematics and its Applications 54, Cambridge University Press, 1995.
- A. Ya. Khinchin, *Continued Fractions*, University of Chicago Press, 1964. (Dover reprint available.)
- Y. Pesin, "Characteristic Lyapunov exponents and smooth ergodic theory," *Russian Mathematical Surveys*, 32(4):55–114, 1977.
- D. Ruelle, "A measure associated with Axiom A attractors," *American Journal of Mathematics*, 98(3):619–654, 1976.
- Ya. G. Sinai, "Markov partitions and C-diffeomorphisms," *Functional Analysis and Its Applications*, 2(1):61–82, 1968.
- S. Smale, "Differentiable dynamical systems," *Bulletin of the American Mathematical Society*, 73(6):747–817, 1967.
- M. Viana, *Stochastic Dynamics of Deterministic Systems*, Lecture Notes, IMPA, 1997.
- L.-S. Young, "What are SRB measures, and which dynamical systems have them?," *Journal of Statistical Physics*, 108(5–6):733–754, 2002.

Recommended Reading

Bowen [1975/2008] is a masterpiece of concision. In under 80 pages, it develops the thermodynamic formalism for Axiom A systems, constructs SRB measures via Markov partitions, and proves the existence of equilibrium states. It is demanding but deeply rewarding, and is the original source for much of Section 12.2.

Katok and Hasselblatt [1995] is the comprehensive reference for Chapters 18–20 on hyperbolic dynamics, structural stability, and the ergodic theory of hyperbolic systems. It is encyclopedic and can serve as both a textbook and a reference.

Young [2002] is a beautifully written survey on SRB measures, accessible to non-specialists, that explains both what these measures are and why they matter. It is the ideal entry point for Section 12.2.

Khinchin [1964] gives a self-contained treatment of the metric theory of continued fractions, including the ergodic-theoretic results of Section 12.5, at a level accessible to undergraduates.

For the statistical properties of dynamical systems (Section 12.3), the lecture notes of Viana [1997] provide an excellent introduction to the ideas around decay of correlations, central limit theorems, and large deviations in the dynamical context.

Einsiedler and Ward [2011] is a modern textbook on ergodic theory with extensive connections to number theory, covering the Gauss map and continued fractions among many other topics. It is thorough and clearly written.

Part III

Reservoir Computing

Chapter 13

From Dynamical Systems to Computation

"The purpose of computing is insight, not numbers." — Richard Hamming

13.1 Part III: Reservoir Computing

With Parts I and II behind us, the reader now commands a substantial toolkit: topological dynamics, stability theory, invariant measures, ergodicity, mixing, Lyapunov exponents, and the reconstruction theorems of Takens. These were developed in their own right, as the mathematical theory of systems that evolve in time. But dynamical systems theory is not merely descriptive. In Part III, we turn to a question that may at first seem surprising:

Can a dynamical system compute?

The answer is yes, and the framework that makes this precise — **reservoir computing** — draws directly on the mathematics we have built. This chapter provides the bridge.

13.2 Computation with Dynamics: A Historical Perspective

The idea that physical processes can perform computation is as old as computation itself. Before exploring reservoir computing in mathematical detail, it is worth recalling this history, because it reveals that the connection between dynamics and computation is not an analogy — it is an identity.

13.2.1 Mechanical and Analog Computers

The earliest computers were dynamical systems in the most literal sense. Charles Babbage's Difference Engine (1822) and Analytical Engine (1837) were mechanical devices: gears rotated, levers translated, and the physical state of the machine encoded the state of a computation. The tide-predicting machines of Lord Kelvin (1876) used systems of pulleys and gears to sum

harmonic components — they were analog dynamical systems whose trajectories traced the solution to a computational problem.

The differential analyzers of Vannevar Bush (1931) solved ordinary differential equations by constructing a physical system whose dynamics were governed by those same equations. Here the connection is transparent: the machine does not *simulate* the dynamical system; the machine *is* the dynamical system, and its trajectory *is* the solution.

These examples suggest a principle: **any sufficiently rich dynamical system can, in some sense, compute**. The question is what "sufficiently rich" means.

13.2.2 Turing Machines as Dynamical Systems

Even the most abstract model of computation — the Turing machine — is a dynamical system in disguise. Recall the definition: a Turing machine consists of a finite set of states Q, a tape alphabet Γ, a transition function $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$, and distinguished start, accept, and reject states. At each step, the machine reads a symbol, writes a symbol, changes state, and moves the tape head.

Let us formalize this. The **configuration** of a Turing machine at time t is a triple $c(t) = (q(t), \tau(t), p(t))$, where $q(t) \in Q$ is the current state, $\tau(t) : \mathbb{Z} \to \Gamma$ is the tape contents, and $p(t) \in \mathbb{Z}$ is the head position. The set of all configurations forms a (countably infinite) space $\mathcal{C}$, and the transition function induces a map

$$F : \mathcal{C} \to \mathcal{C}, \qquad c(t+1) = F(c(t)).$$

This is a discrete deterministic dynamical system on $\mathcal{C}$. The orbit $\{c(0), c(1), c(2), \ldots\}$ is the computation. A halting computation corresponds to a trajectory that reaches a fixed point (or more precisely, a subset of configurations from which F is undefined or trivially the identity). A non-halting computation corresponds to a trajectory that never enters this set — the orbit wanders indefinitely through $\mathcal{C}$.

From this viewpoint, the halting problem asks whether a given initial condition leads to a trajectory that converges to a fixed point. Undecidability, in dynamical terms, is the statement that there is no algorithm to determine this for all initial conditions. The reader may find this reminiscent of the difficulties we encountered in Part I when asking whether a given point lies in the basin of attraction of an attractor.

13.2.3 Neural Networks as Dynamical Systems

The modern resurgence of interest in computation-through-dynamics comes from neural networks. We will treat these in detail in Section 13.2, but the essential observation is this: a recurrent neural network is a parameterized discrete dynamical system on $\mathbb{R}^N$. The parameters (weights, biases) determine the map, and the network's behavior on a given input is the trajectory of this system. Training the network — adjusting the parameters so that the trajectories produce desired outputs — is a problem in the control of dynamical systems.

13.2.4 The Central Question

The history above leads us to a central question:

> If dynamical systems can exhibit rich, complex behavior — chaos, sensitive dependence, mixing, strange attractors — can we *harness* that complexity for computation?

The intuition is appealing. A chaotic system explores its state space in an intricate, structured way. A mixing system, as we saw in Part II, forgets initial conditions but retains statistical structure. An ergodic system visits every part of its state space. These properties sound like they could be useful for processing information — but the intuition needs to be made precise.

Reservoir computing is the framework that does this. Before we can introduce it, we need to understand the computational systems it grew out of: neural networks.

13.3 Neural Networks as Dynamical Systems

We now introduce neural networks with enough precision for mathematical analysis. The reader who has encountered neural networks in a machine learning course will find a different emphasis here: we care about the *dynamics*, not the *implementation*.

13.3.1 Feedforward Networks: Static Maps

A **feedforward neural network** is a parameterized function $f_\theta : \mathbb{R}^m \to \mathbb{R}^p$ constructed by composing affine maps with pointwise nonlinearities. A network with L layers takes the form

$$f_\theta(\mathbf{x}) = W_L\, \sigma\big(W_{L-1}\, \sigma\big(\cdots \sigma(W_1 \mathbf{x} + \mathbf{b}_1) \cdots\big) + \mathbf{b}_{L-1}\big) + \mathbf{b}_L,$$

where $W_\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$ are weight matrices, $\mathbf{b}_\ell \in \mathbb{R}^{n_\ell}$ are bias vectors, and $\sigma : \mathbb{R} \to \mathbb{R}$ is a nonlinear **activation function** applied componentwise. Common choices include:

- The **hyperbolic tangent**: $\sigma(x) = \tanh(x)$.
- The **logistic sigmoid**: $\sigma(x) = (1 + e^{-x})^{-1}$.
- The **ReLU** (rectified linear unit): $\sigma(x) = \max(0, x)$.

The parameter vector θ collects all weights and biases. The universal approximation theorem (Cybenko 1989, Hornik et al. 1989) guarantees that, under mild conditions on σ, such networks can approximate any continuous function on a compact set to arbitrary accuracy, provided n_1 (the width of the hidden layer) is sufficiently large.

Crucially, a feedforward network is not a dynamical system. It is a static map: given an input $\mathbf{x}$, it produces an output $f_\theta(\mathbf{x})$. There is no temporal evolution, no state that persists from one input to the next. Feedforward networks have no memory. If we present a sequence of inputs $\mathbf{x}(1), \mathbf{x}(2), \ldots$, the output at time t depends only on $\mathbf{x}(t)$, not on any previous input. For many tasks — image classification, for instance — this is sufficient. But for tasks that involve temporal structure — speech, time series, control — it is not.

13.3.2 Recurrent Neural Networks: Dynamics on State Space

A **recurrent neural network** (RNN) introduces a state vector $\mathbf{h}(t) \in \mathbb{R}^N$ that evolves over discrete time:

$$\boxed{\mathbf{h}(t+1) = \sigma\big(W\mathbf{h}(t) + W_{\text{in}}\mathbf{u}(t) + \mathbf{b}\big)}$$

where:

- $\mathbf{h}(t) \in \mathbb{R}^N$ is the **hidden state** (or simply the *state*) at time t,
- $\mathbf{u}(t) \in \mathbb{R}^m$ is the **input** at time t,
- $W \in \mathbb{R}^{N\times N}$ is the **recurrent weight matrix**,
- $W_{\text{in}} \in \mathbb{R}^{N\times m}$ is the **input weight matrix**,
- $\mathbf{b} \in \mathbb{R}^N$ is a **bias vector**,
- $\sigma : \mathbb{R} \to \mathbb{R}$ is a nonlinear activation function, applied componentwise.

The output is typically produced by a **readout** (or output) layer:

$$\mathbf{y}(t) = g\big(W_{\text{out}}\mathbf{h}(t) + \mathbf{b}_{\text{out}}\big),$$

where $W_{\text{out}} \in \mathbb{R}^{p\times N}$, and g is an output activation (often the identity for regression tasks).

This is a dynamical system. Specifically, it is a **nonautonomous discrete dynamical system** (or, equivalently, a skew-product system driven by the input signal). For each fixed input sequence $\{\mathbf{u}(t)\}_{t\geq 0}$, the state evolves according to time-dependent maps

$$F_t : \mathbb{R}^N \to \mathbb{R}^N, \qquad F_t(\mathbf{h}) = \sigma(W\mathbf{h} + W_{\text{in}}\mathbf{u}(t) + \mathbf{b}).$$

The state at time t is obtained by composition:

$$\mathbf{h}(t) = F_{t-1} \circ F_{t-2} \circ \cdots \circ F_0(\mathbf{h}(0)).$$

When the input is absent ($\mathbf{u}(t) = \mathbf{0}$ for all t), we recover an **autonomous** system

$$\mathbf{h}(t+1) = F(\mathbf{h}(t)), \qquad F(\mathbf{h}) = \sigma(W\mathbf{h} + \mathbf{b}),$$

which is exactly the kind of iterated map we studied in Part I. Let us make some observations.

State space. If $\sigma = \tanh$, then the image of F lies in $(-1,1)^N$, so the state space is effectively the open hypercube. For σ equal to the logistic sigmoid, the state space is $(0,1)^N$. For ReLU activations, the state space is $[0,\infty)^N$, and trajectories can be unbounded.

Fixed points. A fixed point of the autonomous system satisfies $\mathbf{h}^* = \sigma(W\mathbf{h}^* + \mathbf{b})$. For $\sigma = \tanh$ and $\mathbf{b} = \mathbf{0}$, the origin $\mathbf{h}^* = \mathbf{0}$ is always a fixed point. Its stability is determined by the Jacobian at $\mathbf{0}$:

$$DF(\mathbf{0}) = \operatorname{diag}(\sigma'(W \cdot \mathbf{0} + \mathbf{b})) \cdot W = \operatorname{diag}(\sigma'(\mathbf{b})) \cdot W.$$

When $\mathbf{b} = \mathbf{0}$ and $\sigma = \tanh$, we have $\sigma'(0) = 1$, so $DF(\mathbf{0}) = W$. By our stability theorems from Chapter 3, the origin is asymptotically stable if and only if the **spectral radius** $\rho(W) < 1$. If $\rho(W) > 1$, the origin is unstable, and the dynamics may exhibit more complex behavior —

limit cycles, quasi-periodic orbits, or chaos.

Dependence on W. The recurrent weight matrix W is the single most important parameter of the system. It determines the topology and geometry of the dynamics. This will become a recurring theme in Part III.

13.3.3 A Remark on Continuous-Time Formulations

The discrete-time RNN described above has a natural continuous-time counterpart. Consider the ODE

$$\tau \dot{\mathbf{h}}(t) = -\mathbf{h}(t) + \sigma\big(W\mathbf{h}(t) + W_{\text{in}}\mathbf{u}(t) + \mathbf{b}\big),$$

where $\tau > 0$ is a time constant. This is a system of N coupled first-order ODEs — a continuous-time dynamical system on $\mathbb{R}^N$, driven by the input $\mathbf{u}(t)$. The discrete-time RNN can be viewed as a forward Euler discretization of this ODE with step size $\Delta t = \tau$. This connection to ODEs will be important when we discuss Liquid State Machines in Section 13.5.

13.4 The Problem of Training Recurrent Neural Networks

To use an RNN for a computational task, one must choose the parameters $\theta = (W, W_{\text{in}}, \mathbf{b}, W_{\text{out}}, \mathbf{b}_{\text{out}})$ so that the network produces desired outputs. This is the **training problem**, and for RNNs it is notoriously difficult. Understanding *why* it is difficult requires dynamical systems thinking — and this understanding will motivate reservoir computing.

13.4.1 The Optimization Problem

Suppose we have a training sequence: inputs $\{\mathbf{u}(t)\}_{t=0}^{T-1}$ and desired outputs $\{\mathbf{d}(t)\}_{t=0}^{T-1}$. Define a loss function

$$\mathcal{L}(\theta) = \sum_{t=0}^{T-1} \ell\big(\mathbf{y}(t),\, \mathbf{d}(t)\big),$$

where ℓ is a per-step loss (e.g., $\ell(\mathbf{y}, \mathbf{d}) = \|\mathbf{y} - \mathbf{d}\|^2$) and $\mathbf{y}(t)$ depends on θ through the entire trajectory $\mathbf{h}(0), \mathbf{h}(1), \ldots, \mathbf{h}(t)$. The training problem is to minimize $\mathcal{L}(\theta)$ over θ.

13.4.2 Backpropagation Through Time

The standard approach is gradient descent: compute $\nabla_\theta \mathcal{L}$ and update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$ for some learning rate $\eta > 0$. To compute the gradient, one "unrolls" the recurrence in time, treating the RNN as a very deep feedforward network with shared weights, and applies the chain rule. This is **backpropagation through time** (BPTT).

Consider the gradient of $\mathcal{L}$ with respect to the recurrent weights W. By the chain rule:

$$\frac{\partial \mathcal{L}}{\partial W} = \sum_{t=0}^{T-1} \frac{\partial \ell(t)}{\partial \mathbf{y}(t)} \frac{\partial \mathbf{y}(t)}{\partial \mathbf{h}(t)} \sum_{k=0}^{t} \left(\prod_{j=k}^{t-1} \frac{\partial \mathbf{h}(j+1)}{\partial \mathbf{h}(j)} \right) \frac{\partial^+ \mathbf{h}(k)}{\partial W},$$

where $\frac{\partial^+ \mathbf{h}(k)}{\partial W}$ denotes the "immediate" (direct) partial derivative at time k, treating $\mathbf{h}(k-1)$ as fixed. The key term is the product of Jacobians:

$$\prod_{j=k}^{t-1} J(j), \qquad \text{where } J(j) = \frac{\partial \mathbf{h}(j+1)}{\partial \mathbf{h}(j)} = D_j \cdot W,$$

and $D_j = \mathrm{diag}\big(\sigma'(W\mathbf{h}(j) + W_{\text{in}}\mathbf{u}(j) + \mathbf{b})\big)$ is the diagonal matrix of activation derivatives at time j.

13.4.3 The Vanishing and Exploding Gradient Problem

The gradient signal propagating from time t back to time k is scaled by the matrix product $\prod_{j=k}^{t-1} J(j)$. This is a product of $(t-k)$ matrices. The behavior of such products is exactly the subject of **multiplicative ergodic theory** (Chapter 11).

By the Oseledets multiplicative ergodic theorem, for an ergodic system the product $\prod_{j=0}^{n-1} J(j)$ grows (or decays) exponentially at rates determined by the **Lyapunov exponents** $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$. More precisely, the norm of the product acting on a "generic" vector $\mathbf{v}$ satisfies

$$\lim_{n\to\infty} \frac{1}{n} \log \left\| \prod_{j=0}^{n-1} J(j)\, \mathbf{v} \right\| = \lambda_1.$$

This has immediate consequences for gradient computation:

- If $\lambda_1 > 0$ (the maximal Lyapunov exponent is positive), then the gradient products grow exponentially. This is the **exploding gradient problem**: the gradient $\partial\mathcal{L}/\partial W$ diverges as T increases, making optimization unstable.
- If $\lambda_1 < 0$ (all Lyapunov exponents are negative), then the gradient products decay exponentially. This is the **vanishing gradient problem**: the gradient contribution from distant time steps is exponentially suppressed, and the network cannot learn long-range temporal dependencies.

The connection to dynamics is direct. Recall from Part I that $\lambda_1 > 0$ characterizes sensitive dependence on initial conditions — chaos. An RNN operating in the chaotic regime has exploding gradients. An RNN in the stable (contracting) regime has vanishing gradients. The boundary between these regimes — the **edge of chaos** — will turn out to be precisely where reservoir computing is most effective.

Remark 13.1. This connection was recognized early, though not in the language of Lyapunov exponents. Hochreiter's 1991 diploma thesis identified the fundamental difficulty: the error signal flowing backwards through time either vanishes or explodes. Bengio, Simard, and Frasconi (1994) provided a more detailed analysis and coined the term "vanishing gradient problem." Their analysis was essentially a study of the spectral properties of the Jacobian products — multiplicative ergodic theory in all but name.

13.4.4 Attempts to Solve the Problem

The vanishing/exploding gradient problem has driven much of the development of recurrent architectures.

Gradient clipping. A pragmatic fix: if $\|\nabla_\theta \mathcal{L}\|$ exceeds a threshold, rescale it. This prevents explosion but does not address vanishing.

Long Short-Term Memory (LSTM). Hochreiter and Schmidhuber (1997) introduced a carefully designed architecture with **gating mechanisms** that allow the network to maintain a roughly constant error flow over time. The LSTM cell contains a "memory cell" $c(t)$ governed by

$$\begin{aligned}
\mathbf{f}(t) &= \sigma_g\big(W_f \mathbf{h}(t-1) + W_{f,\text{in}}\mathbf{u}(t) + \mathbf{b}_f\big),\\
\mathbf{i}(t) &= \sigma_g\big(W_i \mathbf{h}(t-1) + W_{i,\text{in}}\mathbf{u}(t) + \mathbf{b}_i\big),\\
\mathbf{c}(t) &= \mathbf{f}(t) \odot \mathbf{c}(t-1) + \mathbf{i}(t) \odot \tanh\big(W_c \mathbf{h}(t-1) + W_{c,\text{in}}\mathbf{u}(t) + \mathbf{b}_c\big),
\end{aligned}$$

where σ_g is the logistic sigmoid, $\odot$ denotes the Hadamard (elementwise) product, and $\mathbf{f}(t)$, $\mathbf{i}(t)$ are the "forget" and "input" gates, respectively. The key idea, in dynamical terms, is that when $\mathbf{f}(t) \approx \mathbf{1}$ and $\mathbf{i}(t) \approx \mathbf{0}$, the cell state satisfies $\mathbf{c}(t) \approx \mathbf{c}(t-1)$ — the Jacobian $\partial\mathbf{c}(t)/\partial\mathbf{c}(t-1)$ is close to the identity. This ensures that the Lyapunov exponent along the cell state direction is close to zero, preventing both vanishing and explosion.

Gated Recurrent Units (GRU). Cho et al. (2014) proposed a simplified gating architecture with similar properties.

These architectures are effective in practice but come at a cost: they are complex, require careful initialization, and training still involves gradient descent through long unrolled computation graphs. The computational cost scales with $T \times N \times |\theta|$, and convergence can be slow.

This motivates a radical question: *do we need to train the recurrent dynamics at all?*

13.5 The Key Insight: Separate Dynamics from Readout

The central idea of reservoir computing emerges from a simple but powerful observation.

13.5.1 The Separation Principle

Consider again the RNN equation

$$\mathbf{h}(t+1) = \sigma\big(W\mathbf{h}(t) + W_{\text{in}}\mathbf{u}(t) + \mathbf{b}\big), \qquad \mathbf{y}(t) = W_{\text{out}}\mathbf{h}(t).$$

The system has two parts:

1. The **dynamics**: the recurrent evolution $\mathbf{h}(t) \mapsto \mathbf{h}(t+1)$, governed by $(W, W_{\text{in}}, \mathbf{b})$.
2. The **readout**: the linear map $\mathbf{h}(t) \mapsto \mathbf{y}(t)$, governed by W_{out}.

All the difficulty of training lies in part (1) — the recurrent dynamics. The readout is a simple linear map. The insight is:

> **Fix the dynamics. Train only the readout.**

That is, choose W, W_{in}, and $\mathbf{b}$ randomly (according to certain principles we will develop in Chapter 14), and never modify them. Use the fixed recurrent system as a **reservoir** — a rich source of dynamical features. Then find W_{out} by solving a linear regression problem:

$$W_{\text{out}} = \arg\min_{W_{\text{out}}} \sum_t \|\mathbf{y}(t) - \mathbf{d}(t)\|^2 = \arg\min_{W_{\text{out}}} \sum_t \|W_{\text{out}}\mathbf{h}(t) - \mathbf{d}(t)\|^2.$$

This is a standard linear least-squares problem with a closed-form solution. Let $H \in \mathbb{R}^{T\times N}$ be the matrix whose rows are $\mathbf{h}(t)^\top$, and let $D \in \mathbb{R}^{T\times p}$ have rows $\mathbf{d}(t)^\top$. Then

$$W_{\text{out}}^\top = (H^\top H)^{-1} H^\top D = H^+ D,$$

where H^+ denotes the Moore-Penrose pseudoinverse. In practice one often uses **Tikhonov regularization** (ridge regression):

$$W_{\text{out}}^\top = (H^\top H + \alpha I)^{-1} H^\top D,$$

for a small $\alpha > 0$, which improves numerical stability and generalization.

13.5.2 Why This Might Work

At first glance, this seems too simple. How can a random dynamical system, with no task-specific training, produce useful computations?

The answer lies in the nature of the reservoir dynamics. The state $\mathbf{h}(t)$ depends, through the recurrence, on the entire history of inputs $\mathbf{u}(0), \mathbf{u}(1), \ldots, \mathbf{u}(t)$. The nonlinear recurrence creates a rich, high-dimensional representation of this input history. The reservoir performs a **nonlinear expansion**: it maps the input history into a high-dimensional state space $\mathbb{R}^N$ where different features of the input sequence become linearly separable.

Consider an analogy from linear algebra. The **kernel trick** in machine learning maps data into a high-dimensional feature space where nonlinear relationships become linear. The reservoir does something similar, but *in time*: it maps a temporal input signal into a high-dimensional state space where temporal relationships become accessible to a linear readout.

For this to work, the reservoir must satisfy certain properties:

1. **The input must influence the state.** The reservoir must be driven by the input — it cannot ignore it.
2. **The state must depend on the input history, not just the current input.** The reservoir must have memory.
3. **The state must eventually forget the distant past.** Otherwise the representation is dominated by ancient history and cannot adapt to current inputs.
4. **Different input histories must produce different states.** The reservoir must be able to distinguish inputs.
5. **The state must be sufficiently high-dimensional and nonlinear.** A linear reservoir can only compute linear functionals of the input.

These desiderata translate directly into dynamical systems properties. Property (3) is a form of asymptotic stability — specifically, the **echo state property** that we will formalize in Chapter 14. Properties (1)–(4) together relate to the concept of **observability** from control theory. Property (5) relates to the dimension and nonlinearity of the dynamics.

13.5.3 The Computational Advantage

The practical advantages of this approach are significant:

- **Training cost.** Solving a linear regression problem costs $O(N^2T + N^3)$, compared to $O(N^2T \cdot \text{epochs})$ for BPTT. For large T, this is a dramatic saving.
- **No vanishing/exploding gradients.** There is no backpropagation through time. The gradient problem simply does not arise.
- **Stability.** Linear regression is a convex optimization problem with a unique global minimum (when $H^\top H + \alpha I$ is invertible, which is always the case for $\alpha > 0$). There are no local minima, saddle points, or other pathologies of the loss landscape.
- **Simplicity.** The implementation is straightforward.

The cost, of course, is that we have given up the ability to optimize the dynamics. The reservoir must be "good enough" as drawn from a random distribution. This places the burden on understanding *which* random dynamical systems make good reservoirs — a question that our mathematical toolkit is well-equipped to address.

13.6 Two Independent Inventions

The reservoir computing paradigm was discovered independently, and nearly simultaneously, by two research groups working in different communities with different motivations. This convergence lends credibility to the idea: the same core principle emerged from different starting points.

13.6.1 Echo State Networks (Jaeger, 2001)

Herbert Jaeger, working at the German National Research Center for Information Technology (GMD), introduced **Echo State Networks** (ESNs) in a technical report in 2001. The framework is exactly the discrete-time system described in Section 13.4:

$$\mathbf{h}(t+1) = \tanh\big(W\mathbf{h}(t) + W_{\text{in}}\mathbf{u}(t)\big), \qquad \mathbf{y}(t) = W_{\text{out}}\mathbf{h}(t).$$

Jaeger's key contributions were:

1. The formulation of the **echo state property** (ESP): the requirement that the reservoir state $\mathbf{h}(t)$ be asymptotically independent of the initial condition $\mathbf{h}(0)$, depending only on the input history. This is a fading-memory or input-forgetting property.
2. A practical sufficient condition: the ESP holds if the spectral radius $\rho(W) < 1$. (As we shall see in Chapter 14, this condition is sufficient but not necessary.)
3. Empirical demonstrations that ESNs, despite their simplicity, could match or exceed the performance of fully-trained RNNs on benchmark time-series prediction tasks — at a fraction of the computational cost.

The name "echo state" reflects the fading-memory property: the current state is an "echo" of past inputs, with recent inputs echoing more loudly than distant ones.

13.6.2 Liquid State Machines (Maass, Natschla0̆308ger, Markram, 2002)

Independently, Wolfgang Maass, Thomas Natschla0̆308ger, and Henry Markram, working at the intersection of computational neuroscience and theoretical computer science, introduced **Liquid State Machines** (LSMs) in 2002. Their motivation was biological: the neocortex appears to process information using networks of spiking neurons that are not precisely tuned, yet are capable of remarkable computational feats. How?

The LSM framework uses a continuous-time dynamical system as the reservoir:

$$\tau_i \dot{v}_i(t) = -v_i(t) + \sum_j w_{ij}\, s_j(t) + \sum_k w_{ik}^{\text{in}}\, u_k(t),$$

where $v_i(t)$ is the membrane potential of neuron i, $s_j(t)$ represents the (nonlinear) output of neuron j, and the system is driven by the input signal $\{u_k(t)\}$. The neurons may be spiking (integrate-and-fire models) rather than rate-based, and the connections may include synaptic dynamics and delays. Despite the differences in formulation, the underlying principle is identical: a fixed, random, high-dimensional dynamical system driven by input, with a trained linear readout.

Maass et al. introduced the notion of the **separation property**: the requirement that different input streams produce sufficiently different internal states. They proved, under certain conditions, a form of computational universality: any time-invariant filter with fading memory can be approximated by an LSM with a sufficiently large reservoir and an appropriate readout.

13.6.3 Unification: Reservoir Computing

For several years, the ESN and LSM communities developed largely in parallel, with different terminology, different implementation details, and different theoretical frameworks. In 2007, Verstraeten, Schrauwen, D'Haene, and Stroobandt proposed the term **reservoir computing** as a unifying label for both approaches, identifying their shared computational principle:

> A fixed, high-dimensional, nonlinear dynamical system (the *reservoir*) transforms input signals into a rich state representation, from which a trained linear readout extracts the desired computation.

This unification was significant because it revealed the essential structure: the choice of reservoir (discrete-time vs. continuous-time, rate-based vs. spiking, small vs. large) is a design choice. The mathematical principles governing performance — the echo state property, separation, approximation capacity — are shared.

The 2009 survey by Lukoševičius and Jaeger provides a comprehensive account of the state of the field at that time and remains an excellent reference.

13.7 Why This Connects to Parts I and II

We close this chapter by making explicit the connections between reservoir computing and the mathematical theory developed in the first two parts of this book. These connections are not

superficial analogies — they are the theoretical foundations on which reservoir computing rests.

13.7.1 Stability and the Echo State Property (Part I)

The echo state property requires that the reservoir dynamics be, in a certain sense, **asymptotically stable** with respect to initial conditions. This connects directly to Chapter 3 (stability of fixed points), Chapter 4 (Lyapunov stability), and Chapter 5 (structural stability). The spectral radius condition $\rho(W) < 1$ is a linearization-based stability criterion, just as in our analysis of discrete maps. In Chapter 14, we will formulate the echo state property precisely and prove it using contraction mapping arguments that generalize the stability theory of Part I.

13.7.2 Chaos and the Edge of Chaos (Part I)

The vanishing/exploding gradient dichotomy reflects the dichotomy between stable and chaotic dynamics. As we discussed in Chapter 7, the transition to chaos in iterated maps is characterized by the maximal Lyapunov exponent crossing zero. Empirically, reservoirs perform best near this transition — the **edge of chaos** — where the dynamics are rich enough to generate useful features but stable enough to maintain the echo state property. We will make this precise in Chapter 15 using the Lyapunov exponent theory from Part I.

13.7.3 Ergodicity and Information Processing (Part II)

The ergodic theory of Part II provides the language for discussing how reservoirs process information statistically. When driven by a stationary ergodic input, the reservoir state process $\{\mathbf{h}(t)\}$ is itself a stationary process (assuming the echo state property holds). The **mixing properties** studied in Chapter 9 determine how quickly the reservoir forgets its past and how effectively it integrates new information. The **entropy** concepts from Chapter 10 quantify the information content of the reservoir state. The **Oseledets theorem** from Chapter 11 governs the growth rates of perturbations and hence the sensitivity of the reservoir to its input.

13.7.4 Takens' Theorem and Reconstruction (Part I)

Perhaps the deepest connection is to **Takens' embedding theorem** (Chapter 6). Recall the setup: given a dynamical system on a manifold M with a map $\Phi : M \to M$ and an observation function $h : M \to \mathbb{R}$, the delay-coordinate map

$$\Psi(\mathbf{x}) = \big(h(\mathbf{x}),\, h(\Phi(\mathbf{x})),\, \ldots,\, h(\Phi^d(\mathbf{x}))\big)$$

is generically an embedding of M into $\mathbb{R}^{d+1}$ when $d \geq 2\dim(M)$.

Now consider a reservoir driven by scalar observations $u(t) = h(\Phi^t(\mathbf{x}_0))$ of an unknown dynamical system. The reservoir state $\mathbf{h}(t)$ depends on $u(t), u(t-1), \ldots$ — it is a nonlinear function of the delay coordinates of the observed system. If the reservoir has the echo state property and is sufficiently high-dimensional, then $\mathbf{h}(t)$ provides a (nonlinear, learned) embedding of the underlying attractor.

This observation, first made rigorous by several authors in the 2010s, provides a theoretical justification for why reservoir computing is particularly effective for tasks involving dynamical systems: time-series prediction, system identification, attractor reconstruction, and Lyapunov exponent estimation. The reservoir, in effect, performs a generalized Takens embedding.

13.7.5 A Roadmap for Part III

The connections identified above will be developed rigorously in the chapters that follow:

- **Chapter 14** defines the echo state property precisely and establishes conditions under which it holds, using the language of contractions and stability from Part I.
- **Chapter 15** analyzes the dynamics of reservoirs — Lyapunov exponents, information dimension, and the edge of chaos — using tools from both Parts I and II.
- **Chapter 16** develops the approximation theory of reservoir computing: what functions can a reservoir compute, and how does this depend on the reservoir's dynamical properties?
- **Chapter 17** connects reservoir computing to Takens' theorem and explores applications to dynamical systems tasks.

The stage is set. We have seen that dynamical systems can compute, that recurrent neural networks are dynamical systems, that training these dynamics is hard, and that one can sidestep the difficulty by fixing the dynamics and training only a linear readout. The remainder of Part III develops the mathematical theory of this idea.

Recommended Reading

1. **H. Jaeger** (2001). "The 'echo state' approach to analysing and training recurrent neural networks." *GMD Technical Report 148*. The foundational paper on Echo State Networks. Remarkably clear and still highly readable.
2. **W. Maass, T. Natschläger, and H. Markram** (2002). "Real-time computing without stable states: A new framework for neural computation based on perturbations." *Neural Computation*, 14(11):2531–2560. The Liquid State Machine paper. More theoretically ambitious than Jaeger's report, with results on computational universality.
3. **S. Hochreiter and J. Schmidhuber** (1997). "Long short-term memory." *Neural Computation*, 9(8):1735–1780. The LSTM paper, essential background for understanding what reservoir computing is an alternative to.
4. **Y. Bengio, P. Simard, and P. Frasconi** (1994). "Learning long-term dependencies with gradient descent is difficult." *IEEE Transactions on Neural Networks*, 5(2):157–166. The definitive early analysis of the vanishing gradient problem.
5. **D. Verstraeten, B. Schrauwen, M. D'Haene, and D. Stroobandt** (2007). "An experimental unification of reservoir computing methods." *Neural Networks*, 20(3):391–403. The paper that coined the term "reservoir computing" and unified ESNs and LSMs.
6. **M. Lukoševičius and H. Jaeger** (2009). "Reservoir computing approaches to recurrent neural network training." *Computer Science Review*, 3(3):127–149. An excellent survey of the field as of 2009, accessible and comprehensive.

7. **S. Hochreiter** (1991). "Untersuchungen zu dynamischen neuronalen Netzen." Diploma thesis, Technische Universität München. The first identification of the vanishing gradient problem, predating the Bengio et al. paper by three years.

For mathematical background on nonautonomous dynamical systems, the reader may consult:

1. **P. Kloeden and M. Rasmussen** (2011). *Nonautonomous Dynamical Systems.* American Mathematical Society. A rigorous treatment of the theory needed for input-driven systems.

Exercises

Exercise 13.1. (Fixed points and stability). Consider an autonomous RNN with $N = 2$, activation $\sigma = \tanh$, bias $\mathbf{b} = \mathbf{0}$, and recurrent weight matrix

$$W = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}$$

for $a > 0$.

(a) Show that $\mathbf{h}^* = \mathbf{0}$ is a fixed point.

(b) Compute the Jacobian $DF(\mathbf{0})$ and determine the eigenvalues. For which values of a is the fixed point asymptotically stable?

(c) The eigenvalues of W are $\pm ia$. These are complex with modulus a. Explain, in terms of the dynamics, what you expect to observe for a slightly less than 1 versus a slightly greater than 1. Verify your prediction numerically by iterating the map for both cases.

—

Exercise 13.2. (Vanishing gradients and Lyapunov exponents). Consider an autonomous RNN with $\sigma = \tanh$ and $\mathbf{b} = \mathbf{0}$. Let $\mathbf{h}^* = \mathbf{0}$ be the fixed point and assume $\rho(W) < 1$ so that $\mathbf{h}^*$ is asymptotically stable.

(a) Show that the Jacobian at any point $\mathbf{h}$ satisfies $\|J(\mathbf{h})\|_2 \leq \|W\|_2$, where $\|\cdot\|_2$ denotes the spectral norm (largest singular value). *Hint:* $|\tanh'(x)| \leq 1$ for all x.

(b) Conclude that for any trajectory $\mathbf{h}(0), \mathbf{h}(1), \ldots$ converging to the fixed point,

$$\left\| \prod_{j=0}^{n-1} J(j) \right\|_2 \leq \|W\|_2^n.$$

(c) What does this imply about the maximal Lyapunov exponent? What does it imply about the gradient $\partial \mathbf{h}(t)/\partial \mathbf{h}(0)$ as $t \to \infty$?

(d) Explain why the condition $\rho(W) < 1$ is insufficient to guarantee rapid vanishing of gradients, while $\|W\|_2 < 1$ is sufficient. Give an example of W with $\rho(W) < 1$ but $\|W\|_2 > 1$.

—

Exercise 13.3. (Linear reservoirs and limitations). Suppose we replace tanh with the identity function, obtaining a linear reservoir:

$$\mathbf{h}(t+1) = W\mathbf{h}(t) + W_{\text{in}}\mathbf{u}(t).$$

(a) Assuming $\rho(W) < 1$ and $\mathbf{h}(0) = \mathbf{0}$, show that

$$\mathbf{h}(t) = \sum_{k=0}^{t-1} W^{t-1-k} W_{\text{in}}\mathbf{u}(k).$$

(b) Show that the output $\mathbf{y}(t) = W_{\text{out}}\mathbf{h}(t)$ is a *linear* functional of the input history $\{\mathbf{u}(k)\}_{k=0}^{t-1}$. Specifically, $\mathbf{y}(t) = \sum_{k=0}^{t-1} C_{t-1-k}\,\mathbf{u}(k)$ for matrices C_j that you should identify.

(c) Explain why a linear reservoir cannot approximate a nonlinear input-output map, no matter how large N is. Why does this demonstrate the necessity of nonlinear activation functions?

—

Exercise 13.4. (Reservoir computing by hand). Consider a scalar reservoir ($N = 1$) with $\sigma = \tanh$, weight $w = 0.9$, input weight $w_{\text{in}} = 0.5$, and no bias:

$$h(t+1) = \tanh(0.9\,h(t) + 0.5\,u(t)).$$

The task is to learn the identity map $d(t) = u(t)$ from the reservoir state.

(a) For the input sequence $u(t) = \sin(0.3\,t)$ with $t = 0, 1, \ldots, 50$ and $h(0) = 0$, compute the reservoir states $h(0), h(1), \ldots, h(50)$ numerically.

(b) Find w_{out} by linear regression: $w_{\text{out}} = \arg\min_w \sum_{t=1}^{50} (w \cdot h(t) - u(t))^2$.

(c) How well does the trained readout approximate $u(t)$ on the training data? What about for $t = 51, \ldots, 100$ (generalization)?

(d) Repeat with $u(t) = \sin(0.3\,t) + 0.5\sin(0.7\,t)$. Does performance degrade? Why might a single-neuron reservoir struggle with more complex signals?

—

Exercise 13.5. (Turing machines and dynamics). Let M be a Turing machine with state set $Q = \{q_0, q_1, q_H\}$ (where q_H is the halt state), tape alphabet $\Gamma = \{0, 1, \sqcup\}$ (where $\sqcup$ is blank), and transition function:

| State | Read | Write | Move | Next State | |——-|——|——-|——|————| | q_0 | 0 | 1 | R | q_1 | | q_0 | 1 | 0 | R | q_0 | | q_0 | $\sqcup$ | $\sqcup$ | L | q_H | | q_1 | 0 | 0 | R | q_0 | | q_1 | 1 | 1 | R | q_1 | | q_1 | $\sqcup$ | $\sqcup$ | L | q_H |

(a) Starting with tape contents $\ldots \sqcup \sqcup 0110 \sqcup \sqcup \ldots$, head at the leftmost 0, and state q_0, trace the trajectory of the dynamical system $F : \mathcal{C} \to \mathcal{C}$ until the machine halts. Record the configuration at each step.

(b) What does this Turing machine compute? Describe its function in terms of the tape contents at start and at halt.

(c) Describe the "phase space" of this dynamical system. Is it finite-dimensional? Is it compact? Compare with the phase spaces of the continuous dynamical systems in Part I.

—

Exercise 13.6. (Separation property). Consider two input sequences $\mathbf{u}^{(1)}(t)$ and $\mathbf{u}^{(2)}(t)$ that are identical for $t < T_0$ and differ for $t \geq T_0$. Let $\mathbf{h}^{(1)}(t)$ and $\mathbf{h}^{(2)}(t)$ be the corresponding reservoir states, starting from the same initial condition.

(a) Show that $\mathbf{h}^{(1)}(t) = \mathbf{h}^{(2)}(t)$ for all $t \leq T_0$.

(b) For $t > T_0$, define the separation $\delta(t) = \|\mathbf{h}^{(1)}(t) - \mathbf{h}^{(2)}(t)\|$. Show that for $\sigma = \tanh$,

$$\delta(t+1) \leq \|W\|_2 \, \delta(t) + \|W_{\text{in}}\|_2 \, \|\mathbf{u}^{(1)}(t) - \mathbf{u}^{(2)}(t)\|.$$

(c) Discuss the implications: if $\|W\|_2 < 1$, does the reservoir amplify or suppress differences between input signals? What if $\|W\|_2 > 1$? Why is this relevant to the reservoir's ability to distinguish different inputs?

(d) Argue that a good reservoir must balance two competing requirements: separation of different inputs (which benefits from larger $\|W\|_2$) and the echo state property (which requires the dynamics to be contractive in some sense). Relate this to the edge-of-chaos hypothesis.

Chapter 14

Echo State Networks

In the preceding chapters, we developed the mathematical foundations of dynamical systems, ergodic theory, and the general framework of reservoir computing. We now turn to the most widely studied concrete instantiation of the reservoir computing paradigm: the **Echo State Network** (ESN), introduced by Jaeger (2001). The ESN makes the abstract ideas of Chapter 13 concrete and computationally tractable. A large, randomly generated recurrent network serves as the reservoir, and only a simple linear readout is trained. The theoretical justification for this architecture rests on a property called the *echo state property*, which we will analyze in detail using the contraction and Lyapunov theory developed in Part I.

14.1 Architecture of an Echo State Network

An echo state network consists of three components: an **input layer**, a **reservoir** (also called the *dynamical reservoir* or *hidden layer*), and a **readout layer**. We describe each in turn.

14.1.1 The Three Components

Input layer. At each discrete time step t, the network receives an input vector $\mathbf{u}(t) \in \mathbb{R}^K$. The input is connected to the reservoir through a weight matrix $W_{\text{in}} \in \mathbb{R}^{N\times K}$, where N is the number of reservoir neurons. The matrix W_{in} is generated randomly and held fixed throughout training.

Reservoir. The reservoir consists of N neurons whose activations form the state vector $\mathbf{x}(t) \in \mathbb{R}^N$. The neurons are recurrently connected through a weight matrix $W \in \mathbb{R}^{N\times N}$. Like W_{in}, the matrix W is generated randomly at initialization and is never modified. The reservoir is the computational core of the network: it transforms the input into a high-dimensional, nonlinear, dynamical representation.

Readout layer. The output $\mathbf{y}(t) \in \mathbb{R}^L$ is a linear function of the reservoir state:

$$\mathbf{y}(t) = W_{\text{out}}\mathbf{x}(t),$$

where $W_{\text{out}} \in \mathbb{R}^{L\times N}$ is the readout weight matrix. This is the *only* part of the network that is trained.

The architecture can be visualized schematically as follows:

```
  Input Reservoir Output
u(t) in R^K x(t) in R^N y(t) in R^L

[u_1] ----\ /--[x_1]---[x_2]--\ /--- [y_1]
[u_2] -----W_in-->/ | \ / | \ \--W_out--/---- [y_2]
 ... | / [x_3]--[x_4] | \ ...
[u_K] ----/ / | \ / | \ [x_5] \--- [y_L]
               / [x_6]--[x_7]--/ |
               \ ... ... /
                \---- recurrent W ---/
```

The key structural feature is the separation of concerns: the reservoir provides a rich dynamical representation, and the readout extracts the desired computation from that representation. Since W_{out} is a linear map, training reduces to solving a linear regression problem.

14.1.2 State Update Equation

The reservoir state evolves according to the update rule

$$\mathbf{x}(t+1) = f(W\mathbf{x}(t) + W_{\text{in}}\mathbf{u}(t+1) + \mathbf{b}), \tag{14.1}$$

where $f : \mathbb{R}^N \to \mathbb{R}^N$ is a componentwise nonlinear activation function, typically tanh, and $\mathbf{b} \in \mathbb{R}^N$ is a bias vector. The tanh function is applied elementwise:

$$f(\mathbf{z})_i = \tanh(z_i), \quad i = 1, \ldots, N.$$

The choice of tanh is natural: it is a bounded, monotone, smooth sigmoidal function with $\tanh(0) = 0$ and $|\tanh(z)| < 1$ for all $z \in \mathbb{R}$. Its derivative satisfies $\tanh'(z) = 1 - \tanh^2(z) \leq 1$, a fact we will use repeatedly.

Remark 14.1. Equation (14.1) defines a nonautonomous discrete dynamical system on $\mathbb{R}^N$. The input sequence $\{\mathbf{u}(t)\}$ acts as a time-dependent driving signal. With no input ($\mathbf{u} \equiv \mathbf{0}$ and $\mathbf{b} = \mathbf{0}$), the system reduces to the autonomous map $\mathbf{x} \mapsto \tanh(W\mathbf{x})$, whose dynamics we studied in Chapter 5.

14.1.3 Leaky Integrator Variant

A common and important generalization replaces the update rule (14.1) with a *leaky integrator*:

$$\mathbf{x}(t+1) = (1-\alpha)\,\mathbf{x}(t) + \alpha\, f(W\mathbf{x}(t) + W_{\text{in}}\mathbf{u}(t+1)), \tag{14.2}$$

where $\alpha \in (0, 1]$ is the **leak rate**. When $\alpha = 1$, we recover the standard update (14.1) (with $\mathbf{b} = \mathbf{0}$). For $\alpha < 1$, the reservoir state changes more slowly: the current state is a weighted average of the previous state and the nonlinear update. This introduces a form of exponential smoothing that allows the reservoir to operate on slower timescales.

The leak rate α can be interpreted as a discrete-time approximation of a continuous-time system. If we write $\alpha = \Delta t/\tau$ for a time step Δt and time constant τ, then (14.2) approximates the ODE

$$\tau\,\dot{\mathbf{x}} = -\mathbf{x} + f(W\mathbf{x} + W_{\text{in}}\mathbf{u}).$$

Smaller values of α correspond to longer time constants and slower dynamics, which can be beneficial when the input signal varies slowly relative to the discrete time step.

14.2 The Echo State Property

The central theoretical concept underlying ESNs is the **echo state property** (ESP). Informally, the ESP asserts that the reservoir "forgets" its initial state: the current state $\mathbf{x}(t)$ is determined, in the long run, by the history of inputs alone. This is what allows the reservoir to function as a well-defined operator on input sequences.

14.2.1 Formal Definition

We work with the state update equation (14.1) and consider input sequences drawn from a compact set $\mathcal{U} \subset \mathbb{R}^K$. Let $\mathbf{u} = (\ldots, \mathbf{u}(-1), \mathbf{u}(0), \mathbf{u}(1), \ldots)$ denote a bi-infinite input sequence with $\mathbf{u}(t) \in \mathcal{U}$ for all t.

Definition 14.1 (Echo State Property)**.** The reservoir (14.1) has the **echo state property** with respect to the input set $\mathcal{U}$ if, for every input sequence $\mathbf{u}$ with values in $\mathcal{U}$ and for any two initial states $\mathbf{x}(0), \mathbf{x}'(0) \in \mathbb{R}^N$, the corresponding state trajectories satisfy

$$\lim_{t\to\infty} \|\mathbf{x}(t) - \mathbf{x}'(t)\| = 0. \tag{14.3}$$

When the ESP holds, the reservoir state $\mathbf{x}(t)$ is asymptotically independent of the initial condition. This means there exists, for each input sequence, a unique trajectory that all initial conditions converge to. More precisely, we obtain the following.

Proposition 14.1. **If the reservoir has the ESP, then for each bi-infinite input sequence* $\mathbf{u}$ *there exists a unique trajectory* $\{\bar{\mathbf{x}}(t)\}_{t\in\mathbb{Z}}$ *such that for every initial condition* $\mathbf{x}(0)$*, the driven trajectory satisfies* $\|\mathbf{x}(t) - \bar{\mathbf{x}}(t)\| \to 0$ *as* $t \to \infty$*. Moreover,* $\bar{\mathbf{x}}(t)$ *depends only on the input history* $(\ldots, \mathbf{u}(t-1), \mathbf{u}(t))$*.**

Proof. * Fix an input sequence $\mathbf{u}$. For each $s \le t$, let $\mathbf{x}^{(s)}(t)$ denote the state at time t obtained by starting from $\mathbf{x}(s) = \mathbf{0}$ at time s and driving with the input. By the ESP, for any $s_1 < s_2$, we have $\|\mathbf{x}^{(s_1)}(t) - \mathbf{x}^{(s_2)}(t)\| \to 0$ as $t \to \infty$. For fixed t, consider the sequence $\{\mathbf{x}^{(s)}(t)\}_{s\le t}$ as $s \to -\infty$. Since all states lie in the compact set $[-1,1]^N$ (the range of tanh), we can extract a convergent subsequence. The ESP guarantees that the limit is unique (any two accumulation points are driven by the same input from time $-\infty$ and must coincide). Define $\bar{\mathbf{x}}(t) = \lim_{s\to-\infty} \mathbf{x}^{(s)}(t)$. This trajectory depends only on the input history up to time t, and the ESP ensures that every initial condition converges to it. □

□

Remark 14.2. The ESP is precisely the property needed for the reservoir to implement a well-defined **filter**: a causal, time-invariant functional mapping from input histories to states. In the language of Chapter 13, the reservoir with the ESP defines an *echo state map* $E : \mathcal{U}^{(-\infty,t]} \to \mathbb{R}^N$ that sends the input history to the current reservoir state.

14.2.2 Connection to Contraction Theory

The ESP is fundamentally a **contraction** property. Recall from Chapter 4 that a map $F : \mathbb{R}^N \to \mathbb{R}^N$ is a contraction with respect to a norm $\|\cdot\|$ if there exists $\gamma \in [0,1)$ such that $\|F(\mathbf{x}) - F(\mathbf{x}')\| \leq \gamma\|\mathbf{x} - \mathbf{x}')\|$ for all $\mathbf{x}, \mathbf{x}'$.

In our setting, for a fixed input $\mathbf{u}$, the one-step update defines a map

$$F_{\mathbf{u}}(\mathbf{x}) = f(W\mathbf{x} + W_{\text{in}}\mathbf{u} + \mathbf{b}).$$

If $F_{\mathbf{u}}$ is a contraction for every $\mathbf{u} \in \mathcal{U}$, then the Banach fixed-point theorem guarantees that repeated application forgets the initial state, and the ESP follows. This is the strategy behind the spectral radius condition of the next section.

14.3 Sufficient Conditions for the Echo State Property

14.3.1 The Spectral Radius Condition

The most widely used sufficient condition for the ESP involves the **spectral radius** of the reservoir weight matrix W.

Theorem 14.2 (Spectral Radius Condition for ESP)**.** **Let $f = \tanh$ applied componentwise. If $\rho(W) < 1$, where $\rho(W)$ denotes the spectral radius of W, then the reservoir (14.1) has the echo state property for any compact input set $\mathcal{U} \subset \mathbb{R}^K$.**

Proof. * Let $\mathbf{x}(t)$ and $\mathbf{x}'(t)$ be two state trajectories driven by the same input sequence $\{\mathbf{u}(t)\}$, starting from different initial conditions. Define $\mathbf{e}(t) = \mathbf{x}(t) - \mathbf{x}'(t)$. We have

$$\mathbf{e}(t+1) = f(W\mathbf{x}(t) + W_{\text{in}}\mathbf{u}(t+1) + \mathbf{b}) - f\big(W\mathbf{x}'(t) + W_{\text{in}}\mathbf{u}(t+1) + \mathbf{b}\big)\,.$$

Since tanh is applied componentwise and $|\tanh'(z)| = 1 - \tanh^2(z) \leq 1$ for all z, the mean value theorem gives

$$|f(\mathbf{a})_i - f(\mathbf{a}')_i| = |\tanh(a_i) - \tanh(a_i')| \leq |a_i - a_i'|$$

for each component i. In vector notation, writing $D(t)$ for the diagonal matrix with entries $D_{ii}(t) = \tanh'(\xi_i(t)) \in [0,1]$ (where $\xi_i(t)$ is the appropriate intermediate value), we have

$$\mathbf{e}(t+1) = D(t)\,W\,\mathbf{e}(t).$$

Therefore,

$$\|\mathbf{e}(t+1)\| \leq \|D(t)\, W\, \mathbf{e}(t)\| \leq \|D(t)\| \cdot \|W\| \cdot \|\mathbf{e}(t)\| \leq \|W\| \cdot \|\mathbf{e}(t)\|, \tag{14.4}$$

since $\|D(t)\| \leq 1$ (as $D(t)$ is diagonal with entries in $[0,1]$) in any operator norm induced by a monotone vector norm.

Now, inequality (14.4) gives $\|\mathbf{e}(t)\| \leq \|W\|^t \|\mathbf{e}(0)\|$. However, $\|W\|$ is the operator norm, which may exceed 1 even when $\rho(W) < 1$ (since $\rho(W) \leq \|W\|$ for any operator norm, with equality not guaranteed). We therefore need a more refined argument.

Since $\rho(W) < 1$, Gelfand's formula gives $\rho(W) = \lim_{n\to\infty} \|W^n\|^{1/n}$. For any γ with $\rho(W) < \gamma < 1$, there exists a matrix norm $\|\cdot\|_*$ (an operator norm induced by an appropriate vector norm on $\mathbb{R}^N$) such that $\|W\|_* \leq \gamma$ (see, e.g., Horn and Johnson, *Matrix Analysis*, Lemma 5.6.10). This norm can be constructed explicitly: for any $\epsilon > 0$ with $\rho(W) + \epsilon < 1$, there exists an invertible matrix P such that $\|P^{-1}WP\| \leq \rho(W)+\epsilon$, which defines the norm $\|\mathbf{z}\|_* = \|P^{-1}\mathbf{z}\|$.

In this adapted norm, the diagonal matrix $D(t)$ still satisfies $\|D(t)\|_* \leq 1$ (since $D(t)$ has entries in $[0,1]$, and a change of basis does not affect this bound when we account for the equivalence of norms, though we must be slightly careful). Alternatively, and more cleanly: we iterate the bound. We have

$$\mathbf{e}(t) = D(t-1)W \cdot D(t-2)W \cdots D(0)W \cdot \mathbf{e}(0).$$

Since $\|D(s)\| \leq 1$ for each s,

$$\|\mathbf{e}(t)\| \leq \|W^t\| \cdot \|\mathbf{e}(0)\|,$$

where the inequality follows because the $D(s)$ factors, being contractive, can only reduce the norm. More precisely, for any vector $\mathbf{v}$,

$$\|D(s)W\mathbf{v}\| \leq \|W\mathbf{v}\|$$

so by induction $\|\mathbf{e}(t)\| \leq \|W^t\| \cdot \|\mathbf{e}(0)\|$. Now by Gelfand's formula, $\|W^t\|^{1/t} \to \rho(W) < 1$, so there exist $C > 0$ and $\gamma \in (\rho(W), 1)$ such that $\|W^t\| \leq C\gamma^t$ for all $t \geq 0$. Hence

$$\|\mathbf{e}(t)\| \leq C\gamma^t \|\mathbf{e}(0)\| \to 0 \quad \text{as } t \to \infty.$$

This establishes the ESP. □

Remark 14.3. The proof reveals that the convergence is exponentially fast, with rate governed by $\rho(W)$. Smaller spectral radius means faster forgetting of initial conditions, but also faster forgetting of inputs – a tradeoff we will discuss in Section 14.4.

Remark 14.4. The converse is not true: $\rho(W) < 1$ is sufficient but not necessary for the ESP. Yildiz, Jaeger, and Kiebel (2012) showed that the ESP can hold even when $\rho(W) > 1$, because the input-driven system may be contractive even if the autonomous system ($\mathbf{u} \equiv \mathbf{0}$) is not. The nonlinearity tanh saturates for large arguments, and strong input driving can keep the reservoir in the contractive regime. This is an important subtlety for practice.

14.3.2 The Role of Input Scaling

The spectral radius condition considers W in isolation, but the effective dynamics depend on the interplay between W and W_{in}. To see this, note that the Jacobian of the state update (14.1) with respect to $\mathbf{x}$ is

$$J(t) = D(t)\, W, \tag{14.5}$$

where $D(t) = \operatorname{diag}\Big(1 - \tanh^2(W\mathbf{x}(t) + W_{\text{in}}\mathbf{u}(t+1) + \mathbf{b})\Big)$. The diagonal entries of $D(t)$ are the derivatives $\tanh'(\cdot)$ evaluated at the pre-activation values. When the pre-activations are large in magnitude (which happens when $\|W_{\text{in}}\|$ is large, pushing neurons into saturation), the entries of $D(t)$ become small. This means strong input scaling effectively makes the reservoir *more contractive*, even if $\rho(W) \geq 1$.

Conversely, weak input scaling keeps neurons in the approximately linear regime near $\tanh(z) \approx z$ for $|z| \ll 1$, where $D(t) \approx I$. In this regime, the effective dynamics are close to the linear map $\mathbf{x} \mapsto W\mathbf{x}$, and the spectral radius condition $\rho(W) < 1$ becomes essentially necessary.

14.3.3 Necessary Condition: Lyapunov Exponents

The ESP is connected to the Lyapunov exponents of the driven reservoir system, which we studied in Chapter 6.

Definition 14.2. The **largest (maximal) Lyapunov exponent** of the driven reservoir system is

$$\lambda_{\max} = \lim_{t\to\infty} \frac{1}{t} \ln \left\| \prod_{s=0}^{t-1} J(s) \right\|, \tag{14.6}$$

where $J(s) = D(s)\, W$ is the Jacobian (14.5), and the limit is taken along a typical trajectory.

By the multiplicative ergodic theorem (Theorem 6.3, Oseledets' theorem), under suitable ergodicity assumptions on the input process, this limit exists almost surely and is independent of the initial condition.

Theorem 14.3 (Necessary Condition for ESP)**.** **If the echo state property holds, then the largest Lyapunov exponent of the driven system satisfies $\lambda_{\max} < 0$.**

Proof sketch. sketch.* If $\lambda_{\max} \geq 0$, then there exist perturbations that do not decay (or grow) exponentially under the linearized dynamics. This means that nearby trajectories do not converge, contradicting the ESP. The rigorous argument uses the Oseledets decomposition: if $\lambda_{\max} \geq 0$, the corresponding Oseledets subspace yields directions along which the distance between trajectories does not shrink to zero. □

The condition $\lambda_{\max} < 0$ is both necessary and (generically) sufficient for the ESP. It provides a more refined characterization than the spectral radius condition, but it is harder to verify in practice because it depends on the input statistics and the resulting reservoir dynamics.

□

Remark 14.5. The spectral radius condition $\rho(W) < 1$ can now be understood as implying $\lambda_{\max} < 0$ for all inputs. Indeed, from the proof of Theorem 14.1, we saw that $\|D(s)W\| \leq \|W\|$ for all s, so

$$\lambda_{\max} = \lim_{t\to\infty} \frac{1}{t} \ln \left\| \prod_{s=0}^{t-1} D(s)W \right\| \leq \lim_{t\to\infty} \frac{1}{t} \ln \|W^t\| = \ln \rho(W) < 0.$$

—

14.4 Reservoir Design and Hyperparameters

The performance of an ESN depends critically on several hyperparameters that govern the reservoir dynamics. We discuss the most important ones.

14.4.1 Spectral Radius

The spectral radius $\rho(W)$ is the single most important hyperparameter. It controls the *timescale of memory* in the reservoir.

- **Small $\rho(W)$ (e.g., $\rho(W) \ll 1$).** The reservoir is strongly contractive. States are dominated by recent inputs; the memory of past inputs decays rapidly. The reservoir dynamics are essentially those of a feedforward nonlinear filter with short memory.
- **$\rho(W)$ near 1.** The reservoir retains information about past inputs for longer. The dynamics are richer, with longer transients and more complex temporal patterns in the state. This is typically the regime of best performance for tasks requiring memory.
- **$\rho(W) > 1$.** The autonomous system ($\mathbf{u} \equiv \mathbf{0}$) may be unstable, but the driven system can still possess the ESP if the input keeps the dynamics in the contractive regime (Remark 14.4). However, the risk of losing the ESP increases.

In practice, the reservoir weight matrix is often constructed as follows:

1. Generate a random matrix $\widetilde{W}$ (e.g., entries drawn from a uniform or normal distribution).
2. Compute its spectral radius $\rho(\widetilde{W})$.
3. Rescale: $W = \frac{\rho_{\text{desired}}}{\rho(\widetilde{W})}\widetilde{W}$.

This ensures that $\rho(W) = \rho_{\text{desired}}$.

14.4.2 Reservoir Size

The reservoir size N (number of neurons) determines the dimensionality of the state space and hence the *computational capacity* of the reservoir. A larger reservoir can represent more complex functions of the input history. However, there are diminishing returns: doubling N does not double performance. Moreover, larger reservoirs require more training data (to avoid overfitting the readout) and more computation.

A rough guideline from practice: N should be at least as large as the number of independent "features" needed to solve the task, but the exact relationship depends on the task complexity and the reservoir dynamics.

14.4.3 Input Scaling

The input weight matrix W_{in} is typically generated randomly (e.g., uniform on $[-\sigma_{\text{in}}, \sigma_{\text{in}}]$), and the parameter σ_{in} controls the **input scaling**.

- **Small σ_{in}**: neurons operate in the approximately linear regime $\tanh(z) \approx z$. The reservoir behaves approximately as a linear dynamical system.
- **Large σ_{in}**: neurons are driven into saturation ($\tanh(z) \approx \pm 1$). The reservoir behaves more like a binary, highly nonlinear system.

The input scaling thus controls the *degree of nonlinearity* in the reservoir's response. Tasks that require strong nonlinear processing of the input benefit from larger input scaling; tasks that are approximately linear benefit from smaller input scaling.

14.4.4 Sparsity and Topology

In many implementations, W is a **sparse** matrix: most entries are zero. Sparse reservoirs often perform comparably to dense ones, and they are computationally cheaper (both for state updates and for eigenvalue computations). A typical choice is to have each neuron connected to roughly 10–20 others, regardless of N.

Beyond sparsity, the **topology** of the reservoir connections can be varied:

- **Random (Erdos-Renyi)**: each possible connection is present independently with some probability. This is the standard choice.
- **Ring**: neurons are arranged in a cycle, with each neuron connected to its nearest neighbors. This creates a simple, structured delay line.
- **Small-world**: a ring topology with a few random long-range connections. This combines local structure with global communication.
- **Scale-free**: connection degrees follow a power law distribution.

Empirically, the choice of topology has a secondary effect compared to the spectral radius and input scaling. The ring topology, however, has been shown to produce surprisingly good results with minimal randomness (Rodan and Tino, 2011).

14.4.5 The Edge of Chaos Hypothesis

An influential conjecture in reservoir computing is that reservoirs operating near the **edge of chaos** — the boundary between ordered (contractive) and chaotic dynamics — have optimal computational properties. The idea, borrowed from the theory of cellular automata and random Boolean networks (Langton, 1990), is that:

- In the ordered regime, the reservoir rapidly forgets inputs and cannot maintain complex computations.
- In the chaotic regime, the reservoir is too sensitive to perturbations, and the ESP is lost.
- At the boundary, the reservoir achieves a balance between memory and sensitivity, enabling the richest possible computation.

In the ESN setting, this boundary corresponds roughly to $\lambda_{\max} = 0$, or in the autonomous case, to $\rho(W) = 1$. The hypothesis has received both theoretical and empirical support (Bertschinger and Natschlager, 2004; Boedecker et al., 2012), though the picture is not en-

tirely clear-cut. For some tasks, the optimal spectral radius is well below 1; for others, values slightly above 1 perform best.

Remark 14.6. The edge of chaos hypothesis connects reservoir computing to the broader theory of computation at phase transitions, which has deep connections to statistical physics and the theory of critical phenomena. From the dynamical systems perspective, the transition at $\lambda_{\max} = 0$ is a bifurcation of the driven system, and the computational properties of the reservoir can be understood through the lens of bifurcation theory (Chapter 8).

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14.5 Training the Readout

The defining feature of the ESN approach is that training is restricted to the readout weights W_{out}. Since the readout is linear, this is a linear regression problem with a closed-form solution.

14.5.1 Collecting Reservoir States

Given a training input sequence $\mathbf{u}(1), \mathbf{u}(2), \ldots, \mathbf{u}(T_{\text{total}})$ and corresponding target outputs $\mathbf{d}(1), \mathbf{d}(2), \ldots, \mathbf{d}(T_{\text{tota}}$ the training procedure is:

1. **Initialize** the reservoir state $\mathbf{x}(0)$ (e.g., to $\mathbf{0}$).
2. **Drive** the reservoir with the training input using the update equation (14.1), producing states $\mathbf{x}(1), \mathbf{x}(2), \ldots, \mathbf{x}(T_{\text{total}})$.
3. **Discard** the first T_0 states (the *washout* or *transient* period) to eliminate the effect of the arbitrary initial condition. The ESP guarantees that after sufficiently many steps, the state is effectively independent of $\mathbf{x}(0)$.
4. **Collect** the remaining states $\mathbf{x}(T_0 + 1), \ldots, \mathbf{x}(T_{\text{total}})$ and corresponding targets $\mathbf{d}(T_0 + 1), \ldots, \mathbf{d}(T_{\text{total}})$.

For notational convenience, relabel the collected data so that the training set consists of T pairs $\{(\mathbf{x}(t), \mathbf{d}(t))\}_{t=1}^{T}$, where $T = T_{\text{total}} - T_0$.

14.5.2 The Regression Problem

We seek $W_{\text{out}} \in \mathbb{R}^{L \times N}$ to minimize the squared error between the network output $\mathbf{y}(t) = W_{\text{out}}\mathbf{x}(t)$ and the target $\mathbf{d}(t)$:

$$\min_{W_{\text{out}}} \sum_{t=1}^{T} \|\mathbf{d}(t) - W_{\text{out}}\mathbf{x}(t)\|^2.$$

Form the **state matrix** $X \in \mathbb{R}^{N \times T}$ whose columns are the collected states:

$$X = \begin{bmatrix} \mathbf{x}(1) & \mathbf{x}(2) & \cdots & \mathbf{x}(T) \end{bmatrix},$$

and the **target matrix** $D \in \mathbb{R}^{L \times T}$:

$$D = \begin{bmatrix} \mathbf{d}(1) & \mathbf{d}(2) & \cdots & \mathbf{d}(T) \end{bmatrix}.$$

The regression problem becomes

$$\min_{W_{\text{out}}} \|D - W_{\text{out}} X\|_F^2, \tag{14.7}$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

14.5.3 Ridge Regression

In principle, the solution to (14.7) is given by the pseudoinverse: $W_{\text{out}} = DX^+$. In practice, this is ill-conditioned because the reservoir states are often highly correlated (the state matrix X has nearly collinear columns). Instead, we use **ridge regression** (Tikhonov regularization), adding a penalty on $\|W_{\text{out}}\|_F^2$:

$$\min_{W_{\text{out}}} \left\{ \|D - W_{\text{out}} X\|_F^2 + \beta \|W_{\text{out}}\|_F^2 \right\}, \tag{14.8}$$

where $\beta > 0$ is the **regularization parameter**.

Proposition 14.4. *The unique solution to (14.8) is*

$$W_{out} = DX^\top (XX^\top + \beta I_N)^{-1}. \tag{14.9}$$

Equivalently, $W_{out} = (DX^\top)(XX^\top + \beta I_N)^{-1}$, where $XX^\top \in \mathbb{R}^{N\times N}$ is the state correlation matrix.

Proof. * The objective function (14.8) can be written, treating the rows of W_{out} independently, as L separate scalar ridge regression problems. For a single output dimension, we minimize

$$\|\mathbf{d}^\top - \mathbf{w}^\top X\|^2 + \beta \|\mathbf{w}\|^2$$

over $\mathbf{w} \in \mathbb{R}^N$, where $\mathbf{d}^\top \in \mathbb{R}^{1\times T}$ is one row of D and $\mathbf{w}^\top$ is the corresponding row of W_{out}. Setting the gradient to zero:

$$-2X(\mathbf{d} - X^\top \mathbf{w}) + 2\beta \mathbf{w} = \mathbf{0},$$

which gives

$$XX^\top \mathbf{w} + \beta \mathbf{w} = X\mathbf{d},$$

$$\mathbf{w} = (XX^\top + \beta I)^{-1} X\mathbf{d}.$$

The matrix $XX^\top + \beta I$ is symmetric positive definite (since $XX^\top$ is positive semidefinite and $\beta > 0$), so the inverse exists and the solution is unique. Assembling all L rows gives (14.9). □

Remark 14.7. An equivalent formulation uses the "dual" form. Define $\tilde{X} = X^\top \in \mathbb{R}^{T\times N}$ (the state matrix with time steps as rows). Then $W_{\text{out}}^\top = \tilde{X}^\top (\tilde{X}\tilde{X}^\top + \beta I_T)^{-1} \tilde{D}^\top$, which involves inverting a $T \times T$ matrix instead of $N \times N$. When $T \gg N$ (many time steps, moderate reservoir), the $N \times N$ form (14.9) is preferred; when $N \gg T$, the $T \times T$ form may be more efficient.

14.5.4 Why Ridge Regression?

There are several reasons to prefer ridge regression over ordinary least squares ($\beta = 0$):

1. **Numerical stability.** The matrix $XX^\top$ is often ill-conditioned because the reservoir states are correlated. Adding βI improves the condition number from $\kappa(XX^\top)$ to at most $(\sigma_{\max}^2 + \beta)/\beta$, where $\sigma_{\max}$ is the largest singular value of X.
2. **Regularization (overfitting prevention).** The penalty $\beta\|W_{\text{out}}\|_F^2$ shrinks the readout weights toward zero, preventing the readout from fitting noise in the training data. This is especially important when N is large relative to T.
3. **Bias-variance tradeoff.** Ridge regression introduces a small bias (the readout does not perfectly fit the training data) but substantially reduces variance (the readout generalizes better to new data). The parameter β controls this tradeoff and is typically chosen by cross-validation.
4. **Convexity.** The objective (14.8) is strictly convex (for $\beta > 0$), ensuring a unique global minimum with a closed-form solution. No iterative optimization or gradient descent is needed.

This last point is a major practical advantage of ESNs over fully trained recurrent networks: the training is a single matrix computation, not an iterative optimization over a non-convex loss landscape.

14.6 Worked Example: Mackey-Glass Time Series Prediction

We now work through a complete example to illustrate the ESN methodology. The task is to predict a chaotic time series generated by the Mackey-Glass delay differential equation.

14.6.1 The Mackey-Glass System

The Mackey-Glass equation is the delay differential equation

$$\frac{dx}{dt} = \frac{a\,x(t-\tau_d)}{1 + x(t-\tau_d)^{10}} - b\,x(t), \tag{14.10}$$

with standard parameters $a = 0.2$, $b = 0.1$, and delay $\tau_d = 17$. For $\tau_d > 16.8$, the system exhibits chaotic behavior. We discretize the solution with time step Δt to obtain a scalar time series $s(1), s(2), \ldots$

The prediction task is: given $s(t)$, predict $s(t+h)$ for some prediction horizon h (say $h = 1$ for one-step-ahead prediction). We frame this as an ESN problem with scalar input $u(t) = s(t)$ ($K = 1$) and scalar target $d(t) = s(t+h)$ ($L = 1$).

14.6.2 Reservoir Setup

We choose the following hyperparameters:

- Reservoir size: $N = 500$.
- Spectral radius: $\rho(W) = 0.9$.
- Input scaling: $\sigma_{\text{in}} = 0.1$.

- Bias: $\mathbf{b} = \mathbf{0}$ (no bias, for simplicity).
- Regularization: $\beta = 10^{-6}$.
- Washout: $T_0 = 500$ time steps.
- Training length: $T = 5000$ time steps (after washout).

The reservoir weight matrix is constructed as:

1. Generate $\widetilde{W} \in \mathbb{R}^{500\times 500}$ with entries drawn uniformly from $\{-1, 0, 0, 0, 0, 1\}$ (resulting in approximately 1/3 nonzero entries).
2. Compute $\rho(\widetilde{W})$ and rescale: $W = 0.9 \cdot \widetilde{W}/\rho(\widetilde{W})$.

The input weights are drawn uniformly: $(W_{\text{in}})_i \sim \text{Uniform}(-0.1, 0.1)$ for $i = 1, \ldots, 500$, giving a column vector $W_{\text{in}} \in \mathbb{R}^{500\times 1}$.

14.6.3 Running the Reservoir

Starting from $\mathbf{x}(0) = \mathbf{0} \in \mathbb{R}^{500}$, we drive the reservoir with the Mackey-Glass time series:

$$\mathbf{x}(t+1) = \tanh(W\mathbf{x}(t) + W_{\text{in}}\, s(t+1))\,, \quad t = 0, 1, 2, \ldots \tag{14.11}$$

After $T_0 = 500$ washout steps, we collect the states. The state at time $t > T_0$ encodes, through the reservoir dynamics, a nonlinear function of the input history $(s(1), s(2), \ldots, s(t))$. By the ESP (guaranteed since $\rho(W) = 0.9 < 1$), the dependence on the initial condition $\mathbf{x}(0)$ has decayed to negligible levels after the washout.

14.6.4 Forming the Regression Problem

We assemble the state matrix and target vector. Since $L = 1$ (scalar output), the target matrix D is a row vector:

$$X = \begin{bmatrix} \mathbf{x}(T_0+1) & \mathbf{x}(T_0+2) & \cdots & \mathbf{x}(T_0+T) \end{bmatrix} \in \mathbb{R}^{500\times 5000},$$

$$D = \begin{bmatrix} s(T_0+1+h) & s(T_0+2+h) & \cdots & s(T_0+T+h) \end{bmatrix} \in \mathbb{R}^{1\times 5000}.$$

The readout is a row vector $W_{\text{out}} \in \mathbb{R}^{1\times 500}$.

14.6.5 Solving for the Readout

Applying formula (14.9):

$$W_{\text{out}} = DX^\top (XX^\top + \beta I_{500})^{-1}. \tag{14.12}$$

Here $XX^\top \in \mathbb{R}^{500\times 500}$ is the empirical correlation matrix of the reservoir states, $DX^\top \in \mathbb{R}^{1\times 500}$ is the cross-correlation between states and targets, and $\beta = 10^{-6}$ provides regularization. The computation requires:

1. Computing $XX^\top$: this is a sum of $T = 5000$ outer products, $XX^\top = \sum_{t=1}^{T} \mathbf{x}(t)\mathbf{x}(t)^\top$.
2. Computing $DX^\top$: this is $\sum_{t=1}^{T} d(t)\mathbf{x}(t)^\top$.
3. Solving the 500×500 linear system $(XX^\top + \beta I)\mathbf{w} = (DX^\top)^\top$ for $\mathbf{w} = W_{\text{out}}^\top$.

14.6.6 Evaluation

On a test input sequence $s(T_{\text{total}}+1), s(T_{\text{total}}+2), \ldots$, we continue driving the reservoir (maintaining the state from the end of training) and compute the predicted output:

$$\hat{s}(t+h) = W_{\text{out}}\,\mathbf{x}(t).$$

The prediction quality is measured by the normalized root mean square error (NRMSE):

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{T_{\text{test}}}\sum_t (s(t+h)-\hat{s}(t+h))^2}}{\sigma_s},$$

where σ_s is the standard deviation of the target time series. With the parameters above, one typically obtains NRMSE $\approx 10^{-3}$ to 10^{-2} for one-step-ahead prediction, demonstrating that the ESN can learn the short-term dynamics of the chaotic Mackey-Glass system with high accuracy.

14.6.7 Why Does This Work?

The reservoir creates a **high-dimensional nonlinear embedding** of the input history. Each reservoir neuron $x_i(t)$ is, by virtue of the recurrent dynamics, a nonlinear function of the past inputs $s(t), s(t-1), s(t-2), \ldots$ The collection of $N = 500$ such functions spans a rich function space. The linear readout then selects the appropriate linear combination of these nonlinear features to approximate the target function $s(t) \mapsto s(t+h)$.

This is analogous to kernel methods in machine learning: the reservoir plays the role of a (random, temporal) feature map that lifts the input into a high-dimensional space where the desired function becomes approximately linear. The key difference is that the reservoir's features are *temporal*: they depend not on a single input, but on the entire input history, weighted by the reservoir's fading memory.

From the dynamical systems perspective, the reservoir implements a nonlinear observer of the Mackey-Glass system. The Takens embedding theorem (Chapter 7) guarantees that, under generic conditions, a sufficiently long delay embedding of a scalar observable can reconstruct the attractor of the underlying dynamical system. The reservoir states provide a richer, nonlinear version of such a delay embedding, which is why the ESN can capture the dynamics even of chaotic systems.

14.7 Worked Example: Nonlinear Function Approximation

We now consider a different type of task: computing a nonlinear function of past inputs. This illustrates the ESN's role as a *nonlinear filter with memory.*

14.7.1 Task: Polynomial of Past Inputs

Let $u(t)$ be a scalar input drawn i.i.d. from a uniform distribution on $[-1, 1]$. The target output is

$$d(t) = 0.4\,u(t) + 0.4\,u(t)\,u(t-1) + 0.1\,u(t-2)^2 + 0.05\,u(t-3). \tag{14.13}$$

This target depends on the current input and three past inputs, combined through nonlinear (quadratic) operations. The task tests both the reservoir's **memory** (it must retain information about $u(t-1), u(t-2), u(t-3)$) and its **nonlinear processing** (it must compute products of past inputs).

14.7.2 Setup

We use a smaller reservoir for this simpler task:

- $N = 100$, $\rho(W) = 0.95$, $\sigma_{\text{in}} = 0.5$, $\beta = 10^{-4}$.
- Training: $T = 3000$ steps after $T_0 = 200$ washout steps.
- Input: $u(t) \sim \text{Uniform}(-1, 1)$, i.i.d.

14.7.3 Analysis

The state update is

$$\mathbf{x}(t+1) = \tanh(W\mathbf{x}(t) + W_{\text{in}}\,u(t+1)).$$

Each component $x_i(t)$ can be formally expanded as a Volterra-like series in the past inputs. For example, in the regime where $\|W\|$ and $\|W_{\text{in}}\|$ are moderate, a Taylor expansion of tanh gives, to leading order:

$$x_i(t) \approx \sum_j (W_{\text{in}})_j\,u(t) + \sum_{j,k} W_{ij}(W_{\text{in}})_k\,u(t)\,u(t-1) + \cdots$$

This expansion shows that the reservoir states naturally contain monomials of past inputs. With $N = 100$ neurons and random weights, the reservoir generates a large number of such monomials (and more complex nonlinear combinations), providing a rich basis from which the linear readout can approximate the target (14.13).

14.7.4 Training and Evaluation

The training proceeds exactly as in Section 14.6: collect states, form X and D, solve the ridge regression (14.9). On a held-out test set of $T_{\text{test}} = 1000$ i.i.d. input steps, we compute the NRMSE. With the parameters above, one typically obtains NRMSE $\approx 10^{-2}$ or better.

To understand the result, we can examine the readout weights $W_{\text{out}} \in \mathbb{R}^{1\times 100}$. Typically, only a fraction of the 100 weights are large; the readout selects a relatively sparse linear combination of the reservoir states that best approximates the target polynomial (14.13).

14.7.5 Delayed XOR Task

As a second example within this section, consider the **delayed XOR** task: the input $u(t) \in \{0, 1\}$ is binary, and the target is

$$d(t) = u(t - \delta_1) \oplus u(t - \delta_2), \tag{14.14}$$

where $\oplus$ denotes XOR (exclusive or) and δ_1, δ_2 are delays. Since XOR is a nonlinear Boolean function, this task requires both nonlinearity and memory (with depth $\max(\delta_1, \delta_2)$).

To apply an ESN, we encode $\{0, 1\}$ as real values (e.g., $0 \mapsto -0.5$, $1 \mapsto +0.5$) and use the reservoir as before. The XOR function can be expressed algebraically as

$$u_1 \oplus u_2 = u_1 + u_2 - 2u_1u_2 \quad (\text{on } \{0, 1\}),$$

so the target is a polynomial of past inputs, and the analysis of Section 14.7.3 applies. The reservoir must generate features that include (or approximate) the product $u(t - \delta_1) \cdot u(t - \delta_2)$; the linear readout then combines them to produce the XOR.

For small delays ($\delta_1, \delta_2 \leq 5$, say), a reservoir of size $N = 100$ with $\rho(W) \approx 0.9$ can learn this task with high accuracy. As the delays increase, a larger spectral radius (longer memory) and more neurons are needed.

14.8 Theoretical Perspective

We close the chapter with some broader theoretical observations connecting ESNs to the themes of the book.

14.8.1 ESNs as Nonautonomous Dynamical Systems

An ESN driven by an input sequence is a nonautonomous discrete dynamical system:

$$\mathbf{x}(t + 1) = F(\mathbf{x}(t), \mathbf{u}(t + 1)), \quad F(\mathbf{x}, \mathbf{u}) = \tanh(W\mathbf{x} + W_{\text{in}}\mathbf{u} + \mathbf{b}).$$

The ESP states that this system has a globally attracting trajectory for each input sequence. In the language of nonautonomous dynamics (Chapter 10), the system possesses a **pullback attractor** that is a single trajectory. This is the strongest possible form of stability for a nonautonomous system.

14.8.2 Fading Memory and the Stone-Weierstrass Theorem

Under the ESP, the reservoir implements a **fading memory filter**: a functional H that maps the input history $(\ldots, \mathbf{u}(t-1), \mathbf{u}(t))$ to the output $\mathbf{y}(t)$, with the property that recent inputs have more influence than distant ones. Boyd and Chua (1985) showed that fading memory filters on compact input sets can be uniformly approximated by Volterra series (finite-degree polynomials of the input history). Since an ESN with sufficiently many neurons can approximate any finite collection of such polynomial terms (as argued informally in Section 14.7.3), this provides a universal approximation justification: ESNs with large enough reservoirs can approximate any fading memory filter to arbitrary accuracy.

This result is the temporal analog of the universal approximation theorem for feedforward neural networks. It can be made rigorous using the Stone-Weierstrass theorem on the space of continuous fading memory functionals, as shown by Grigoryeva and Ortega (2018).

14.8.3 Ergodic Inputs and Stationary Reservoirs

When the input $\{\mathbf{u}(t)\}$ is a stationary ergodic process, and the reservoir has the ESP, the reservoir state $\{\mathbf{x}(t)\}$ is also a stationary ergodic process (since it is a deterministic function of the input history). This connects the ESP to the ergodic theory of Part II: time averages of reservoir states converge to ensemble averages, which is essential for the training procedure (Section 14.5) to work consistently. The empirical correlation matrix $\frac{1}{T}XX^\top$ converges, by the ergodic theorem, to the true correlation matrix $\mathbb{E}[\mathbf{x}(t)\mathbf{x}(t)^\top]$ as $T \to \infty$.

Exercises

Exercise 14.1. (Contraction with leaky integration). Consider the leaky integrator ESN (14.2) with $f = \tanh$. Show that if $(1-\alpha)+\alpha\,\rho(W) < 1$, then the system has the echo state property. *Hint*: bound the Jacobian of the right-hand side of (14.2) with respect to $\mathbf{x}$ and apply an argument analogous to the proof of Theorem 14.1.

Exercise 14.2. (Linear reservoir). Suppose the activation function f is the identity ($f(\mathbf{z}) = \mathbf{z}$), so the reservoir is the linear system $\mathbf{x}(t+1) = W\mathbf{x}(t) + W_{\text{in}}\mathbf{u}(t+1)$. (a) Show that the echo state property holds if and only if $\rho(W) < 1$. (b) Derive an explicit formula for the reservoir state as a function of the input history: $\mathbf{x}(t) = \sum_{k=0}^{\infty} W^k W_{\text{in}}\mathbf{u}(t-k)$. (c) Show that this is a vector-valued infinite moving average (IMA) filter, and relate the memory length to $\rho(W)$.

Exercise 14.3. (Spectral radius is not necessary). Construct a 2×2 matrix W with $\rho(W) = 1.5$ and an input sequence $\{u(t)\}$ such that the ESN with $f = \tanh$ and $W_{\text{in}} = \begin{pmatrix}1\\1\end{pmatrix}$ still has the echo state property. *Hint*: use a constant input with large magnitude, so that tanh saturates and the effective Jacobian is strongly contractive.

Exercise 14.4. (Ridge regression as Bayesian inference). Consider the readout training problem with scalar output ($L = 1$). Suppose the prior on $\mathbf{w} = W_{\text{out}}^\top \in \mathbb{R}^N$ is $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \beta^{-1}I)$ and the likelihood is $d(t)|\mathbf{w}, \mathbf{x}(t) \sim \mathcal{N}(\mathbf{w}^\top\mathbf{x}(t), \sigma^2)$. Show that the maximum a posteriori (MAP) estimate of $\mathbf{w}$ given the training data $\{(\mathbf{x}(t), d(t))\}_{t=1}^T$ is the ridge regression solution (14.9) with regularization parameter $\beta\sigma^2$.

Exercise 14.5. (Memory capacity). Consider a linear ESN ($f = \text{id}$) with scalar input ($K = 1$) and the task of recovering the input from k time steps ago: $d(t) = u(t-k)$. (a) Show that the readout trained by ordinary least squares achieves zero error if $k < N$ (assuming W is chosen appropriately), i.e., the reservoir can perfectly remember the last $N-1$ inputs. (b) Using the orthogonality of delayed inputs (for i.i.d. input), show that the total memory capacity $\sum_{k=1}^{\infty} \text{Corr}^2(\hat{d}_k(t), u(t-k))$ is at most N for any readout. This is the result of Jaeger (2001).

Exercise 14.6. (Lyapunov exponent computation). Consider an ESN with $N = 2$, $W = \begin{pmatrix} 0 & w \\ -w & 0 \end{pmatrix}$, and a constant scalar input $u(t) = c$ with $W_{\text{in}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. (a) Find the fixed point $\mathbf{x}^*$ of the driven system $\mathbf{x} \mapsto \tanh(W\mathbf{x} + W_{\text{in}}c)$. (b) Compute the Jacobian $J = D^*W$ at the fixed point, where $D^* = \text{diag}(1 - (x_1^*)^2,\ 1 - (x_2^*)^2)$. (c) Express the Lyapunov exponents in terms of the eigenvalues of J. For what values of w and c is the largest Lyapunov exponent negative?

Exercise 14.7. (Topology experiment, conceptual). Consider two ESNs of size $N = 100$: one with a random sparse reservoir matrix W (Erdos-Renyi with connection probability 0.1) and one with a ring topology (each neuron connected to its two nearest neighbors on a cycle). Both are scaled to spectral radius $\rho(W) = 0.9$. (a) Describe qualitatively how the ring topology creates a delay-line structure. What is the maximum delay that the ring can represent? (b) Argue that for a task requiring memory of exactly k steps, the ring topology is more efficient (requires smaller N) than a random topology when k is not too large. (c) Conversely, argue that the random topology may be superior for tasks requiring nonlinear combinations of inputs at many different delays, because of the richer mixing of information.

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Recommended Reading

1. **H. Jaeger** (2001). *The "echo state" approach to analysing and training recurrent neural networks.* GMD Technical Report 148. The foundational paper introducing ESNs and the echo state property. Essential reading.
2. **H. Jaeger and H. Haas** (2004). *Harnessing nonlinearity: predicting chaotic systems and saving energy in wireless communication.* Science, 304(5667), 78–80. Demonstrated ESNs on chaotic time series prediction (including Mackey-Glass) and a practical engineering application, bringing ESNs to broad scientific attention.
3. **M. Lukoševičius and H. Jaeger** (2009). *Reservoir computing approaches to recurrent neural network training.* Computer Science Review, 3(3), 127–149. A comprehensive survey of reservoir computing, including ESNs and liquid state machines. Covers practical guidelines for hyperparameter selection.
4. **I. B. Yildiz, H. Jaeger, and S. J. Kiebel** (2012). *Re-visiting the echo state property.* Neural Networks, 35, 1–9. A careful re-examination of the ESP, showing that the spectral radius condition is sufficient but not necessary, and providing refined conditions based on the driven dynamics.
5. **M. Buehner and P. Young** (2006). *A tighter bound for the echo state property.* IEEE Transactions on Neural Networks, 17(3), 820–824. Provides tighter sufficient conditions for the ESP using the singular values of W.
6. **S. Grigoryeva and J.-P. Ortega** (2018). *Echo state networks are universal.* Neural Networks, 108, 495–508. Proves the universal approximation property of ESNs for filters with fading memory, connecting to the Stone-Weierstrass theorem.
7. **N. Bertschinger and T. Natschlager** (2004). *Real-time computation at the edge of chaos in recurrent neural networks.* Neural Computation, 16(7), 1413–1436. Theoretical

and empirical investigation of the edge-of-chaos hypothesis in recurrent networks.

Chapter 15

Theoretical Foundations of Reservoir Computing

In the preceding chapters, we constructed reservoir computers and demonstrated their empirical effectiveness on time-series tasks. We now turn to the theoretical questions that any mathematician should demand answers to: What, precisely, can reservoir computers compute? Why do they work? And what are their fundamental limitations?

This chapter develops the mathematical theory of reservoir computing from first principles. We will see that the framework connects naturally to classical functional analysis, approximation theory, and the theory of nonlinear systems. The central results — the universal approximation theorem for reservoir computers, the information processing capacity bound, and the edge-of-chaos hypothesis — reveal reservoir computing not as an ad hoc engineering trick but as a principled computational architecture grounded in the mathematics of dynamical systems.

Throughout this chapter, we work with discrete-time systems. The input is a sequence $u = (\ldots, u(t-2), u(t-1), u(t))$ taking values in a compact set $\mathcal{U} \subset \mathbb{R}^d$, the reservoir state is $\mathbf{x}(t) \in \mathbb{R}^N$, and the output is $y(t) \in \mathbb{R}^m$. The reservoir update and readout are

$$\mathbf{x}(t+1) = F(\mathbf{x}(t), u(t+1)), \qquad y(t) = h(\mathbf{x}(t)),$$

where $F : \mathbb{R}^N \times \mathcal{U} \to \mathbb{R}^N$ is the reservoir map and $h : \mathbb{R}^N \to \mathbb{R}^m$ is the readout function.

15.1 Reservoir Computing as Function Approximation

15.1.1 From Sequences to Outputs

Consider the reservoir system driven by an input sequence. If the echo state property holds (Chapter 14), the reservoir state $\mathbf{x}(t)$ is uniquely determined by the left-infinite input history $\mathbf{u}_t = (\ldots, u(t-2), u(t-1), u(t))$. That is, there exists a map

$$\mathcal{E} : \mathcal{U}^{-\mathbb{N}} \to \mathbb{R}^N$$

such that $\mathbf{x}(t) = \mathcal{E}(\mathbf{u}_t)$, where $\mathcal{U}^{-\mathbb{N}}$ denotes the space of left-infinite sequences with values in $\mathcal{U}$.

The reservoir computer thus implements a composition:

$$\mathbf{u}_t \overset{\mathcal{E}}{\mapsto} \mathbf{x}(t) \overset{h}{\mapsto} y(t).$$

The entire system realizes a *functional*

$$\mathcal{H} = h \circ \mathcal{E} : \mathcal{U}^{-\mathbb{N}} \to \mathbb{R}^m$$

that maps input histories to outputs. This is the object we must study. Note the distinction: a function maps finite-dimensional vectors to vectors; a functional (in our usage) maps sequences — elements of an infinite-dimensional space — to finite-dimensional outputs.

15.1.2 The Approximation Question

The fundamental question of reservoir computing theory is:

> **What class of functionals $\mathcal{H} : \mathcal{U}^{-\mathbb{N}} \to \mathbb{R}^m$ can be approximated by reservoir computers?**

This parallels the classical universal approximation question for feedforward neural networks. There, the answer (Cybenko 1989, Hornik et al. 1989) is that feedforward networks with a single hidden layer can approximate any continuous function on a compact set. Here, we seek an analogous result for the richer setting of sequence-to-output maps.

The answer, as we shall see, involves the *fading memory property*: reservoir computers can approximate precisely those functionals that depend continuously on the input history, with geometrically decaying sensitivity to the distant past.

15.1.3 The Role of the Two Components

It is useful to separate the computational work into two parts:

1. **The reservoir** $\mathcal{E}$ maps the infinite-dimensional input history into a finite-dimensional state. It performs a *nonlinear, history-dependent embedding.*
1. **The readout** h maps the finite-dimensional state to the output. In standard reservoir computing, h is a simple function (linear or polynomial).

The power of the reservoir computer depends on both components: the reservoir must produce states that are *sufficiently informative* about the input history, and the readout must be *sufficiently expressive* to extract the desired output from those states. We will formalize these requirements as the *separation property* and the *approximation property* in Section 15.4.

15.2 Fading Memory and the Echo State Property

15.2.1 The Fading Memory Property

Not all functionals on input sequences are reasonable computational targets. A functional that depends sensitively on the input received a million time steps ago — with no decay in that

sensitivity — is not physically realizable by any finite system. The correct class of functionals for reservoir computing is captured by the *fading memory property*, introduced by Boyd and Chua (1985).

Definition 15.1 (Weighted norm on input sequences)**.** Let $w : \mathbb{N} \to (0, \infty)$ be a *weight function* satisfying: 1. $w(0) = 1$, 2. w is monotonically decreasing, 3. $\lim_{k\to\infty} w(k) = 0$.

For an input sequence $\mathbf{u} = (\ldots, u(-1), u(0)) \in \mathcal{U}^{-\mathbb{N}}$, define the *weighted supremum norm*

$$\|\mathbf{u}\|_w = \sup_{k\geq 0} w(k)\, \|u(-k)\|,$$

where $\|u(-k)\|$ is the Euclidean norm on $\mathbb{R}^d$.

The weight function w encodes the principle that recent inputs (k small) matter more than distant ones (k large). Because $w(k) \to 0$, differences in the distant past are suppressed. Common choices include exponential weights $w(k) = r^k$ for $r \in (0, 1)$ and polynomial weights $w(k) = (1 + k)^{-\alpha}$ for $\alpha > 0$.

Definition 15.2 (Fading memory)**.** A time-invariant functional $\mathcal{H} : \mathcal{U}^{-\mathbb{N}} \to \mathbb{R}^m$ has the *fading memory property* if there exists a weight function w such that $\mathcal{H}$ is continuous with respect to the topology induced by $\|\cdot\|_w$. That is, for every $\mathbf{u} \in \mathcal{U}^{-\mathbb{N}}$ and every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\|\mathbf{u} - \mathbf{v}\|_w < \delta \implies \|\mathcal{H}(\mathbf{u}) - \mathcal{H}(\mathbf{v})\| < \varepsilon.$$

The fading memory property says: if two input histories agree approximately on the recent past (and the weight function suppresses the distant past), their outputs are close. This is the mathematically precise version of the intuition that "the system eventually forgets."

Remark 15.1. The weighted topology on $\mathcal{U}^{-\mathbb{N}}$ is strictly coarser than the product topology (where two sequences are close only if they agree on *all* coordinates). By requiring continuity only in this coarser topology, the fading memory property is a weaker — and hence more permissive — condition than continuity in the product topology.

15.2.2 The Echo State Property Implies Fading Memory

We now prove the key connection between the echo state property (ESP) and fading memory. Recall from Chapter 14 that a reservoir has the ESP if, for every input sequence, the reservoir state is asymptotically independent of the initial condition.

Theorem 15.4. *Let the reservoir map $F : \mathbb{R}^N \times \mathcal{U} \to \mathbb{R}^N$ be such that the reservoir system*

$$\mathbf{x}(t + 1) = F(\mathbf{x}(t), u(t + 1))$$

has the echo state property on a compact set $\mathcal{U}$ of inputs. Suppose further that F is Lipschitz continuous in its first argument uniformly over $\mathcal{U}$: there exists $L < 1$ such that

$$\|F(\mathbf{x}, u) - F(\mathbf{y}, u)\| \leq L\|\mathbf{x} - \mathbf{y}\|$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ and $u \in \mathcal{U}$. Then the reservoir-to-state functional $\mathcal{E}$ has the fading memory property with respect to the exponential weight $w(k) = L^k$.

Proof. * Since $L < 1$, the map $F(\cdot, u)$ is a contraction for each u. By the echo state property, for any input sequence $\mathbf{u}$, the state converges to a unique trajectory independent of the initial condition. Denote this state by $\mathbf{x}(t) = \mathcal{E}(\mathbf{u}_t)$.

Consider two input sequences $\mathbf{u}$ and $\mathbf{v}$ and let $\mathbf{x}(t) = \mathcal{E}(\mathbf{u}_t)$ and $\tilde{\mathbf{x}}(t) = \mathcal{E}(\mathbf{v}_t)$ be the corresponding reservoir states at time $t = 0$. These states are obtained as limits of the iterated compositions. Define the sequences of states starting from a common initial condition $\mathbf{x}_0$ at time $-T$:

$$\mathbf{x}^{(T)}(k+1) = F(\mathbf{x}^{(T)}(k), u(k+1)), \quad k = -T, \ldots, -1,$$
$$\tilde{\mathbf{x}}^{(T)}(k+1) = F(\tilde{\mathbf{x}}^{(T)}(k), v(k+1)), \quad k = -T, \ldots, -1,$$

with $\mathbf{x}^{(T)}(-T) = \tilde{\mathbf{x}}^{(T)}(-T) = \mathbf{x}_0$. By the ESP, $\mathbf{x}^{(T)}(0) \to \mathcal{E}(\mathbf{u}_0)$ and $\tilde{\mathbf{x}}^{(T)}(0) \to \mathcal{E}(\mathbf{v}_0)$ as $T \to \infty$.

Now we bound the difference at time 0. Let M_F be the Lipschitz constant of F with respect to its second argument (which exists since F is continuous on the compact set). At each step k from $-T$ to -1:

$$\|\mathbf{x}^{(T)}(k+1) - \tilde{\mathbf{x}}^{(T)}(k+1)\| = \|F(\mathbf{x}^{(T)}(k), u(k+1)) - F(\tilde{\mathbf{x}}^{(T)}(k), v(k+1))\|.$$

Adding and subtracting $F(\mathbf{x}^{(T)}(k), v(k+1))$ and applying the triangle inequality:

$$\leq L\|\mathbf{x}^{(T)}(k) - \tilde{\mathbf{x}}^{(T)}(k)\| + M_F\|u(k+1) - v(k+1)\|.$$

Let $e_k = \|\mathbf{x}^{(T)}(k) - \tilde{\mathbf{x}}^{(T)}(k)\|$. We have the recurrence $e_{k+1} \leq L \cdot e_k + M_F\|u(k+1) - v(k+1)\|$ with $e_{-T} = 0$. Unrolling:

$$e_0 \leq M_F \sum_{j=0}^{T-1} L^j \|u(-j) - v(-j)\|.$$

Now we use the weighted norm. Since $\|u(-j) - v(-j)\| \leq \frac{1}{w(j)}\|\mathbf{u} - \mathbf{v}\|_w = L^{-j}\|\mathbf{u} - \mathbf{v}\|_w$, we get

$$e_0 \leq M_F \sum_{j=0}^{T-1} L^j \cdot L^{-j}\|\mathbf{u} - \mathbf{v}\|_w = M_F \cdot T \cdot \|\mathbf{u} - \mathbf{v}\|_w.$$

This bound grows with T, which is not immediately useful. However, we can improve it by using a weight function that decays *faster* than L^k. Choose $w(k) = \rho^k$ with $L < \rho < 1$. Then $\|u(-j) - v(-j)\| \leq \rho^{-j}\|\mathbf{u} - \mathbf{v}\|_w$, and

$$e_0 \leq M_F \sum_{j=0}^{T-1} L^j \rho^{-j}\|\mathbf{u} - \mathbf{v}\|_w = M_F\|\mathbf{u} - \mathbf{v}\|_w \sum_{j=0}^{T-1} \left(\frac{L}{\rho}\right)^j.$$

Since $L/\rho < 1$, the geometric series converges as $T \to \infty$:

$$\|\mathcal{E}(\mathbf{u}_0) - \mathcal{E}(\mathbf{v}_0)\| \leq \frac{M_F}{1 - L/\rho}\|\mathbf{u} - \mathbf{v}\|_w.$$

This shows that $\mathcal{E}$ is Lipschitz continuous — and hence continuous — with respect to $\|\cdot\|_w$. Since the readout h is continuous, the composite $\mathcal{H} = h \circ \mathcal{E}$ is continuous in the weighted norm, establishing the fading memory property. □

Remark 15.2. The contraction condition $L < 1$ is stronger than the echo state property per se. The ESP requires only that the state be asymptotically independent of initial conditions, not that the state map be a uniform contraction. However, the contraction condition is the standard sufficient condition used in practice and is equivalent to the ESP for many commonly used reservoir architectures (see Yildiz, Jaeger, and Kiebel 2012 for a thorough discussion).

15.2.3 Why Fading Memory Is the Right Condition

The restriction to fading-memory functionals is not merely a technical convenience. There are fundamental reasons why this is the correct class:

1. **Physical realizability.** Any physical system with bounded energy and finite degrees of freedom can only implement fading-memory computations. A system that retains perfect information about inputs from the arbitrarily distant past would require unbounded resources (e.g., infinite precision or infinite memory).

1. **Stability.** Fading memory implies a form of input-output stability: bounded perturbations to the input produce bounded perturbations to the output.

1. **Learnability.** Fading-memory functionals form a tractable class for learning. As we shall see, they admit approximation by polynomials and truncated Volterra series, providing concrete approximation schemes.

1. **Generality.** Despite being a restriction, the class of fading-memory functionals is large. It includes all ARMA filters, all stable nonlinear filters, Volterra systems with decaying kernels, and many other systems of practical interest.

Boyd and Chua (1985) showed that fading memory is equivalent to the requirement that the functional be approximable by finite Volterra series — a result we will revisit in Section 15.5.

15.3 Universal Approximation

We now come to the central theoretical result: reservoir computers are *universal approximators* of fading-memory functionals. This is the analogue of the universal approximation theorem (Cybenko 1989) for feedforward neural networks, lifted from functions to functionals.

15.3.1 The Theorem

Theorem 15.6 (Universal Approximation for Reservoir Computers; Maass, Natschläger, and Markram 2002; Grigoryeva and Ortega 2018). *Let $\mathcal{K} \subset \mathcal{U}^{-\mathbb{N}}$ be a compact set of input sequences (in the product topology on $\mathcal{U}^{-\mathbb{N}}$, with $\mathcal{U}$ compact). Let $\mathcal{H} : \mathcal{K} \to \mathbb{R}$ be a continuous

functional with the fading memory property. Then for every $\varepsilon > 0$, there exists a reservoir system with the echo state property and a polynomial readout $p : \mathbb{R}^N \to \mathbb{R}$ such that*

$$\sup_{\mathbf{u}\in\mathcal{K}} |\mathcal{H}(\mathbf{u}) - p(\mathcal{E}(\mathbf{u}))| < \varepsilon.$$

In other words: the set of functionals realizable by (ESP reservoir + polynomial readout) is dense in the space of fading-memory functionals on compact input sets, under the uniform norm.

15.3.2 Proof Sketch

The proof proceeds in several stages. We present a streamlined version following Grigoryeva and Ortega (2018).

Step 1: The space of input sequences as a compact set.

Equip $\mathcal{U}^{-\mathbb{N}}$ with the product topology. Since $\mathcal{U}$ is compact and Hausdorff, Tychonoff's theorem implies that $\mathcal{U}^{-\mathbb{N}}$ is compact in the product topology. Let $\mathcal{K} \subseteq \mathcal{U}^{-\mathbb{N}}$ be a closed (hence compact) subset.

Step 2: Fading-memory functionals are continuous in the product topology on compact sets.

On a compact subset of $\mathcal{U}^{-\mathbb{N}}$, the weighted topology (for any weight function w with $w(k) \to 0$) coincides with the product topology. This is because both topologies are metrizable and agree on convergent sequences when restricted to $\mathcal{K}$: in both topologies, $\mathbf{u}_n \to \mathbf{u}$ if and only if $u_n(k) \to u(k)$ for each k. (The key point is that $\mathcal{U}$ is compact, so the sequences are uniformly bounded, and the decay of w handles the tail.)

Therefore, a fading-memory functional restricted to $\mathcal{K}$ is a continuous real-valued function on a compact Hausdorff space.

Step 3: Applying the Stone-Weierstrass theorem.

The Stone-Weierstrass theorem states: if X is a compact Hausdorff space and $\mathcal{A} \subset C(X, \mathbb{R})$ is a subalgebra that *separates points* and *contains the constants*, then $\mathcal{A}$ is dense in $C(X, \mathbb{R})$ in the uniform norm.

We will show that the set of functionals of the form $p \circ \mathcal{E}$, where p ranges over polynomials $\mathbb{R}^N \to \mathbb{R}$ and $\mathcal{E}$ is the reservoir-to-state map, forms a subalgebra of $C(\mathcal{K}, \mathbb{R})$ satisfying the hypotheses of Stone-Weierstrass.

Step 3a: Algebra structure. The set of polynomial functions in the components $x_1, \ldots, x_N$ of $\mathcal{E}(\mathbf{u})$ forms an algebra: it is closed under addition, scalar multiplication, and pointwise multiplication (since the product of two polynomials is a polynomial). It contains the constants (degree-0 polynomials).

Step 3b: Separation of points. We need: if $\mathbf{u} \neq \mathbf{v}$ in $\mathcal{K}$, then there exists a polynomial p such that $p(\mathcal{E}(\mathbf{u})) \neq p(\mathcal{E}(\mathbf{v}))$. It suffices to show that $\mathcal{E}(\mathbf{u}) \neq \mathcal{E}(\mathbf{v})$, for then a coordinate projection (a linear polynomial) separates them.

The requirement $\mathcal{E}(\mathbf{u}) \neq \mathcal{E}(\mathbf{v})$ whenever $\mathbf{u} \neq \mathbf{v}$ is exactly the *separation property* of the reservoir, which we discuss in detail in Section 15.4. It does not hold for arbitrary reservoirs — a reservoir with $N = 1$ neuron cannot separate all distinct input histories. But it holds for

"sufficiently rich" reservoirs, and we can always ensure it by taking N large enough.

More precisely, we do not need $\mathcal{E}$ to separate *all* input histories — only those in $\mathcal{K}$ that are distinguished by the target functional. Since a fading-memory functional $\mathcal{H}$ depends on inputs only up to the decay of the weight function, it suffices for $\mathcal{E}$ to separate histories that differ in any finite window. Any ESP reservoir with a non-degenerate state map achieves this.

Step 3c: Conclusion. By Stone-Weierstrass, the algebra $\{p \circ \mathcal{E} : p \text{ polynomial}\}$ is dense in $C(\mathcal{K}, \mathbb{R})$. Since every fading-memory functional on $\mathcal{K}$ lies in $C(\mathcal{K}, \mathbb{R})$, it can be uniformly approximated by $p \circ \mathcal{E}$ for a suitable polynomial p. □

Remark 15.3. The proof is *non-constructive*: it tells us that an approximating reservoir exists but does not tell us how to find it. In practice, we fix the reservoir (randomly) and train only the readout. The theorem guarantees that if the reservoir is rich enough and the readout is expressive enough, the approximation can be made arbitrarily good.

Remark 15.4. The requirement for polynomial readouts can be relaxed. Linear readouts suffice if the reservoir itself provides sufficiently nonlinear features. In practice, linear readouts trained by ridge regression work remarkably well because typical reservoir architectures (with tanh nonlinearities and recurrent connections) already produce highly nonlinear functions of the input history.

15.3.3 Connection to Classical Universal Approximation

It is instructive to compare this result with the classical universal approximation theorem for feedforward neural networks.

| | Feedforward Networks | Reservoir Computers | |—|—|—| | **Input** | Vectors $\mathbf{x} \in \mathbb{R}^d$ | Sequences $\mathbf{u} \in \mathcal{U}^{-\mathbb{N}}$ | | **Target class** | Continuous functions | Fading-memory functionals | | **Approximation tool** | Hidden layer features | Reservoir state map $\mathcal{E}$ | | **Readout** | Linear combination | Polynomial (or linear) | | **Key theorem** | Cybenko 1989, Hornik et al. 1989 | Maass et al. 2002, Grigoryeva & Ortega 2018 | | **Proof technique** | Hahn-Banach / Stone-Weierstrass | Stone-Weierstrass |

The parallel is striking. In both cases, a fixed nonlinear feature map (hidden layer or reservoir) combined with a simple readout achieves universal approximation. The reservoir setting is more complex because the input space is infinite-dimensional, requiring the additional structure of fading memory to make the problem tractable.

15.4 Separation Property and Approximation Property

Grigoryeva and Ortega (2018) identified two independent properties that together characterize the computational power of a reservoir computer. Decomposing universality into these two components clarifies the distinct roles of the reservoir and the readout.

15.4.1 The Separation Property

Definition 15.3 (Separation property)**.** A reservoir with echo state map $\mathcal{E} : \mathcal{U}^{-\mathbb{N}} \to \mathbb{R}^N$ has the *separation property* on $\mathcal{K} \subseteq \mathcal{U}^{-\mathbb{N}}$ if $\mathcal{E}$ is injective on $\mathcal{K}$: for all $\mathbf{u}, \mathbf{v} \in \mathcal{K}$,

$$\mathbf{u} \neq \mathbf{v} \implies \mathcal{E}(\mathbf{u}) \neq \mathcal{E}(\mathbf{v}).$$

The separation property says that the reservoir preserves all the information in the input history — no two distinct histories are "confused" by being mapped to the same state. This is an *injectivity* requirement on the encoding.

Example 15.1. Consider a linear reservoir $\mathbf{x}(t+1) = A\mathbf{x}(t) + Bu(t+1)$ with $\|A\| < 1$. The echo state map is

$$\mathcal{E}(\mathbf{u}_0) = \sum_{k=0}^{\infty} A^k Bu(-k).$$

This is injective (on appropriate input sets) if and only if the pair (A, B) is *observable* in the control-theoretic sense, meaning that the matrix $[B, AB, A^2B, \ldots, A^{N-1}B]$ has rank N. For generic random choices of A and B, this condition holds.

15.4.2 The Approximation Property

Definition 15.4 (Approximation property)**.** A family of readout functions $\mathcal{R} \subseteq \{h : \mathbb{R}^N \to \mathbb{R}\}$ has the *approximation property* if for every compact set $K \subset \mathbb{R}^N$ and every continuous function $g : K \to \mathbb{R}$, the restriction of $\mathcal{R}$ to K is dense in $C(K, \mathbb{R})$ in the uniform norm.

The approximation property says that the readout class is expressive enough to approximate any continuous function of the reservoir state. Polynomials have this property (by the Weierstrass approximation theorem). So do feedforward neural networks with one hidden layer (by the classical UAT). Linear functions do *not* have this property in general, though they may suffice when the reservoir states already encode the relevant nonlinear features.

15.4.3 Universality = Separation + Approximation

Theorem 15.12. *A reservoir computer with echo state map $\mathcal{E}$ and readout class $\mathcal{R}$ is a universal approximator of fading-memory functionals on a compact set $\mathcal{K}$ if and only if:*

1. *$\mathcal{E}$ has the separation property on $\mathcal{K}$, and*
2. *$\mathcal{R}$ has the approximation property.*

Proof sketch. sketch.* ($\Leftarrow$) The separation property makes $\mathcal{E}$ a homeomorphism onto its image $\mathcal{E}(\mathcal{K})$ (since $\mathcal{E}$ is a continuous injection from a compact space to a Hausdorff space, it is a homeomorphism onto its image). Any fading-memory functional $\mathcal{H}$ on $\mathcal{K}$ induces a continuous function $g = \mathcal{H} \circ \mathcal{E}^{-1}$ on the compact set $\mathcal{E}(\mathcal{K})$. By the approximation property, there exists $h \in \mathcal{R}$ with $\|h - g\|_\infty < \varepsilon$ on $\mathcal{E}(\mathcal{K})$. Then $\|h \circ \mathcal{E} - \mathcal{H}\|_\infty < \varepsilon$ on $\mathcal{K}$.

($\Rightarrow$) If $\mathcal{E}$ does not separate points, then there exist distinct $\mathbf{u}, \mathbf{v} \in \mathcal{K}$ with $\mathcal{E}(\mathbf{u}) = \mathcal{E}(\mathbf{v})$. Any functional of the form $h \circ \mathcal{E}$ must assign the same value to both, so any fading-memory functional with $\mathcal{H}(\mathbf{u}) \neq \mathcal{H}(\mathbf{v})$ cannot be approximated. Similarly, if $\mathcal{R}$ lacks the approximation property, there exist continuous functions of the state that cannot be approximated by the readout. □

This decomposition has important practical implications:

- **If approximation fails**, one should enrich the readout (e.g., use polynomial

features of the reservoir state, or a neural network readout).

- **If separation fails**, one should increase the reservoir size or change its architecture to make its dynamics richer.

The two failures require different interventions, and conflating them leads to ineffective troubleshooting.

15.5 Volterra Series and Reservoir Computing

15.5.1 Volterra Series Representation

The Volterra series provides an explicit representation of fading-memory functionals, analogous to the Taylor series for smooth functions.

Definition 15.5 (Volterra series)**.** A *Volterra series* of a time-invariant functional $\mathcal{H}$ is the expansion

$$y(t) = h_0 + \sum_{n=1}^{\infty} \int_0^{\infty} \cdots \int_0^{\infty} h_n(\tau_1, \ldots, \tau_n) \prod_{k=1}^{n} u(t - \tau_k)\, d\tau_1 \cdots d\tau_n,$$

where $h_0 \in \mathbb{R}$ is the zeroth-order term and $h_n : [0, \infty)^n \to \mathbb{R}$ is the *n-th order Volterra kernel*.

In the discrete-time setting, the integrals become sums:

$$y(t) = h_0 + \sum_{n=1}^{\infty} \sum_{\tau_1=0}^{\infty} \cdots \sum_{\tau_n=0}^{\infty} h_n(\tau_1, \ldots, \tau_n) \prod_{k=1}^{n} u(t - \tau_k).$$

The n-th order term captures n-th order nonlinear interactions between inputs at different times. The first-order term is a linear convolution; the second-order term captures pairwise interactions; and so on.

15.5.2 Boyd-Chua Theorem

The following classical result, due to Boyd and Chua (1985), establishes that Volterra series are the "right" tool for fading-memory functionals.

Theorem 15.1 (Boyd and Chua 1985)**.** **A time-invariant functional $\mathcal{H} : \mathcal{U}^{-\mathbb{N}} \to \mathbb{R}$ has the fading memory property if and only if it can be uniformly approximated on compact input sets by finite-degree Volterra series with decaying kernels.**

The "if" direction is straightforward: a finite Volterra series with decaying kernels depends continuously on the input in any weighted norm with sufficiently slow decay. The "only if" direction uses the Stone-Weierstrass theorem applied to the algebra generated by the time-delay functionals $\mathbf{u} \mapsto u(t - k)$ — the same essential argument as in the proof of Theorem 15.6.

15.5.3 Reservoir States and Volterra Kernels

The connection between reservoir computing and Volterra series is direct and illuminating. Consider a reservoir with nonlinear activation σ and update rule

$$\mathbf{x}(t+1) = \sigma(W\mathbf{x}(t) + W_{\text{in}}u(t+1)).$$

For small inputs and spectral radius $\rho(W) < 1$, expand σ as a power series $\sigma(z) = z + \alpha_2 z^2 + \alpha_3 z^3 + \cdots$ and iterate. The reservoir state at time t becomes a sum of terms involving products of delayed inputs — which is precisely a (truncated) Volterra series.

More concretely, consider the linear reservoir $\mathbf{x}(t) = \sum_{k=0}^{\infty} A^k Bu(t-k)$. Each component $x_i(t)$ is a *first-order* Volterra functional of the input (a linear filter). Taking products of components $x_i(t)x_j(t)$ gives *second-order* terms. This is why polynomial readouts are natural: a degree-d polynomial in N reservoir states computes Volterra terms up to order d (with the specific kernel structure determined by the reservoir weights).

Proposition 15.2. **A reservoir computer with N neurons, echo state property, and a polynomial readout of degree d implements a functional that is a finite Volterra series of order at most d, with kernels determined by the reservoir parameters.**

Proof sketch. sketch.* Each reservoir state component is a convergent power series in delayed inputs (by iterating the reservoir equation and expanding the nonlinearity). A polynomial of degree d in these components yields products of at most d such series. Each such product is a sum of terms of the form $c \cdot u(t-\tau_1) \cdots u(t-\tau_n)$ with $n \leq d$, which are Volterra terms of order at most d. □

15.5.4 Connection to Wiener's Theory

Norbert Wiener developed the theory of nonlinear functionals of stochastic processes in the 1940s and 1950s, using what are now called *Wiener series* — orthogonalized versions of Volterra series with respect to Gaussian white noise inputs. The Wiener kernels are related to the Volterra kernels by a triangular transformation.

The relevance to reservoir computing is that Wiener's framework provides an orthogonal decomposition of the input-output map, facilitating the analysis of which components of the functional the reservoir captures. The information processing capacity theory (Section 15.6) can be seen as a modern, finite-dimensional version of this decomposition.

15.6 Information Processing Capacity

15.6.1 Definition

How much computational work can a reservoir actually perform? The *information processing capacity* (IPC), introduced by Dambre, Verstraeten, Schrauwen, and Massar (2012), provides a precise answer.

The setup: consider a reservoir with N neurons driven by i.i.d. inputs $u(t)$ drawn from a distribution with zero mean. Let $\{f_i\}_{i=1}^{\infty}$ be a complete orthonormal system of functions on the input space (e.g., Legendre polynomials if the input is uniformly distributed on $[-1, 1]$). Define the *target functionals*

$$z_{\mathbf{d}}(t) = \prod_{k=1}^{\infty} f_{d_k}(u(t-k+1)),$$

where $\mathbf{d} = (d_1, d_2, \ldots)$ is a multi-index with finitely many nonzero entries. These targets form a complete orthonormal system for functionals of the input history (with respect to the product measure).

Definition 15.6 (Capacity for a target)**.** The *capacity* of the reservoir for target $z_{\mathbf{d}}$ is

$$C_{\mathbf{d}} = \frac{\mathrm{Cov}(\hat{z}_{\mathbf{d}}, z_{\mathbf{d}})^2}{\mathrm{Var}(\hat{z}_{\mathbf{d}}) \cdot \mathrm{Var}(z_{\mathbf{d}})},$$

where $\hat{z}_{\mathbf{d}}(t) = \mathbf{w}^T \mathbf{x}(t)$ is the best linear readout approximation (obtained by linear regression of $z_{\mathbf{d}}$ on $\mathbf{x}$). This is the squared correlation coefficient (R^2) between the optimal linear reconstruction and the target.

Definition 15.7 (Total information processing capacity)**.** The *total IPC* is

$$C_{\text{total}} = \sum_{\mathbf{d}} C_{\mathbf{d}},$$

where the sum is over all multi-indices $\mathbf{d}$ with at least one nonzero entry.

15.6.2 The Capacity Bound

Theorem 15.3 (Dambre et al. 2012)**.** **For a reservoir with N neurons and i.i.d. inputs, the total information processing capacity satisfies**

$$C_{total} \leq N.$$

Proof. * The proof is elegant and rests on linear algebra. The reservoir state at time t is $\mathbf{x}(t) \in \mathbb{R}^N$. For a given target $z_{\mathbf{d}}$, the optimal linear reconstruction is $\hat{z}_{\mathbf{d}}(t) = \mathbf{w}_{\mathbf{d}}^T \mathbf{x}(t)$, where $\mathbf{w}_{\mathbf{d}}$ minimizes the mean squared error. The capacity is

$$C_{\mathbf{d}} = \frac{\mathbf{w}_{\mathbf{d}}^T \Sigma_{xz_{\mathbf{d}}} \Sigma_{xz_{\mathbf{d}}}^T \mathbf{w}_{\mathbf{d}}}{\mathbf{w}_{\mathbf{d}}^T \Sigma_{xx} \mathbf{w}_{\mathbf{d}} \cdot \mathrm{Var}(z_{\mathbf{d}})},$$

where $\Sigma_{xx} = \mathbb{E}[\mathbf{x}\mathbf{x}^T]$ is the state covariance and $\Sigma_{xz_{\mathbf{d}}} = \mathbb{E}[\mathbf{x} z_{\mathbf{d}}]$ is the cross-covariance. For the optimal readout:

$$C_{\mathbf{d}} = \frac{\Sigma_{xz_{\mathbf{d}}}^T \Sigma_{xx}^{-1} \Sigma_{xz_{\mathbf{d}}}}{\mathrm{Var}(z_{\mathbf{d}})}.$$

Now sum over all targets. Since the $z_{\mathbf{d}}$ are orthonormal (under the product measure), and the reservoir state lives in an N-dimensional space, we can expand each $x_i(t)$ in the basis $\{z_{\mathbf{d}}\}$:

$$x_i(t) = \sum_{\mathbf{d}} a_{i,\mathbf{d}}\, z_{\mathbf{d}}(t) + \text{noise independent of inputs.}$$

By Parseval's theorem:

$$\sum_{\mathbf{d}} C_{\mathbf{d}} = \sum_{\mathbf{d}} \Sigma_{xz_{\mathbf{d}}}^T \Sigma_{xx}^{-1} \Sigma_{xz_{\mathbf{d}}} / \mathrm{Var}(z_{\mathbf{d}}) = \mathrm{tr}(\Sigma_{xx}^{-1} \cdot AA^T),$$

where A is the matrix with columns $\Sigma_{xz_{\mathbf{d}}}/\sqrt{\mathrm{Var}(z_{\mathbf{d}})}$.

The critical observation is that the $z_{\mathbf{d}}$ are orthonormal, so the coefficients $a_{i,\mathbf{d}}$ satisfy Bessel's inequality. A direct calculation using the Cauchy-Schwarz inequality and the fact that $\mathbf{x}(t)$ has N components shows:

$$C_{\text{total}} = \sum_{\mathbf{d}} C_{\mathbf{d}} \leq \mathrm{rank}(\Sigma_{xx}) \leq N.$$

The bound is achieved when the reservoir states span N linearly independent directions in the function space of the input history, i.e., when the reservoir makes maximal use of its N degrees of freedom. □

Remark 15.5. The bound $C_{\text{total}} \leq N$ is tight. It is achieved, for instance, by a linear reservoir with an orthogonal weight matrix and appropriately chosen input weights.

15.6.3 Memory-Nonlinearity Trade-off

The total IPC decomposes naturally:

$$C_{\text{total}} = C_{\text{lin}} + C_{\text{nonlin}},$$

where C_{lin} is the *linear memory capacity* (the sum of capacities for first-order targets $z_{(0,\ldots,0,1,0,\ldots)}(t) = u(t-k)$) and C_{nonlin} is the sum over all higher-order targets.

Since $C_{\text{total}} \leq N$ is fixed, there is a fundamental *trade-off* between linear memory and nonlinear processing:

- A reservoir with high linear memory capacity devotes most of its N degrees of freedom to remembering past inputs, leaving little room for nonlinear computation.
- A reservoir with high nonlinear capacity computes complex functions of recent inputs but forgets the distant past quickly.

Jaeger (2002) proved that for a linear reservoir with N neurons, the linear memory capacity is exactly N (all capacity is devoted to linear memory, as one would expect). For nonlinear reservoirs, the linear memory capacity is strictly less than N, and the deficit is taken up by nonlinear terms.

Proposition 15.4. *(Jaeger 2002). *For a linear echo state network with N neurons and i.i.d. scalar inputs,**

$$C_{lin} = \sum_{k=1}^{\infty} C_k^{lin} = N,$$

where C_k^{lin} is the capacity for reconstructing $u(t-k)$.

This result says that a linear reservoir with N neurons can linearly reconstruct the last N inputs with perfect fidelity (summing over all delays), but no more. Each additional neuron adds exactly one unit of memory capacity.

15.6.4 Connection to Information Theory

The IPC has a natural information-theoretic interpretation. Each capacity $C_{\mathbf{d}} \in [0, 1]$ measures the fraction of variance in the target $z_{\mathbf{d}}$ that is captured by the reservoir states through a linear readout. The total C_{total} counts the total number of "independent pieces of information" about the input history that the reservoir retains.

This connects to the entropy of the reservoir state distribution: a reservoir with IPC close to N has states that are maximally informative about the input history, while a reservoir with IPC much less than N "wastes" degrees of freedom (e.g., by having redundant or constant state components).

15.7 The Edge of Chaos

15.7.1 The Hypothesis

Perhaps the most provocative idea in reservoir computing theory is the *edge of chaos* hypothesis:

> **Edge of Chaos Hypothesis.** Computational performance of a dynamical system used as a reservoir is maximized when its dynamics are poised at the boundary between ordered (contracting) and chaotic (expanding) behavior.

This hypothesis has its origins in the work of Langton (1990) on cellular automata, who observed that the most complex computational behavior occurred at the transition between ordered and chaotic rule classes. Kauffman (1993) extended this to Boolean networks. The idea was brought into the reservoir computing context by Bertschinger and Natschläger (2004) and Legenstein and Maass (2007).

15.7.2 Mathematical Characterization

The "edge of chaos" can be characterized in terms of the *maximum Lyapunov exponent* $\lambda_{\max}$ of the reservoir dynamics. Recall from Part I that the Lyapunov exponent measures the average exponential rate of divergence of nearby trajectories.

For the reservoir system $\mathbf{x}(t+1) = F(\mathbf{x}(t), u(t+1))$, driven by a stationary ergodic input, the maximum Lyapunov exponent is

$$\lambda_{\max} = \lim_{T\to\infty} \frac{1}{T} \ln \|J_T J_{T-1} \cdots J_1 \mathbf{v}\|$$

for a generic initial vector $\mathbf{v}$, where $J_t = D_{\mathbf{x}} F(\mathbf{x}(t-1), u(t))$ is the Jacobian of the reservoir map with respect to the state.

The three regimes are:

- **Ordered regime** ($\lambda_{\max} < 0$): The dynamics are contracting.

Perturbations to the state decay exponentially. The echo state property holds robustly. The reservoir has strong fading memory but limited computational richness.

- **Chaotic regime** ($\lambda_{\max} > 0$): The dynamics are expanding.

Perturbations grow exponentially. The echo state property fails: the reservoir state depends sensitively on initial conditions, not just the input. Fading memory is lost.

- **Edge of chaos** ($\lambda_{\max} \approx 0$): The dynamics are neither contracting

nor expanding. The reservoir is maximally sensitive to inputs without being unstable. This is conjectured to be the optimal operating point.

15.7.3 Evidence

Bertschinger and Natschläger (2004) studied random Boolean networks used as reservoirs and found that computational metrics (mutual information between reservoir state and input history) were maximized at the critical point between ordered and chaotic phases.

Legenstein and Maass (2007) proved that for a class of recurrent neural networks, the *kernel quality* (a measure of the reservoir's ability to separate different input streams) and the *generalization ability* exhibit a peak at the edge of chaos. Specifically, they showed:

Theorem 15.5 (Legenstein and Maass 2007, informal)**.** **For randomly connected threshold networks, the expected separation between reservoir states driven by different inputs is maximized when the network is at the critical point (edge of chaos).**

The proof examines the propagation of perturbations through the network. In the ordered regime, perturbations shrink and inputs become indistinguishable. In the chaotic regime, perturbations from noise dominate and the state is unreliable. At the critical point, perturbations neither grow nor shrink, allowing maximal discrimination.

15.7.4 Criticisms and Nuances

The edge-of-chaos hypothesis, while influential, is not universally valid. Several important qualifications are necessary:

1. **Task dependence.** The optimal dynamical regime depends on the task. Tasks

requiring long memory (e.g., reconstructing $u(t-100)$) benefit from dynamics deep in the ordered regime, where memory is long but nonlinear processing is limited. Tasks requiring complex nonlinear transformations of recent inputs benefit from dynamics closer to the chaotic regime. There is no single "best" operating point.

1. **Input dependence.** The effective Lyapunov exponent depends on the input

statistics. A reservoir that is stable for one class of inputs may be chaotic for another. The edge of chaos is not a fixed property of the reservoir but depends on the input-reservoir interaction.

1. **Finite-size effects.** The Lyapunov exponent is an asymptotic quantity. For

finite reservoirs and finite time horizons, the relevant quantity is the *finite-time Lyapunov exponent*, which may fluctuate significantly.

1. **Non-universality.** Verstraeten et al. (2007) showed empirically that for

certain tasks, optimal performance occurs well within the ordered regime, not at the edge of chaos. The hypothesis is better understood as a useful heuristic than a theorem.

1. **Spectral radius is not the whole story.** A common practical heuristic is to

set the spectral radius $\rho(W)$ to be near 1 (the "edge" for linear systems). But the effective Lyapunov exponent depends on the nonlinearity, input scaling, and other factors. A spectral radius of 1 does not necessarily place the system at the edge of chaos for a nonlinear reservoir.

15.7.5 Connection to the Memory-Nonlinearity Trade-off

The edge-of-chaos hypothesis is related to the IPC framework of Section 15.6. At the edge of chaos:

- The linear memory capacity is moderate (the reservoir remembers the past, but not indefinitely).
- The nonlinear capacity is also substantial (the reservoir performs nontrivial computations).
- The total IPC is close to the maximum N.

Moving into the ordered regime increases linear memory at the expense of nonlinear capacity. Moving into the chaotic regime destroys both (the ESP fails, and the state becomes unreliable). The edge of chaos represents the Pareto frontier of the memory-nonlinearity trade-off.

15.8 Generalization Bounds

15.8.1 The Generalization Question

The theoretical results above address *expressiveness*: what can reservoir computers represent? An equally important question is *generalization*: how well does a reservoir computer trained on finite data perform on unseen inputs?

In a typical setup, we observe a training sequence $(u(1), y^*(1)), \ldots, (u(T), y^*(T))$ and fit the readout weights $\mathbf{w}$ by minimizing the empirical loss

$$\hat{L}(\mathbf{w}) = \frac{1}{T} \sum_{t=1}^{T} \|y^*(t) - \mathbf{w}^T \mathbf{x}(t)\|^2.$$

We want to bound the *expected loss* $L(\mathbf{w}) = \mathbb{E}[\|y^*(t) - \mathbf{w}^T \mathbf{x}(t)\|^2]$.

15.8.2 The Reservoir as Implicit Regularizer

The reservoir's dynamics provide a form of implicit regularization. Because the echo state property implies fading memory, the reservoir state $\mathbf{x}(t)$ is a smooth function of the input history. This smoothness constrains the hypothesis class: the set of functionals $\mathbf{u} \mapsto \mathbf{w}^T \mathcal{E}(\mathbf{u})$ is a subset of the fading-memory functionals, and this subset has controlled complexity.

More precisely, the contraction property of the reservoir map bounds the sensitivity of the state to input perturbations (as we showed in Theorem 15.4). This Lipschitz bound on $\mathcal{E}$ translates into a bound on the complexity of the hypothesis class.

15.8.3 Ridge Regression and Regularization

In practice, the readout is trained by *ridge regression* (Tikhonov regularization):

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \frac{1}{T} \sum_{t=1}^{T} \|y^*(t) - \mathbf{w}^T \mathbf{x}(t)\|^2 + \lambda \|\mathbf{w}\|^2.$$

The regularization parameter $\lambda > 0$ controls the bias-variance trade-off:

- **Large λ**: The readout weights are small, the model is smooth, bias is high, variance is low. The model underfits.
- **Small λ**: The readout weights can be large, the model is flexible, bias is low, variance is high. The model overfits.

The closed-form solution is

$$\hat{\mathbf{w}} = (X^T X + \lambda T I)^{-1} X^T \mathbf{y}^*,$$

where $X \in \mathbb{R}^{T \times N}$ is the matrix of reservoir states.

The following result quantifies the generalization performance.

Proposition 15.6. **Let the reservoir have the echo state property with contraction rate $L < 1$, and let the readout be trained by ridge regression with parameter λ. Assume the inputs are drawn from a stationary ergodic process and the target functional has fading memory. Then the expected generalization error satisfies**

$$L(\hat{\mathbf{w}}) - L(\mathbf{w}^*) \leq \mathcal{O}\left(\frac{N}{T\lambda} + \lambda \|\mathbf{w}^*\|^2\right),$$

where $\mathbf{w}^*$ is the best linear readout in the population sense.

This is a standard bias-variance decomposition for ridge regression, but with the important feature that N (the reservoir dimension) appears in the bound. The optimal λ balances the two terms, yielding a generalization error of order $\mathcal{O}(\sqrt{N/T} \cdot \|\mathbf{w}^*\|)$, which vanishes as $T \to \infty$ for fixed N.

15.8.4 Connections to Statistical Learning Theory

The reservoir computing setup fits into the framework of statistical learning theory, though with some complications due to the temporal dependence of the data.

VC dimension and Rademacher complexity. The hypothesis class $\mathcal{F} = \{\mathbf{u} \mapsto \mathbf{w}^T \mathcal{E}(\mathbf{u}) : \|\mathbf{w}\| \leq B\}$ is a class of linear functions of N variables (the reservoir states), constrained to a ball of radius B. For linear classes in N dimensions, the VC dimension is $N+1$ and the Rademacher complexity is $\mathcal{O}(B\sqrt{N/T})$.

Standard learning-theoretic bounds (e.g., Bartlett and Mendelson 2002) then give, for i.i.d. data:

$$L(\hat{\mathbf{w}}) \leq \hat{L}(\hat{\mathbf{w}}) + \mathcal{O}\left(B\sqrt{\frac{N}{T}} + \sqrt{\frac{\ln(1/\delta)}{T}}\right)$$

with probability at least $1 - \delta$.

Temporal dependence. The i.i.d. assumption is too strong for time series. However, for stationary mixing processes, analogous bounds hold with modified constants that depend

on the mixing rate. The fading memory property of the reservoir ensures that the effective dependence between $\mathbf{x}(t)$ and $\mathbf{x}(t+k)$ decays exponentially with k (at rate L^k), which enables concentration inequalities for dependent data (see, e.g., Mohri and Rostamizadeh 2010).

Gonon and Ortega (2020) provided the most rigorous treatment to date, establishing generalization bounds for reservoir computing that account for the temporal structure, the reservoir's contraction properties, and the regularization of the readout. Their results show that the generalization error decreases as $\mathcal{O}(T^{-1/2})$ under mild conditions, matching the rate for i.i.d. settings up to logarithmic factors.

15.8.5 Practical Implications

The theoretical analysis yields several practical guidelines:

1. **Reservoir size N should be matched to data size T.** A very large reservoir with limited data will overfit unless strongly regularized.

1. **Ridge regression is not optional.** The regularization parameter λ is essential for generalization, not merely a numerical convenience. It should be selected by cross-validation.

1. **Stronger contraction (smaller L) improves generalization** at the cost of reduced expressiveness. This is another manifestation of the bias-variance trade-off.

1. **The reservoir provides a "free" form of regularization** through its dynamics: by mapping inputs to a fading-memory representation, it automatically limits the complexity of the hypothesis class.

15.9 Summary

The theoretical foundations of reservoir computing rest on several pillars:

| Concept | Role | Key Result | |—|—|—| | Fading memory | Defines the target class | Boyd-Chua theorem (Theorem 15.14) | | Echo state property | Ensures well-defined reservoir map | Implies fading memory (Theorem 15.4) | | Separation property | Reservoir preserves input information | Necessary for universality (Theorem 15.12) | | Approximation property | Readout is expressive enough | Sufficient with separation (Theorem 15.12) | | Universal approximation | RC can approximate any FM functional | Grigoryeva-Ortega theorem (Theorem 15.6) | | IPC bound | Limits on what N neurons can compute | $C_{\text{total}} \leq N$ (Theorem 15.18) | | Edge of chaos | Heuristic for optimal dynamics | Task-dependent, not a theorem | | Generalization | Performance on unseen data | $\mathcal{O}(T^{-1/2})$ rates with regularization |

Together, these results show that reservoir computing is a principled framework for approximating temporal functionals, with rigorous guarantees on expressiveness, capacity, and generalization. The key insight is that the *dynamics* of the reservoir provide the computational power — the readout merely extracts the answer.

Exercises

Exercise 15.1. (Fading memory and linear filters). Let $\mathcal{H}$ be a causal linear time-invariant filter with impulse response $\{h_k\}_{k=0}^{\infty}$:

$$y(t) = \sum_{k=0}^{\infty} h_k\, u(t-k).$$

(a) Show that $\mathcal{H}$ has the fading memory property if and only if $\sum_{k=0}^{\infty} |h_k|/w(k) < \infty$ for some weight function w with $w(k) \to 0$.

(b) Show that this condition is satisfied whenever $|h_k| \le Cr^k$ for some $C > 0$ and $r \in (0,1)$.

(c) Give an example of a linear filter that does *not* have the fading memory property.

Exercise 15.2. (Separation and observability). Consider the linear reservoir $\mathbf{x}(t+1) = A\mathbf{x}(t) + \mathbf{b}u(t+1)$ with $A \in \mathbb{R}^{N\times N}$, $\|A\| < 1$, and $\mathbf{b} \in \mathbb{R}^N$.

(a) Show that the echo state map is $\mathcal{E}(\mathbf{u}) = \sum_{k=0}^{\infty} A^k \mathbf{b}\, u(-k)$.

(b) Show that $\mathcal{E}$ is injective on the space of bounded input sequences if and only if the observability matrix $\mathcal{O} = [\mathbf{b}, A\mathbf{b}, A^2\mathbf{b}, \ldots, A^{N-1}\mathbf{b}]$ has rank N.

(c) For $N = 2$, $A = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.3 \end{pmatrix}$, and $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, verify that the separation property holds.

Exercise 15.3. (IPC of a linear reservoir). Consider a linear reservoir $x(t+1) = a\,x(t) + u(t+1)$ with a single neuron and $|a| < 1$, driven by i.i.d. inputs $u(t) \sim \text{Uniform}[-1, 1]$.

(a) Compute the echo state map $\mathcal{E}(\mathbf{u}) = \sum_{k=0}^{\infty} a^k u(-k)$.

(b) Compute the linear memory capacity C_k^{lin} for target $z_k(t) = u(t-k)$ for each delay $k \ge 1$.

(c) Verify that $\sum_{k=1}^{\infty} C_k^{\text{lin}} = 1 = N$.

(d) What happens to the distribution of capacity across delays as $|a| \to 1^-$? Interpret this in terms of the memory-nonlinearity trade-off.

Exercise 15.4. (Stone-Weierstrass and reservoir universality). Let $\mathcal{K} = \{0,1\}^{-\mathbb{N}}$ be the space of binary input sequences (equipped with the product topology, making it homeomorphic to the Cantor set).

(a) Show that $\mathcal{K}$ is compact and metrizable.

(b) Let $\mathcal{E} : \mathcal{K} \to \mathbb{R}^N$ be a continuous injection. Show that the algebra generated by the coordinate functions $\{x_1 \circ \mathcal{E}, \ldots, x_N \circ \mathcal{E}\}$ separates points of $\mathcal{K}$.

(c) Conclude that every continuous function on $\mathcal{K}$ can be uniformly approximated by polynomials in the reservoir states.

(d) Why does this argument fail if $\mathcal{E}$ is not injective? Give a concrete example.

Exercise 15.5. (Volterra series truncation). Consider the second-order Volterra system

$$y(t) = \sum_{\tau_1=0}^{\infty} h_1(\tau_1)\, u(t-\tau_1) + \sum_{\tau_1=0}^{\infty}\sum_{\tau_2=0}^{\infty} h_2(\tau_1,\tau_2)\, u(t-\tau_1)\, u(t-\tau_2),$$

with $h_1(\tau) = \alpha^\tau$ and $h_2(\tau_1,\tau_2) = \beta^{\tau_1+\tau_2}$, where $|\alpha|, |\beta| < 1$.

(a) Show that this functional has the fading memory property. Identify a suitable weight function.

(b) Construct a reservoir with $N = 2$ neurons that computes this functional exactly (with a polynomial readout of degree 2). *Hint:* use one neuron as a linear filter with decay rate α and another with decay rate β.

(c) Compute the approximation error when the Volterra series is truncated to delays $\tau_1, \tau_2 \leq M$, as a function of M.

Exercise 15.6. (Edge of chaos in a simple reservoir). Consider the scalar reservoir

$$x(t+1) = \tanh(\gamma\, x(t) + \sigma\, u(t+1)),$$

where $\gamma > 0$ is the feedback gain and $u(t)$ is i.i.d. with $u(t) \sim \mathcal{N}(0, 1)$.

(a) Show that the echo state property holds for $\gamma < 1$ (assuming σ is sufficiently small). *Hint:* compute $|\partial F/\partial x|$ and show it is less than 1 on average.

(b) Compute the Lyapunov exponent $\lambda = \mathbb{E}[\ln|\gamma(1 - x^2)|]$ where x is drawn from the stationary distribution. (This may require numerical computation.)

(c) Find the critical value γ_c at which $\lambda = 0$ (the "edge of chaos") for $\sigma = 0.1$. You may use numerical methods.

(d) Empirically investigate the linear memory capacity C_1^{lin} (capacity for reconstructing $u(t-1)$) as a function of γ for $\gamma \in (0, \gamma_c)$. What do you observe near the edge of chaos?

Exercise 15.7. (Generalization error). Consider a reservoir with N neurons trained on T time steps with ridge regression parameter λ.

(a) Write down the expected generalization error as a function of N, T, λ, and the norm $\|\mathbf{w}^*\|$ of the optimal readout weights.

(b) Find the optimal λ^* that minimizes this bound, as a function of N, T, and $\|\mathbf{w}^*\|$.

(c) Substitute λ^* back to find the optimal generalization error rate. Verify that it scales as $\mathcal{O}(\sqrt{N/T})$.

(d) Discuss the implications: if you double the reservoir size N, how much more training data T do you need to maintain the same generalization error?

—

Recommended Reading

The following references provide deeper treatments of the topics in this chapter.

Foundational:

- **Boyd, S. and Chua, L.O.** (1985). "Fading memory and the problem of approximating nonlinear operators with Volterra series." *IEEE Transactions on Circuits and Systems*, 32(11), 1150–1161. The paper that introduced fading memory as the correct notion for nonlinear system approximation. Essential reading.
- **Jaeger, H.** (2002). "Short term memory in echo state networks." *GMD Report 152*. The technical report that introduced the echo state property and proved the linear memory capacity result. Contains many results that were ahead of their time.

Universal Approximation:

- **Maass, W., Natschläger, T., and Markram, H.** (2002). "Real-time computing without stable states: A new framework for neural computation based on perturbations." *Neural Computation*, 14(11), 2531–2560. Introduced liquid state machines and the separation/approximation framework.
- **Grigoryeva, L. and Ortega, J.-P.** (2018). "Echo state networks are universal." *Neural Networks*, 108, 495–508. The definitive treatment of universal approximation for reservoir computers, using the Stone-Weierstrass approach presented in this chapter.

Information Processing Capacity:

- **Dambre, J., Verstraeten, D., Schrauwen, B., and Massar, S.** (2012). "Information processing capacity of dynamical systems." *Scientific Reports*, 2, 514. Introduced the IPC framework and proved the capacity bound.

Edge of Chaos:

- **Bertschinger, N. and Natschläger, T.** (2004). "Real-time computation at the edge of chaos in recurrent neural networks." *Neural Computation*, 16(7), 1413–1436.
- **Legenstein, R. and Maass, W.** (2007). "Edge of chaos and prediction of computational performance for neural circuit models." *Neural Networks*, 20(3), 323–334.

Generalization:

- **Gonon, L. and Ortega, J.-P.** (2020). "Reservoir computing universality with stochastic inputs." *IEEE Transactions on Neural Networks and Learning Systems*, 31(1), 100–112. Rigorous generalization bounds accounting for temporal dependence.

Further Theory:

- **Gonon, L. and Ortega, J.-P.** (2021). "Fading memory echo state networks are universal." *Neural Networks*, 138, 10–13. A concise proof refining the 2018 result.
- **Verzelli, P., Alippi, C., and Livi, L.** (2021). "Learn to synchronize, synchronize to learn." *Chaos*, 31(8), 083119. Connects reservoir computing to synchronization theory in dynamical systems.
- **Yildiz, I.B., Jaeger, H., and Kiebel, S.J.** (2012). "Re-visiting the echo state property." *Neural Networks*, 35, 1–9. A careful analysis of the relationship between the echo state property and the contraction condition.

Chapter 16

Reservoir Computing Meets Ergodic Theory

In Parts I and II of this book, we developed the theory of dynamical systems and ergodic theory as self-contained mathematical disciplines. In Part III, we introduced reservoir computing as a computational framework built on top of recurrent dynamics. Now we bring these threads together.

The central claim of this chapter is that reservoir computing is not merely *inspired by* dynamical systems — it *is* a dynamical system, and its computational properties are *determined by* its ergodic-theoretic characteristics. The echo state property is a statement about Lyapunov exponents. Generalization from finite training data is a consequence of the Birkhoff ergodic theorem. The ability of reservoirs to reconstruct chaotic attractors follows from Takens' embedding theorem. These are not loose analogies; they are precise mathematical relationships.

We assume the reader is familiar with the reservoir computing framework introduced in Chapter 14, particularly the state update equation

$$\mathbf{x}(t+1) = f\big(W_{\text{in}}\, u(t+1) + W\,\mathbf{x}(t)\big), \tag{16.1}$$

the echo state property (Chapter 14), information processing capacity (Chapter 15), Lyapunov exponents and the multiplicative ergodic theorem (Chapter 6), and the Birkhoff ergodic theorem (Chapter 8).

16.1 The Reservoir as a Dynamical System — Revisited

16.1.1 The Autonomous Case

Consider first a reservoir with no input: $W_{\text{in}} = 0$. The state update reduces to

$$\mathbf{x}(t+1) = f\big(W\,\mathbf{x}(t)\big) = T(\mathbf{x}(t)),$$

where $T : \mathbb{R}^N \to \mathbb{R}^N$ is a smooth map. This is precisely a discrete-time dynamical system of the kind studied in Part I. If $f = \tanh$, then T maps the cube $[-1,1]^N$ into itself, so by the

Brouwer fixed point theorem, T has at least one fixed point. Depending on W, the system may have attracting fixed points, periodic orbits, or chaotic attractors — the full zoo of nonlinear dynamics.

Everything from Part I applies directly: we can compute the Lyapunov exponents of T, characterize its invariant sets, study its bifurcations as the spectral radius $\rho(W)$ varies, and so on.

16.1.2 The Driven Case: Nonautonomous and Random Dynamical Systems

With input, the situation is richer. The state update

$$\mathbf{x}(t+1) = f\big(W_{\text{in}}\, u(t+1) + W\,\mathbf{x}(t)\big)$$

defines a *nonautonomous* dynamical system: the map changes at every time step because the input $u(t)$ changes. If the input is a realization of a stationary stochastic process, we enter the realm of *random dynamical systems.*

Definition 16.1 (Skew-product formulation). Let $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ be a measure-preserving dynamical system modeling the input process, where $\omega \in \Omega$ represents an input sequence and $\theta : \Omega \to \Omega$ is the left shift. Define the *cocycle*

$$\varphi(t, \omega, \mathbf{x}_0) = T_{\theta^{t-1}\omega} \circ \cdots \circ T_{\theta\omega} \circ T_\omega(\mathbf{x}_0),$$

where $T_\omega(\mathbf{x}) = f(W_{\text{in}}\, u_\omega(1) + W\,\mathbf{x})$ and $u_\omega(t)$ is the input at time t under realization ω. The *skew-product* system on $\Omega \times \mathbb{R}^N$ is

$$\Theta(\omega, \mathbf{x}) = \big(\theta(\omega),\, T_\omega(\mathbf{x})\big).$$

This skew-product is the correct mathematical framework for an input-driven reservoir. The input process θ drives the dynamics; the reservoir state $\mathbf{x}$ is "slaved" to the input. The key question is: *does the skew-product system have a well-defined invariant measure, and is it ergodic?*

16.1.3 Invariant Measures of the Driven Reservoir

Under the echo state property (ESP), for each input realization ω, the reservoir state converges to a unique trajectory $\bar{\mathbf{x}}(t, \omega)$ regardless of the initial condition $\mathbf{x}_0$. This means there exists a measurable map $h : \Omega \to \mathbb{R}^N$ such that

$$\bar{\mathbf{x}}(0, \omega) = h(\omega), \quad \bar{\mathbf{x}}(t, \omega) = \varphi(t, \omega, h(\omega)).$$

The map h *pushes forward* the input measure $\mathbb{P}$ to a measure $\mu = h_*\mathbb{P}$ on the reservoir state space $\mathbb{R}^N$. That is, for any measurable set $A \subseteq \mathbb{R}^N$,

$$\mu(A) = \mathbb{P}\big(\{\omega \in \Omega : h(\omega) \in A\}\big).$$

The measure $\nu = \mathbb{P} \times_h \delta$ on $\Omega \times \mathbb{R}^N$, concentrated on the graph of h, is invariant under the

skew-product Θ. Thus the ESP guarantees existence and uniqueness of the invariant measure for the driven reservoir.

Remark 16.1. If the ESP fails, multiple invariant measures may coexist, and the reservoir's behavior depends on the initial condition — a serious problem for reproducibility of computation. —

16.2 Generalization and Ergodic Averages

16.2.1 Training as Time Averaging

Recall from Chapter 14 that training a reservoir amounts to:

1. Driving the reservoir with a training input sequence $u(1), \ldots, u(T)$.
2. Collecting the state vectors $\mathbf{x}(1), \ldots, \mathbf{x}(T)$ into a state matrix X.
3. Solving the linear regression $W_{\text{out}} = YX^\dagger$ for the readout weights, where Y contains the target outputs.

The regression computes sample statistics from the trajectory. For instance, the matrix $XX^\top/T$ is a time average:

$$\frac{1}{T}XX^\top = \frac{1}{T}\sum_{t=1}^{T}\mathbf{x}(t)\,\mathbf{x}(t)^\top.$$

This is a time average of the observable $g(\mathbf{x}) = \mathbf{x}\,\mathbf{x}^\top$ along the reservoir trajectory.

16.2.2 The Birkhoff Theorem Applied

Let us now bring the Birkhoff ergodic theorem (Theorem 8.4) to bear. Working in the skew-product framework, define the observable $g : \Omega \times \mathbb{R}^N \to \mathbb{R}$ by $g(\omega, \mathbf{x}) = \phi(\mathbf{x})$ for some function ϕ of the reservoir state (e.g., a component of $\mathbf{x}\mathbf{x}^\top$, or a cross-correlation between state and target).

Theorem 16.2 (Ergodic convergence of training statistics). Suppose the input-driven reservoir system $(\Omega \times \mathbb{R}^N, \Theta, \nu)$ is ergodic, and $g \in L^1(\nu)$. Then for ν-almost every $(\omega, \mathbf{x}_0)$,

$$\lim_{T\to\infty}\frac{1}{T}\sum_{t=0}^{T-1} g\big(\Theta^t(\omega, \mathbf{x}_0)\big) = \int_{\Omega\times\mathbb{R}^N} g\, d\nu.$$

Proof. * This is a direct application of Birkhoff's theorem to the measure-preserving system $(\Omega \times \mathbb{R}^N, \Theta, \nu)$. □

The practical consequences are significant.

A single long training sequence suffices. If the system is ergodic, the time average along one trajectory converges to the space average. We do not need multiple independent training runs or i.i.d. samples — a single sufficiently long input sequence will do. This is precisely why reservoir computing works with a single training pass over the data.

If the system is not ergodic, trouble ensues. If the state space decomposes into multiple ergodic components, a single training trajectory explores only one component. The trained readout will generalize only within that component, potentially failing on inputs that

drive the reservoir into a different region of state space. This can occur when the reservoir has multiple attractors — a situation the ESP is designed to prevent.

16.2.3 Generalization and Mixing Rates

The Birkhoff theorem guarantees convergence, but says nothing about the *rate*. For practical reservoir computing, we need to know: how long must the training sequence be?

The rate of convergence is controlled by the *mixing properties* of the system (Chapter 10). Recall:

- **Strong mixing** implies the correlations $\mathrm{Cov}(g \circ \Theta^n, g)$ decay to zero.
- **Exponential mixing** (which holds for many hyperbolic systems) gives correlation decay at a rate $|C(n)| \leq C_0\, e^{-\alpha n}$ for some $\alpha > 0$.

For an exponentially mixing system, the central limit theorem (Theorem 10.9) yields

$$\frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} [g(\Theta^t(\omega, \mathbf{x}_0)) - \bar{g}] \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

where $\sigma^2 = \sum_{n=-\infty}^{\infty} C(n)$ and $\bar{g} = \int g\, d\nu$. This tells us:

$$\text{Training error} \approx \frac{\sigma}{\sqrt{T}},$$

with the constant σ determined by the mixing rate. Faster mixing means smaller σ and better generalization from shorter training sequences. Slow mixing (as occurs near bifurcation points or in systems with long memory) requires longer training.

Proposition 16.1. *Let the driven reservoir system be exponentially mixing with rate $\alpha > 0$. Then the mean squared error of the time-averaged state correlation matrix satisfies*

$$\mathbb{E}\left[\left\| \frac{1}{T} \sum_{t=1}^{T} \mathbf{x}(t)\mathbf{x}(t)^\top - \mathbb{E}_\nu[\mathbf{x}\mathbf{x}^\top] \right\|_F^2\right] = O\left(\frac{1}{T}\right),$$

where the implied constant depends on α, N, and the moments of $\mathbf{x}$ under ν.

This connects generalization in reservoir computing to a fundamental quantity in ergodic theory. Improving generalization — a machine learning goal — reduces to estimating mixing rates — an ergodic-theoretic problem.

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16.3 Lyapunov Exponents and the Echo State Property

16.3.1 The Multiplicative Ergodic Theorem Applied to Reservoirs

The echo state property asserts that initial conditions are forgotten. Lyapunov exponents quantify the rate at which nearby trajectories converge or diverge. The connection between the two is direct and precise.

The Jacobian of the reservoir map at time t is

$$D_{\mathbf{x}} T_{\theta^t \omega}(\mathbf{x}(t)) = \operatorname{diag}\big(f'(W_{\text{in}} u(t+1) + W\mathbf{x}(t))\big) \cdot W \stackrel{\text{def}}{=} A(t, \omega),$$

where the diagonal matrix contains the derivatives of the activation function evaluated at the pre-activation values. For $f = \tanh$, we have $f'(z) = 1 - \tanh^2(z) \in (0, 1]$.

The product of these Jacobians,

$$M(T, \omega) = A(T-1, \omega) \cdots A(1, \omega) \cdot A(0, \omega),$$

governs the evolution of infinitesimal perturbations. By Oseledets' multiplicative ergodic theorem (Theorem 6.12):

Theorem 16.4 (Lyapunov spectrum of the driven reservoir). Suppose the skew-product system $(\Omega \times \mathbb{R}^N, \Theta, \nu)$ is ergodic and $\log^+ \|A(0, \omega)\| \in L^1(\nu)$. Then there exist numbers $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$ (the *Lyapunov exponents*) such that for ν-almost every $(\omega, \mathbf{x})$,

$$\lim_{T \to \infty} \frac{1}{T} \log \|M(T, \omega)\, \mathbf{v}\| = \lambda_i$$

for all $\mathbf{v}$ in the i-th Oseledets subspace (but not in the $(i+1)$-th).

The integrability condition is easily verified: since $f' \in (0, 1]$ for tanh, we have $\|A(t, \omega)\| \leq \|W\|$, so $\log^+ \|A(0, \omega)\| \leq \log^+ \|W\| < \infty$.

16.3.2 ESP as Negativity of the Maximum Lyapunov Exponent

Theorem 16.5. If the maximum Lyapunov exponent $\lambda_1 < 0$, then the echo state property holds: for every input realization ω and any two initial conditions $\mathbf{x}_0, \mathbf{x}_0'$,

$$\|\mathbf{x}(t, \omega, \mathbf{x}_0) - \mathbf{x}(t, \omega, \mathbf{x}_0')\| \to 0 \quad \text{as } t \to \infty,$$

at an exponential rate governed by λ_1.

Sketch of proof. For nearby initial conditions, linearization gives

$$\|\delta \mathbf{x}(T)\| \approx \|M(T, \omega)\, \delta \mathbf{x}(0)\| \sim e^{\lambda_1 T} \|\delta \mathbf{x}(0)\|.$$

If $\lambda_1 < 0$, perturbations shrink exponentially. The extension to non-infinitesimal perturbations uses the fact that tanh is a global contraction (it is 1-Lipschitz), combined with a cone-field argument. The full proof appears in Manjunath and Jaeger (2013). □

The converse is not quite true: the ESP can hold even when $\lambda_1 = 0$ in degenerate cases. But generically, $\lambda_1 < 0$ and the ESP are equivalent.

Corollary 16.6. A sufficient condition for $\lambda_1 < 0$ is that

$$\sup_{\mathbf{x} \in [-1,1]^N} \|D_{\mathbf{x}} T_\omega(\mathbf{x})\| < 1 \quad \text{for } \mathbb{P}\text{-a.e. } \omega.$$

Since $\|D_{\mathbf{x}} T_\omega(\mathbf{x})\| \leq \|W\|$ for $f = \tanh$, a crude sufficient condition is $\|W\| < 1$. The more commonly used sufficient condition $\rho(W) < 1$ (spectral radius less than 1) from Chapter 14 is

weaker but not always sufficient; the relationship between the spectral radius condition and the Lyapunov exponent condition is subtle and input-dependent.

16.3.3 The Edge of Chaos

The most interesting regime is when $\lambda_1 \approx 0$: the *edge of chaos*.

- When $\lambda_1 < 0$: the reservoir is *contracting*. It forgets initial conditions (good for the ESP), but it also forgets the input history quickly. The system has short memory.
- When $\lambda_1 > 0$: the reservoir is *expanding*. Small differences in input are amplified exponentially. The ESP fails, and computation becomes unreliable.
- When $\lambda_1 \approx 0$: the reservoir is *critical*. It has the longest memory and the richest dynamics, but is on the verge of instability.

This can be made precise. The *memory capacity* (Chapter 15) of a linear reservoir is exactly N (the number of neurons), independent of the spectral radius. For nonlinear reservoirs, memory capacity depends on λ_1: it is maximized near $\lambda_1 = 0$ and decreases as λ_1 becomes more negative.

Proposition 16.7 (Memory-stability tradeoff). For a reservoir with state update (16.1) and i.i.d. input, the fading memory norm

$$\|F\|_{\text{mem}} = \sum_{k=0}^{\infty} \sup_u \left\| \frac{\partial \mathbf{x}(t)}{\partial u(t-k)} \right\|$$

is finite if and only if $\lambda_1 < 0$. Moreover, the characteristic decay length of the memory scales as $\ell \sim |\lambda_1|^{-1}$.

Thus $|\lambda_1|^{-1}$ controls the effective memory horizon. Near the edge of chaos, $|\lambda_1|^{-1} \to \infty$, and the reservoir remembers the distant past — at the cost of approaching instability.

16.4 Information Processing Capacity and Entropy

16.4.1 IPC Revisited

Recall from Chapter 15 that the total information processing capacity (IPC) of an N-dimensional reservoir is bounded by N, and this bound is achieved by linear reservoirs. The IPC decomposes into contributions from different input functionals:

$$C_{\text{total}} = \sum_i C_i \leq N, \quad C_i = \frac{\text{Cov}^2(y_i, \hat{y}_i)}{\text{Var}(y_i)\,\text{Var}(\hat{y}_i)},$$

where y_i is the i-th target functional of the input and $\hat{y}_i$ is the reservoir's best linear reconstruction.

The distribution of this capacity across different functionals (linear memory, nonlinear transformations, cross-terms) is determined by the reservoir's dynamics — and this is where ergodic theory enters.

16.4.2 Entropy and Computational Capacity

The Kolmogorov–Sinai (KS) entropy (Definition 9.7) of the driven reservoir system measures the rate at which the system produces new information. By the Pesin formula (Theorem 9.14), for a smooth system with an SRB measure,

$$h_{\mathrm{KS}} = \sum_{\lambda_i > 0} \lambda_i,$$

where the sum is over the positive Lyapunov exponents.

For a reservoir satisfying the ESP, all Lyapunov exponents are negative, so $h_{\mathrm{KS}} = 0$ for the *autonomous* part of the dynamics. This makes sense: a contracting system produces no information on its own. All information in the reservoir states comes from the *input*. The reservoir is a transducer, not a generator, of information.

Now consider the *joint* system (input + reservoir). The KS entropy of the skew-product Θ satisfies

$$h_{\mathrm{KS}}(\Theta) = h_{\mathrm{KS}}(\theta) + h_{\mathrm{KS}}(\Theta \mid \theta),$$

where $h_{\mathrm{KS}}(\theta)$ is the entropy of the input process and $h_{\mathrm{KS}}(\Theta \mid \theta)$ is the conditional entropy of the reservoir given the input. Under the ESP, the reservoir state is a deterministic function of the input history, so $h_{\mathrm{KS}}(\Theta \mid \theta) = 0$, and

$$h_{\mathrm{KS}}(\Theta) = h_{\mathrm{KS}}(\theta).$$

The reservoir preserves the entropy of the input without adding to it. But the *distribution* of information across the reservoir state components $x_1, \ldots, x_N$ is determined by the reservoir dynamics. A well-designed reservoir spreads the input information across its N dimensions in a way that makes it accessible to the linear readout.

16.4.3 The Computational Regime Diagram

These considerations lead to a three-regime picture, parametrized by a control parameter such as the spectral radius $\rho(W)$ or, more precisely, by λ_1.

| Regime | λ_1 | ESP | KS entropy | Behavior | |—|—|—|—|—| | Ordered | $\ll 0$ | Yes | 0 (autonomous) | Short memory, limited nonlinear capacity | | Edge of chaos | ≈ 0 | Marginal | 0 (autonomous) | Long memory, maximum capacity | | Chaotic | > 0 | No | > 0 (autonomous) | Input sensitivity, unreliable |

In the ordered regime, the reservoir state entropy (under the invariant measure μ) is low: the states cluster in a small region of $[-1, 1]^N$. In the chaotic regime, the reservoir generates its own entropy, overwhelming the input signal. At the edge of chaos, the reservoir's state entropy is high (it uses its state space efficiently) while remaining deterministically controlled by the input.

Proposition 16.2. *Let $H(\mathbf{x})$ denote the differential entropy of the reservoir state distribution μ, and let $I(u^{(-\infty,0]}; \mathbf{x}(0))$ denote the mutual information between the input history and the*

current state. Under the ESP,

$$I(u^{(-\infty,0]}; \mathbf{x}(0)) = H(\mathbf{x}(0)),$$

since the state is a deterministic function of the input history (the conditional entropy $H(\mathbf{x}(0) \mid u^{(-\infty,0]})$ *vanishes). Maximizing computational capacity thus requires maximizing* $H(\mathbf{x})$ *subject to the ESP constraint* $\lambda_1 < 0$*.*

This is the information-theoretic formulation of the edge-of-chaos hypothesis for reservoir computing.

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16.5 Takens' Embedding Theorem and Reservoir Computing

16.5.1 Delay Embeddings Revisited

Recall Takens' theorem (Theorem 6.15): if M is a compact manifold, $T : M \to M$ is a C^2 diffeomorphism, and $h : M \to \mathbb{R}$ is a C^2 observation function, then for generic (T, h), the delay map

$$\Phi_{T,h}^{(d)}(\mathbf{z}) = \big(h(\mathbf{z}),\, h(T(\mathbf{z})),\, \ldots,\, h(T^{d-1}(\mathbf{z}))\big)$$

is an embedding of M into $\mathbb{R}^d$ provided $d \geq 2\dim(M) + 1$.

Now consider a reservoir driven by a scalar time series $u(t) = h(S^t(\mathbf{z}_0))$ generated by a dynamical system (M, S) with observation function h. The reservoir state at time t depends on the input history:

$$\mathbf{x}(t) = \Psi\big(u(t), u(t-1), u(t-2), \ldots\big),$$

where Ψ is the *echo state map* (the function guaranteed to exist by the ESP). In practice, the dependence on the distant past decays exponentially (at rate $|\lambda_1|$), so effectively

$$\mathbf{x}(t) \approx \Psi_K\big(u(t), u(t-1), \ldots, u(t-K)\big)$$

for some memory horizon K.

16.5.2 Reservoirs as Generalized Embeddings

The reservoir state $\mathbf{x}(t) \in \mathbb{R}^N$ is a function of the delay vector $(u(t), u(t-1), \ldots)$, which by Takens' theorem is (generically) an embedding of the source system's attractor. Therefore, the reservoir state is a *nonlinear transformation of a delay embedding.*

This observation was made rigorous by Hart, Hook, and Dawes (2020):

Theorem 16.9 (Hart, Hook, Dawes 2020). Let (M, S) be a C^1 dynamical system on a compact d-dimensional manifold, and let $h : M \to \mathbb{R}$ be a C^1 observation function. Consider a reservoir with N neurons and echo state map Ψ. If $N \geq 2d + 1$ and the reservoir satisfies certain genericity conditions, then the map

$$E : M \to \mathbb{R}^N, \quad \mathbf{z} \mapsto \Psi\big(h(\mathbf{z}), h(S^{-1}(\mathbf{z})), h(S^{-2}(\mathbf{z})), \ldots\big)$$

is an embedding: it is injective and its derivative has full rank.

In other words, a sufficiently large reservoir driven by a scalar observable of a dynamical system generically reconstructs the underlying attractor, up to diffeomorphism. The reservoir state space contains a faithful copy of the source system's geometry.

16.5.3 Why Reservoirs Can Predict Chaos

This theorem explains one of the most striking empirical results in reservoir computing: the ability to predict chaotic systems (such as the Lorenz attractor) from a single scalar time series.

The argument proceeds in three steps:

1. **Embedding**: The reservoir state $\mathbf{x}(t)$ is diffeomorphic to the state $\mathbf{z}(t)$ of the source system. Thus $\mathbf{z}(t) = E^{-1}(\mathbf{x}(t))$ can be recovered from the reservoir state.
2. **Dynamics on the embedding**: Since E is a diffeomorphism onto its image, the dynamics S on M are conjugate to dynamics $\tilde{S} = E \circ S \circ E^{-1}$ on the image $E(M) \subset \mathbb{R}^N$. In particular,

$$\mathbf{x}(t+1) = \tilde{S}(\mathbf{x}(t)).$$

1. **Prediction via readout**: To predict the future observation $u(t+1) = h(S(\mathbf{z}(t)))$, we need the function $h \circ S \circ E^{-1} : \mathbb{R}^N \to \mathbb{R}$. The linear readout $W_{\text{out}}\mathbf{x}(t)$ is trained to approximate this function. If the function is sufficiently regular and the reservoir state has sufficient dimension, the approximation can be made arbitrarily good.

Remark 16.2. The reservoir does more than a simple delay embedding. The delay embedding produces d coordinates that are all linear functionals of the input history (just shifted copies). The reservoir produces N coordinates that are *nonlinear* functionals of the input history. This richer feature set can make the readout problem easier — the target function $h \circ S \circ E^{-1}$ may be closer to linear in the reservoir coordinates than in the delay coordinates. This is the computational advantage of reservoir computing over classical delay-coordinate methods.

16.5.4 An Illustrative Example

Consider the Lorenz system

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z$$

with the standard parameters $\sigma = 10$, $\rho = 28$, $\beta = 8/3$. The attractor has dimension approximately 2.06.

A delay embedding of the scalar observable $x(t)$ requires at least $d = 2 \cdot 2.06 + 1 \approx 6$ coordinates. A reservoir with $N = 300$ neurons, driven by $x(t)$, creates a 300-dimensional embedding. This is vastly over-determined, which means:

- The embedding is robust (small perturbations do not destroy it).

- Many different linear functionals of $\mathbf{x}(t)$ can approximate the target, so the readout problem is well-conditioned.
- The reservoir can simultaneously predict multiple observables (x, y, z) of the source system from the same state $\mathbf{x}(t)$.

This is precisely what is observed in practice: reservoirs with a few hundred neurons can predict the Lorenz system for several Lyapunov times (the characteristic timescale $1/\lambda_1^{(\text{Lorenz})}$ of the source system's chaos).

16.6 Physical Reservoirs and Their Dynamics

16.6.1 The Universality of the Framework

Nothing in the preceding analysis requires the reservoir to be an artificial recurrent neural network. The mathematical framework applies to *any* dynamical system used as a computational substrate. What matters is:

1. The system is driven by an input signal.
2. The system has a high-dimensional state that depends on the input history (fading memory / ESP).
3. The state can be observed (read out).

Physical dynamical systems satisfying these conditions can serve as reservoirs. This idea, known as *physical reservoir computing*, has been explored extensively since about 2010.

16.6.2 Photonic Reservoirs

Larger et al. (2012) demonstrated that a single nonlinear node with delayed feedback — a semiconductor laser with an optical feedback loop — can serve as a reservoir. The dynamics are governed by a delay-differential equation:

$$\tau \dot{x}(t) = -x(t) + \eta \sin^2(x(t - \tau_D) + \rho u(t)),$$

where τ_D is the feedback delay and η is the feedback strength.

The delay line effectively creates $N = \tau_D/\delta$ virtual neurons, where δ is the spacing between samples. In the language of dynamical systems, the infinite-dimensional state space of the delay-differential equation is discretized into a finite-dimensional reservoir.

The ergodic-theoretic analysis applies directly:

- The Lyapunov exponents determine the echo state property.
- The system exhibits a transition from ordered to chaotic behavior as η increases.
- Optimal performance occurs at the edge of chaos.

The key advantage is *speed*: optical systems can process data at GHz rates, orders of magnitude faster than electronic implementations.

16.6.3 Mechanical Reservoirs

A mass-spring network with nonlinear springs, driven by an external force, is a Hamiltonian (or dissipative, if damping is present) dynamical system. Its state — the positions and velocities of all masses — constitutes a high-dimensional reservoir.

Vibrating plates and membranes have also been used. An elastic plate driven at one point and measured at multiple points creates a spatially distributed reservoir. The partial differential equation governing the plate's dynamics replaces the state update equation (16.1), but the computational framework is identical.

16.6.4 Quantum Reservoirs

A quantum system driven by classical inputs evolves according to a sequence of quantum channels:

$$\rho(t+1) = \mathcal{E}_{u(t+1)}(\rho(t)),$$

where $\rho(t)$ is the density matrix and $\mathcal{E}_u$ is a completely positive, trace-preserving map depending on the input. The reservoir state is extracted via measurements on $\rho(t)$.

The echo state property in the quantum setting requires that the quantum channel be *forgetful*: the system must lose memory of its initial state. This is the quantum analogue of contractivity and is related to the mixing properties of the quantum channel. The Lyapunov exponents of the quantum dynamics — defined via the multiplicative ergodic theorem applied to the superoperator $\mathcal{E}$ — determine whether this condition holds.

16.6.5 Biological Reservoirs

Maass, Natschläger, and Markram (2002) introduced the *liquid state machine*, which models a cortical microcircuit as a reservoir. The "liquid" is a recurrent network of spiking neurons; the "state" is the high-dimensional pattern of membrane potentials and synaptic currents.

This was, in fact, the original biological motivation for reservoir computing. The cortex receives sensory input, transforms it through complex recurrent dynamics, and reads out task-relevant information via downstream neurons. The mathematical framework of this chapter applies: the cortical circuit is a dynamical system, the echo state property is necessary for reliable computation, and the edge-of-chaos hypothesis predicts that cortical circuits should operate near criticality — a prediction with empirical support from neuroscience.

16.6.6 A Unifying Table

The following table summarizes how the general theory specializes to different physical substrates. In each case, the core mathematical structure is the same.

Substrate	State space	Dynamics	ESP condition	Observables
RNN	$\mathbb{R}^N$	Difference equation (16.1)	$\lambda_1 < 0$	Direct readout of $\mathbf{x}$
Photonic	$C^0([-\tau_D, 0])$	Delay-differential eq.	Negative max. LE	Photodetector signal
Mechanical	$\mathbb{R}^{2N}$ (positions, velocities)	Newton's equations	Damping + $\lambda_1 < 0$	Displacement sensors
Quantum				

Density matrices $\mathcal{S}(\mathcal{H})$ | Quantum channel iteration | Forgetfulness of $\mathcal{E}$ | Measurement outcomes | | Biological | Membrane potentials, currents | Spiking neuron models | Decay of perturbations | Spike rates, LFP |

16.7 The Bigger Picture: Dynamics, Statistics, and Computation

16.7.1 Three Perspectives on One System

We have now seen the same reservoir from three angles:

Dynamical systems theory tells us about the *geometry* of the reservoir's state space. It describes the attractors, their dimensions, the sensitive dependence on initial conditions (or lack thereof), and the embedding properties that allow reconstruction of input dynamics. The key tools are Takens' theorem, Lyapunov exponents, and bifurcation theory.

Ergodic theory tells us about the *statistics* of the reservoir's trajectories. It guarantees that time averages converge to space averages (Birkhoff), quantifies the rate of this convergence (mixing), measures the information content of the dynamics (entropy), and provides the multiplicative ergodic theorem that underpins the Lyapunov analysis. The key tools are the Birkhoff theorem, mixing rates, and KS entropy.

Computation theory (in the reservoir computing sense) tells us what *functions* the reservoir can approximate and how well. It quantifies memory (IPC), expressiveness (nonlinear capacity), and learnability (generalization from finite data). The key tools are approximation theory, linear algebra, and statistical learning theory.

The deep insight of this chapter is that these three perspectives are not independent — they are facets of a single mathematical structure.

16.7.2 A Dictionary

The following correspondences summarize the connections:

| Dynamical Systems | Ergodic Theory | Reservoir Computing | |—|—|—| | Attractor | Support of invariant measure | Accessible state space | | Lyapunov exponents | Multiplicative ergodic theorem | Echo state property, memory | | Contraction / expansion | Entropy production | Reliability / chaos | | Embedding dimension | — | Required reservoir size | | Bifurcation | Phase transition | Edge of chaos | | Conjugacy | Measure isomorphism | Equivalent computation | | Delay embedding (Takens) | — | Attractor reconstruction | | — | Birkhoff's theorem | Training convergence | | — | Mixing rate | Generalization rate | | — | KS entropy | Information processing |

16.7.3 Open Questions

Several fundamental questions remain open at the interface of these three fields:

1. **Optimal reservoir design.** Given a computational task, what reservoir dynamics are optimal? The edge-of-chaos heuristic provides qualitative guidance, but a precise opti-

mization theory — one that selects the best invariant measure for a given class of target functions — does not yet exist.

2. **Non-stationary inputs.** Most of the ergodic-theoretic analysis assumes stationary (or at least ergodic) input processes. Real-world signals are often non-stationary. Extending the theory to non-stationary inputs requires tools from non-stationary ergodic theory and time-varying dynamical systems.
3. **Finite-time ergodic theory.** The Birkhoff theorem is an asymptotic result. For practical training with finite T, we need finite-time concentration inequalities. While these exist for some classes of mixing processes, sharp bounds for reservoir computing remain an active area of research.
4. **Beyond the echo state property.** Some recent work suggests that reservoirs *without* the ESP — operating in the chaotic regime — can be useful for certain tasks, particularly those involving chaotic source systems. Understanding this requires going beyond the contractive framework and into the theory of chaotic synchronization.

16.7.4 A Closing Reflection

Ergodic theory was developed by Boltzmann, Birkhoff, and von Neumann to understand the foundations of statistical mechanics: why do time averages of physical observables agree with ensemble averages? The answer — ergodicity — was a profound insight about the relationship between dynamics and statistics.

A century later, the same mathematics explains why a machine learning algorithm trained on a single time series can generalize to unseen data. The reservoir's trajectory through state space is ergodic; the time average computed during training converges to the space average that determines performance on new inputs. Birkhoff's theorem, proved in 1931, guarantees the generalization of an algorithm conceived in 2001.

This is the nature of mathematics: theorems proved for one purpose illuminate phenomena in entirely different domains. The convergence of dynamical systems, ergodic theory, and reservoir computing is not a coincidence — it reflects the fact that these fields study the same underlying structures from different vantage points.

Exercises

Exercise 16.1. (Lyapunov exponents of a linear reservoir). Consider a linear reservoir with no input: $\mathbf{x}(t+1) = W\mathbf{x}(t)$, where $W \in \mathbb{R}^{N\times N}$.

(a) Show that the Lyapunov exponents are $\lambda_i = \log|\mu_i|$, where $\mu_1, \ldots, \mu_N$ are the eigenvalues of W (counted with multiplicity).

(b) Conclude that the echo state property holds if and only if $\rho(W) < 1$.

(c) Now let W be a random matrix drawn from the Gaussian Orthogonal Ensemble, scaled so that $\rho(W) \approx r$. Numerically compute λ_1 as a function of r for $N = 100, 500, 1000$. What do you observe as $N \to \infty$?

Exercise 16.2. (Birkhoff's theorem and training convergence). Let $\mathbf{x}(t) \in \mathbb{R}^N$ be the state of an ergodic reservoir driven by an i.i.d. input $u(t) \sim \text{Uniform}[-1, 1]$.

(a) Define $\bar{x}_i(T) = \frac{1}{T}\sum_{t=1}^{T} x_i(t)$ and $\bar{C}_{ij}(T) = \frac{1}{T}\sum_{t=1}^{T} x_i(t)x_j(t)$. By Birkhoff's theorem, these converge almost surely. To what limits?

(b) Implement a reservoir with $N = 100$, $\rho(W) = 0.9$, and i.i.d. uniform input. Compute $\bar{x}_1(T)$ and $\bar{C}_{12}(T)$ for $T = 10^2, 10^3, 10^4, 10^5$. Plot the convergence. How does the rate compare to the $O(1/\sqrt{T})$ rate predicted by the central limit theorem?

(c) Repeat with $\rho(W) = 0.99$. Is the convergence faster or slower? Explain in terms of mixing rates.

Exercise 16.3. (Takens' theorem and reservoir prediction). Consider the logistic map $z(t+1) = 4z(t)(1 - z(t))$ on $[0, 1]$.

(a) The attractor is the entire interval $[0, 1]$, which is one-dimensional. By Takens' theorem, how many delay coordinates are needed to embed it?

(b) Build a reservoir with $N = 50$ neurons driven by $u(t) = z(t)$. Train it to predict $z(t+1)$ from $\mathbf{x}(t)$. What prediction accuracy do you achieve?

(c) Now train the same reservoir to predict $z(t+k)$ for $k = 1, 2, 5, 10, 20$. How does accuracy degrade with k? Relate this to the positive Lyapunov exponent of the logistic map ($\lambda = \log 2$).

(d) Compare the reservoir's prediction to that obtained from a simple delay embedding: predicting $z(t + 1)$ from $(z(t), z(t - 1), z(t - 2))$ using linear regression. Which is better, and why?

Exercise 16.4. (ESP and Lyapunov exponents, numerical). Implement a reservoir with $N = 200$ and tanh activation.

(a) For spectral radii $\rho(W) \in \{0.5, 0.8, 0.95, 1.0, 1.2, 1.5\}$, numerically estimate the maximum Lyapunov exponent λ_1 by tracking the evolution of a small perturbation $\delta\mathbf{x}$ along a trajectory of length $T = 10000$, with i.i.d. Gaussian input. Use the standard algorithm: periodically renormalize $\delta\mathbf{x}$ to prevent overflow/underflow, and average $\log \|\delta\mathbf{x}\|$ over time.

(b) For each $\rho(W)$, test the ESP by driving the reservoir with the same input from two different random initial conditions. Plot $\|\mathbf{x}(t)-\mathbf{x}'(t)\|$ as a function of t. Verify that convergence occurs when $\lambda_1 < 0$ and divergence occurs when $\lambda_1 > 0$.

(c) Find the critical spectral radius ρ^* at which $\lambda_1 = 0$ (approximately). Is $\rho^* = 1$? If not, explain why.

Exercise 16.5. (Entropy and computational capacity). Consider a reservoir with $N = 100$.

(a) For spectral radii $\rho(W) \in \{0.1, 0.5, 0.9, 0.99, 1.05\}$ and i.i.d. uniform input on $[-1, 1]$, estimate the differential entropy $H(\mathbf{x})$ of the stationary state distribution. One approach: compute the covariance matrix Σ of the stationary states and use the Gaussian entropy formula $H_{\text{Gauss}} = \frac{1}{2}\log\det(2\pi e\,\Sigma)$ as an upper bound.

(b) For each spectral radius, compute the linear memory capacity $MC = \sum_{k=1}^{100} r^2(k)$, where $r(k)$ is the correlation between $u(t - k)$ and its best linear reconstruction from $\mathbf{x}(t)$.

(c) Plot $H(\mathbf{x})$ and MC as functions of $\rho(W)$ on the same axes. At what spectral radius is each maximized? Discuss the relationship.

Exercise 16.6. (Physical reservoir simulation). Simulate a "mechanical reservoir" consisting of $N = 20$ masses connected by nonlinear springs (Duffing-type: force $= -kx - \beta x^3$) with damping, arranged in a random network.

(a) Drive mass 1 with an input signal $u(t)$. Read out the positions of all 20 masses. Train a linear readout to perform a nonlinear transformation of the input (e.g., $y(t) = u(t)^2$).

(b) Vary the damping coefficient and the spring nonlinearity β. Is there an "edge of chaos" where performance is optimized?

(c) Compare performance to a standard tanh reservoir with $N = 20$. Discuss.

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Recommended Reading

- **Hart, A. G., Hook, J. L., & Dawes, J. H. P.** (2020). Embedding and approximation theorems for echo state networks. *Neural Networks*, 128, 234–247. The rigorous connection between Takens' embedding theorem and reservoir computing.
- **Grigoryeva, L. & Ortega, J.-P.** (2018). Echo state networks are universal. *Neural Networks*, 108, 495–508. Proves that echo state networks with linear readout are universal approximators of filters with fading memory.
- **Jaeger, H.** (2001). The "echo state" approach to analysing and training recurrent neural networks. *GMD Report 148*. The foundational paper introducing echo state networks and the echo state property.
- **Manjunath, G. & Jaeger, H.** (2013). Echo state property linked to an input: exploring a fundamental characteristic of recurrent neural networks. *Neural Computation*, 25(3), 671–696. Rigorous analysis of the echo state property in terms of contractive maps and its dependence on the input.
- **Larger, L., Soriano, M. C., Brunner, D., Appeltant, L., Gutiérrez, J. M., Pesquera, L., Mirasso, C. R., & Fischer, I.** (2012). Photonic information processing beyond Turing: an optoelectronic implementation of reservoir computing. *Optics Express*, 20(3), 3241–3249. Demonstration of reservoir computing with a single delay-coupled photonic node.
- **Tanaka, G., Yamane, T., Héroux, J. B., Nakane, R., Kanazawa, N., Takeda, S., Numata, H., Nakano, D., & Hirose, A.** (2019). Recent advances in physical reservoir computing: a review. *Neural Networks*, 115, 100–123. Comprehensive survey of physical reservoir computing across multiple substrates.
- **Arnold, L.** (1998). *Random Dynamical Systems*. Springer. The definitive reference on the mathematical theory of random dynamical systems, including the skew-product formulation and multiplicative ergodic theorem.
- **Lukoševičius, M. & Jaeger, H.** (2009). Reservoir computing approaches to recurrent neural network training. *Computer Science Review*, 3(3), 127–149. Accessible survey connecting the practical and theoretical aspects of reservoir computing.

Chapter 17

Applications, Frontiers, and Open Problems

In the preceding chapters, we built a theoretical edifice connecting dynamical systems, ergodic theory, and reservoir computing. We now turn to applications, surveying how these ideas play out in practice, what frontiers are being explored, and what remains unknown. This chapter is necessarily less rigorous than its predecessors — it is a guided tour of a rapidly evolving landscape rather than a sequence of theorems — but we ground each topic in the mathematics developed earlier.

17.1 Time Series Prediction

The most natural application of reservoir computing is predicting the future of a time series from its past. The mathematical setup is as follows.

17.1.1 The Prediction Problem

Let $\mathbf{s}(t) \in \mathbb{R}^d$ be the state of a deterministic dynamical system $\dot{\mathbf{s}} = F(\mathbf{s})$, and let $x(t) = h(\mathbf{s}(t))$ be a scalar observable. Given a sequence of observations $x(1), x(2), \ldots, x(T)$, we wish to predict $x(T + \tau)$ for some lead time $\tau > 0$.

By Takens' embedding theorem (Chapter 6), the delay vector

$$\mathbf{d}(t) = (x(t), x(t - \Delta), x(t - 2\Delta), \ldots, x(t - (m - 1)\Delta))$$

generically provides a diffeomorphic embedding of the attractor when $m \geq 2d + 1$. In particular, there exists a smooth function $G : \mathbb{R}^m \to \mathbb{R}$ such that

$$x(t + \tau) = G(\mathbf{d}(t)).$$

The reservoir, driven by the scalar input $x(t)$, creates a high-dimensional nonlinear embedding of the input history — a generalized, learned version of the delay embedding. The readout layer then approximates G.

17.1.2 Benchmark Systems

The Mackey-Glass equation. This delay-differential equation,

$$\dot{x}(t) = \beta \frac{x(t-\tau_d)}{1 + x(t-\tau_d)^n} - \gamma x(t),$$

with standard parameters $\beta = 0.2$, $\gamma = 0.1$, $n = 10$, $\tau_d = 17$, produces a chaotic time series with a strange attractor of dimension approximately 2.1. It has been a standard benchmark since Farmer and Sidorowich (1987). Reservoir computers achieve normalized mean square errors (NMSE) on the order of 10^{-8} for one-step prediction on this system (Jaeger & Haas, 2004).

NARMA systems. The k-th order nonlinear autoregressive moving average (NARMA) system is defined by

$$y(t+1) = \alpha y(t) + \beta y(t) \sum_{i=0}^{k-1} y(t-i) + \gamma u(t-k+1)u(t) + \delta,$$

where $u(t)$ is a random input. NARMA-10 and NARMA-30 are standard benchmarks that test a reservoir's memory capacity and nonlinear processing simultaneously.

The Lorenz system. Short-term prediction of the Lorenz attractor is straightforward for reservoir computers. The more interesting challenge is long-term: can the reservoir reproduce the *climate* (the invariant measure) of the Lorenz system? We address this in Section 17.2.

17.1.3 Performance Metrics

The normalized mean square error is defined as

$$\text{NMSE} = \frac{\sum_t (y(t) - \hat{y}(t))^2}{\sum_t (y(t) - \bar{y})^2},$$

where $\hat{y}(t)$ is the prediction and $\bar{y}$ is the mean of the target. An NMSE of 0 is perfect prediction; an NMSE of 1 is no better than predicting the mean.

For chaotic systems, a more informative metric is the **valid prediction time** (or prediction horizon): the time until the prediction error exceeds a threshold, typically one attractor diameter. For the Lorenz system with standard parameters, reservoir computers achieve prediction horizons of several Lyapunov times, where one Lyapunov time is $1/\lambda_{\max} \approx 1/0.9 \approx 1.1$ time units.

17.2 Chaotic System Modeling and Climate Replication

17.2.1 Beyond Short-Term Prediction

For chaotic systems, pointwise prediction necessarily fails after a few Lyapunov times — this is the essence of sensitive dependence on initial conditions. But a more ambitious and subtle goal is to replicate the **climate** of the system: its long-term statistical properties.

In the language of ergodic theory, this means reproducing the system's invariant measure (or SRB measure). If the autonomous reservoir (running without teacher forcing, feeding its own predictions back as input) generates a trajectory whose time averages match the space averages of the true system, then by the ergodic theorem, the reservoir has effectively learned the invariant measure.

17.2.2 The Pathak et al. Results

Pathak, Hunt, Girvan, Lu, and Ott (2017, 2018) demonstrated that reservoir computers can:

1. **Predict** the short-term evolution of chaotic systems (Lorenz, Rössler, Kuramoto-Sivashinsky).
2. **Replicate the climate**: the autonomous reservoir, running freely, produces trajectories with the correct Lyapunov exponents, attractor dimension, and invariant measure — even though pointwise prediction has long since diverged.

This is a striking result. The reservoir does not merely memorize the training data; it learns the underlying dynamics well enough to reproduce the correct statistical properties. In ergodic-theoretic terms, the reservoir's autonomous dynamics support an invariant measure that closely approximates the SRB measure of the true system.

17.2.3 Hybrid Reservoir Computing

When partial knowledge of the governing equations is available, it can be incorporated. Lu, Pathak, Hunt, Girvan, Brockett, and Ott (2017) proposed a hybrid approach:

$$\mathbf{x}(t+1) = f(W\mathbf{x}(t) + W_{in}\mathbf{u}(t+1)),$$
$$\hat{\mathbf{y}}(t) = W_{out}\begin{pmatrix}\mathbf{x}(t)\\ \mathbf{k}(t)\end{pmatrix},$$

where $\mathbf{k}(t)$ is the output of a knowledge-based model (e.g., a coarse or imperfect simulation). The reservoir learns to correct the model's errors. This consistently outperforms either the model or the reservoir alone.

17.3 Signal Classification

Reservoir computing is naturally suited to temporal pattern recognition. The reservoir transforms a time-varying input signal into a trajectory in a high-dimensional state space. Different input classes produce different trajectories, and the readout layer acts as a classifier on the terminal state (or a time-averaged state).

Spoken digit recognition was an early benchmark (Verstraeten et al., 2005). The input is a cochlear-filtered speech signal; the reservoir state at the end of the utterance encodes enough information for a linear classifier to distinguish digits. Performance is competitive with hidden Markov models and recurrent neural networks trained with backpropagation through time, at a fraction of the training cost.

The mathematical picture: the reservoir implements a nonlinear embedding of the space of input signals into $\mathbb{R}^N$. The separation property (Chapter 15) ensures that distinct signal classes are mapped to distinct regions of state space. The linear readout then separates these regions with a hyperplane.

17.4 Physical Reservoir Computing

One of the most distinctive features of reservoir computing is that the reservoir need not be a simulated neural network. Any sufficiently rich dynamical system can serve as a reservoir. This opens the door to **physical reservoir computing**: using the native dynamics of physical systems for computation.

17.4.1 Photonic Reservoirs

Appeltant et al. (2011) demonstrated reservoir computing using a single semiconductor laser with delayed optical feedback. The key idea is **time-multiplexing**: the delay line is divided into N virtual nodes, each corresponding to a time slot within one delay period. The state of these virtual nodes collectively acts as the reservoir state.

The dynamics are governed by a delay-differential equation of the form

$$\epsilon \dot{x}(t) = -x(t) + f\big(\beta x(t - \tau_D) + \gamma J(t)\big),$$

where τ_D is the delay time, $J(t)$ is the input, and f is a nonlinear function determined by the laser physics. The delay creates effective high-dimensional dynamics from a single physical node.

Photonic reservoirs can operate at GHz rates, orders of magnitude faster than electronic implementations. Larger et al. (2012) and Brunner et al. (2013) further developed this approach.

17.4.2 Mechanical Reservoirs

Nakajima and colleagues have demonstrated that soft bodies — silicone arms, mass-spring networks, vibrating plates — can serve as reservoirs. The mechanical response of these systems to input forces creates a rich, high-dimensional state that can be read out for computation.

This has implications for robotics: the body of a robot is itself a dynamical system, and its physics can be harnessed for computation rather than being an obstacle to overcome.

17.4.3 Spintronic Reservoirs

Magnetic systems exhibit complex nonlinear dynamics. Torrejon et al. (2017) demonstrated reservoir computing using a single magnetic tunnel junction, exploiting the nonlinear magnetization dynamics governed by the Landau-Lifshitz-Gilbert equation.

17.4.4 Quantum Reservoir Computing

Quantum systems offer exponentially large Hilbert spaces, suggesting enormous computational capacity. Fujii and Nakajima (2017) proposed quantum reservoir computing, where a quantum system driven by classical inputs serves as the reservoir, and measurements of observables provide the readout.

The measurement problem adds an interesting twist: each measurement partially collapses the quantum state, so the readout strategy becomes part of the reservoir design. The mathematical framework extends naturally — the echo state property becomes a condition on the quantum channel's contractivity.

17.4.5 The Unifying Principle

In every case, the mathematical requirements are the same:

1. **Echo state property**: the physical dynamics must be contracting, so that the current state depends on the input history, not the initial condition.
2. **Sufficient dimensionality**: the state space must be rich enough (high-dimensional, nonlinear) to separate different input histories.
3. **Readability**: one must be able to observe enough of the state to train a readout.

The theory from Chapters 14–16 applies uniformly: Lyapunov exponents characterize stability, ergodic properties determine generalization, and information processing capacity quantifies computational power.

17.5 Next-Generation Reservoir Computing

17.5.1 The NGRC Framework

Gauthier, Bollt, Griffith, and Barbosa (2021) introduced **next-generation reservoir computing (NGRC)**, which replaces the random recurrent network with a structured, deterministic feature map based on time-delay embeddings.

The state vector is constructed directly from delayed observations:

$$\mathbf{O}(t) = \big(x(t),\ x(t-\Delta),\ x(t-2\Delta),\ \ldots,\ x(t-(k-1)\Delta)\big).$$

Nonlinear features are then formed as products:

$$\mathbf{p}(t) = \mathrm{vec}\big(\mathbf{O}(t) \otimes \mathbf{O}(t)\big),$$

where $\otimes$ denotes the outer product (or higher-order tensor products for higher-degree nonlinearities). The combined feature vector $(\mathbf{O}(t), \mathbf{p}(t))$ serves as the "reservoir state," and a linear readout is trained by ridge regression as before.

17.5.2 Connection to Takens' Theorem

The connection to Takens' embedding theorem is now explicit and direct. The linear part $\mathbf{O}(t)$ is precisely a delay embedding. The nonlinear part $\mathbf{p}(t)$ enriches this embedding with polynomial features, enabling the readout to approximate nonlinear functions of the attractor coordinates.

In the language of Chapter 15, NGRC constructs the separation and approximation properties by design rather than relying on random connectivity to provide them. The theoretical justification is cleaner, though the universality results are more restrictive (finite polynomial degree).

17.5.3 Advantages and Limitations

Advantages:

- No random matrices, hence no sensitivity to random seeds.
- Fewer hyperparameters (delay Δ, number of delays k, polynomial degree).
- Often superior performance on standard benchmarks.
- Faster training (the feature matrix is structured and typically smaller).

Limitations:

- The polynomial feature space grows combinatorially with the degree and number of delays.
- For multivariate inputs, the approach becomes expensive.
- The theoretical understanding of when NGRC outperforms classical ESN is incomplete.

17.6 Deep Reservoirs

Gallicchio and Micheli (2017) proposed **deep echo state networks**: stacking multiple reservoir layers, where each layer processes the output of the previous one:

$$\mathbf{x}^{(\ell)}(t+1) = f\big(W^{(\ell)}\mathbf{x}^{(\ell)}(t) + W_{in}^{(\ell)}\mathbf{x}^{(\ell-1)}(t+1)\big), \quad \ell = 1, \ldots, L,$$

with $\mathbf{x}^{(0)}(t) = \mathbf{u}(t)$ being the input. The readout is trained on the concatenation of states from all layers.

The analogy with deep neural networks suggests that deeper reservoirs create increasingly abstract representations. Empirically, deep reservoirs sometimes outperform single-layer reservoirs of the same total size, particularly on tasks requiring hierarchical temporal features.

Open questions: What is the optimal depth? How do the echo state property and Lyapunov exponents behave as a function of depth? Is there a "deep" version of the universal approximation results from Chapter 15? These remain largely unanswered.

17.7 Spatiotemporal Systems

Many real-world systems — fluid flows, cardiac dynamics, climate — are spatially extended, governed by partial differential equations. Extending reservoir computing to such systems requires handling high-dimensional, spatially structured data.

17.7.1 Parallel Reservoir Architectures

Pathak et al. (2018) proposed a divide-and-conquer approach for the Kuramoto-Sivashinsky (KS) equation,

$$\partial_t u + u\partial_x u + \partial_x^2 u + \partial_x^4 u = 0,$$

which exhibits spatiotemporal chaos. The spatial domain is divided into overlapping regions, each assigned its own reservoir. Each reservoir receives input from its region and its neighbors. The readout produces predictions for the local region.

This architecture scales linearly with domain size (rather than quadratically, as a single large reservoir would). Pathak et al. demonstrated both short-term prediction and long-term climate replication of the KS system with this approach — a striking application of the ideas from Chapter 16 to a PDE setting.

17.7.2 Mathematical Perspective

From the dynamical systems viewpoint, the KS equation defines an infinite-dimensional dynamical system on a function space. Its attractor, however, is finite-dimensional (for a finite domain). The parallel reservoir architecture implicitly exploits the spatial locality of the dynamics: the state at a given point depends primarily on nearby points, at least over short times.

The ergodic-theoretic perspective applies as well: the KS equation has an SRB-like measure on its attractor, and the reservoir's ability to replicate the climate corresponds to approximating this measure.

17.8 Open Theoretical Questions

We collect here the most significant open problems at the interface of dynamical systems, ergodic theory, and reservoir computing.

17.8.1 Optimal Reservoir Design

Problem. Given a computational task (defined as a target functional on input sequences), what reservoir architecture (topology, weights, nonlinearity) is optimal?

No general theory exists. Current practice relies on hyperparameter search, guided by heuristics (spectral radius near 1, sparse connectivity, etc.). A theory of optimal reservoir design would be a major advance.

17.8.2 The Edge of Chaos

Problem. Is computation truly optimized at the edge of chaos? If so, for which tasks and in what precise sense?

The evidence is mixed. Bertschinger and Natschläger (2004) showed that the edge of chaos maximizes certain information-theoretic measures. But Verstraeten et al. (2007) found that

the optimal spectral radius depends strongly on the task. A rigorous understanding of when and why the edge of chaos is beneficial remains elusive.

17.8.3 Capacity and Approximation Bounds

Problem. Can we obtain tighter bounds on the information processing capacity and approximation error of specific reservoir architectures?

Dambre et al. (2012) showed that the total IPC is at most N (the reservoir size), but this bound is often not tight. For structured tasks, much smaller reservoirs may suffice. A task-dependent capacity theory would be valuable.

17.8.4 Generalization Theory

Problem. What are the proper generalization bounds for reservoir computing?

The connection to ergodic theory (Chapter 16) suggests that mixing rates should control generalization. Faster mixing means the training data are "more independent," so generalization should be better. Making this precise — connecting mixing coefficients to PAC-style bounds — is an active area.

Gonon and Ortega (2020) have made progress using tools from statistical learning theory adapted to the time-series setting, but a complete theory is lacking.

17.8.5 Ergodicity of Driven Reservoirs

Problem. Under what conditions on the input process and reservoir architecture is the driven reservoir system ergodic?

If the input is i.i.d. or a stationary ergodic process, and the reservoir has the echo state property, one expects the driven system to be ergodic. But precise conditions — especially for non-stationary or weakly dependent inputs — are not fully established.

17.8.6 Connections to Kernel Methods

The reservoir implicitly defines a kernel on input sequences:

$$K(\mathbf{u}, \mathbf{v}) = \langle \mathbf{x}(\mathbf{u}), \mathbf{x}(\mathbf{v}) \rangle,$$

where $\mathbf{x}(\mathbf{u})$ is the reservoir state driven by input $\mathbf{u}$. This connects reservoir computing to the theory of reproducing kernel Hilbert spaces (RKHS).

Problem. Can kernel theory provide new insights into reservoir computing? Can we characterize the RKHS induced by specific reservoir architectures?

Some progress has been made (see Hermans & Schrauwen, 2012), but the full potential of this connection is unexplored.

17.8.7 Beyond Fading Memory

Problem. Can reservoir computers perform computations that require long-term memory?

By definition, fading-memory systems eventually forget the past. Near the edge of chaos, memory can be very long but still finite. For tasks requiring truly long-term dependencies (e.g., context-free grammars, long-range correlations), standard reservoirs are theoretically limited. Understanding these limitations precisely, and whether they can be overcome by architectural modifications, is an open question.

17.9 Conclusion: Looking Back and Forward

This textbook has traced a path through three deeply interconnected subjects.

In **Part I**, we developed the theory of dynamical systems: the geometry of phase spaces, the stability of fixed points and periodic orbits, the structure of bifurcations, and the emergence of chaos. We saw that even simple deterministic systems can exhibit bewilderingly complex behavior, and we developed tools — Lyapunov exponents, Poincaré maps, symbolic dynamics — to make sense of this complexity.

In **Part II**, we brought measure theory and ergodic theory to bear on dynamical systems. The ergodic theorems gave us a rigorous framework for the statistical properties of deterministic systems: time averages equal space averages for ergodic systems, mixing quantifies how fast correlations decay, and entropy measures the rate of information creation. These are not merely abstract tools — they provide the "right" language for understanding chaotic dynamics.

In **Part III**, we saw how these mathematical ideas converge in reservoir computing. A reservoir is a dynamical system used as a computational substrate. Its echo state property is a stability condition (Part I). Its ability to generalize from finite training data is governed by ergodic theory (Part II). Its computational capacity is measured by information-theoretic quantities rooted in entropy. The deep connections are not accidental: reservoir computing works *because* of the rich structure that dynamical systems and ergodic theory reveal.

The field is young and full of open problems. The questions listed in Section 17.8 are not idle curiosities — they are fundamental challenges whose resolution would deepen our understanding of computation, dynamics, and their interplay. We hope that the mathematical foundations laid in this book will equip the reader to engage with these questions.

Exercises

Exercise 17.1. Consider the Mackey-Glass system with $\tau_d = 17$, sampled at intervals of $\Delta t = 6$. The attractor dimension is approximately 2.1. What is the minimum delay embedding dimension guaranteed by Takens' theorem to reconstruct the attractor? In practice, reservoir computers with $N = 300$ neurons achieve excellent predictions. Discuss why the practical requirement is so different from the Takens bound.

Exercise 17.2. In the hybrid reservoir computing approach (Section 17.2.3), suppose the knowledge-based model perfectly captures the linear dynamics but misses the nonlinear terms. Argue, using the universal approximation results of Chapter 15, that the reservoir can in principle learn the correction term. What properties must the reservoir have for this to work?

Exercise 17.3. Consider a delay-based photonic reservoir with delay time τ_D and N virtual nodes separated by $\theta = \tau_D/N$. The input is held constant over each interval θ.

(a) Write the state update equation for the virtual nodes $x_i(t)$, $i = 1, \ldots, N$, in terms of $x_{i-1}(t)$ and $x_N(t-1)$ (the delayed feedback).

(b) Identify the recurrent weight matrix W implicitly defined by this architecture. What is its structure?

(c) Under what conditions on the nonlinearity f and the feedback strength β does this system satisfy the echo state property?

Exercise 17.4. The NGRC method uses polynomial features of delay embeddings. Suppose the target system has an attractor of dimension d and the observable is generic.

(a) What is the minimum number of delays k required by Takens' theorem?

(b) If we use polynomial features of degree p, how many total features are there as a function of k and p?

(c) Discuss the curse of dimensionality: for what values of d and p does the feature space become impractically large? How might this be mitigated?

Exercise 17.5. Choose a physical reservoir computing platform from Section 17.4 (photonic, mechanical, spintronic, or quantum). For your chosen platform:

(a) Identify the governing dynamical equations.

(b) Discuss what the echo state property means physically for this system.

(c) Identify the relevant Lyapunov exponents and discuss what regime (ordered, edge of chaos, chaotic) the system should operate in for optimal computation.

(d) What are the practical limitations of this platform (speed, noise, scalability)?

Exercise 17.6. The parallel reservoir architecture for spatiotemporal systems (Section 17.7.1) divides space into overlapping regions. Consider the overlap width w as a parameter.

(a) Argue that w must be at least as large as the spatial correlation length of the system for the architecture to work.

(b) For the Kuramoto-Sivashinsky equation, estimate the spatial correlation length from the known properties of the attractor.

(c) What happens if w is too small? Too large?

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Recommended Reading

This section provides an extended reading list, organized by topic, for readers who wish to go further.

17.9.1 Reservoir Computing: Foundational Papers

- H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Technical Report 148, 2001. *The paper that started echo state networks.*

- W. Maass, T. Natschläger, and H. Markram, "Real-time computing without stable states: A new framework for neural computation based on perturbations," *Neural Computation* 14(11), 2002. *The liquid state machine paper.*
- M. Lukoševičius and H. Jaeger, "Reservoir computing approaches to recurrent neural network training," *Computer Science Review* 3(3), 2009. *The standard survey — an excellent entry point.*

17.9.2 Reservoir Computing: Theory

- L. Grigoryeva and J.-P. Ortega, "Echo state networks are universal," *Neural Networks* 108, 2018. *The definitive universality result.*
- J. Dambre, D. Verstraeten, B. Schrauwen, and S. Massar, "Information processing capacity of dynamical systems," *Scientific Reports* 2, 2012. *Introduces and analyzes IPC.*
- L. Gonon and J.-P. Ortega, "Reservoir computing universality with stochastic inputs," *IEEE Transactions on Neural Networks and Learning Systems* 31(1), 2020. *Generalization bounds.*
- A. Hart, J. Hook, and J. Dawes, "Embedding and approximation theorems for echo state networks," *Neural Networks* 128, 2020. *Connects Takens' theorem to reservoir computing rigorously.*
- G. Manjunath and H. Jaeger, "Echo state property linked to an input: Exploring a fundamental characteristic of recurrent neural networks," *Neural Computation* 25(3), 2013. *Lyapunov exponent characterization of ESP.*

17.9.3 Reservoir Computing: Applications

- J. Pathak, B. Hunt, M. Girvan, Z. Lu, and E. Ott, "Model-free prediction of large spatiotemporally chaotic systems from data: A reservoir computing approach," *Physical Review Letters* 120, 2018. *Climate replication and spatiotemporal prediction.*
- D. J. Gauthier, E. Bollt, A. Griffith, and W. A. S. Barbosa, "Next generation reservoir computing," *Nature Communications* 12, 2021. *The NGRC framework.*
- H. Jaeger and H. Haas, "Harnessing nonlinearity: Predicting chaotic systems and saving energy in wireless communication," *Science* 304, 2004. *An influential early demonstration.*

17.9.4 Physical Reservoir Computing

- L. Appeltant et al., "Information processing using a single dynamical node as complex system," *Nature Communications* 2, 2011. *The first delay-based photonic reservoir.*
- L. Larger et al., "Photonic information processing beyond Turing: an optoelectronic implementation of reservoir computing," *Optics Express* 20(3), 2012. *Photonic RC at high speed.*
- K. Nakajima and I. Fischer (eds.), *Reservoir Computing: Theory, Physical Implementations, and Applications*, Springer, 2021. *A comprehensive edited volume on physical RC.*

- G. Tanaka et al., "Recent advances in physical reservoir computing: A review," *Neural Networks* 115, 2019. *A thorough survey of physical implementations.*

17.9.5 Dynamical Systems and Ergodic Theory (General)

- S. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., Westview Press, 2015. *The standard undergraduate textbook on dynamical systems.*
- A. Katok and B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*, Cambridge University Press, 1995. *The comprehensive graduate reference.*
- P. Walters, *An Introduction to Ergodic Theory*, Springer, 1982. *The standard introduction to ergodic theory.*
- V. I. Arnold, *Ordinary Differential Equations*, MIT Press, 1978. *A classic with geometric insight.*

17.9.6 Further Directions

- N. Bertschinger and T. Natschläger, "Real-time computation at the edge of chaos in recurrent neural networks," *Neural Computation* 16(7), 2004. *The edge-of-chaos hypothesis for reservoir computing.*
- C. Gallicchio and A. Micheli, "Deep echo state network (DeepESN): A brief survey," arXiv:1712.04323, 2017. *Deep reservoir architectures.*
- M. Hermans and B. Schrauwen, "Recurrent kernel machines: Computing with infinite echo state networks," *Neural Computation* 24(1), 2012. *The kernel perspective on reservoir computing.*
- K. Fujii and K. Nakajima, "Harnessing disordered-ensemble quantum dynamics for machine learning," *Physical Review Applied* 8, 2017. *Quantum reservoir computing.*